UNIVERSITY OF COPENHAGEN
FACULTY OF SOCIAL SCIENCES
DEPARTMENT OF ECONOMICS



Meandering Through Labour Economics

Mental Well-being, Misperceptions, and Milieu in the Labour Market

Asker Nygaard Christensen

Ph.D. Dissertation

Ph.D. Dissertation

Meandering Through Labour Economics

Mental Well-being, Misperceptions, and Milieu in the Labour Market

Asker Nygaard Christensen

Department of Economics

University of Copenhagen

Submitted: June 30, 2026

Advisors: Nikolaj Arpe Harmon and Daphné Skandalis

Contents

Acknowledgements iii

Introduction v

Introduktion ix

1 Globalisation and the Gender Gap in Mental Well-Being 1

2 Perceived Wage Differences and Early Career Job Search Choices 51

3 Parental Transmission of Jobs and Non-Wage Amenities 179

Acknowledgements

Writing a dissertation is hard. Therefore, I am forever grateful to my advisors who made it exciting. I have learned much from your different approaches to research, your shared rigour and attention to detail, and observing you battle out ideas. I also want to thank my wonderful co-authors: Beata, Björn, and Sonja, from whom I have learned so much, and of course Jakob, my co-author *and* 'shadow-advisor', your guidance has been invaluable.

I also want to thank Michèle for letting me visit her at Cornell and for taking the time to discuss my crazy research ideas, many of which unfortunately did not have time to mature enough to be included in this thesis, but have laid the groundwork for what I will be doing after the Ph.D. I am also very grateful to all the amazing people at Cornell and in Ithaca that made my visit not only more enlightening but also fun.

Doing a Ph.D. is also an exercise in endurance: handling failures, battling feelings of inadequacy, and navigating unknown waters. Therefore, I am especially grateful for my friends in the Ph.D. programme who helped keep me sane during the tribulations. Likewise, my friends outside the programme reminded me to enjoy life outside work and have made that life much more enjoyable. I also want to thank my family. Parents lay the foundations for the life of their children in known and unknown ways, and I am very grateful for the foundations my parents have given me. I am also grateful for my sister and the amazing family she has built. Birk and Sølvi are fantastic human beings.

The most important person for everything is my partner, Saphir. You make every day a better day, your support and advice are invaluable, and you tell me the truth when I lie to myself. Plus, the number of layers you can pack in a single joke is mesmerising. The attentive reader will notice that all chapters are authored under a different last name than my own. Saphir and I are getting married, so obtaining a Ph.D. will only be the second-best thing to happen in my life this year. Spending the rest of my life with you is the easiest decision I ever made.

Asker Nygaard Christensen

30th June 2026

Introduction

An encompassing and succinct description of the topic of this thesis and its three chapters would be: labour. This is admittedly a very broad and slightly grandiose topic for a thesis. I could tell you this was by design and that the three chapters were planned at the beginning to provide a comprehensive answer to a singular question; but I suppose a good thesis does not start with deception. Rather, the focus of each chapter and my involvement in them have come into being somewhat serendipitously, guided by my mentors and interests.

The three chapters explore different facets of what is—at the time of writing—the primary driver of our economy: human labour. They explore how doing labour affects the labourers; how labourers search for jobs that will remunerate them for their labour; and finally, how that search is affected by the labourers' family background. Any singular conclusion I can give from the thesis is that the labour market is complex: it affects us in ways we do not expect and often do not have the agency to plan for; it is difficult to navigate as we do not have complete information about the possible paths that are available to us; and our decisions along our path in the labour market are shaped by preferences that are influenced by our family background, which we have no control over. This complexity has provided a reliable labour market for economists and other researchers trying to understand it. This thesis is a small brick in the cumulative output of all that research labour.

I hope the reader will excuse my lack of foresight in the planning of this thesis and find each chapter enjoyable and interesting in its individuality. I certainly found the laborious process of producing them enjoyable, not least because of the many co-authors I was fortunate enough to work with.

The Ph.D. thesis was completed as part of the 4+4 Ph.D. programme at the Department of Economics, University of Copenhagen. Please note that Chapter 2 builds on my Master's thesis, which I completed as part of the programme.

The first chapter asks whether globalisation carries a gender-biased cost for workers' mental well-being that falls unevenly on women. Trading with partners in distant time

zones often pushes work outside ordinary hours—early calls, late emails—which is harder to reconcile with household labour, and that labour is more often performed by women. We study this question using Danish data that link individual workers—with their prescription records—to firms, and those firms to their trade with foreign partners. As the overlap between working hours in a Danish firm and its trading partners narrows, we find that women raise their use of antidepressants relative to men. The effect is largest among college-educated workers under 45, and larger among single mothers. The paper speaks to a familiar point regarding globalisation: while most economists agree that globalisation is a net positive for the economy, there can exist distributional and spillover consequences that deserve the attention of researchers and policymakers.

The second chapter turns to information and asks whether early career workers know the labour market they are entering, particularly the relative pay of their possible career choices. Using a survey of early career job seekers linked to administrative data, we ask respondents to report average wage levels in types of jobs that are relevant to their education. Comparing these beliefs with averages in the administrative data, we find that respondents are far from fully informed. By treating some of the respondents with information about the administrative averages, we find that this information is relevant for their job search: they update their beliefs about their own expected wages and change their planned applications. In administrative data, the information treatment also shifts wages in their first job after the survey, further strengthening the conclusion that relative wage misperceptions matter for job search. Relative wage misperceptions at the early career stage are important because they suggest applications are not optimally directed; ill-informed choices early in the career can have long-lasting consequences for labour market outcomes.

The third chapter stays with the early career job seekers and the same survey, but looks at how family background affects job search, asking why many workers end up in the same line of work as their parents. We ask job seekers how likely they would be to apply for, and to accept, different types of jobs relevant to their education—including when we hold pay equal across all options. Using our administrative data, we link the labour market history of each respondent's parent and correlate the respondent's valuation of the jobs with whether their parent has worked in them. Our different measures of job valuation

let us separate three reasons a worker might intend to follow a parent's path: the job may be easier to get, it may pay more, or it may simply be more appealing for some other reason, such as preferences for what the job entails. The survey isolates job seeker beliefs and intended choices, and we find that preferences are important for the transmission of parental employment.

For the reader's convenience, the abstracts of each chapter are printed below.

Chapter 1: Globalization and the Gender Gap in Mental Well-Being

co-authored with Björn Thor Arnarson, Beata S. Javorcik, and Jakob R. Munch

Working with business partners in distant countries often requires operating outside standard working hours, potentially harming workers' mental well-being—especially for women balancing professional and family responsibilities. This hypothesis is investigated using matched employer—employee data from Denmark merged with information on workers' use of prescription antidepressants (AD) and firms' import and export transactions. The analysis exploits variation within job spells and controls for both unobservable firm-year heterogeneity and for the average differential in annual AD use among men and women. The results indicate that a decline in the business hours overlap between a Danish firm and its foreign trading partners leads to an increased AD use among women relative to men. This effect is economically meaningful, largest for college-educated workers under 45 and more pronounced for single mothers. In addition to the gender effect, workers in occupations that requires establishing relationships with others are more strongly affected than those in other occupations.

Chapter 2: Perceived Wage Differences and Early Career Job Search Choices

co-authored with Nikolaj Arpe Harmon, Sonja Settele, and Daphné Skandalis

Workers' early career jobs are a key determinant of their lifetime earnings—and yet workers may misdirect their search if they misperceive wage differences across jobs. We study this using a survey of early career job seekers in Denmark linked to administrative data. The survey (i) elicits beliefs about typical wages in three jobs relevant to each respondent, (ii) provides a randomized information treatment revealing the actual value of these typical

wages, and (iii) elicits beliefs about their own expected wages and planned applications. Comparing beliefs to actual values reveals that gaps in typical wages across jobs are largely misperceived. While some wage gaps are overestimated, most are underestimated, making lower-paying jobs appear overly attractive overall. Using the randomization, we find that typical wage information has large effects on respondents' beliefs about their own expected wages and their planned applications, in the direction predicted by their initial misperceptions. These effects translate into real labor market outcomes: in administrative data, the information shifts the wages respondents receive in their first post-survey job—upward for those who underestimated wage gaps and downward for those who overestimated them. Embedding our results in a discrete choice framework, we estimate that wage misperceptions misdirect about 9.5% of job applications at labor market entry, resulting in the same welfare loss as a 0.8% reduction in all wages.

Chapter 3: Parental Transmission of Jobs and Non-Wage Amenities

co-authored with Nikolaj Arpe Harmon, Sonja Settele, and Daphné Skandalis

Workers tend to follow their parents into similar kinds of jobs. Whether this is a problem depends on why: if children inherit privileged access to their parents' line of work or the pay that comes with it, transmission may reflect a misallocation of talent; if instead they inherit a taste for the work itself—its non-wage amenities—the same sorting may carry welfare gains. The amenity channel is the hardest to observe and the least studied. To fill this gap, we analyse a survey of early career job seekers in Denmark, linked to administrative data. For each respondent we define three job types relevant to them and elicit, for each, the likelihood of (i) applying; (ii) accepting an offer; and (iii) accepting an offer when all jobs pay the same wage. These questions map to a discrete-choice model of job search that lets us distinguish the role of perceived access, pay, and non-wage amenities in shaping applications. Respondents are 29% more likely to apply to a job type where a parent has worked, and 25% more likely to accept an offer from it at equal wages. Perceived access and pay leave little trace on intended parental transmission; meaning preferences over non-wage amenities are an important factor.

Introduktion

En dækkende og kortfattet beskrivelse af emnet for denne afhandling og dens tre kapitler ville være: arbejde. Dette er ganske vist et meget bredt og en anelse storladent emne for en afhandling. Jeg kunne skrive, at det var tilsigtet, og at de tre kapitler fra begyndelsen var planlagt til at give et fyldestgørende svar på ét enkelt spørgsmål; men en god afhandling bør vel ikke begynde med bedrag. I stedet er emnerne i hvert kapitel og min inddragelse i dem opstået noget nær tilfældigt, guidet af mine vejledere og interesser.

De tre kapitler udforsker forskellige facetter af det, der—i skrivende stund—er den primære drivkraft i vores økonomi: menneskeligt arbejde. De udforsker, hvordan det at udføre arbejde påvirker arbejderne; hvordan arbejdere søger efter job, der vil aflønne dem for deres arbejde; og endelig, hvordan denne søgning påvirkes af arbejdernes familiebaggrund. Den eneste samlede konklusion, jeg kan drage af afhandlingen, er, at arbejdsmarkedet er komplekst: det påvirker os på måder, vi ikke forventer, og som vi ofte ikke har autonomien til at planlægge efter; det er vanskeligt at navigere i, da vi ikke kender alle de mulige veje, der er tilgængelige for os; og vores beslutninger undervejs i arbejdsmarkedet formes af præferencer, der er påvirket af vores familiebaggrund, som vi ikke har nogen kontrol over. Denne kompleksitet har skabt meget arbejde for økonomer og andre forskere, der forsøger at forstå det. Denne afhandling er en lille brik i den samlede produktion af alt det forskningsarbejde.

Jeg håber, at læseren vil bære over med mit manglende fremsyn i planlægningen af denne afhandling og finde de individuelle kapitler underholdende og interessante i deres mangfoldighed. Jeg fandt i hvert fald den arbejdskrævende proces med at frembringe dem underholdende, ikke mindst på grund af de mange medforfattere, jeg var så heldig at arbejde sammen med.

Ph.d.-afhandlingen er udarbejdet som en del af 4+4-ph.d.-programmet ved Økonomisk Institut, Københavns Universitet. Bemærk venligst, at kapitel 2 bygger på mit speciale, som jeg færdiggjorde som en del af programmet.

Det første kapitel undersøger, om globaliseringen indebærer en kønsskæv omkostning for arbejdernes mentale trivsel, som rammer kvinder hårdere end mænd. Handel med partnere i fjerne tidszoner skubber ofte arbejdet uden for normal arbejdstid—tidlige opkald, sene e-mails—hvilket er sværere at forene med hjemlige forpligtelser og arbejde, som oftere udføres af kvinder. Vi undersøger dette spørgsmål ved hjælp af danske data, der kobler arbejdere—med data om deres medicinforbrug—til virksomheder, og disse virksomheder til deres handel med udenlandske partnere. Når overlappet mellem arbejdstiderne i en dansk virksomhed og dens handelspartnere bliver mindre, finder vi, at kvinder øger deres brug af antidepressiva i forhold til mænd. Effekten er størst blandt arbejdere med en videregående uddannelse under 45 år og større blandt enlige mødre. Artiklen taler ind i en velkendt pointe om globalisering: selvom de fleste økonomer er enige om, at globalisering samlet set er en fordel for økonomien, er der fordelingsmæssige og afledte konsekvenser, som fortjener forskeres og politikeres opmærksomhed.

Det andet kapitel drejer sig om information og spørger, om arbejdere tidligt i karrieren kender det arbejdsmarked, de er på vej ind på, specifikt den relative løn for deres mulige karrierevalg. Ved hjælp af en spørgeskemaundersøgelse blandt jobsøgende tidligt i karrieren, koblet til administrative data, beder vi respondenterne om at angive de gennemsnitlige lønniveauer i jobtyper, der er relevante for deres uddannelse. Når vi sammenligner disse forventninger med gennemsnittene i de administrative data, finder vi, at respondenterne langtfra er fuldt informerede, og ofte ikke ved, hvilken karrierevej der typisk har de højeste lønninger. Ved at give nogle af respondenterne information om de administrative gennemsnit finder vi, at denne information er relevant for deres jobsøgning: de opdaterer deres forventninger om deres egen forventede løn og ændrer deres planlagte ansøgninger. I de administrative data flytter informationsinterventionen også den løn, de tjener i deres første job efter undersøgelsen, hvilket yderligere styrker konklusionen om, at relative fejlopfattelser af løn har betydning for jobsøgningen. Relative fejlopfattelser af løn tidligt i karrieren er vigtige, fordi de tyder på, at ansøgninger ikke er optimalt rettede; dårligt informerede valg tidligt i karrieren kan have langvarige konsekvenser for jobsøgeres arbejdsliv.

Det tredje kapitel bliver ved de jobsøgende tidligt i karrieren og den samme undersøgelse,

men ser på, hvordan familiebaggrund påvirker jobsøgningen, og undersøger, hvorfor mange arbejdere ender i samme type arbejde som deres forældre. Vi spørger de jobsøgende, hvor sandsynligt det er, at de vil søge og acceptere forskellige jobtyper, der er relevante for deres uddannelse—også når vi holder lønnen ens på tværs af alle muligheder. Ved hjælp af vores administrative data kobler vi arbejdsmarkedshistorikken for hver respondents forælder og korrelerer respondentens påskønnelse af jobbene med, om deres forælder har arbejdet i dem. Vores forskellige mål for jobpåskønnelse gør det muligt for os at adskille tre grunde til, at en arbejder kan have til hensigt at følge en forælders vej: jobbet kan være lettere at få, det kan være højere lønnet, eller det kan ganske enkelt være mere tiltalende af en anden grund, såsom præferencer for, hvad jobbet indebærer. Undersøgelsen isolerer de jobsøgendes forventninger og tilsigtede valg, og vi finder, at præferencer er vigtige for videregivelsen af forældres beskæftigelse.

Til læserens bekvemmelighed er resuméerne af hvert kapitel gengivet nedenfor.

Kapitel 1: Globalization and the Gender Gap in Mental Well-Being

skrevet sammen med Björn Thor Arnarson, Beata S. Javorcik og Jakob R. Munch

At arbejde sammen med forretningspartnere i fjerne lande kræver ofte, at man arbejder uden for normal arbejdstid, hvilket potentielt kan skade arbejdernes mentale trivsel—især for kvinder, der skal balancere professionelle og familiemæssige forpligtelser. Denne hypotese undersøges ved hjælp af linkede arbejdsgiver-arbejdstager-data fra Danmark sammenholdt med information om arbejdernes brug af receptpligtige antidepressiva (AD) og virksomhedernes import- og eksporttransaktioner. Analysen udnytter variation inden for jobforløb og kontrollerer både for uobserverbar virksomheds-år-heterogenitet og for den gennemsnitlige forskel i den årlige AD-brug mellem mænd og kvinder. Resultaterne indikerer, at et fald i overlappet i arbejdstider mellem en dansk virksomhed og dens udenlandske handelspartnere fører til en øget AD-brug blandt kvinder i forhold til mænd. Denne effekt er økonomisk betydningsfuld, størst for arbejdere med en videregående uddannelse under 45 år og mere udtalt for enlige mødre. Ud over kønseffekten påvirkes arbejdere i erhverv med meget relationsarbejde stærkere end dem i andre erhverv.

Kapitel 2: Perceived Wage Differences and Early Career Job Search Choices

skrevet sammen med Nikolaj Arpe Harmon, Sonja Settele og Daphné Skandalis

Arbejderes job tidligt i karrieren er en afgørende faktor for deres livsindkomst—ikke desto mindre kan arbejdere fejldirigere deres jobsøgning, hvis de fejlopfatter lønforskelle på tværs af job. Vi undersøger dette ved hjælp af en spørgeskemaundersøgelse blandt jobsøgende tidligt i karrieren i Danmark, koblet til administrative data. Undersøgelsen (i) afdækker forventninger om gennemsnitlige lønninger i tre job, der er relevante for hver respondent, (ii) giver en randomiseret informationsintervention, der afslører de faktiske gennemsnitlige lønninger, og (iii) afdækker forventninger om deres egen forventede løn og planlagte ansøgninger. Sammenligningen af respondenternes forventninger med gennemsnitlige lønninger viser, at forskelle i typiske lønninger på tværs af job i vid udstrækning fejlopfattes. Selvom nogle lønforskelle overvurderes, undervurderes de fleste, hvilket samlet set får lavtlønnede job til at fremstå overdrevent attraktive. Ved hjælp af randomiseringen finder vi, at information om typiske lønninger har store effekter på respondenternes egen forventede løn og deres planlagte ansøgninger, i den retning, som deres oprindelige fejlopfattelser forudsiger. Disse effekter omsættes til faktiske ændringer: i de administrative data flytter informationen den løn, respondenterne modtager i deres første job efter undersøgelsen—opad for dem, der undervurderede lønforskelle, og nedad for dem, der overvurderede dem. Ved at se vores resultater gennem en diskret-valg-model estimerer vi, at fejlopfattelser af løn fejldirigerer omkring 9,5% af jobansøgningerne ved indtræden på arbejdsmarkedet, hvilket resulterer i det samme velfærdstab som en reduktion på 0,8% i alle lønninger.

Kapitel 3: Parental Transmission of Jobs and Non-Wage Amenities

skrevet sammen med Nikolaj Arpe Harmon, Sonja Settele og Daphné Skandalis

Arbejdere har tendens til at følge deres forældre ind i lignende typer job. Hvorvidt dette er et problem, afhænger af hvorfor: hvis børn arver privilegeret adgang til deres forældres fag eller den løn, der følger med, kan transmissionen afspejle en fejlallokering af talent; hvis de i stedet arver en smag for selve arbejdet—dets jobgoder ud over lønnen—kan den samme sortering medføre velfærdsgevinster. Præferencekanalen er den sværeste at observere og den mindst undersøgte. For at udfylde dette hul analyserer vi en spørgeskemaundersøgelse

blandt jobsøgende tidligt i karrieren i Danmark, koblet til administrative data. For hver respondent definerer vi tre jobtyper, der er relevante for dem, og afdækker for hver enkelt sandsynligheden for (i) at søge; (ii) at acceptere et tilbud; og (iii) at acceptere et tilbud, når alle job har den samme løn. Disse spørgsmål svarer til en diskret-valg-model for jobsøgning, der gør det muligt for os at skelne mellem betydningen af opfattet adgang, løn og jobgoder ud over lønnen for tilsigtede jobansøgninger. Respondenterne er 29% mere tilbøjelige til at søge en jobtype, hvor en forælder har arbejdet, og 25% mere tilbøjelige til at acceptere et tilbud derfra ved samme løn. Opfattet adgang og løn har lille effekt på transmissionen fra forældre på tilsigtede jobansøgninger, hvilket betyder, at præferencer for jobgoder ud over lønnen er en vigtig faktor.

Chapter 1

Globalisation and the Gender Gap in Mental Well-Being

Globalization and the gender gap in mental well-being[*†]

Björn Thor Arnarson

ROCKWOOL foundation and University of Copenhagen

Asker Nygaard David

University of Copenhagen

Beata S. Javorcik

University of Oxford, EBRD and CEPR

Jakob R. Munch

University of Copenhagen, CESifo and IZA

Abstract

Working with business partners in distant countries often requires operating outside standard working hours, potentially harming workers' mental well-being—especially for women balancing professional and family responsibilities. This hypothesis is investigated using matched employer—employee data from Denmark merged with information on workers' use of prescription antidepressants (AD) and firms' import and export transactions. The analysis exploits variation within job spells and controls for both unobservable firm-year heterogeneity and for the average differential in annual AD use among men and women. The results indicate that a decline in the business hours overlap between a Danish firm and its foreign trading partners leads to an increased AD use among women relative to men. This effect is economically meaningful, largest for college-educated workers under 45 and more pronounced for single mothers. In addition to the gender effect, workers in occupations that requires establishing relationships with others are more strongly affected than those in other occupations.

JEL classifications: F10, F66, F16, J16, I10, I30

Keywords: Globalization, Mental well-being, Gender gap, O*NET job characteristics

[*]We thank seminar participants at the University of Copenhagen, the EBRD, the ROCKWOOL Foundation Research Unit, the CEPR-Paris Dauphine Globalisation and Workforce Composition workshop 2024, the DGPE workshop 2022, the Gender Gaps Conference 2023 hosted by GRAPE, the Economics of Working Environment conference (Copenhagen), the CESifo Area Conference on Global Economy (Munich), Laura Alfaro, Mette Gørtz, Nikolaj Harmon, and two anonymous reviewers for helpful comments and suggestions. We gratefully acknowledge funding from the ROCKWOOL Foundation Research Unit.

[†]This article has been published in the Journal of International Economics: https://www.sciencedirect.com/science/article/pii/S0022199626000437.

1 Introduction

Recent decades have seen rapid growth in international trade and foreign direct investment, as firms have created supply chains spanning the globe in order to optimize production costs and reach customers in distant parts of the world. Running these complex production and distribution networks requires interactions with customers and suppliers located in different time zones, and thus may necessitate working outside of standard business hours, thereby disrupting the work-life balance.[1] This can take a particular toll on the mental well-being of women, who need to reconcile work demands with family obligations.

This study aims to shed light on the consequences of working across time zones for the gender gap in mental well-being of workers, proxied by their use of prescription antidepressant medications. We use matched employer–employee data from Denmark for 2008–2017, encompassing the universe of Danish manufacturing firms in the private sector, combined with detailed customs data on exports and imports. Importantly, we observe comprehensive information covering every individual worker's prescription drug purchases, disaggregated by drug type. Our outcome of interest is an indicator variable capturing whether a worker has purchased prescription antidepressants in a given year. We follow Bøler et al. (2018) and proxy working across time zones by using business hours overlap between Denmark and its export and import markets, weighted by the number of products sold or purchased in a given country.[2]

Our identification strategy rests on two pillars. First, we condition on job-spell fixed effects, i.e., we exploit time variation in the firm's business hours overlap with its trading partners within a given worker–firm relationship. This allows us to account for idiosyncratic and time-invariant factors that affect individuals' health, such as pre-natal and early-life environment and genetic differences (e.g. Black et al. (2007)).

Second, we conjecture that working across time zones is particularly difficult for women because of the need to juggle work and family obligations and even more so for young women

[1] For instance, an employee of a global company reported that "[d]epending on what country he is working with, he has to adjust his time for fathering. When he was working with Malaysia he lost time with his children in the mornings, because he had to be online with their [Malaysian] time. When he is working with Austria the time with the kids in the afternoon and evening has to be adjusted to the working hours in Austria." (Kvande, 2009, pp. 68–69). This view is confirmed by Chauvin et al. (2020) who studied internal communications in a large multinational firm and found that an increase in time differential due to the Daylight Saving Time adjustment induced employees in non-routine tasks to shift synchronous communications across the boundary of their workday into leisure time.

[2] Danish health care is free and universal, with every individual having access to health care regardless of income and employment status. This makes Denmark a better setting for our study than, for instance, the U.S., where workers' access to health care is correlated with income and employment status (e.g. Currie and Madrian (1999)) and enables us to separate the direct effect of work characteristics on well-being from an indirect effect operating through access to health care. See section 2 for a more detailed discussion of the healthcare system in Denmark.

(i.e., those under 45 years of age) who are likely to have young children.[3] We focus on the gap in antidepressant usage between women and men working for the same firm in a given year, which allows us to control for unobservable firm-year-specific heterogeneity that may, e.g., drive firms' internationalization, corporate culture, or impact the average level of stress among the workforce. To account for differential propensity to use antidepressants among the genders over time, we control for an interaction of a female indicator with year dummies. We also control for education, experience, experience squared, having children and 2-digit occupational fixed effects.

To summarize, our analysis holds the worker–firm match fixed, controls for unobservable firm-year heterogeneity, and asks how a change in the firms' business hours overlap with its foreign buyers and suppliers translates into a differential antidepressant use among its female and male workers.

Our results can be summarized as follows. First, we find a positive and statistically significant relationship between working across time zones and the gender gap in antidepressant usage. In other words, a lower business hours overlap is associated with a higher incidence of antidepressant usage among women relative to men working for the same company in a given year. This relationship is present among college-educated workers, but not in the general population of workers. This is intuitive, as college-educated workers are more likely to hold jobs in management, sales, marketing, customer support, or other occupations that necessitate interactions with foreign business partners. We also find that the relationship is only found among younger workers (under 45), which is in line with younger workers being more likely to have young children and, therefore, more family obligations. As a consequence, the relationship between working across time zones and the gender gap in antidepressant use is particularly strong among young workers with a college education.

The estimated magnitude is economically meaningful. A decline in business hours overlap by 5 h from an initial level of 8, equivalent to moving all trade from Germany to the eastern part of Canada, is associated with a 1.7% point increase in the likelihood of antidepressant use by women relative to men among college-educated workers under 45. This implies increasing the prevalence of antidepressant use by 57%, which in this subpopulation is equal to 3%.

Our results are robust to controlling for firms' business exposure to less gender-equal countries, thus eliminating the possibility that gender biases and stereotypes among business partners make

[3] As shown by Thielemans et al. (2019), women tend to perform a higher share of household work than men do. This is also in line with the findings of Bøler et al. (2018) who show that working across time zones is associated with a higher gender wage gap in Norway.

the working lives of female employees more stressful. They are robust to using alternative definitions of business hours overlap as well as alternative econometric approaches (linear probability model and logit). They also hold if we drop the period of the Global Financial Crisis (2008–9) to eliminate the possibility of economic downturn affecting our results.

Second, our hypothesis that the need to work outside of standard business hours is especially taxing for employees juggling work and family obligations finds support in an analysis allowing for differential effects for single mothers. This analysis shows that the impact of working across time zones on antidepressant usage is 3–4 times larger for single mothers than for women in general, which suggests that the pressures of maintaining work-life balance are likely to be one of the main factors driving our results. This effect is more than twice as large for college-educated single mothers than for mothers without college education.

Third, we confirm our results using an alternative source of variation stemming from activities of the Danish Trade Council. Caseworkers at Danish embassies and consulates contact Danish firms, often sequentially in alphabetical order, to offer export promotion services in specific foreign markets. Buus et al. (2025) use these calls as a source of plausibly exogenous variation and document that such calls result in new export flows. We show that receiving a call from a country with a lower business hours overlap than the business hours overlap in the current export portfolio leads to an increase in the gender gap in antidepressant usage among college educated workers under 45, but not among the population of all workers.[4]

It is worth reiterating that our results are not driven by self-selection of workers into particular firms or by self-selection of firms with a healthier workforce into greater internationalization, as we hold the work-firm match fixed and control for firm-year fixed effects. Moreover, the inclusion of female-year fixed effects means that our findings cannot be attributed to the changing incidence of antidepressant usage between men and women over time.

Two extensions boost our confidence in the results. We document a similar relationship for business hours overlap with respect to foreign subsidiaries of Danish companies. Again, an increase in the need to work across time zones necessitated by the need to communicate with the firm's foreign subsidiaries is associated with a greater gender gap in antidepressant use. This relationship holds for younger workers with a college education and is *in addition* to the effect driven by the composition of foreign trade.

[4]In an extension, we drop firm-year fixed effects and exploit the shocks stemming from the activities of the Danish Trade Council to examine the impact of working across time zones on the antidepressant use by male workers. We do not find a statistically significant effect.

Finally, following Goldin (2014), we focus on five occupational characteristics (available from O*NET) that limit the temporal flexibility of workers, as they are linked to working particular hours and long hours. We note, though, that these results may be contaminated by the self-selection of men and women into particular occupations. We find that a lower business hours overlap is associated with a greater antidepressant usage among workers in these occupations (relative to those employed in other occupations). This effect holds *in addition* to our finding that women are affected more than their male colleagues. When we zoom in on individual job characteristics, we find that the results are driven by occupations (i) requiring establishing and maintaining interpersonal relationships, (ii) requiring contact with others, (iii) with fewer substitutes for a given worker. These effects are economically meaningful. For instance, workers in the top quarter of occupations where contact with others is required see a 0.9% point increase in antidepressant use when the business hours overlap declines from 8 to 3 h.

Taken together, our results are consistent with the need to work across time zones, and thus outside of standard business hours, increasing the gender gap in mental well-being of workers. The contribution of our study lies in documenting this effect as well as pointing out a new channel through which globalization differentially affects men and women as well as workers in particular occupational groups. Our study also contributes to improving understanding of factors leading to or exacerbating depression. This is interesting in its own right, given that depression alone accounts for 4.3% of the global burden of disease and is among the largest single causes of disability worldwide (accounting for 11% of all years lived with disability globally), particularly for women. This not only impacts the affected individuals and their families but also has large economic consequences in terms of lost days of work and output losses globally (WHO (2021)).

Our paper is related to several strands of the existing literature. The first one encompasses studies on the gender impact of globalization, most of which focus on the gender wage gap (GWG). Juhn et al. (2014) argue that trade liberalization induces prospective exporters to adopt new technology that is more modern and requires less physical strength from workers. This improves women's labour market outcomes in the blue-collar tasks, while leaving them unchanged in the white-collar tasks. They find that the introduction of NAFTA was associated with entry of new firms and adoption of more modern technology, at the same time as firms hired relatively more blue-collar females and increased their share of the wage bill. Bøler et al. (2018) argue that there is a systematic difference in the GWG between exporting firms and non-exporters, as the latter requires working particular hours to communicate with partners in different time zones and may

therefore disproportionately reward employees' flexibility. If women are less flexible, or perceived as such, exporters will exhibit a higher GWG than non-exporters. This hypothesis is confirmed by an analysis of matched employer–employee data from Norway. Their results suggest a firm's entry into exporting increases the GWG by about 3% points for college-educated workers. A lower overlap in business hours between the Norwegian exporter and its foreign markets and thus a greater need for interactions with foreign buyers outside of standard business hours are associated with a higher GWG. The contribution of our study lies in pointing out another channel through which globalization may lead to differential outcomes for men and women, namely mental well-being.[5]

Our paper also contributes to the nascent literature on work-induced health outcomes. Hummels et al. (2025) employ the same Danish data to study the effect of rising workload on health. Using both within-job-spell regression analysis and event studies, they show that large increases in firm sales lead workers to log longer hours and experience higher probabilities of stress, depression, heart disease, and strokes, with more pronounced effects for high-risk groups such as older workers, job-strained workers, and those with long initial work hours. We differ from their paper in that we focus on stress caused by working across time zones necessitated by firms' internationalization rather than demand surges and their differential implications for men and women. In another related study, Belloni et al. (2022) use survey-based assessment of depressive symptoms among UK workers to investigate how EU-wide occupation-level changes in working conditions affect mental health. They consider an index of working time quality which captures working long or atypical hours (i.e., working more than 48 h a week or 10 h a day, night or weekend work, shift work), frequently changing work schedule or work schedule changing at a short notice and the lack of control over work schedule. They document that a deterioration in working time quality is associated with worse mental well-being. Our study improves over their analysis by using an objective measure of mental well-being (namely, use of prescription medication) and by analysing changes to working conditions in the workplace of a given individual (rather than in the EU conditions specific to a given occupation).

Our study also speaks to the broader literature on the impact of globalization on health outcomes. This literature has predominantly focused on the negative effects of import competition,

[5]Another strand of the literature documents the positive effects of foreign ownership on female employment thanks to multinationals transplanting their meritocratic culture and flexible working arrangements to foreign affiliates (Kodama et al. (2018), Tang and Zhang (2021), Choi and Greaney (2022)). Our study contributes to this body of research by demonstrating that by working across time zones necessitated by the presence of foreign affiliates may increase the gender gap in mental well-being.

such as greater incidence of suicides and drug overdoses (Pierce and Schott, 2020), more job-related injuries (McManus and Schaur, 2016), and deterioration in physical and mental health of workers (Adda and Fawaz, 2020; Colantone et al., 2019). However, this literature has also documented a decline in workplace accident rates in the aftermath of foreign acquisitions (Javorcik et al., 2025). Our contribution lies in drawing attention to the taxing effects of working across time zones, which is necessitated by globalization, and their differential effect on men and women.

Finally, our paper is also related to the literature documenting how time zone differential affects the organization of knowledge within multinational firms (Gumpert (2018)), collaboration between inventors (Bircan et al. (2024)) and the gender wage gap (Bøler et al. (2018)). We contribute to this literature by pointing out the negative impact of working across time zones on the gender gap in workers' mental well-being.

The rest of the paper is structured as follows. The next section describes the Danish healthcare system, presents the data sources and the definitions of the main variables. Section 3 discusses the estimating equation. Section 4 presents the empirical results on the gender gap in mental well-being. Section 5 focuses on robustness checks. Section 6 exploits differences in occupational characteristics. Section 7 contains the conclusions.

2 Data and setting

In this section, we first discuss the main features of the Danish healthcare system, with a focus on mental well-being. Secondly, we describe the datasets used and the construction of the regression sample.

2.1 Healthcare in Denmark

In Denmark, all citizens have free and equal access to a high-quality health care system, independent of employment status. Around 84% of all health care expenditures were publicly financed in 2015 while the remaining 16% were financed (mostly) through patient co-payments (The Ministry of Health (2017)).[6]

To get access to health care services, each resident has an assigned general practitioner (GP). Around 5.1 million Danes (out of the population of almost 6 million) met their GP physician at least once in 2022. This was true of 2.4 million men and 2.7 million women (Statistics Denmark (2024)). Visiting the GP is free of charge, and they act as a gatekeeper between the primary level

[6]For a more in-depth overview of the health care system in Denmark see The Ministry of Health (2017).

and specialized healthcare services. GPs can refer individuals further to see a specialist, however, the vast majority of cases are handled without further examination or specialized treatment (The Ministry of Health (2017)).

It is estimated that in any given year, around 20% of the Danish population experiences a problem with their mental health (The Ministry of Health (2017)). As gatekeepers, managing mental well-being is therefore a core task of Danish GPs, and it is estimated that around 25% of all contacts are related to mental well-being (National Board of Health (2010)). GPs can prescribe antidepressive medications to patients and other treatment alternatives. The cost of prescription drugs is subsidized, and the subsidy depends on the individual's prior consumption of medicine within a period of a year. There is an automatic yearly cap on co-payment that is adjusted each year (as an example, in 2023 the maximum co-payment was DKK 4435 or 643 US dollars).

2.2 Data

We draw information from a number of administrative registers maintained by Statistics Denmark and construct a comprehensive matched worker–firm dataset for the period 2008–2017 for the universe of firms in the private sector in Denmark. The data includes information on individuals (e.g., wages, occupation, health outcomes) and firms (sales, foreign ownership, exports, imports etc.).

The sources of the individual level data are: a) the population register (BEF), b) the integrated database for labour market research (IDA) which holds information about, e.g., labour market status, occupation (4-digit International Classification of Occupations (ISCO) codes) and demographics, c) the family and household register (FAM) where individuals are linked through family ties, and d) the education register (UDDA) from where we draw information about highest completed educational degree. We define measures for health outcomes using the prescription drugs register (LMDB), which includes detailed information on the purchases of prescription drugs for all individuals by drug type (ATC codes). The register is comprehensive and includes all purchases of prescription drugs. This allows us to define our measure for mental health: an indicator variable indicating if the individual has purchased antidepressants in a given year.[7] Individuals are in all registers identified by the same personal ID code, which allows us to straightforwardly merge the individual-level data.

The firm-level data come from: a) the firm statistics register (FIRM) which holds information

[7]Antidepressants are identified by the ATC code N06A, more information about the ATC system can be found at the WHO website: https://www.whocc.no/atc_ddd_index/

about sales, employment and industry codes, b) the account statistics register (FIRE) from where we obtain a measure of the capital stock of all firms, c) the foreign trade statistics register (UH), where all import and export transactions are registered by detailed product codes (6-digit harmonized system (HS)) and origin or destination country, and d), outward foreign affiliate statistics (FATO), which has information about employment, sales, and host country for all foreign subsidiaries of Danish firms. The FATO register is only available for 2010–2017. The firms are in all registers identified by the same firm ID code, and we obtain consistent links between individuals and firms from the FIDA database.

As we are interested in the link between globalization and mental well-being, we restrict the sample to workers aged 20–60. The final regression sample includes Danish manufacturing firms with more than 10 employees that both import and export.[8] The final dataset includes just under 2 million worker-year observations from approximately 3,400 unique firms. The impact of our sample restriction on the coverage rate of our sample can be seen in Table A1 in Appendix. The sample covers 81% of total Danish manufacturing employment and 98% of Danish manufacturing exports. Simple descriptive statistics at the worker level are shown in Table 1 by gender and education status. For example, women are more likely to use antidepressants: 4.3% of college-educated women use antidepressants, while the corresponding number for college-educated men is 3%. For workers without a college education, 6.5% of women and 4.3% of men use antidepressants. As illustrated in the Appendix, Figure A1, the antidepressant use has remained quite stable for each group during the period under study. Appendix A.2 shows that there is a lot of within-firm across-time variation in both *bho* and antidepressant usage.

2.3 Business hours overlap

Following Bøler et al. (2018), we define the business hours overlap (*bho*) between a Danish firm and a partner country d as the number of working hours (9:00–17:00) that overlap in Denmark and the time zone of the partner country:

$$bho_d = \max\left\{8 - \left|UTC^d_{\text{DK offset}}\right|, 0\right\}, \tag{1}$$

where $\left|UTC^d_{\text{DK offset}}\right|$ is the absolute value of the time zone difference between Denmark and country d according to the coordinated universal time. We then construct an average *bho* measure

[8] This condition has been implemented such that if a firm in a single year has fewer than 10 employees, but more than 10 in the year before and the year after, it is kept in the sample for that year.

for firm j weighted by the number of 6-digit HS products from each partner country (both exports and imports) in the following way:

$$bho_{jt}^{\text{trade}} = \sum_{d \in D_{jt}} \frac{\text{Products}_{djt}^{exp} + \text{Products}_{djt}^{imp}}{\sum_{d \in D_{jt}} \text{Products}_{djt}^{exp} + \text{Products}_{djt}^{imp}} \cdot bho_d \tag{2}$$

where $\text{Products}_{djt}^{exp}$ is the number of exported products of firm j to country d in year t, and $\text{Products}_{djt}^{imp}$ is the number of imported products of firm j to country d in year t. D_{jt} is the set of countries firm j traded with in year t. That is, the weight on bho_d is country d's share of firm j's total number of imported or exported country-product combinations.

Figure 1 shows the distribution of bho_{jt}^{trade} for the firms in the sample. It is evident that many firms have close to complete business hours overlap with their trade partners, reflecting that partner firms are located in the same time zone as Denmark. However, a sizeable portion of firms have a low *bho* which would tend to put pressure on local employees to work outside normal business hours. In the empirical analysis, we take logs of *bho*, which leads to 0.02% of the sample observations being dropped.

We also construct a *bho* measure based on outward foreign direct investments by Danish firms by utilizing the FATO register. We define this *bho* measure by calculating a simple average of the business hours overlap of each firm's foreign affiliates, accounting for the number of foreign affiliates, Affiliates_{djt}, in each destination d:

$$bho_{jt}^{\text{Affiliates}} = \sum_{d \in D_{jt}} \frac{\text{Affiliates}_{djt}}{\sum_{d \in D_{jt}} \text{Affiliates}_{djt}} \cdot bho_d \tag{3}$$

Figure 2 shows the distribution of the *bho* measure based on foreign affiliates for firms with at least one affiliate. This measure is more skewed towards full overlap in business hours than the trade-based *bho* measure. Figure A2 in Appendix shows the evolution of the two *bho* variables over time.

2.4 Gender gap index

Attitudes towards women may differ across destination countries. For robustness, following Bøler et al. (2018) we control for such differences by employing a Gender Gap Index from the World Economic Forum. It pertains to 2007 and covers 128 countries. The index measures relative gaps between men's and women's outcomes. These outcomes are "related to basic rights such as health,

education, economic participation or political empowerment" (Hausmann et al., 2007, p. 3).[9] For each firm, the index is an average weighted by product–destination country combinations, similar to the business hours overlap.

2.5 O*NET data

We also explore job characteristics that are likely to be associated with working across time zones. Following Goldin (2014), we add O*NET data by linking them to ISCO occupation codes and focus on five occupational characteristics that in her judgement limit temporal flexibility of workers as they are linked to working particular hours and long hours. The five characteristics are: 1) Time pressure, 2) Contact with others, 3) Establishing and maintaining interpersonal relationships, 4) Structured versus unstructured work, 5) Freedom to make decisions. See Appendix B for more detailed information about the underlying questions in O*NET.

3 Empirical approach

Our estimation equation takes the following form:

$$y_{ijt} = \tau Female_i \cdot \ln bho_{jt}^{\text{trade}} + X_{it}'\beta + \delta_t Female_i + \rho_{ij} + \eta_{jt} + \epsilon_{ijt}, \tag{4}$$

where the dependent variable is an indicator for whether worker i employed at firm j purchased prescription antidepressant medication in year t. The explanatory variable of interest is an interaction term between a female indicator for worker i and firm j's business hours overlap with its trade partners in year t. Controls include: experience and its square, an indicator for having children that are younger than 25 and live at home, indicators for education, and occupational fixed effects. Most importantly, this specification controls for worker–firm (spell) and firm-year fixed effects. Finally, it controls for female-year fixed effects. Standard errors are clustered at the firm level.

Our identification strategy rests on two pillars. First, we condition on job-spell fixed effects, which means that we exploit variation over time in the firm's business hours overlap within a given worker–firm relationship. This allows us to account for idiosyncratic and time-invariant factors that affect individuals' health, such as pre-natal and early-life environment and genetic differences.

[9] More information about its construction can be found in the World Economic Forum's Gender Gap report, Hausmann et al. (2007).

Second, we focus on the differential prevalence of antidepressant use between women and men. The underlying hypothesis is that working across time zones is particularly difficult for women because of the need to juggle work and family obligations, and even more so for young women who are likely to have young children. This is in line with the findings of Bøler et al. (2018) who show that working across time zones is associated with a higher gender wage gap for this subpopulation. Focusing on the gender gap allows us to control for unobservable firm-year-specific heterogeneity such as corporate culture, average level of stress among the workforce, etc. In other words, we examine how the differential propensity to take antidepressants of men and women *working for the same firm in the same year* is related to the business hours overlap. The inclusion of firm-year fixed effects controls for factors driving firms' internationalization, and thus eliminates the possibility that firms with healthier populations are more or less likely to internationalize.[10] They also control for demand and productivity shocks to firms. The inclusion of female-year fixed effects takes into account any differential propensity to use antidepressants between genders and its evolution over time.

To reiterate, we hold the worker–firm match fixed, control for unobservable firm-year heterogeneity, and ask how a change in the firms' business hours overlap with its foreign buyers and suppliers translates into a differential antidepressant use among its female and male workers

4 Estimation results: Gender gap in mental well-being

4.1 Hypotheses related to the gender gap

Our hypothesis is that working across time zones requires work outside of standard business hours and disturbs the work-life balance. In turn, this may cause stress, lead to depression, and hence the use of antidepressant medication. However, not all employees are equally exposed to this risk.

Employees with the greatest exposure to this channel are those who directly communicate with business partners abroad. This includes: managers, sales representatives, marketing specialists, logistics and accounting staff, technical product support, supply chain managers, and others. Production workers are unlikely to be affected. Typically, workers in the former group will have a college education. Thus, we posit that the anticipated effects are likely to be seen among employees with college education.[11]

[10] This is a concern to the extent that a healthier workforce makes a firm more productive, and more productive firms tend to serve more distant markets (as shown by Eaton et al. (2011)).

[11] This is in line with the findings on business hours overlap and the gender wage gap in the study by Bøler et al. (2018).

Further, we expect that the need for temporal flexibility is harder to achieve for women (in line with arguments presented by Goldin (2014)), hence our focus on the gender gap in antidepressant usage. Finally, we expect larger effects among younger workers (due to general childcare responsibilities) and in particular among younger college-educated workers.

4.2 Baseline results

The hypotheses outlined above suggest that the largest effects should be found among younger college-educated workers. However, in the interest of transparency, we start by reporting the estimation results for the full sample. We also gradually add various fixed effects to shed light on how much (if at all) they matter. The results for the full sample, reported in the top panel of Table 2, show no statistically significant relationship between the business hours overlap and the gender gap in antidepressant usage. This is regardless of the set of fixed effects included in a given specification.

The next panel shows the results for the subpopulation of college-educated workers. Here, the anticipated relationship is statistically significant at the 10 percent level in all but the first specification. This pattern suggests the importance of including job spell fixed effects, which capture endogenous mobility and time-invariant characteristics of the worker. Once spell and firm-year fixed effects are included, the inclusion of additional fixed effects hardly changes the magnitude or the significance level.

The magnitude of the estimated effects is economically meaningful. A decline in the business hours overlap by 5 h from an initial level of 8, equivalent to moving all trade from Germany to the eastern part of Canada, is associated with a one-percentage-point increase in the likelihood of antidepressant use by women relative to men.[12] This effect is sizeable, given the prevalence of antidepressant use in this subpopulation of 3.4%.

The next panel presents the estimation results for the subsample of workers under 45 years of age. The coefficients of interest are statistically significant at the five percent level in all but the first column, which excludes spell fixed effects. The size of the estimated coefficients suggests a 1.2% point increase in the antidepressant gender gap as a result of a five-hour decline in business hours overlap. This is equivalent to a third of the average use for this subsample.

The bottom panel shows the estimates for the final subsample of young (under 45) workers

[12] A five-hour reduction in business hours overlap from an initial $bho = 8$ to $bho = 3$ implies, $\ln(8) - \ln(3) \approx 0.98$, i.e. approximately one. Hence, coefficients on $\ln(bho)$ can be interpreted approximately as the effect of such a change.

with a college education. The estimated coefficients are statistically significant at the 5% level once spell fixed effects are included and larger than before. In our thought experiment, they imply a 1.7% point increase in the gender gap, which is more than half of the average for this group.

Taken together, our results are consistent with working across time zones causing stress and mental distress among women, relative to men, and manifesting itself in a higher gender gap in the use of antidepressant medication. The effects are present for the subgroups we expect to be more exposed to our proposed mechanism, i.e. younger college educated workers. Appendix Table C2 shows that these effects are not present for the non-college subsample nor for the sample of older workers, two groups which we would not expect to be affected.

4.3 Zooming in on single parents

One potential concern is that women may be more likely to seek medical treatment, including for mental health related conditions. To give us confidence that our results are indeed related to working across time zones putting a greater strain on work-life balance we ask whether a group that would be expected *a priori* to find balancing work and family obligations more challenging is affected by working across time zones to a greater extent. Namely, we focus on single mothers.

A substantial share of the Danish population chooses not to formalize their relationships through marriage. Therefore, we define a single parent as an adult living together with their own child under the age of 19, but not living with a non-related adult with a less than 15-year age difference from themselves. As one would expect, single parents constitute a relatively small share of the sample: 7.85% of college-educated women and 9.27% of women without college education are single mothers in the sample. The corresponding figures for single fathers are 1.95% and 2.77%, respectively.

We estimate an extended version of equation (4), which allows us to compare the effect of business hours overlap on single parents of either gender and cohabitating parents of either gender relative to employees without children, all working for a given company in a given year. As before, we hold the worker–firm match fixed which means that our identification comes from the changes to a firm's business hours overlap with its partners and the changes in an individual's personal situation, such as being a parent or a single parent. The results, presented in Table 3, focus on younger workers with and without college education (see columns 1 and 2 versus 3 and 4, respectively). They include all the controls used earlier, including occupational fixed effects and their interaction with the female indicator.

The results are in line with our priors. We confirm our baseline results as the interaction term *female * bho* remains negative and statistically significant in the sample of college-educated workers. Its magnitude is very similar to that found earlier. More importantly, the relationship between working across time zones and the use of antidepressants appears to be particularly strong for single mothers. The coefficient on the interaction terms *female * single parent * bho* is negative and much larger in magnitude than the coefficient on the *female * bho* interaction. It is statistically significant at the one percent level in the subsample of college-educated workers. Thus, as anticipated, a lower business hours overlap is associated with a much higher antidepressant usage among college-educated single mothers than among other women with college education. In the subsample of workers without a college education, the coefficients point in the same direction, but are much smaller and not statistically significant.

Moving on to other estimates, we find that college-educated single mothers are more likely to use antidepressants, which is consistent with findings from Finland showing that single mothers substantially increase their antidepressant use at the time of separation with only a moderate decline after separation (Kühn et al., 2023). The opposite is true of college-educated single fathers, for whom business hours overlap also has the opposite sign. This is in line with the observation that men are less likely to seek medical help, particularly if they are not in a relationship.[13] Interactions terms with parent indicator variables do not appear to be statistically significant.

In summary, our finding of the impact of working across time zones on antidepressant usage being three times larger for college-educated single mothers than for college-educated women in general suggests that the pressures of maintaining work-life balance are likely to be one of the factors driving our results. The finding that the effect on single mothers is only robust in the subsample of college-educated workers and not for those without a college education is also consistent with our priors that skilled workers are more likely to hold positions requiring contacts with business partners abroad and thus are more likely to be affected the pressures of working across time zones. Finally, our confidence in the findings is further boosted by the fact that they confirm the results from another Nordic country on separation increasing antidepressant use among single mothers.[14]

[13]The positive coefficient on *Single parent * bho* is somewhat puzzling. However, this result is not robust to an alternative estimation method discussed in Section 5.5. Recent research has also highlighted the health vulnerabilities of single fathers. For instance, Chiu et al. (2018) report that single fathers face a three-times higher risk of mortality compared with either single mothers or partnered fathers. Similarly, Chiu et al. (2017) find that although single fathers and single mothers are equally likely to report poor overall health, single fathers are only half as likely to seek professional help.

[14]Recall that the inclusion of spell effects means that our results are identified from changes over time.

5 Robustness checks

We subject our findings to a battery of robustness checks, ranging from exploiting exogenous variation that affects the firm's export portfolio, through gender attitudes in partner countries, to searching for similar effects among Danish multinationals.

5.1 Exogenous shocks to business hours overlap

Although the inclusion of firm-year fixed effects controls for any firm-year-specific unobservables driving the changes in the firm's trade portfolio and hence the business hours overlap with its foreign partners, one might be concerned about the potential correlation between the firm's decisions and its gender gap in mental well-being. Thus ideally our analysis would rely on an exogenous shock shifting the business hours overlap. This section uses the activities of the Trade Council in Denmark as a source of such a shock.

The Trade Council is part of the Ministry of Foreign Affairs and employs caseworkers based at Danish consulates and embassies around the world. Their main task is to sell export promotion services (partner searches, market analyses, commercial assistance, etc.) to Danish firms with the aim of boosting Danish exports to the country where the caseworker is located. Firms may self-select into buying export promotion services, but roughly half of the services sold are initiated by caseworkers reaching out to Danish firms and offering assistance. Caseworkers reach out to firms if they believe there are unexplored export potentials for Danish firms in the destination market, motivated by their annual sales targets that are unlikely to be met by firms pro-actively demanding export promotion services.

Buus et al. (2025) argue that these caseworker calls are quasi-random in nature since caseworkers tend to approach firms sequentially in the alphabetical order. That is, firms with names starting with letters early in the alphabet are overrepresented among the approached firms. They show that at the firm-level, being called by a caseworker strongly predicts export status and volume to the country where the caseworker is located.

We follow Buus et al. (2025) and use data on the caseworker calls to Danish firms, obtained from the Trade Council. For each call, we observe the firm ID, the host country of the caseworker and the year of the call. We create an indicator variable taking on the value of one if a firm is approached by a caseworker located in a country with a lower business overlap than the business hours overlap specific to the firm's export portfolio in the previous year ($\mathbb{I}\left\{bho_{jt}^{\text{call}} < bho_{j,t-1}^{\text{exp}}\right\}$). The underlying logic is that if the call generates an additional export opportunity, it will lower

the firm's overall business hours overlap with its export markets. More specifically, we estimate the following equation:

$$\begin{aligned} y_{ijt} = \nu_1 Female_i \cdot Call_{jt} + \nu_2 Female_i \cdot Call_{jt} \cdot \mathbb{I}\left\{bho_{jt}^{\text{call}} < bho_{j,t-1}^{\text{exp}}\right\} \\ + \nu_3 Female_i \cdot \ln bho_{j,t-1}^{\text{exp}} + X_{it}'\beta + \delta_t Female_i + \rho_{ij} + \eta_{jt} + \epsilon_{ijt}, \end{aligned} \tag{5}$$

where $Call_{jt}$, is a dummy variable indicating if firm j was approached by any Trade Council caseworker at time t, bho_{jt}^{call} is the business hours overlap to the caller country, and $bho_{j,t-1}^{\text{exp}}$ is the lagged average export business hours overlap weighted by product variety. The estimate of interest is ν_2, which captures the differential impact on women in firms being approached by a caseworker located in a country that potentially reduces the firms' business hours overlap. We expect that such calls should lead to an increase in the gender gap in antidepressant use.

Table 4 presents the estimation results for all workers and for the subsample of college educated workers below 45 years. The $Female * Call$ interaction never appears to be statistically significant, which is consistent with our belief that calls are quasi-random. As for the estimate of interest, it is not statistically significant in the full worker population (Panel A) as one would expect. By contrast, for younger college educated workers (Panel B), women increase the antidepressant usage relative to men, if the firm is approached by Trade Council workers from countries that would reduce its business hours overlap. We take this as evidence that an arguably exogenous shift to the firm-level business hours overlap has a qualitatively similar effect as in our baseline results of Section 4.2.

5.2 What is the impact of business hours overlap on men?

Our baseline specification does not lend itself to examining the impact of the business hours overlap on male workers because of the inclusion of firm-year fixed effects. These fixed effects are needed to address the potential reverse causality: firms, whose workforce enjoys better mental well-being, may be more productive, and in turn more productive firms tend to serve more distant markets in both spatial and temporal sense (Eaton et al., 2011). This concern makes any results omitting firm-year fixed effects unreliable.

However, exploiting exogenous variation due to activities of the Danish Trade Council gives us an opportunity to examine the impact of the business hours overlap on the firm's overall workforce. We do this by dropping firm-year fixed effects from equation (5) and adding non-interacted business hours overlap and call terms. The results, presented in Table 5 confirm the pattern found earlier.

Namely, we find a positive and statistically significant coefficient on the interaction term between an indicator for receiving a call from a location distant in temporal sense and the female dummy. As expected, this term is statistically significant only in the subsample of college educated workers under 45 years of age. The coefficient on the indicator for receiving a call from a distant location, which captures the overall effect on the firm's workforce, bears the expected positive coefficient, one third of the size of the coefficient on its interaction with the female dummy, but it is not statistically significant. When it comes to the lagged *bho* term, its negative sign suggests that more trade with temporarily distant markets is associated with a higher antidepressant usage among the firm's workforce. The interaction between lagged *bho* and the female dummy is (as expected) negative and sizeable, but not statistically significant. Although these signs patterns are intuitive, we urge caution when interpreting the results for lagged *bho* due to the lack of firm-year fixed effects.[15]

5.3 Are our findings driven by gender attitudes in partner countries?

A potential concern is that the business hours overlap may be correlated with societal attitudes, such as the position of women in the society in the partner countries, rather than just purely reflecting the time zone differential. For instance, it may be more difficult, less pleasant, and more stressful for female employees of Danish companies to conduct business in countries where women's rights are restricted or women have a lower social status. Countries, where this is typically known to be the case, tend to be far away from Denmark and will thus have a low business hours overlap. If so, our findings could simply be reflecting the mental toll of doing business under such conditions has on female employees, rather than the need for working outside the standard office hours due to time zone differences.

To address this concern, we control for attitudes towards women and gender equality in the trading partner countries. We follow Bøler et al. (2018) and use the Gender Gap Index which is compiled by the World Economic Forum and pertains to the year 2007 (see Section 2.4 for more information about the index). The value of the index relevant to each firm is calculated with an approach that is analogous to the business hours overlap. To save space from now on we only report the results for the most stringent specification (described in equation (4)) and for

[15]The results from a baseline specification omitting firm-year fixed effects, presented in Online Appendix Table O.E1, show a positive correlation between *bho* and antidepressant use. As argued, these results cannot be interpreted as causal as they could reflect the fact that better well-being of the workforce allows firms to perform better and access distant markets. When both contemporaneous and lagged *bho* terms enter the specification, neither appears to be statistically significant (see Table O.E2 in the Online Appendix).

the subsamples of younger workers and younger workers with college education. These are the subsamples where the baseline exercise produced the largest effects.

Columns 1 and 2 in Table 6 present specifications controlling for the Gender Gap Index. Our results are robust to the additional control and remain significant at the five percent level. The size of the estimated coefficients goes up, now implying an almost 2% point effect on the gender gap in mental well-being for college-educated workers resulting from a five-hour decline in business hours overlap. As for the effect of gender attitudes in partner countries, it is in the opposite direction of what one might expect, though it is statistically significant only for workers without college education. Additional robustness checks, presented in Appendix Table C3, show that if the time-varying GGI is used (instead of its 2007 value), its interaction with the female dummy becomes insignificant, while our baseline effect of *bho* remains robust.[16]

5.4 Are the results robust to excluding the Global Financial Crisis?

Our data include 2008 and 2009, which were the years of the Global Financial Crisis. As a small open economy, Denmark experienced large drops in employment and trade (The Ministry of Industry, Business and Financial Affairs (2013)). To make sure that our results are not affected by the Global Financial Crisis, we re-estimated our baseline specification excluding observations for 2008 and 2009. The results, shown in columns 3 and 4 of Table 6, remain robust for the subsample of college-educated workers.

5.5 Are similar effects found in Danish multinationals?

The logic of our exercise would imply that similar effects should be found in Danish firms with subsidiaries abroad. This is the question to which we turn next. We include an interaction between a female dummy and the *bho* associated with foreign affiliates as defined in (3). For firms with no foreign affiliates this *bho* measure is set to 8. We also include a female dummy interacted with a dummy for having at least one foreign affiliate in a given year. This allows us to distinguish the gendered effect of being a multinational firm from the effect of changes in the average *bho* of a

[16]The surprising result is likely due to multicollinearity between our *bho* measure and the Gender Gap Index measure. In Online Appendix Table O.F1, we also show that there is no effect in specifications without the *bho* interaction but including just the interaction term between the female dummy and a gender gap index. We use here three different gender indices: the World Economic Forum index used in Table 6, the UN Gender Inequality Index (GII), and the World Bank's Women Business and the Law index (WBL). GGI is composite metric of gender inequality along three dimensions: reproductive health, empowerment and the labour market. WBL measures legal gender equality by scoring economies on laws affecting women's economic participation along dimensions such as mobility, workplace, pay, marriage, parenthood, and pension. In none of the cases, the estimates are statistically significant.

firm's subsidiaries. In other words, we capture both the extensive as well as the intensive margin.

The results, presented in columns 5 and 6 of Table 6, are in line with our hypothesis. As expected, we find that a lower business hours overlap with foreign subsidiaries of Danish multinationals is associated with an increase in the relative use of antidepressants by women in the subsample of college-educated workers. Our main results for the business hours overlap based on firms' international trade flows remain robust for college-educated workers. The coefficients are very similar to those found earlier, though the significance level is lower. The size of the coefficient on the trade-based business hour overall is more than twice as large as the size of the coefficient based on the *bho* with foreign subsidiaries. Finally, the estimates also suggest that a firm being a multinational has no bearing on the gender gap in antidepressant usage.

5.6 Worker exits

One may be concerned about the possibility that changes to business hours overlap induce some workers to leave the firm where they are employed, biasing our estimates. The direction of bias is not *a priori* clear. On the one hand, workers more prone to mental distress may be more likely to leave. On the other hand, they may be more likely to stay as they may find it more difficult to find alternative employment.

To shed some light on this issue, we examine worker exits from the firm. We observe some workers going to another employer, while others exit our sample. Recall that our dataset encompasses manufacturing firms with more than 10 employees engaged in imports and exports, hence the latter option may mean unemployment or employment in a firm outside our sample. We consider three possibilities, all defined as dummies in year t by observing at $t+1$: (i) exit from the firm regardless of whether we observe the worker in the sample afterwards, (ii) switching to another firm in the sample, or (iii) exit from the sample.

We focus on the differential impact of business hours overlap for exits by female workers. We control for a plethora of fixed effects described earlier. We consider the full sample, the subsample of college-educated workers, the subsample of workers under 45 years of age, and the subsample of workers under 45 with college education. As evident from Appendix Table C4, the coefficient of interest never reaches conventional significance levels, regardless of the subsample and the definition of exit considered. This attenuates our concern about differential exit rates biasing our results.

5.7 Other robustness checks

Source of variation To shed light on the source of variation driving our results, we have augmented our specification with interactions of the female dummy with the number of varieties traded, firm's export intensity and firm's import intensity. While none of these newly introduced interaction terms are statistically significant once spell fixed effects are introduced, our core result remains robust as seen in Table C1. This gives us confidence that what matters for antidepressant usage is the business hours overlap rather than the intensity of engagement in international trade.

Controlling for wages In another robustness check, we add to our baseline specification the worker's wage and its interaction with the female indicator, though noting the potential endogeneity problem. The results are presented in Online Appendix Table O.A1. Among workers under 45, a higher wage is associated with a lower antidepressant usage among men and, to a much lower extent, women. This is consistent with the literature showing that a higher income is associated with better mental health (Thomson et al. (2022), Reed et al. (2025)).

Once we focus on college-educated workers under 45, the wage loses its statistical significance, presumably because there is much less variation in the sample and because college-educated workers enjoy higher salaries. The latter is consistent the diminishing returns found by Reed et al. (2025): for a given increase in income, benefits on health are smaller at higher income levels. Interestingly, among women in this subgroup, higher wages are associated with a higher antidepressant usage. This may be due to positions of greater responsibility being better remunerated but also generating more stress due to longer hours. And indeed, this effect loses its statistical significance once female-year and occupational fixed effects (or gender specific occupational fixed effects) are included.

Raw time zone differential Our measure of *bho*, for obvious reasons, does not distinguish between a time zone differential of 9 and 12 h. Therefore, in a robustness check, we replace *bho* with the raw time difference (i.e., the absolute value of the UTC offset). Our baseline conclusions are robust to this change in the subsample of college educated workers under 45, i.e., the larger the time difference vis-à-vis the foreign markets, the higher the gender gap in antidepressant usage. The results for all college educated workers are much weaker. See Table O.B1 in Online Appendix.

Unlogged bho measure As yet another robustness check, we re-estimated our regressions using an unlogged measure of *bho*. The results, presented in Tables O.C1–O.C3 in the Online

Appendix, support our conclusions.

Logit estimation The use of linear probability models might be problematic for rare events because the tails of the distribution are important (see e.g. Horrace and Oaxaca (2006) and Angrist and Pischke (2009)). Therefore, to test the robustness of our results we have estimated our main empirical specifications using a logistic regression model (logit), rather than a linear probability model. Tables O.D1 and O.D3 in the Online Appendix replicate Tables 2 and 3 with a logit model. The direction and significance of our results replicate, although the magnitudes cannot be compared directly. If we consider college educated workers, the implied marginal effect of a decline in the business hours overlap from 8 to 3 h is associated with an increase in the likelihood of female workers taking antidepressants (relative to their male counterparts doing so) of 7.7% points for a person with the average likelihood of usage for the college sample. This suggests that our linear probability approach understates the magnitude of the effect.

6 Exploiting differences between occupational characteristics

Not all occupations require interactions with foreign business partners and thus working across time zones. For instance, such interactions are crucial for an account manager in charge of important customers abroad, but not for a production worker. In this subsection, we use variation in occupational characteristics to test our hypothesis that working across time zones is taxing for mental well-being.

Following Goldin (2014), we use O*NET data and focus on five occupational characteristics that in her judgment limit the temporal flexibility of workers, as they are linked to working particular hours and long hours. These are: time pressure, contact with others, establishing and maintaining interpersonal relationships, structured versus unstructured work (which captures how easy it is to find a substitute for a given worker), and freedom to make decisions. We use the average of the characteristics for each 4-digit occupational code or an indicator variable for the top 25% of occupations ranked highest in terms of these characteristics. See Appendix B for more details.

As discussed in the data section, each worker is assigned to their first occupation reported in the data set to avoid possible endogeneity of mental well-being either driving workers to change occupations or preventing them from doing so.[17] As pointed out by Kambourov and Manovskii

[17]We also note the possibility of self-selection of men and women into particular occupations clouding this analysis.

(2009), the first-reported occupation code in a job spell tends to be more accurate. Using time-invariant occupation codes means there is no need to include *Female * ONET* interaction, as it will be time-invariant and hence subsumed by the spell fixed effect.

Focusing on occupational characteristics is thus an alternative identification strategy that allows us to link working across time zones to mental well-being. We consider occupational variation in isolation, together with the gender effect as well as allowing for an interaction between the two. We restrict our attention to workers under 45 years of age with a college education, as this is the group for whom the tension between professional and family obligations is likely to be the greatest, which was confirmed by our analysis so far.

We start by focusing on the measure encompassing all characteristics before considering them one by one. The results, presented in Table 7, support our hypothesis. We find a higher response of antidepressant usage to business hours overlap among employees in occupations that *a priori* would be expected to be more affected than among other workers. Both the continuous measure and the indicator variable for highly exposed occupations are negative and statistically significant when interacted with the business hours overlap. The estimated effect is economically meaningful: a decline in the business hours overlap from 8 to 3 h is associated with a 0.9% point increase in antidepressant usage among workers in the highly exposed occupations (column 6). This is large, given the sample average of 3%.

Moreover, controlling for the impact of business hours overlap on occupations with particular characteristics does not affect our baseline findings that working across time zones leads to a greater gender gap in antidepressant usage. The interaction of interest maintains its sign and magnitude and remains statistically significant. This suggests that our baseline findings are not driven by self-selection of women into occupations that may have been more sensitive to working across time zones. This is also true if we allow the identified occupation to vary within a job spell, as can be seen in Appendix Table C5. Finally, the triple interactions are not statistically significant, which suggest that there is no interplay between the two effects.

In Table 8, we consider each of five occupational characteristics one by one. Not surprisingly, we find that business hours overlap differentially affects workers in occupations requiring establishing and maintaining personal relationships, occupations where contact with others is important, and occupations with few close substitutes. Differences in occupations along the time pressure dimension or freedom to make decisions are not statistically significant. As before, our findings on women being affected more than men by the need to work across time zones remain robust.

In summary, two different identification strategies, one based on gender differences and one on occupational characteristics, point in the same direction, suggesting that working across time zones takes a toll on mental well-being of workers.

7 Conclusions

This paper documents some underappreciated consequences of globalization, namely its impact on the gender gap in mental well-being of workers. It argues that trading with business partners located in different time zones puts a greater burden on female workers by necessitating working outside the standard business hours, thus potentially interfering with family obligations.

We use matched employer–employee data from Denmark, combined with information on workers' use of prescription antidepressant medication and transaction-level information on firms' import and export activities. Our identification strategy rests on several pillars. First, we condition on job-spell fixed effects in our estimation, i.e., the source of our variation is the change over time in the firm's business hours overlap within a given worker–firm relationship. Second, we focus on the gap in antidepressant usage between men and women working for the same firm in a given year, which allows us to control for unobservable firm-year-specific heterogeneity that may drive firms' internationalization, corporate culture, the average level of stress among the work-force, etc. Finally, in a robustness check, we exploit plausibly exogenous variation in export flows due to activities of the Danish Trade Council.

Our findings indicate that a decline in the business hours overlap between a Danish firm and its foreign business partners is associated with an increased incidence of antidepressant use among women relative to men. A decline in business hours overlap by 5 h, equivalent to moving all trade from Germany to the eastern part of Canada, leads to a 1.7% point increase in the likelihood of antidepressant use by women relative to men among college-educated workers under 45 years of age. This implies increasing the prevalence of antidepressant use by 57%, which in this subpopulation is equal to 3%.

Our analysis further shows that the impact of working across time zones on antidepressant usage is three times larger for single mothers than for women in general, which suggests that the pressures of maintaining work-life balance are likely to be one of the key factors driving our findings.

Finally, we show that *in addition* to the gender effect, occupational characteristics matter: a decline in the business hours overlap from 8 to 3 h is associated with a 0.9% point increase

in antidepressant usage among workers in the highly exposed occupations. This is large, given the sample average of 3%. Workers in occupations where contact with others or establishing interpersonal relationships is important are affected to a greater extent.

Goldin (2014) convincingly argued that enhancing the temporal flexibility of jobs is what is needed to achieve greater convergence in labour market outcomes of men and women. Our results suggest that globalization may make this task more challenging.

References

Adda, J. and Y. Fawaz (2020, 05). The Health Toll of Import Competition. *The Economic Journal 130*(630), 1501–1540.

Angrist, J. D. and J.-S. Pischke (2009). *Mostly harmless econometrics: an empiricist's companion*. Princeton: Princeton University Press. OCLC: ocn231586808.

Belloni, M., L. Carrino, and E. Meschi (2022). The impact of working conditions on mental health: Novel evidence from the UK. *Labour Economics 76*, 102176.

Bircan, Ç., B. Javorcik, and S. Pauly (2024). Barriers to Knowledge Creation in the Multinational Firm. SSRN Scholarly Paper 4833686.

Black, S. E., P. J. Devereux, and K. G. Salvanes (2007). From the Cradle to the Labor Market? The Effect of Birth Weight on Adult Outcomes. *The Quarterly Journal of Economics 122*(1), 409–439.

Buus, M. T., J. R. Munch, J. Rodrigue, and G. Schaur (2025). Do Export Support Programs Affect Prices, Quality, Markups, and Marginal Costs? Evidence From a Natural Policy Experiment. *Review of Economics and Statistics 107*(1), 172–187.

Bøler, E. A., B. Javorcik, and K. H. Ulltveit-Moe (2018). Working across time zones: Exporters and the gender wage gap. *Journal of International Economics 111*, 122–133.

Chauvin, J., P. Choudhury, and T. P. Fang (2020). The effects of temporal distance on intra-firm communication: Evidence from daylight savings time. *Harvard Business School Technology & Operations Mgt. Unit Working Paper* (21-052).

Chiu, M., F. Rahman, P. Kurdyak, J. Cairney, N. Jembere, and S. Vigod (2017). Self-rated health and mental health of lone fathers compared with lone mothers and partnered fathers: a population-based cross-sectional study. *Journal of Epidemiology & Community Health 71*(5), 417–423.

Chiu, M., F. Rahman, S. Vigod, C. Lau, J. Cairney, and P. Kurdyak (2018). Mortality in single fathers compared with single mothers and partnered parents: a population-based cohort study. *The Lancet Public Health 3*(3), e115–e123.

Choi, J. and T. M. Greaney (2022). Global influences on gender inequality: Evidence from female employment in Korea. *International Economic Review 63*(1), 291–328.

Colantone, I., R. Crinò, and L. Ogliari (2019). Globalization and mental distress. *Journal of International Economics 119*, 181–207.

Currie, J. and B. C. Madrian (1999). Chapter 50 Health, health insurance and the labor market. In *Handbook of Labor Economics*, Volume 3, pp. 3309–3416. Elsevier.

Eaton, J., S. Kortum, and F. Kramarz (2011). An anatomy of international trade: Evidence from French firms. *Econometrica 79*(5), 1453–1498.

Goldin, C. (2014). A Grand Gender Convergence: Its Last Chapter. *American Economic Review 104*(4), 1091–1119.

Gumpert, A. (2018). The Organization of Knowledge in Multinational Firms. *Journal of the European Economic Association 16*(6), 1929–1976.

Hausmann, R., L. D. Tyson, and S. Zahidi (2007). *The Global Gender Gap Report 2007*. Number 112006. World Economic Forum, https://www3.weforum.org/docs/WEF_GenderGap_Report_2007.pdf.

Horrace, W. C. and R. L. Oaxaca (2006, March). Results on the bias and inconsistency of ordinary least squares for the linear probability model. *Economics Letters 90*(3), 321–327.

Hummels, D., J. R. Munch, and C. Xiang (2025). No Pain, No Gain: Work Demand, Work Effort, and Worker Health. *The Review of Economics and Statistics 107*, 898–913.

Javorcik, B., A. Lo Turco, D. Maggioni, and G. Santoni (2025). Foreign ownership and occupational safety: Evidence from france. Working paper.

Juhn, C., G. Ujhelyi, and C. Villegas-Sanchez (2014). Men, women, and machines: How trade impacts gender inequality. *Journal of Development Economics 106*, 179–193.

Kodama, N., B. S. Javorcik, and Y. Abe (2018). Transplanting corporate culture across international borders: Foreign direct investment and female employment in japan. *The World Economy 41*(5), 1148–1165.

Kvande, E. (2009). Work–Life Balance for Fathers in Globalized Knowledge Work. Some Insights from the Norwegian Context. *Gender, Work & Organization 16*(1), 58–72.

Kühn, M., N. Metsä-Simola, and P. Martikainen (2023). Pathways into single motherhood, re-partnering, and trajectories of antidepressant medication purchases. *Social Psychiatry and Psychiatric Epidemiology 58*(1), 409–420.

McManus, T. C. and G. Schaur (2016). The effects of import competition on worker health. *Journal of International Economics 102*, 160–172.

National Board of Health (2010). Specialevejledning for psykiatri. Technical Report j.nr. 7-203-01-90/19.

Pierce, J. R. and P. K. Schott (2020, March). Trade Liberalization and Mortality: Evidence from US Counties. *American Economic Review: Insights 2*(1), 47–64.

Reed, H. R., D. Nettle, F. Parra-Mujica, G. Stark, R. Wilkinson, M. T. Johnson, and E. A. Johnson (2025). Examining the relationship between income and both mental and physical health among adults in the uk: Analysis of 12 waves (2009-2022) of understanding society. *PLoS ONE 20*, e0316792.

Statistics Denmark (2024). SYGK2. Technical report, Statistics Denmark.

Tang, H. and Y. Zhang (2021). Do multinationals transfer culture? Evidence on female employment in China. *Journal of International Economics 133*, 103518.

The Ministry of Health (2017). Healthcare in Denmark - An Overview. Technical report.

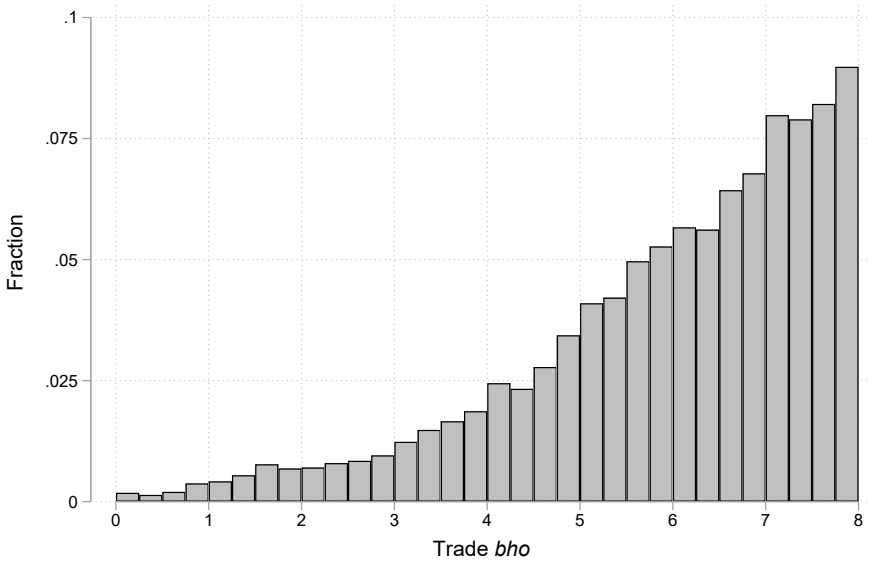
The Ministry of Industry, Business and Financial Affairs (2013). Den Finansielle Krise i Danmark – årsager, konsekvenser og læring. Technical report, The Ministry of Industry, Business and Financial Affairs.

Thielemans, G., P. Fallesen, and D. Mortelmans (2019). Division of houshold labor and relationship dissolution in Denmark. *The Rockwool Foundation Research*.

Thomson, R. M., E. Igelström, A. K. Purba, M. Shimonovich, H. Thomson, G. McCartney, A. Reeves, A. Leyland, A. Pearce, and S. V. Katikireddi (2022). How do income changes impact on mental health and wellbeing for working-age adults? a systematic review and meta-analysis. *The Lancet Public Health 7*(7), e557–e569.

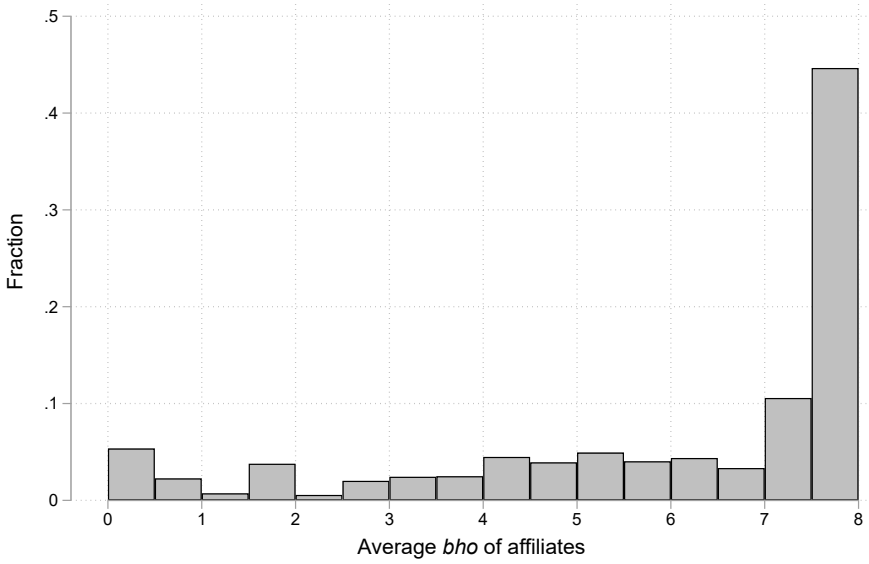
WHO (2021). *Comprehensive Mental Health Action Plan 2013–2030*. WHO, Geneva.

Figure 1: Distribution of *bho*, 2008–2017



Note: This figure shows the distribution of average *bho* weighted by the variety of products imported and exported, for Danish manufacturing firms that export and import and have more than 10 employees in 2008–2017.

Figure 2: Distribution of *bho* for foreign affiliates, 2010–2017



Note: This figure shows the distribution of average *bho* weighted by number of foreign affiliates in each country, for Danish firms that import and export and have more than 10 employees in 2010–2017, with at least 1 affiliate.

Table 1: Means for workers in 2008–2017

	Men		Women	
	College	No college	College	No college
ln wage	5.645	5.330	5.462	5.191
Age	42.178	42.965	40.492	43.881
Experience	18.188	22.901	16.218	21.282
Children	1.205	0.900	1.197	0.990
Goldin (2014) O*NET measure	0.530	−0.256	0.583	−0.147
Firm's trade bho	5.570	6.152	5.596	5.888
Parents	58.96%	43.55%	61.17%	49.61%
Single parents	1.61%	1.87%	7.14%	8.05%
Antidepressant users	3.01%	4.29%	4.30%	6.53%
Any AD use in 2008–2017	6.69%	10.15%	9.47%	14.59%
Observations	356,405	1,005,479	220,488	404,694

The sample includes workers aged 20–60, in manufacturing firms that export and import and have more than 10 employees. The years included are 2008–2017. Parental status is defined as having a child younger than 19, living at home, and single status is defined as not living with a partner. Firm's trade *bho* is defined in section 2.3. The O*NET measure of Goldin (2014) is the average of the O*NET variables: Time Pressure, Contact with others, Structured versus Unstructured Work, Freedom to Make Decisions, Establishing and Maintaining Interpersonal Relationships. These variables characterize the typical related to an occupation. Their values are standardized on the full sample. Means of each O*NET characteristic across groups can be seen in Table A3.

Table 2: Effect of business hours overlap on the gender gap in antidepressant use

	(1)	(2)	(3)	(4)	(5)
Panel A: All workers					
Female × ln(bho trade)	-0.0015	-0.0035	-0.0033	-0.0030	-0.0030
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
N	1,897,559	1,834,681	1,827,209	1,827,209	1,827,209
Firms	3,268	2,848	2,847	2,847	2,847
Women	97,569	95,735	95,022	95,022	95,022
Men	218,198	212,266	210,937	210,937	210,937
Average AD usage	0.0441	0.0436	0.0435	0.0435	0.0435
Panel B: Only college workers					
Female × ln(bho trade)	-0.0012	-0.0092*	-0.0098*	-0.0096*	-0.0095*
	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)
N	552,585	536,609	534,832	534,832	534,831
Firms	2,607	2,370	2,356	2,356	2,356
Women	34,845	34,299	34,137	34,137	34,137
Men	57,004	55,980	55,677	55,677	55,677
Average AD usage	0.0345	0.0342	0.0341	0.0341	0.0341
Panel C: Only workers younger than 45					
Female × ln(bho trade)	-0.0038	-0.0123**	-0.0120**	-0.0116**	-0.0115**
	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)
N	995,362	954,326	950,138	950,138	950,138
Firms	3,160	2,817	2,812	2,812	2,812
Women	63,189	61,827	61,389	61,389	61,389
Men	134,147	129,639	128,815	128,815	128,815
Average AD usage	0.0410	0.0402	0.0401	0.0401	0.0401
Panel D: Only college workers younger than 45					
Female × ln(bho trade)	-0.0044	-0.0168**	-0.0176**	-0.0170**	-0.0168**
	(0.006)	(0.007)	(0.007)	(0.008)	(0.008)
N	333,191	322,374	321,283	321,283	321,282
Firms	2,230	2,041	2,029	2,029	2,029
Women	26,810	26,339	26,224	26,224	26,224
Men	38,806	38,039	37,852	37,852	37,852
Average AD usage	0.0308	0.0304	0.0304	0.0304	0.0304
Worker FE	✓				
Firm-year FE	✓	✓	✓	✓	✓
Spell FE		✓	✓	✓	✓
Occupational FE			✓	✓	✓
Female-year FE				✓	✓
Occupational-Female FE					✓
Education FE	✓	✓	✓	✓	✓

Further controls are experience, experience squared and having children. Occupational FE are based on 2-digit ISCO-code. Educational FE have 12 levels. The sample includes workers aged 20–60, in manufacturing firms that export and import and have more than 10 employees, in the years 2008–2017. Standard errors, clustered by firm, in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Focusing on single parents under 45

	(1) No college	(2) No college	(3) College	(4) College
Female × ln(bho trade)	-0.0076	-0.0075	-0.0183*	-0.0179*
	(0.009)	(0.009)	(0.010)	(0.010)
Female × Single parent × ln(bho trade)	-0.0178	-0.0179	-0.0807***	-0.0815***
	(0.017)	(0.017)	(0.026)	(0.026)
Single parent × ln(bho trade)	-0.0028	-0.0029	0.0363**	0.0365**
	(0.015)	(0.015)	(0.017)	(0.017)
Female × Single parent	0.0390	0.0391	0.1446***	0.1461***
	(0.031)	(0.031)	(0.044)	(0.044)
Single parent	-0.0007	-0.0006	-0.0659**	-0.0663**
	(0.028)	(0.028)	(0.027)	(0.027)
Parent × ln(bho trade)	0.0019	0.0019	0.0007	0.0006
	(0.003)	(0.003)	(0.004)	(0.004)
Female × Parent × ln(bho trade)	0.0013	0.0013	0.0068	0.0067
	(0.009)	(0.009)	(0.009)	(0.009)
Female × Parent	-0.0084	-0.0085	-0.0137	-0.0134
	(0.016)	(0.016)	(0.016)	(0.016)
Parent	-0.0003	-0.0002	-0.0005	-0.0004
	(0.006)	(0.006)	(0.007)	(0.007)
Spell FE	✓	✓	✓	✓
Firm-year FE	✓	✓	✓	✓
Female-year FE	✓	✓	✓	✓
Education FE	✓	✓	✓	✓
Occupational FE	✓	✓	✓	✓
Occupational-Female FE		✓		✓
N	622,379	622,378	321,283	321,282
Firms	2,757	2,757	2,029	2,029
Women	35,168	35,168	26,224	26,224
Men	91,116	91,116	37,852	37,852
Average AD usage	0.0453	0.0453	0.0304	0.0304

Further controls are experience and experience squared. Occupational FE are based on 2-digit ISCO-code. Educational FE have 12 levels. The sample includes workers aged 20–44, in manufacturing firms that export and import and have more than 10 employees, in the years 2008–2017. Parental status is defined by having a child younger than 19, living at home, and single status is defined by not living with a partner. Standard errors, clustered by firm, in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effect of export promotion on the gender gap in antidepressant use

	(1)	(2)	(3)	(4)	(5)
Panel A: All workers					
Female × Call	-0.00117	-0.00170	-0.00166	-0.00114	-0.00107
	(0.0013)	(0.0013)	(0.0013)	(0.0014)	(0.0014)
Female × Call × bho(call country) lag diff. < 0	0.00063	0.00117	0.00105	0.00046	0.00044
	(0.0013)	(0.0012)	(0.0012)	(0.0013)	(0.0013)
Female× ln(bho exports) lagged	0.00093	0.00173	0.00195	0.00229	0.00238
	(0.0022)	(0.0024)	(0.0024)	(0.0024)	(0.0024)
N	1,436,595	1,391,734	1,386,890	1,386,890	1,386,888
Firms	2,676	2,365	2,365	2,365	2,365
Women	85,145	83,674	83,168	83,168	83,167
Men	186,530	181,973	181,028	181,028	181,028
Average AD usage	0.0448	0.0443	0.0443	0.0443	0.0443
Panel B: Only college workers younger than 45					
Female × Call	-0.00325	-0.00407	-0.00405	-0.00400	-0.00391
	(0.0027)	(0.0026)	(0.0026)	(0.0026)	(0.0026)
Female × Call × bho(call country) lag diff. < 0	0.00459**	0.00470**	0.00451**	0.00407*	0.00390*
	(0.0022)	(0.0021)	(0.0021)	(0.0022)	(0.0022)
Female× ln(bho exports) lagged	-0.00002	-0.00305	-0.00286	-0.00178	-0.00173
	(0.0050)	(0.0053)	(0.0053)	(0.0056)	(0.0055)
N	256,903	248,694	248,014	248,014	248,012
Firms	1,891	1,758	1,751	1,751	1,751
Women	22,092	21,707	21,640	21,640	21,640
Men	32,381	31,755	31,623	31,623	31,623
Average AD usage	0.0316	0.0312	0.0312	0.0312	0.0312
Firm-year FE	✓	✓	✓	✓	✓
Worker FE	✓				
Spell FE		✓	✓	✓	✓
Occupational FE			✓	✓	✓
Occupational-Female FE					✓
Female-year FE				✓	✓
Education FE	✓	✓	✓	✓	✓

The variable (bho(call country) lag diff. < 0) is a dummy for the difference between the *bho* of the caller country and the lagged export *bho* of the firm being negative. If a country is called by countries in multiple timezones, the average is used. Further controls are experience, experience squared and having children. Occupational FE are based on 2-digit ISCO-code. Educational FE have 12 levels. The sample includes workers aged 20–60, in manufacturing firms that export and import and have more than 10 employees, in the years 2008–2015. Standard errors, clustered by firm, in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effect of export promotion on antidepressant usage

	All workers		College, younger than 45	
	(1)	(2)	(3)	(4)
Call	-0.00030	-0.00054	-0.00135	-0.00096
	(0.0007)	(0.0007)	(0.0014)	(0.0013)
Female × Call	-0.00128	-0.00112	-0.00278	-0.00307
	(0.0013)	(0.0013)	(0.0025)	(0.0024)
Call × bho(call country) lag diff. < 0	0.00009	0.00040	0.00049	0.00106
	(0.0008)	(0.0007)	(0.0015)	(0.0013)
Female × Call × bho(call country) lag diff. < 0	0.00084	0.00064	0.00460**	0.00394*
	(0.0013)	(0.0013)	(0.0021)	(0.0020)
ln(bho exports) lagged	0.00081	0.00016	-0.00566**	-0.00505*
	(0.0013)	(0.0013)	(0.0028)	(0.0026)
Female× ln(bho exports) lagged	0.00108	0.00268	-0.00131	-0.00217
	(0.0022)	(0.0024)	(0.0045)	(0.0050)
Spell FE		✓		✓
Occupational FE		✓		✓
Occupational-Female FE		✓		✓
Female-year FE		✓		✓
Industry-year FE	✓	✓	✓	✓
Education FE	✓	✓	✓	✓
Firm FE	✓		✓	
Worker FE	✓		✓	
N	1,436,609	1,386,904	259,016	250,145
Firms	2,676	2,370	2,166	2,062
Women	85,146	83,169	22,211	21,766
Men	186,533	181,030	32,624	31,865
Average AD usage	0.0448	0.0443	0.0316	0.0312

The variable (bho(call country) lag diff. < 0) is a dummy for the difference between the *bho* of the caller country and the lagged export *bho* of the firm being negative. If a country is called by countries in multiple time zones, the average *bho* is used. Further controls are experience, experience squared and having children. Occupational FE are based on 2-digit ISCO-code. Educational FE have 12 levels. The sample includes workers aged 20–60, in manufacturing firms that export and import and have more than 10 employees, in the years 2008–2015. Standard errors, clustered by firm, in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Robustness checks and extensions: workers under 45

	GGI controls		Drop Financial Crisis		Foreign affiliates	
	(1) All	(2) College	(3) All	(4) College	(5) All	(6) College
Female × ln(bho trade)	-0.0167***	-0.0190**	-0.0081	-0.0183*	-0.0075	-0.0174*
	(0.005)	(0.008)	(0.006)	(0.010)	(0.006)	(0.010)
Female × ln(GGI trade)	0.0976**	0.0494				
	(0.049)	(0.086)				
Female × ln(bho foreign affiliates)					-0.0042	-0.0076**
					(0.003)	(0.003)
Female ×**1** {any foreign affiliates}					-0.0004	-0.0068
					(0.003)	(0.006)
Spell FE	✓	✓	✓	✓	✓	✓
Firm-year FE	✓	✓	✓	✓	✓	✓
Occupational FE	✓	✓	✓	✓	✓	✓
Occupational-Female FE	✓	✓	✓	✓	✓	✓
Female-year FE	✓	✓	✓	✓	✓	✓
Education FE	✓	✓	✓	✓	✓	✓
N	942,297	320,122	724,411	253,008	719,979	251,607
Firms	2,729	1,992	2,552	1,809	2,540	1,797
Women	61,060	26,160	50,814	23,240	50,649	23,172
Men	127,683	37,718	109,278	32,892	108,809	32,734
Average AD usage	0.0401	0.0303	0.0397	0.0300	0.0397	0.0300
Years	2008–2017	2008–2017	2010–2017	2010–2017	2010–2017	2010–2017

GGI trade is World Economic Forum's Gender Gap Index in the year 2007, weighted by the firm's trade variety. Bho foreign affiliates is average bho for a firm weighted by the number of foreign affiliates in each country. Further controls are experience, experience squared and having children. Occupational FE are based on 2-digit ISCO-code. Educational FE have 12 levels. The sample includes workers aged 20–44, in manufacturing firms that export and import and have more than 10 employees. Std. errors, clustered by firm in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Antidepressant use and job characteristics: college workers under 45

	(1)	(2)	(3)	(4)	(5)	(6)
O*NET × ln(bho trade)	-0.0065**		-0.0063**		-0.0071**	
	(0.003)		(0.003)		(0.003)	
D(O*NET) × ln(bho trade)		-0.0092*		-0.0088*		-0.0085*
		(0.005)		(0.005)		(0.005)
Female × ln(bho trade)			-0.0168**	-0.0167**	-0.0183**	-0.0162*
			(0.008)	(0.008)	(0.008)	(0.009)
Female × O*NET × ln(bho trade)					0.0025	
					(0.008)	
Female × D(O*NET) × ln(bho trade)						-0.0011
						(0.013)
Spell FE	✓	✓	✓	✓	✓	✓
Firm-year FE	✓	✓	✓	✓	✓	✓
Female-year FE	✓	✓	✓	✓	✓	✓
Education FE	✓	✓	✓	✓	✓	✓
N	319,086	319,086	319,086	319,086	319,086	319,086
Firms	2,028	2,028	2,028	2,028	2,028	2,028
Women	26,061	26,061	26,061	26,061	26,061	26,061
Men	37,560	37,560	37,560	37,560	37,560	37,560
Average AD usage	0.0304	0.0304	0.0304	0.0304	0.0304	0.0304

The O*NET variable is defined as the occupation-level average of Time Pressure, Contact with others, Structured versus Unstructured Work, Freedom to Make Decisions, Establishing and Maintaining Interpersonal Relationships. It has been standardized on the full sample. D() denotes an indicator for the variable being above the 75th percentile. Further controls are experience, experience squared and having children. Educational FE are based on 12 categories. The sample includes workers aged 20–44 with a college degree in manufacturing firms that export and import and have more than 10 employees. D() denotes a dummy for the variable being above the 75th percentile. The first occupation in the job-spell is used for determining the O*NET characteristic The years included are 2008–2017. Std. errors, clustered by firm in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Antidepressant use and specific job characteristics: college workers under 45

	O*NET variable				
	(1) Time Pressure	(2) Contact With Others	(3) Unstructured Work (few close substitutes)	(4) Freedom to Make Decisions	(5) Establishing and Maintaining Interpersonal Relationships
ln(bho trade) × O*NET	-0.0002	-0.0081**	-0.0058*	-0.0048	-0.0058*
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Female × ln(bho trade)	-0.0165**	-0.0167**	-0.0189**	-0.0197**	-0.0153*
	(0.008)	(0.008)	(0.009)	(0.008)	(0.009)
Female × ln(bho trade) × O*NET	0.0023	0.0022	0.0024	0.0042	-0.0018
	(0.007)	(0.008)	(0.008)	(0.008)	(0.009)
Spell FE	✓	✓	✓	✓	✓
Firm-year FE	✓	✓	✓	✓	✓
Female-year FE	✓	✓	✓	✓	✓
Education FE	✓	✓	✓	✓	✓
N	319,086	319,086	319,086	319,086	319,086
Firms	2,028	2,028	2,028	2,028	2,028
Women	26,061	26,061	26,061	26,061	26,061
Men	37,560	37,560	37,560	37,560	37,560
Average AD usage	0.0304	0.0304	0.0304	0.0304	0.0304

Further controls are experience, experience squared and having children. Educational FE are based on 12 categories. The sample includes workers aged 20–44 with a college degree in manufacturing firms that export and import and have more than 10 employees. The years included are 2008–2017. Std. errors, clustered by firm in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendices

A Descriptive statistics

A.1 Sample descriptive statistics

Table A1: Proportion of overall Danish economic activity covered by our sample after restrictions

	Employment	Sales	Imports	Exports
Manufacturing	12.1%	21.2%	31.0%	59.5%
10 employees	11.1%	20.3%	30.4%	58.9%
Share of manufacturing	92.0%	95.5%	98.0%	98.9%
Imports and exports	9.9%	19.0%	30.1%	58.5%
Share of manufacturing	81.4%	89.4%	96.8%	98.2%

The sample conditions are cumulative, such that the 10 employees condition also includes the manufacturing condition, and the imports and exports condition, includes both of the above. The employee condition has been implemented such that if a firm in a single year has fewer employees, but more than 10 in the year before and the year after, it is kept in the sample for that year.

Table A2: Means for workers with any AD usage 2008–2017

	Men		Women	
	College	No college	College	No college
Years in sample	5.858	5.447	5.702	5.293
Years of AD use	2.631	2.300	2.581	2.372
Consecutive years in sample after first year	4.133	3.716	4.060	3.715
Distinct spells in sample after first year	1.141	1.161	1.105	1.114
Consecutive years of AD use after first year	2.476	2.151	2.436	2.235
Distinct spells of AD use after first year	1.155	1.149	1.145	1.137
Unique persons	4,138	18,666	3,730	11,081

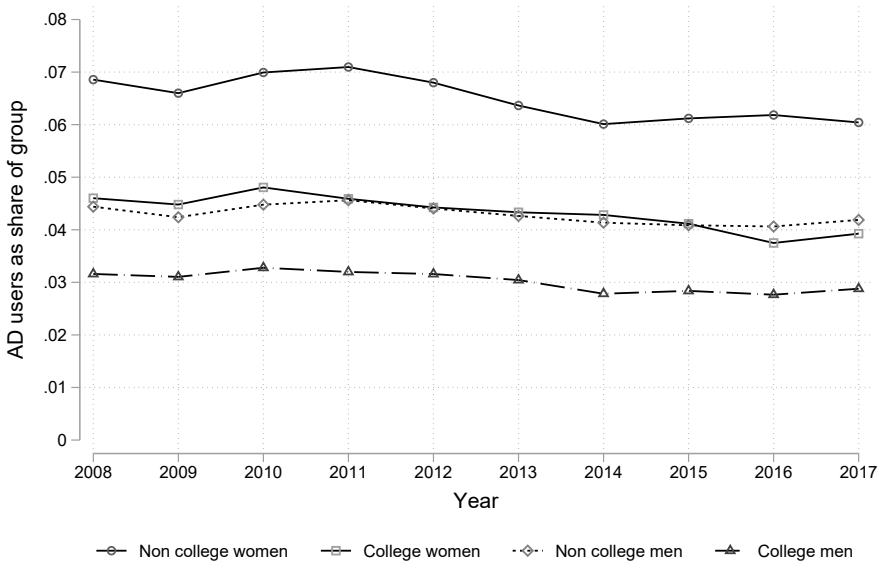
The sample includes workers aged 20–60, in manufacturing firms that export and import and have more than 10 employees, that have bought prescription antidepressants at any point in 2008–2017. The years included are 2008–2017.

Table A3: Means of O*NET characteristics for workers in 2008–2017

	Men		Women	
	College	No college	College	No college
Goldin (2014) O*NET measure	0.530	−0.256	0.583	−0.147
Time Pressure	−0.442	0.210	−0.290	0.023
Contact With Others	0.337	−0.222	0.499	−0.016
Structured versus Unstructured Work	0.659	−0.292	0.588	−0.174
Freedom to Make Decisions	0.618	−0.221	0.532	−0.285
Establishing and Maintaining Interpersonal Relationships	0.670	−0.366	0.697	−0.059
Observations	356,405	1,005,479	220,488	404,694

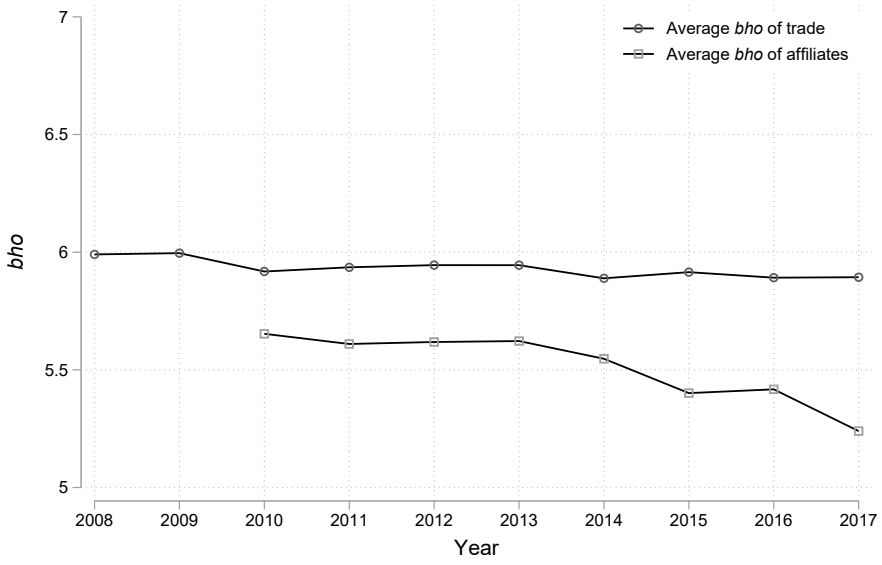
The sample includes workers aged 20–60, in manufacturing firms that export and import and have more than 10 employees. The years included are 2008–2017. The O*NET measure of Goldin (2014) is the average of the O*NET variables: Time Pressure, Contact with others, Structured versus Unstructured Work, Freedom to Make Decisions, Establishing and Maintaining Interpersonal Relationships. These variables characterize the typical related to an occupation. Their values are standardized on the full sample.

Figure A1: Proportion of AD users in subsamples of the population, 2008–2017



Note: This figure shows the evolution in the shares of workers aged 20–60, in international manufacturing firms, that have bought prescription antidepressants in a given year.

Figure A2: Average *bho*, 2008–2017



Note: This figure shows firms' average trade business hours overlap weighted by both exports and imports, and for firms engaged in FDI, their average business hours overlap weighted by the number of foreign affiliates in 2010–2017. The average is weighted by persons in the sample, which are workers aged 20–60, in international manufacturing firms.

A.2 Variation in AD usage and business hours overlap at the firm level

Figure A3: Distribution of within-firm log changes in trade *bho*, 2008–2017

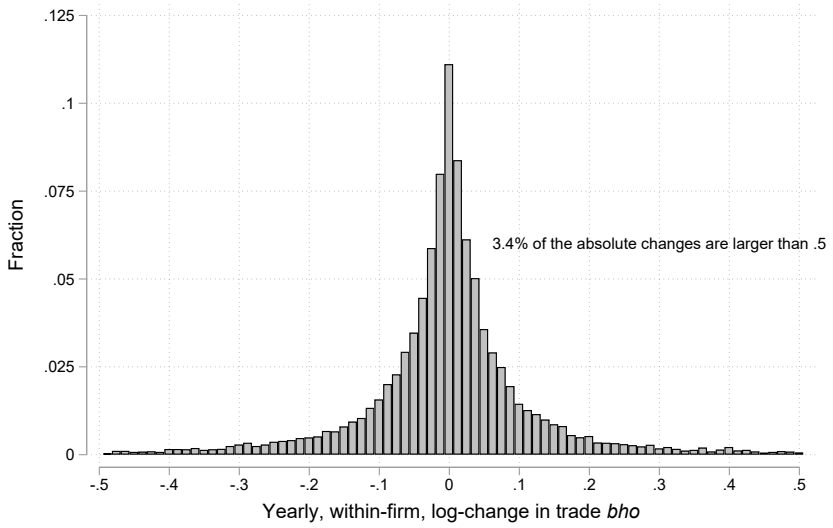
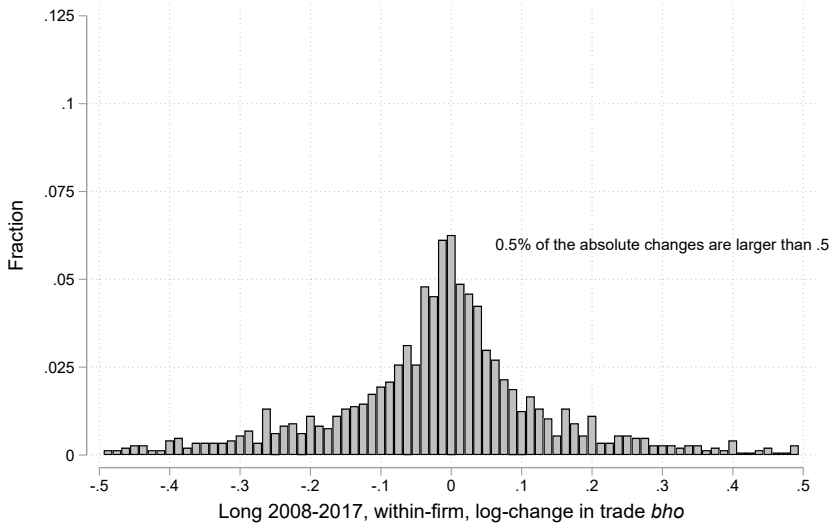


Figure A4: Distribution of within-firm log changes between 2008 and 2017 in trade *bho*



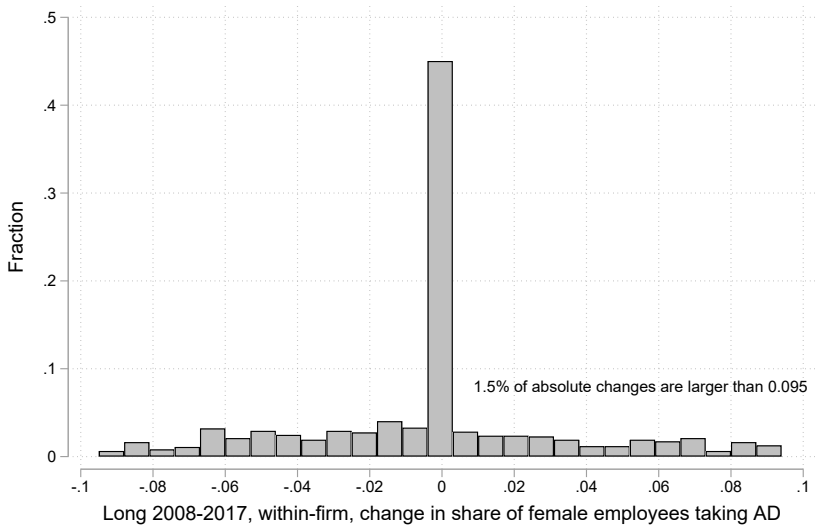
Note: Only firms present in both 2008 and 2017 are included

Table A4: Statistics on the absolute value of within-firm yearly log changes in *bho*

	Affiliate *bho*	Trade *bho*
%-share equal to zero	79.20	1.81
80th percentile	0.00	0.14
90th percentile	0.07	0.26
95th percentile	0.18	0.43
99th percentile	0.69	1.02

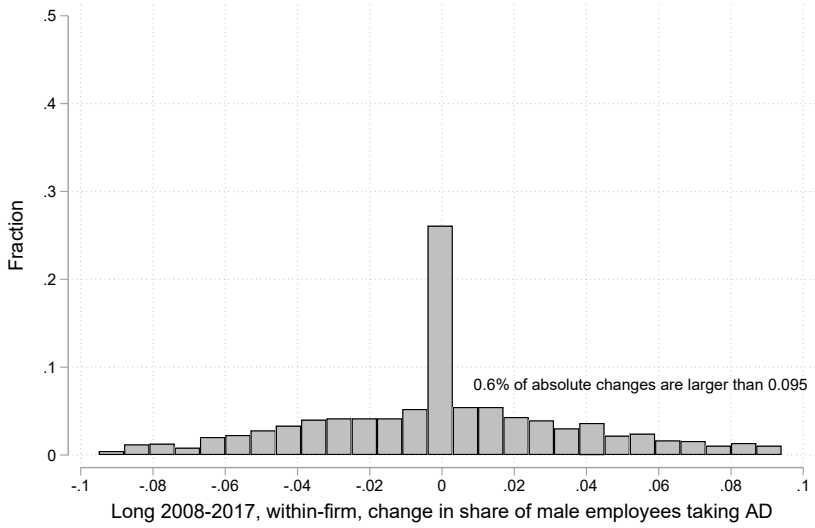
The changes for affiliate *bho*, only include the internal margin, so changes for firms that have foreign affiliates

Figure A5: Distribution of long 2008 to 2017, within-firm, change in share of female workers taking AD



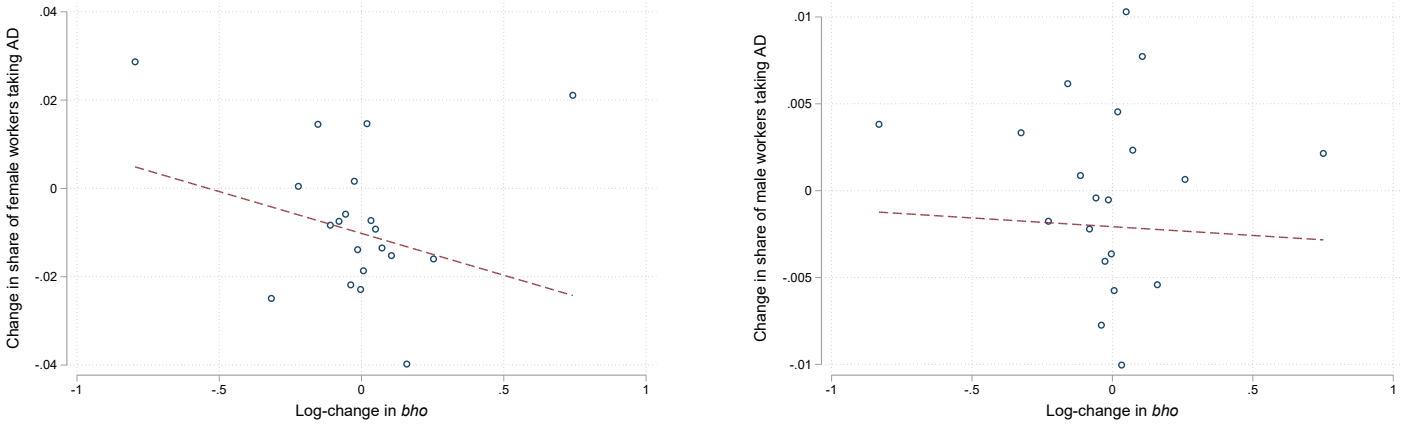
Note: Only firms with female workers in both 2008 and 2017 are included

Figure A6: Distribution of long 2008 to 2017, within-firm, change in share of male workers taking AD



Note: Only firms with male workers in both 2008 and 2017 are included

Figure A7: Within-firm change in parental leave take-up and firm *bho*, 2008–2017



(a) Female workers **(b)** Male workers

Note: Binned scatter plots of the long difference (2008–2017) in within-firm share of workers taking AD against log *bho*. Panel (a) shows female workers and Panel (b) shows male workers. Only firms with workers of the given gender in both 2008 and 2017 are included. The fitted lines are regression lines.

B O*NET survey

Below we reproduce the questions used in the O*NET survey and Goldin's interpretation (see notes to Table 2 in Goldin (2014)). These are:

- Time pressure: *How often does this job require the worker to meet strict deadlines?* Lower pressure means worker does not have to be around at particular times.
- Contact with others: *How much does this job require the worker to be in contact with others (face-to-face, by telephone, or otherwise) in order to perform it?* Less contact means greater flexibility.
- Establishing and maintaining interpersonal relationships: *Developing constructive and cooperative working relationships with others, and maintaining them over time.* The more working relationships, the more workers and clients the employee must be around.
- Structured versus unstructured work: *To what extent is this job structured for the worker, rather than allowing the worker to determine tasks, priorities, and goals?* If the job is highly structured to the worker, there would be a lower chance that the worker would have close substitutes.
- Freedom to make decisions: *How much decision-making freedom, without supervision, does the job offer.* Generally means that the worker determines what each client should receive, rather than being given a specific project, and thus workers are poorer substitutes for each other the greater are these freedoms.

C Extensions and robustness checks

Table C1: The effect AD usage, controlling for trade shares and varieties

	(1)	(2)	(3)	(4)	(5)
Panel A: Only workers younger than 45					
Female × ln(bho trade)	-0.0041	-0.0128**	-0.0124**	-0.0112**	-0.0111**
	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)
Female × ln(varieties)	0.0017**	0.0016	0.0015	0.0002	0.0002
	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
Female × $\frac{Exports}{Sales}$	0.0033	0.0032	0.0035	0.0025	0.0026
	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)
Female × $\frac{Imports}{Purchases}$	-0.0055	-0.0079	-0.0080	-0.0073	-0.0074
	(0.005)	(0.006)	(0.006)	(0.006)	(0.006)
N	995321	954291	950103	950103	950103
Average AD usage	0.0410	0.0402	0.0401	0.0401	0.0401
Panel B: Only college workers younger than 45					
Female × ln(bho trade)	-0.0047	-0.0183**	-0.0192**	-0.0173**	-0.0171**
	(0.006)	(0.008)	(0.008)	(0.008)	(0.008)
Female × ln(varieties)	0.0027**	0.0021	0.0022	-0.0001	-0.0002
	(0.001)	(0.002)	(0.002)	(0.003)	(0.003)
Female × $\frac{Exports}{Sales}$	0.0044	0.0063	0.0064	0.0074	0.0075
	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)
Female × $\frac{Imports}{Purchases}$	-0.0118*	-0.0047	-0.0049	-0.0028	-0.0025
	(0.007)	(0.009)	(0.009)	(0.009)	(0.009)
N	333183	322365	321274	321274	321273
Average AD usage	0.0308	0.0304	0.0304	0.0304	0.0304
Firm-year FE	✓	✓	✓	✓	✓
Worker FE	✓				
Spell FE		✓	✓	✓	✓
Occupational FE			✓	✓	✓
Occupational-Female FE					✓
Female-year FE				✓	✓
Education FE	✓	✓	✓	✓	✓

Further controls are experience, experience squared and having children. Occupational FE are based on 2-digit ISCO-code. Educational FE have 12 levels. The sample includes workers aged 20–60, in manufacturing firms that export and import and have more than 10 employees, in the years 2008–2017. Standard errors, clustered by firm, in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C2: Effect of business hours overlap on gender gap in antidepressant use: alternative subsamples

	(1)	(2)	(3)	(4)	(5)
Panel A: Only non-college workers					
Female × ln(bho trade)	-0.0008	-0.0016	-0.0012	-0.0007	-0.0007
	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)
N	1338693	1292939	1287290	1287290	1287289
Average AD usage	0.0482	0.0475	0.0475	0.0475	0.0475
Panel B: Only non-college workers younger than 45					
Female × ln(bho trade)	-0.0026	-0.0101	-0.0093	-0.0088	-0.0087
	(0.005)	(0.007)	(0.007)	(0.007)	(0.007)
N	654642	625459	622379	622379	622378
Average AD usage	0.0462	0.0453	0.0453	0.0453	0.0453
Panel C: Only workers older than 45					
Female × ln(bho trade)	-0.0021	0.0009	0.0009	0.0011	0.0011
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
N	876335	854173	851017	851017	851017
Average AD usage	0.0474	0.0471	0.0471	0.0471	0.0471
Panel D: Only college workers older than 45					
Female × ln(bho trade)	0.0027	-0.0010	-0.0001	-0.0005	-0.0015
	(0.009)	(0.012)	(0.012)	(0.012)	(0.012)
N	206903	201382	200748	200748	200748
Average AD usage	0.0400	0.0398	0.0397	0.0397	0.0397
Panel E: Only non-college workers older than 45					
Female × ln(bho trade)	-0.0059	0.0001	0.0001	0.0003	0.0005
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
N	664970	648507	645996	645996	645996
Average AD usage	0.0497	0.0495	0.0495	0.0495	0.0495
Worker FE	✓				
Firm-year FE	✓	✓	✓	✓	✓
Spell FE		✓	✓	✓	✓
Occupational FE			✓	✓	✓
Female-year FE				✓	✓
Occupational-Female FE					✓
Education FE	✓	✓	✓	✓	✓

Further controls are experience, experience squared and having children. Occupational FE are based on 2-digit ISCO-code. Educational FE have 12 levels. The sample includes workers, in manufacturing firms that export and import and have more than 10 employees, in the years 2008–2017. Standard errors, clustered by firm, in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C3: Controlling for time-varying GGI

	(1) All	(2) No college	(3) College
Female × ln(bho trade)	-0.0166***	-0.0158**	-0.0202**
	(0.006)	(0.008)	(0.008)
Female × ln(GGI trade)	0.0787	0.1020	0.0661
	(0.050)	(0.065)	(0.087)
Spell FE	✓	✓	✓
Firm-year FE	✓	✓	✓
Occupational FE	✓	✓	✓
Occupational-Female FE	✓	✓	✓
Female-year FE	✓	✓	✓
Education FE	✓	✓	✓
N	942,297	615,847	320,122
Firms	2,729	2,676	1,992
Women	61,060	34,915	26,160
Men	127,683	90,127	37,718
Average AD usage	0.0401	0.0453	0.0303

GGI trade is World Economic Forum's Gender Gap Index, weighted by the firm's trade variety. In each year the GGI for each country is the closest available value. Occupational FE are based on 2-digit ISCO-code. Educational FE have 12 levels. The sample includes workers aged 20–44, in manufacturing firms that export and import and have more than 10 employees, in the years 2008–2017. Std. errors, clustered by firm in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C4: Effect of business hours overlap on gender gap in exit

	(1) Exit sample	(2) Exit firm	(3) Switch firm
Panel A: All workers			
Female × ln(bho trade)	-0.0048	-0.0022	0.0026
	(0.007)	(0.008)	(0.004)
N	1630865	1630865	1630865
Average outcome	0.1023	0.1267	0.0244
Panel B: Only college workers			
Female × ln(bho trade)	0.0129	0.0081	-0.0048
	(0.014)	(0.017)	(0.008)
N	470480	470480	470480
Average outcome	0.0853	0.1155	0.0302
Panel C: Only workers younger than 45			
Female × ln(bho trade)	-0.0076	-0.0028	0.0048
	(0.009)	(0.010)	(0.005)
N	857030	857030	857030
Average outcome	0.0956	0.1247	0.0291
Panel D: Only college workers younger than 45			
Female × ln(bho trade)	0.0023	0.0097	0.0074
	(0.020)	(0.023)	(0.012)
N	285765	285765	285765
Average outcome	0.0800	0.1137	0.0337
Firm-year FE	✓	✓	✓
Spell FE	✓	✓	✓
Occupational FE	✓	✓	✓
Occupational-Female FE	✓	✓	✓
Female-year FE	✓	✓	✓
Education FE	✓	✓	✓

Exit sample means leaving the dataset either through unemployment or switching to a firm not included in the sample. Exit firm means exiting the firm either by leaving the sample or switching to another firm in our sample Further controls are experience, experience squared and having children. Occupational FE are based on 2-digit ISCO-code. Educational FE have 12 levels. The sample includes workers aged 20–60, in manufacturing firms that export and import and have more than 10 employees, in the years are 2008–2016. Standard errors, clustered by firm, in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C5: Antidepressant use and job characteristics: college workers under 45 with within job spell varying occupations

	(1)	(2)	(3)	(4)	(5)	(6)
Female × ln(bho trade)			-0.0168**	-0.0168**	-0.0144*	-0.0155**
			(0.008)	(0.008)	(0.007)	(0.008)
O*NET × ln(bho trade)	-0.0007		0.0004		0.0005	
	(0.000)		(0.001)		(0.001)	
D(O*NET) × ln(bho trade)		-0.0014*		-0.0000		0.0000
		(0.001)		(0.001)		(0.001)
Female × O*NET			-0.0045**		0.0021	
			(0.002)		(0.008)	
Female × D(O*NET)				-0.0056*		-0.0012
				(0.003)		(0.016)
Female × O*NET × ln(bho trade)					-0.0041	
					(0.005)	
Female × D(O*NET) × ln(bho trade)						-0.0026
						(0.009)
Spell FE	✓	✓	✓	✓	✓	✓
Firm-year FE	✓	✓	✓	✓	✓	✓
Occupational FE	✓	✓	✓	✓	✓	✓
Occupational-Female FE	✓	✓	✓	✓	✓	✓
Female-year FE	✓	✓	✓	✓	✓	✓
Education FE	✓	✓	✓	✓	✓	✓
N	318,746	318,746	318,746	318,746	318,746	318,746
Firms	2,028	2,028	2,028	2,028	2,028	2,028
Women	26,056	26,056	26,056	26,056	26,056	26,056
Men	37,564	37,564	37,564	37,564	37,564	37,564
Average AD usage	0.0303	0.0303	0.0303	0.0303	0.0303	0.0303

O*NET variable is Avg. of Time Pressure, Contact with others, Structured versus Unstructured Work, Freedom to Make Decisions, Establishing and Maintaining Interpersonal Relationships. Further controls are experience, experience squared and having children. Educational FE are based on 12 categories. The sample includes workersages 20–44 with a college degree in manufacturing firms that export and import and have more than 10 employees. D() denotes a dummy for the variable being above the 75th percentile. The years included are 2008–2017. Std. errors, clustered by firm in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

D Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jinteco.2026.104253.

The Online Appendix is available at https://ars.els-cdn.com/content/image/1-s2.0-S0022199626000437-mmc1.pdf.

The replication files are available at https://data.mendeley.com/preview/9pnfhj998z?a=2797150c-8baa-4aa9-9aee-4d3dcc1cd618.

Chapter 2

Perceived Wage Differences and Early Career Job Search Choices

Chapter 2

Perceived Wage Differences and Early Career Job Search Choices

Asker Nygaard David Nikolaj Arpe Harmon
Sonja Settele Daphné Skandalis

Abstract

Workers' early career jobs are a key determinant of their lifetime earnings—and yet workers may misdirect their search if they misperceive wage differences across jobs. We study this using a survey of early career job seekers in Denmark linked to administrative data. The survey (i) elicits beliefs about typical wages in three jobs relevant to each respondent, (ii) provides a randomized information treatment revealing the actual value of these typical wages, and (iii) elicits beliefs about their own expected wages and planned applications. Comparing beliefs to actual values reveals that gaps in typical wages across jobs are largely misperceived. While some wage gaps are overestimated, most are underestimated, making lower-paying jobs appear overly attractive overall. Using the randomization, we find that typical wage information has large effects on respondents' beliefs about their own expected wages and their planned applications, in the direction predicted by their initial misperceptions. These effects translate into real labor market outcomes: in administrative data, the information shifts the wages respondents receive in their first post-survey job—upward for those who underestimated wage gaps and downward for those who overestimated them. Embedding our results in a discrete choice framework, we estimate that wage misperceptions misdirect about 9.5% of job applications at labor market entry, resulting in the same welfare loss as a 0.8% reduction in all wages.

JEL-Codes: J31, J62, J64, D83, D84

Contact: Asker Nygaard David, University of Copenhagen, anc@econ.ku.dk. Nikolaj Harmon, University of Copenhagen, nikolaj.harmon@econ.ku.dk. Sonja Settele, University of Cologne, ECONtribute and MPI for Behavioral Economics, settele@wiso.uni-koeln.de. Daphné Skandalis, University of Copenhagen, daphne.skandalis@econ.ku.dk. **Acknowledgements**: We thank Sydnee Caldwell, Michele Belot, Teodora Boneva, Ingrid Haegele, Philipp Kircher, Roland Rathelot, Jeremias Klaeui and Michael Siegenthaler. Georgios Louvaris, Annasofie Marckstrøm Olesen, Sophie Laura Bune, Dánjal Hátún Augustinussen, Carl Benedikt Huels, and Emma Thevissen provided excellent research assistance. **Funding**: Financial support through the Rockwool Foundation (project: 3039) is gratefully acknowledged. Settele: Funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy – EXC 2126/2-390838866. The activities of CEBI are funded by the Danish National Research Foundation (Grant DNRF134) **Ethics approval:** Ethics approval was received from the IRB of the University of Copenhagen. **Pre-registration:** The data collection was pre-registered at the AEA RCT Registry (https://www.socialscienceregistry.org/trials/11592).

1 Introduction

Where young workers start their careers has lasting effects on their trajectories.[1] Yet choosing an entry-level job requires comparing career options without much experience to judge. These options can differ sharply in both pay and non-wage amenities. If workers misperceive how pay differs across careers, they sort into the wrong ones: some settle for needlessly low-paying careers, while others chase high pay into careers they would have traded for better non-wage amenities. A large literature shows that workers misperceive labor market conditions, but it focuses on the effects on how fast they find a job, not which career they pursue. Here, we investigate whether misperceptions steer workers into the wrong careers—at labor market entry, when mistakes are costliest to reverse.

We answer this question by bringing together three ingredients rarely combined: a discrete-choice model of career entry, a tailored survey experiment that manipulates wage beliefs, and administrative records of the jobs respondents actually take. Concretely, we designed a new survey of 1,902 early career job seekers in Denmark, which we link to administrative data on individual job outcomes. We structure the survey to map into a discrete choice model of realistic career paths: for each respondent we construct three personalized job categories—defined by salient dimensions such as sector (private vs. public), industry, occupation, and firm size—that represent genuine career options given their education major. We first elicit incentivized beliefs about the average wage earned by prior cohorts from the respondent's own major, which we compare to ground-truth administrative wages to obtain a clean measure of misperception. We then embed a randomized information treatment that shows treated job seekers the actual average wages of prior cohorts in each category, allowing us to identify the causal effect of correcting these wage misperceptions. We subsequently elicit job seekers' expectations about the wages *they themselves* would earn in each category—the beliefs relevant for their choices—along with their planned applications. Finally, we link respondents to administrative records on realized job outcomes, testing whether the treatment's effects persist beyond stated responses into actual choices

[1] The literature highlights the persistent effects of graduating in a bad labor market (e.g. von Wachter and Bender (2006), Kahn (2010), Cockx and Ghirelli (2016), Wachter (2020)) and of getting a first job at a large firm Arellano-Bover (2024).

and addressing the concern that survey responses may not reflect real decisions.

The first part of our analysis provides novel evidence about the extent and nature of relative wage misperceptions—i.e. misperceptions about how offered wages differ across jobs. Using survey responses about average wages of prior cohorts, we compute job seekers' perceptions of the typical wage gap between each of the three jobs in the survey. We then compare these to the actual gaps in wages among prior cohorts in administrative data. Perceived and actual wage gaps are clearly related (regression slope 0.54; correlation 0.27), suggesting that job seekers are broadly aware of wage differences across jobs. At the same time, substantial relative wage misperceptions exist: In 80% of cases, the perceived wage gap between two jobs differs from the truth by more than 50%. While there is substantial heterogeneity, the typical pattern is that job seekers understate the true gap in wages so that lower-paying jobs appear relatively more attractive. Approximately 65% of wage gaps are underestimated, with 55% underestimated by more than 50% and 36% underestimated by more than 100%—meaning respondents actually perceive the lower-paying job to pay higher wages. This pattern of predominantly underestimated wage gaps holds at the individual level as well: two-thirds of job seekers in our sample are underestimators, in the sense that they underestimate the gap between the highest-paying job and the other two jobs.

We substantiate that these measured misperceptions are practically meaningful in a number of ways. Since we focus on beliefs about an objective benchmark—realized average wages of prior cohorts—the observed misperceptions cannot be explained by respondents having private information about their own situation or about labor market conditions. Yet, we verify in the data that prior cohort wages are indeed relevant for early career workers: beliefs about typical wages for prior cohorts are strongly correlated with beliefs about the potential wages job seekers themselves expect to face. Examining alternative measures and subsamples, we also find no evidence that measured misperceptions are driven by job seekers misunderstanding the three jobs asked about in the survey or not viewing these jobs as relevant for themselves. In line with theories of rational inattention, we find some evidence that job seekers are better informed when true wage gaps are larger and thus more consequential. However, misperceptions are large in these cases as well.

In the second part of our analysis, we leverage the randomized information treatment to establish the causal effects of relative wage information on job search. After job seekers have reported their beliefs about wages of prior cohorts, treated job seekers are shown the actual wages for prior cohorts allowing them to infer their misperceptions about the different jobs. Using post-treatment survey questions, we show that this causes job seekers to update beliefs about the wage differences they themselves expect to face and also changes their planned application behavior accordingly. When job seekers are informed that the wage gap between two jobs for prior cohorts is in fact $X\%$ larger than they thought, the perceived gap in wages that the job seekers themselves expect to face increases by $0.3X\%$. In turn, the relative likelihood of applying to the two jobs changes by $1.2X\%$.

Combining the reduced form treatment effects on wage beliefs and application behavior, we estimate directly how beliefs about own potential wages affect job search. Formally, this amounts to a 2SLS estimator where the information shock from our treatment serves as the instrument for the perceived gap in potential wages between jobs. We find a large effect: a 1% change in the perceived wage gap between two jobs changes the relative likelihood of applying by 4.2%. The 2SLS estimate can be interpreted as the pure effect of wage beliefs under an exclusion restriction: our information treatment should only affect applications by changing beliefs about potential wages. Importantly, we can test this assumption across a range of post-treatment survey questions. The treatment does not significantly affect beliefs about the likelihood of getting a job conditional on applying; the estimates are small and, if anything, go against the usual concern that learning a job pays more makes workers view it as harder to get. Nor does it significantly affect perceived non-wage amenities, where the estimates are again small. Given these results, we interpret the 2SLS estimate as capturing the effect of wage beliefs throughout the rest of the paper.

The next key step of our analysis is to verify that our estimated treatment effects on planned applications also translate into actual labor market behavior, leveraging our linkage to individual outcomes in administrative data. This analysis does not aim to evaluate a specific information policy—our light-touch intervention is not designed to mimic any realistic policy. Rather, it is crucial for establishing that our estimated effects are not confined to survey responses, a common concern with survey experiments that often goes

unaddressed. By design, our information treatment yields clear, directional predictions for post-survey job outcomes that can be tested in administrative data. For the majority of job seekers, who tend to underestimate true wage gaps, misperceptions make lower-paying jobs appear relatively more attractive. The information treatment should therefore shift such job seekers to apply for and get hired into higher-paying jobs, with the reverse holding for those who overestimate true gaps. We confirm these predictions. Among the two-thirds of job seekers who are underestimators, the information treatment significantly increases the actual wages earned in the first post-survey job, while we see evidence of a negative effect for overestimators. As expected, these wage effects are driven by job seekers still actively searching at the time of the survey, who can adjust their behavior to new information without incurring switching or search costs.

Finally, we provide a quantification of the overall costs of relative wage misperceptions among early career workers. To this aim, our survey and reduced form analyses were deliberately set up to identify a simple discrete choice model of job search and hiring. We use this to compute counterfactual application and hiring outcomes in a situation without relative wage misperceptions. We find that existing relative wage misperceptions distort 9.5% of job applications and lower welfare by the same amount as an across-the-board wage decrease of 0.8%. The nature of this welfare loss is highly heterogeneous however. For underestimators, it reflects that misperceptions push them to apply for too many lower-paying jobs. For overestimators, it reflects that misperceptions push them to apply for too many high-paying jobs with low hiring likelihoods and/or poor non-wage amenities.

This paper contributes to a literature on workers' misperceptions about the labor market. A first strand studies misperceptions about the rate of job finding or the overall level of reemployment wages, and shows that such misperceptions prolong unemployment spells (e.g. Krueger and Mueller (2016), Spinnewijn (2015), Mueller et al. (2021), Belot et al. (2019), Belot et al. (2025), Behaghel et al. (2024), Altmann et al. (2018), Ben Dhia et al. (2022), Altmann et al. (2022), Harmon et al. (2024), Altmann et al. (2025)). We depart from this literature by studying a different margin: rather than how long workers search, we examine which jobs they take, and how misperceptions about wage dispersion distort that choice. A second strand asks whether such misperceptions affect mobility among *em-*

ployed workers (Jäger et al., 2024) and finds that high switching costs restrict mobility even when workers know of outside jobs paying higher wages (Caldwell et al., 2025). We focus instead on the *initial* allocation of workers across jobs at labor market entry—a primary determinant of lifetime earnings. Here misperceptions likely play a larger role: because early career job seekers face no switching costs, wage information bears more directly on the decision in front of them.

We also relate to a literature studying the own wage elasticity of job applications, where prior studies analyzing wage variation in online job ads typically find elasticities below 1 (Holzer et al., 1991; Dal Bó et al., 2013; Banfi and Villena-Roldán, 2019; Marinescu and Wolthoff, 2020; Dube et al., 2020; Belot et al., 2022; Azar et al., 2022). Our 2SLS estimate yields a larger elasticity of around 3. One suggestive interpretation is that job seekers in online job-ad settings may interpret wage premiums as signals of reduced hiring probabilities or lower non-wage amenities; post-treatment survey questions in our setting show no such signaling effects, which may help explain the larger elasticity we obtain.

Finally, we connect to the literature on college major choice by studying students' next career step (Beffy et al., 2012; Wiswall and Zafar, 2015; Baker et al., 2018; Kerr et al., 2020). Students' beliefs about major-specific earnings are a significant determinant of major choice, but the estimated responsiveness is small.[2] This suggests that wage information bears far more directly on job choices at entry than on the educational choices made years earlier. Moreover, whereas the wage consequences of major choice realize only years later, the short horizon at labor market entry lets us link respondents to administrative data and verify that the treatment shifts not only stated choices but realized wages. From a policy perspective, labor market entry may therefore be a particularly promising moment to provide students with information about wages across careers.

The rest of the paper is organized as follows. Section 2 discusses the survey design and data. Section 3 provides novel descriptive evidence about misperceptions of wage differences across jobs. Section 4 establishes the causal effects of these misperceptions on job search. Section 5 quantifies aggregate implications using a discrete choice model.

[2] Wiswall and Zafar (2015) and Beffy et al. (2012) report elasticities of major choice to expected earnings well below 0.1, two orders of magnitude smaller than the elasticity of job applications we estimate at entry.

Section 6 concludes.

2 Data and survey design

To measure relative wage misperceptions, we fielded a survey with early career job seekers in Denmark in the summer of 2023. Below, we first describe our setting and the linked administrative data that we use. We then describe the survey data in detail. At the end, we discuss the reliability of our survey measures and present a range of validity checks.

2.1 Empirical setting

Our analysis focuses on graduates from higher education in Denmark. We focus on graduates early in their career since we expect relative wage misperception to be particularly prevalent and consequential when workers have limited labor market experience and may face persistent effects from early job choices. Our focus on Denmark allows us to leverage rich administrative registers in the design, sampling and analysis of the survey. A few other features of the Danish setting are important as well. While most jobs in Denmark are covered by collective bargaining agreements, wage setting is highly decentralized, thus generating substantial wage dispersion across jobs (Dahl et al., 2013). As in most countries, higher education degrees in Denmark typically qualify candidates for a number of distinct careers, meaning graduates face a choice between a set of different jobs and careers, often offering very different terms and wages. Whether workers correctly perceive these wage differences across jobs is an empirical question: On the one hand, workers have limited experience with wages early in their career and—as is the case in many countries—direct information about wages is rarely included in job postings. On the other hand, there are many sources of wage information that active job seekers could rely on in the absence of information costs or behavioral biases.[3] Finally, an important feature of the Danish setting is that graduates can claim unemployment benefits immediately upon graduation. As we discuss below, high-frequency claims data therefore help us survey workers right at the

[3]For example, many Danish worker unions publicize information about typical wages in various categories of jobs.

start of their job search.

2.2 Administrative data

We use administrative data for two main purposes in this project. First, we use it in the design of our survey to identify which types of jobs are relevant for graduates from different degrees, and to construct ground truth measures of the typical wages that these jobs have paid in the past. This is based on the transitions of higher-education graduates into their first jobs, which we observe using individual-level data on labor market outcomes (BFL) and educations (UDDA) for individuals who graduated between 2010 and 2018. Second, we link our survey sample to administrative data to obtain background characteristics and post-survey labor market outcomes. Specifically, we link the Danish registers on individual labor market outcomes (BFL), educations (UDDA), government transfers (DREAM) up to December 2024.

2.3 Definition of job types

Throughout the survey, we ask each respondent about three types of jobs relevant for young workers with their educational background. Below we summarize how we define these job types. Additional details are given in Appendix C.1.

A strength of our research design is that we analyze graduates not from a single degree but from a wide range of majors. Since different jobs and career paths are relevant for graduates from different majors however, this requires us to create job types that are specific to each major. For each major (measured using 6-digit Danish ISCED codes), we construct these tailored job types to achieve three objectives: First, the job types must be relevant and capture a substantial share of respondents' realistic employment opportunities. Second, they should differ in the wages they offer to ensure meaningful trade-offs. Third, they must correspond directly to categories in administrative records so that we can compute objective benchmarks for typical wages using administrative data.

In practice, for each major, we define the three job types by grouping jobs based on some combination of occupation, industry, firm sector, and firm size. We implement a data-

driven procedure to select the appropriate job groupings using administrative data on the first jobs of graduates from the same major between 2010-2018: For each major, we consider all possible ways of grouping jobs by firm sector and size (public/private, above or below 50 employees), occupation codes (using either 3, 4, or 6 digit Danish ISCO codes), and/or industry codes (using either 1- or 2-digit Danish NACE codes). Among all such possible groupings, we then select those that satisfy two conditions: i) the 3 most common job types must cover more than 40% of the transitions into first jobs for prior graduates and ii) the 3rd most common job type must cover at least 5% of transitions. These conditions ensure that the three most common job types altogether cover a substantial share of relevant jobs and that none of them are too narrow. If multiple groupings satisfy the two conditions, we choose the grouping that best predicts entry-level wages.

Some smaller majors cannot be associated with three meaningful job types using this procedure, either because they concern too few graduates or because their graduates all end up in virtually the same type of job. Graduates from these majors were automatically excluded from our study.[4] In the end, our analyses focus on 88 higher-education majors, covering 76% of higher-education graduates in 2010-2018 (see Appendix Table A1). We refer to these as the "selected" majors.

After hand-checking the resulting job types and making marginal adjustments, this procedure gives us three job types for each major that we can describe intuitively to the survey respondents. For example, for respondents with a Master in Economics, the procedure selects a grouping by sector (public vs. private) and 1-digit industry, producing three job types: (1) private-sector jobs in financial and insurance activities, (2) public-sector jobs in public administration and defense, and (3) private-sector jobs in business services. For respondents with a Master in Physics, it instead selects a grouping by sector, firm size, and 3-digit occupation, producing: (1) research and teaching in higher education, (2) teaching at the upper-secondary level, and (3) software development in large private firms. The two examples illustrate how the procedure adapts to each major's labor market: industry distinctions matter more for economists, while occupational distinctions matter

[4]Graduates from these majors could receive a survey invitation but their version of the survey did not ask about detailed job types and did not contain an information treatment.

more for physicists. Table C1 lists the exact job descriptions shown to respondents and Appendix C.1.1 provides the full list of majors and job types. Because the procedure is designed to produce job types that are meaningful to respondents, we included several questions in the survey to assess how well respondents understood them. As discussed in Section 2.6, the responses confirm that the job types were generally well understood, and also allow us to provide robustness analyses restricted to the subsample of respondents for whom comprehension was strongest.

2.4 Sampling procedure

It is important for us to survey workers immediately around the time when they actively search. Otherwise, we may overstate the extent of misperceptions, if workers only seek out detailed wage information right when they need it. Using two sampling procedures, our survey targeted individuals who are either looking for their first job after graduation or are entering unemployed job search within a few years following their graduation. First, we collaborated with the Danish Agency for Labor Market and Recruitment (STAR) to contact all individuals below age 40 who initiated a claim for unemployment benefits during the summer of 2023. Since most education programs end in the summer and many graduates immediately claim benefits, sampling during these months allows us to reach a large fraction of new graduates.[5] Second, we collaborated with the University of Copenhagen (UCPH) to contact all UCPH students about to finish their Master's degree in the summer of 2023. We invited them into the survey in mid-June 2023, i.e. right before typical Master graduation dates. This second sampling procedure allows us to also survey graduates who enter the labor market without going through unemployment.

Sampled individuals were invited to the survey in three waves via the so-called "e-Boks" e-mail system—the official channel through which government agencies in Denmark communicate with citizens.[6] The invitation contained an individualized link to the online

[5] Økonomi og Indenrigsministeriet (2018) report that about half of Danish graduates receive unemployment benefits within 6 months of graduation.

[6] The first wave in mid-June included all individuals sampled from UCPH, as well as new unemployment benefit claimants from the past three weeks. The second and third wave were conducted three and eight weeks later and each included only benefit claimants from the preceding three weeks. Due to an error by our "e-Boks" provider, some observations were lost in the second wave, as explained in Appendix D.1.

survey and provided a monetary incentive to participate.[7] Appendix Table A2 provides descriptive statistics for our sample of invitees and participants. 22,461 people from our selected majors were invited to participate in our survey (Column (2)), 1,902 completed the survey and are included in our study sample (Column (1)). Appendix Table A2 shows that the sample of all invited individuals is quite similar to our study sample in their demographic characteristics, although individuals in the study sample do tend to have higher educational attainment (63% have completed a Master's degree) than the invited population (46% with a Master's degree)—a common phenomenon in online surveys (Haaland et al., 2023). Moreover, survey participants are slightly more likely to have graduated less than a year before the survey (66%, compared to 44% in the invited population). A similar picture emerges when we compare our study sample to the full population of young job seekers. As we show in Table 3 and Appendix Table A4, our conclusions are unchanged if we reweight the study sample to account for non-participation.

2.5 Survey questions

This section describes the key elements of the survey. Additional details, including the full survey instructions, are provided in Appendix C.2.

Belief about prior-cohort average wages The first part of the survey aims to measure how well informed respondents are about typical wage differences across their possible career paths. Chiefly, we elicit respondents' belief about the entry-level *average wage received by prior cohorts* in the three different job types. For each job type, we ask respondents to consider people who graduated in 2010-2018 from the same major as themselves and started working full-time in this job type. We then ask them to report the average pretax monthly wages over the first year of employment for these individuals. To help respondents understand the wage concept we use, we show them the average wage received by prior cohorts with *any* higher education diploma and in *any* job type. This anchor contains no information about wage differences across jobs but clarifies the scale on which

[7] Among those who finished the survey, we raffled off 20 gift cards worth DKK1,000 (approx. USD160) each.

we measure wages (e.g. monthly vs. hourly/yearly wages) and thereby reduces potential measurement error in beliefs (Ansolabehere et al., 2013). Appendix Figure A5 shows that there is no bunching around the anchor. To incentivize effort and truthful reporting, we provide a monetary incentive to get close to the true value of the average wages computed in administrative data: the 3 closest answers received a gift card worth DKK1,000.

In a similar way, we also elicit respondents' beliefs about two other statistics for prior cohorts: the average wage earned 5 years after starting in each job type, and the rate of successful applications sent to each job type. Along with a range of socioeconomic questions at the start of the survey, these beliefs elicited before the information treatment serve as predetermined covariates in various parts of our analysis (cf. Table 1).

Randomized information about prior-cohort average wages Next, the survey includes a randomly assigned information treatment. The treatment group is informed of the true average prior-cohort wages in each of the three job types. This information is illustrated by a figure listing the true average prior-cohort wages in the three job types alongside the respondent's own answers. The control group is shown a corresponding figure only containing their own answers (see Appendix Figures C1-C2).

Planned search behavior and beliefs about own potential wages After the information treatment, we ask respondents to think of three specific jobs that they view as representative of each of the three job types asked about in the survey. We then elicit their planned search behavior and beliefs about what they themselves expect to earn in these jobs as described below.

To measure search behavior, we ask respondents to consider a situation in which they have found job postings for each of three jobs and are able to apply to at most one of them. We then ask them to report the likelihood that they would end up applying to each job posting or to none of them. Answers are reported as probabilities, recorded using four sliders that range from 0.01-0.97 and automatically adjust to make the probabilities sum to 1. Eliciting self-assessed probabilities instead of a single deterministic choice offers two advantages here (Wiswall and Zafar, 2018). First, it increases the information content of

the data because respondents' answers express the strength of their preferences for applying to different jobs rather than just their highest ranked option. Second, it implies that the respondents' answers can be mapped directly to a standard discrete choice framework with decision uncertainty. We use this to compute counterfactuals in Section 5. As an alternative measure of job attractiveness, respondents are also presented with a scenario in which they simultaneously received mutually exclusive job offers from all three jobs and are then asked to report the likelihood of accepting each offer.

Job seekers' application decisions should depend on the potential wage they expect to face in each job type. To elicit beliefs about such *own potential wages*, we next ask respondents to report what starting monthly pretax wage they expect they would earn if hired into each of the three jobs.

Finally, we ask a range of questions about other job characteristics the job seeker expected to face in each of the three jobs. These include work hours, earnings 5 years in the future, the probability of receiving an offer if applying, and how well they expected to perform and get along with colleagues (both on 6-item Likert scales). As a summary measure of the perceived attractiveness of non-wage amenities in the three jobs, respondents are also presented with a scenario in which they simultaneously received mutually exclusive job offers from all three jobs but where the jobs all involved the same wage. As above, job seekers are then asked to report the likelihood of accepting each job offer. As we expand on in Section 4.4, we use these additional questions to assess belief spillovers from the information treatment.

Winsorization Typed numerical survey responses are prone to generating outliers. We winsorize all such variables at the 2.5th and 97.5th percentiles prior to our analyses. This includes survey-reported wages, hours and application success probabilities.

2.6 Discussion of survey data reliability

Understanding of job types A crucial question for our survey design is whether respondents properly understood the three job types they were asked about. We validate this in two ways. First, we directly asked respondents how well they understood the job types:

75% of respondents reported that their understanding was good or very good (Appendix Figure A1). Second, we tested their understanding at the end of the survey by asking some of them to place a concrete job example into one of the three types: 76% of the respondents answered correctly (Appendix Figure A2).[8] As we show later, our results are also robust to excluding respondents showing poor understanding in these questions (Tables 3 and A4).

Relevance of job types Another premise for our analysis is that the three survey job types are relevant to respondents and represent a trade-off in the wages they offer. In Table 2 we provide detailed descriptives on the job types based on both administrative and survey data. Columns correspond to different variables and data sources while panels correspond to different ways of grouping the job types across respondents. In Panel A, we group job types according to the share of prior cohorts who started their first job within each type. As expected, the job types cover a substantial share of starting jobs: the most common job type on average covers 26% of starting jobs for prior cohorts, the second most common 15% and the least common one 10% (Column (1)). In Panel B, we instead group jobs according to each respondent's stated likelihood of applying to the jobs in our survey. Answers confirm that the job types capture relevant employment options considered by job seekers. The reported likelihood of applying for their most preferred job is 62% on average, suggesting that most job seekers have a clear leaning towards one of the job types (Column (2)). The likelihood of applying for their second and third most preferred job types are 22% and 7%, showing that job seekers still have substantial uncertainty about their choice across the three jobs. The generally high reported likelihoods of applying also underscore that the job types were perceived as relevant by respondents; the average reported likelihood of not applying to any of the three job types was only 8%. Finally, in Panel C we group job types according to the average wage paid to prior cohorts and see substantial differences. For the average respondent, typical wages in the highest-paying job are 14% (4,180 DKK) higher than in the lowest paying one (Column (3)).

[8]For details on the creation of the examples, see Appendix Section C.1.2.

Prior cohort wages versus own potential wages Our survey elicits respondents' beliefs about two distinct wage concepts for each of the three job types: average entry-level wages for prior cohorts (elicited before the information treatment) and respondents' own potential entry-level wages (elicited after the information treatment). We focus on prior cohort entry-level wages because they constitute a useful benchmark for the typical entry-level wages that respondents themselves should face which we can compute objectively from administrative data. Contrasting administrative data with elicited beliefs about prior cohorts therefore allows us to construct an individual-level measure of wage misperceptions which is not confounded by job seekers having correct private information about their own labor market opportunities.[9] When making search decisions about which jobs to target, what should matter to job seekers is the potential wages they themselves would face in the different jobs. We therefore elicit these beliefs after the information treatment. This allows us to study the causal effect of wage beliefs on job search decisions using randomized treatment variation.

Planned applications versus real transitions Whenever survey respondents report their planned behavior, a critical question is how likely it is to reflect their actual future behavior. With the linked administrative data, we can validate our survey measure of planned job applications by comparing it to respondents' actual job transitions after the survey. We present these validations in Appendix Figure A3 and Appendix Table A8 and find a strong association. Comparing job seekers from the same major, a 1 percentage point increase in the reported likelihood of applying to some job types is associated with a 0.25 percentage point increase in the likelihood of being hired into this job type later. Conditional on being hired into one of the three job types covered by the survey, this association increases further to 0.66. Moreover, we show in Section 4.5 that the estimated effects of our information treatment on planned applications also translate to changes in actual hiring outcomes in administrative data.

[9]Measuring misperceptions using elicited beliefs about own potential wages raises a fundamental identification problem: the actual wage a job seeker *would* earn in a given job is only observed if the job seeker in fact ends up in this job. Trying to infer these unobserved actual wages from data on other workers will lead to erroneous measures of misperceptions if job seekers have correct private information about how their wage prospects differ from other workers.

Randomization and timing of survey Table 1 shows descriptive statistics for all pre-treatment covariates elicited in the survey, both for the full study sample and for the treatment and control group separately. Comparing treatment and control groups, treatment randomization appears to have been successful; across the 30 pre-determined covariates, we find statistically significant treatment-control differences for two variables at the 10% level. As shown later, all results are also robust to the inclusion of pretreatment controls. A key objective of our survey design was to measure wage beliefs around the time of active job search. Table 1 confirms that we were successful in reaching respondents at this time. 65% of respondents report actively searching at the time of the survey and another 24% report having just secured a new job or being about to start searching. In total 89% of respondents were thus either actively searching, had recently finished, or were about to start searching. As we show later, all results are also robust if we restrict the data to only include respondents who report actively searching at the time of the survey. Appendix Table A3 shows descriptive statistics and balance tests for this sample.

3 Measuring relative wage misperceptions

In this section, we provide descriptive evidence on the extent and nature of wage misperceptions. We do this by contrasting workers' beliefs about the wages of prior cohorts in different jobs with administrative data on the true wages earned by these cohorts.

3.1 Misperception about wage levels in individual jobs

While the focus of this paper is on misperceptions about relative wages across jobs, we begin by briefly considering general misperceptions about the level of wages. The analysis sample consists of 1,902 job seekers indexed by i, each of whom answered questions about three different jobs indexed by j. For each individual-by-job, i, j, our survey data contains the belief about the average wage earned by prior cohorts $\widetilde{V_{i,j}}$, which we compare to the true value $V_{i,j}$ computed from administrative data. Note that throughout the paper, we use tilde (e.g. $\widetilde{V_{i,j}}$) to denote beliefs and lowercase letters to denote logged variables (e.g. $v_{i,j} = \log V_{i,j}$).

For each job seeker and job, Figure 1 compares respondents' beliefs about the average wage earned by prior cohorts to the true values. Panel (a) shows a binscatter plotting the perceived log wage ($\widetilde{v_{i,j}}$) against the truth ($v_{i,j}$). There is a clear positive relationship but it is not one-to-one; the corresponding regression slope is 0.65. This is consistent with recent evidence in Caldwell et al. (2025): people's beliefs are highly but not perfectly correlated with true wages. Second, Panel (b) shows the full distribution of wage misperceptions, measured as the difference between the true and perceived wage in percent of the true wage, i.e., $\left(V_{i,j} - \widetilde{V_{i,j}}\right)/V_{i,j}$. Misperceptions about wages are substantial and heterogeneous. For some individuals and jobs, wages are overestimated ($V_{i,j} - \widetilde{V_{i,j}} < 0$), while for others they are underestimated ($V_{i,j} - \widetilde{V_{i,j}} > 0$). Wage overestimation dominates overall: wages are overestimated in 60% of cases, and the mean overestimation rate is 3.3%. This is in line with findings from the literature on misperceptions about general wage levels. Altmann et al. (2025) find that unemployed Danish job seekers overestimate re-employment wages of comparable workers by about 2% on average.

3.2 Relative wage misperceptions

Next, we turn to the key focus of our paper: misperceptions about wage *differences* across jobs. When job seekers decide which jobs to target in their search, it is these misperceptions about *relative wages* that should matter. Importantly, results on misperceptions about wage levels (such as those in Figure 1) say little about relative misperceptions: If a worker overestimates wages in all jobs to the same extent for example, misperceptions about wage levels can be large while there are no misperceptions about wage differences.[10] To measure these relative wage misperceptions, we focus on wage gaps between pairs of jobs. For each job seeker, we form all possible pairwise comparisons between the three jobs they were asked about, resulting in dyadic data where observations are individuals-by-job pairs, i, j, j'. Letting Δ denote differences in variables across jobs, our survey data allows us to compute job seekers' perceived wage gap between jobs j and j' for prior cohorts, either in

[10] Conversely, if a worker overestimates the wage in low paying jobs and underestimates wages in high paying jobs, misperceptions about wage differences can be large while misperceptions about wage levels are modest.

levels, $\Delta\widetilde{V_{i,j,j'}} = \widetilde{V_{i,j}} - \widetilde{V_{i,j'}}$, or in logs, $\Delta\widetilde{v_{i,j,j'}} = \widetilde{v_{i,j}} - \widetilde{v_{i,j'}}$. We compare these to the true gaps from administrative data, $\Delta V_{i,j,j'} = V_{i,j} - V_{i,j'}$ and $\Delta v_{i,j,j'} = v_{i,j} - v_{i,j'}$.

Figure 2 compares respondents' beliefs about prior cohort wage gaps across jobs to the true prior cohort gaps. Panel (a) shows a binscatter plotting perceived gaps in log wages ($\Delta\widetilde{v_{i,j,j'}}$) against true gaps ($\Delta v_{i,j,j'}$), based on respondent-by-job pairs (i, j, j') ordered such that $\Delta\widetilde{v_{i,j,j'}}$ is positive. Again, we see that beliefs are strongly but not perfectly correlated with the truth; the corresponding regression slope is 0.54.

Panel (b) characterizes the full distribution of relative wage misperceptions. While we focus on log wage gaps both in Panel (a) and the regression analyses presented later, our preferred measure when describing the distribution of misperceptions is the *underestimation rate*, i.e., the difference between the true and perceived gap in percent of the true gap:

$$\frac{\Delta V_{i,j,j'} - \Delta\widetilde{V_{i,j,j'}}}{\Delta V_{i,j,j'}} \tag{1}$$

The advantage of this measure is that it succinctly captures several different qualitative features of misperceptions: An underestimation rate of 0 implies correct beliefs. An underestimation rate between 0 and 100% means that the size of the true gap is being understated. An underestimation rate of 100% implies that the jobs are perceived to pay exactly the same, while an underestimation rate above 100% implies getting the relative wage ranking wrong. Finally, a negative underestimation rate means that the size of the wage gap is in fact being *over*estimated. Panel (b) shows the distribution of the underestimation rate across all individuals and job pairs.[11] In green, we present as a benchmark the CDF that we would observe under perfect information. In blue, the actual CDF highlights that in about 65% of cases, the wage gap is underestimated. Further, we see that 55% of wage gaps are underestimated by more than 50%, while 36% are underestimated by strictly more than 100%—meaning that respondents actually rank the two jobs wrong. We also see that 12% of wage gaps are underestimated by exactly 100%. This corresponds to situations where respondents attributed the exact same wage to two job types even though

[11] Appendix Figure A4 instead shows the raw distribution for the underestimation of the log wage gap, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$, Motivated by theory, this is the misperception measure emphasized in the regression analysis of Section 4.

the jobs in fact paid different wages. Finally, we also see that some wage gaps are instead overestimated, sometimes by a large amount. 25% of wage gaps are overestimated by more than 50% and 20% by more than 100%. Overall, in 80% of cases, the perceived wage gap between two jobs differs from the truth by more than 50%.

In Table 3, we show additional descriptive statistics regarding relative wage misperceptions using different measures and different subsamples of our data. The first row corresponds to the full sample and Columns (1)-(4) simply summarize key statistics regarding the wage gap underestimation rate. In Columns (5) and (6), we consider a different measure: the underestimation of the log wage gap, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$. Motivated by theory, this is the main misperception measure used in the regression analyses in Section 4.[12] Unsurprisingly, we also find large misperceptions using this measure; its mean absolute value is 0.13 log points implying that on average job seekers' beliefs about wage gaps are off by 13%. Appendix Figure A4 shows the distribution of this misperception measure across the sample of all job pairs. In Column (7), we show that the pattern of wage gaps being mostly underestimated holds also if we look across individuals instead of job pairs. Classifying each job seeker as either an *overestimator* or *underestimator* depending on whether they overestimate or underestimate the log gap between the highest-paying job and the average of the two other jobs, we find that 67% of job seekers in our sample are underestimators.

Heterogeneity and robustness To substantiate that the measured misperceptions are meaningful in practice, the additional rows of Table 3 describe the distribution of relative wage misperceptions, with various sample restrictions and changes in variable definitions.

A possible explanation for the large misperception we observe is that some true wage differences may be relatively small; if tracking wage differences is costly for job seekers, theories of rational inattention suggest that they may optimally decide to remain uninformed in these cases.[13] To test this, we restrict our sample to job pairs where the true wage gaps for prior cohorts are above 5% or 10%, respectively. Consistent with rational inattention,

[12] As we return to in Section 5, our reduced form specifications involving log wage gaps can be derived from a simple discrete choice job search model.

[13] For jobs where the true difference in typical wages is small, we may also worry that measured misperceptions stem simply from rounding in responses.

we find evidence that relative misperceptions are indeed smaller for job pairs with larger gaps. However, we continue to see very substantial misperceptions even in these samples. Even restricting to job pairs where the true gap is above 10%, we find that in 63% of cases, the perceived wage gap between two jobs differs from the truth by more than 50%.

The remaining rows dispel a number of other concerns. Measured misperceptions are not driven by rare jobs or by jobs that job seekers do not view as relevant; we see similar levels of misperceptions if we restrict to the two job types where prior cohorts were most frequently hired, or if we restrict to job types where respondents have a high reported likelihood of applying in the control group.[14] Applying a simple procedure to correct obvious reporting errors suggests that such errors do not explain the measured misperceptions.[15] Restricting only to job seekers actively searching at the time of answering the survey, we see that misperceptions are not driven by possible inattention among respondents who have already completed their job search or among graduates who have not started searching actively. Finally, using reweighting or restricting the sample based on measured understanding of job types, we see no evidence that measured misperceptions are affected by survey non-participation or by respondents not understanding the job types in the survey.

A final concern is that relative misperceptions about prior cohort wages could be large simply because they are unrelated to the wages job seekers themselves expect to face and thus are not of relevance for job seekers' decision-making. Using our survey data on own potential wages for the control group, Figure 3 plots beliefs about the gap in wages that job seekers *themselves* expect to face across two jobs, $\Delta\widetilde{w_{i,j,j'}}$, against their belief about the gap in prior cohort wages for these jobs, $\Delta\widetilde{v_{i,j,j'}}$. The plot confirms that prior cohort wages are a relevant benchmark for job seekers; there is a clear positive relationship with an estimated regression slope of 0.63. Moreover, as we show in Section 4 below, receiving information about wages of prior cohorts leads job seekers to update their beliefs about the potential wages they will face.

Summing up, we find that young job seekers have substantial misperceptions about wage differences across jobs that they consider in their job search. Most commonly, mis-

[14] Since the likelihood of applying is elicited after the information treatment, we only include the control group when imposing this sample restriction.

[15] Appendix Section D.2 explains the procedure we use to correct errors.

perceptions take the form of underestimating true wage gaps, although overestimation of wage gaps occurs about a third of the time. An implication of this is that relative wage misperceptions cause lower-paying jobs to appear overly attractive for most job seekers but that the opposite also occurs. Over the next sections, we examine the extent to which these misperceptions distort search behavior and job outcomes.

4 Effects of relative wage misperceptions on job search

In this section, we examine to what extent relative wage misperceptions causally affect job search behavior. We do this by leveraging exogenous variation generated by our information treatment: treated job seekers were shown actual wages for prior cohorts in the different jobs, which should influence their perceptions about what they themselves would receive in each job. We proceed in four steps. First, we show nonparametric treatment-control comparisons to establish the existence of a causal effect. Second, we estimate how strongly job seekers respond by scaling these effects by the size of the misperception revealed by the treatment. Third, we collapse these results into an elasticity that captures the responsiveness of job applications to perceived relative wages, using a 2SLS framework. Throughout these analyses, we use the same individual-by-job pair survey data as in Section 3.2. Finally, we use individual-level administrative data to validate the treatment effects on post-survey hiring outcomes.

4.1 Treatment-control comparisons

Our treatment provided information about actual wages for prior cohorts in the different jobs, and therefore about the actual *wage gaps* in prior cohorts between each job pair (j, j'). The expected effect therefore depends on job seekers' baseline misperception about these gaps. We measure the baseline misperception of the wage gap between job j and j' as the difference between the actual log wage gap and the perceived log wage gap for prior cohorts (elicited *before* the information treatment), $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$. If this misperception measure is positive, the information treatment reveals to the job seeker that they underestimated the wage gap between j and j' for prior cohorts. In response, we expect the job seeker

to upward-adjust the potential wage they expect to receive in job j relative to job j'. Assuming that beliefs about wage gaps affect job search, we also expect them to become more likely to apply for job j relative to job j'.

Column (1) of Table 4 confirms these predictions using simple treatment-control comparisons. Reordering our job pairs so that $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}} \geq 0$ for all included observations, we regress our outcomes of interest on a treatment dummy, T_i.[16] Since treatment was assigned at the individual level, we use cluster-robust inference clustered on individuals throughout all analyses. In Panel A, the outcome is the gap in (log) potential wages that the respondent expects to face, $\Delta\widetilde{w_{i,j,j'}}$ (elicited *after* the treatment). We find a significant effect: informing job seekers that they underestimated prior cohort wages in job j relative to job j' increases the potential wage they themselves expect to face in j relative to j' by 4.2% on average. In Panel B the outcome is the gap in the (log) reported likelihood of applying to the two jobs, which we denote $\Delta\pi_{i,j,j'} = \pi_{i,j} - \pi_{i,j'}$. We again find a significant effect: informing job seekers that they underestimated the prior cohort wages in job j relative to job j' increases their relative likelihood of applying to job j by 19.5% on average. As robustness checks, Columns (2)-(4) repeat the analyses after controlling for all predetermined survey responses and/or restricting only to respondents who report actively searching at the time of the survey. Results are virtually identical to those in Column (1). In addition, Appendix Table A9 shows that we find similar positive treatment effects if we use the likelihood of accepting job offers as the outcome instead of the likelihood of applying.

Heterogeneity by baseline misperceptions In addition to the treatment effect depending on the *sign* of $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$, we of course also expect it to depend on its *magnitude*: we expect a bigger effect if the treatment reveals that the job seeker underestimated the wage gap for prior cohorts by a larger amount. In Figure 4 we group our sample of job-pairs into three equal width-bins based on the value of $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$. Within each

[16] Note that this sample definition does not exclude any unique job pairs from the analysis. For any individual, and given pair of jobs A and B, either the wage in A is underestimated relative to B, or the wage in B is underestimated relative to A (no respondents are exactly correct about wage gaps in our survey).

bin, we then estimate treatment effects using simple treatment-control comparisons, both for the relative perceived gap in own potential wages (Panel (a)) and for the relative application likelihood (Panel (b)). We see the expected patterns. In cases where the information treatment revealed larger underestimation of the prior cohort wage gap, we estimate larger upward-adjustments of the perceived gap in potential wages, as well as larger increases in the relative likelihood of applying. Conducting formal tests for pairwise equality, all estimates in Panel (a) are significantly different from each other at the 1 percent level. Given the lower precision in Panel (b), only the first and third estimates are significantly different here and only at the 10 percent level.

4.2 Quantifying the response to corrected misperceptions

The treatment-control comparisons above establish that correcting relative wage misperceptions shifts beliefs and applications, with larger effects when the treatment reveals a larger misperception. We now scale these effects by the size of the revealed misperception, yielding a per-unit coefficient that does not depend on the particular distribution of misperception sizes in our sample. Both panels of Figure 4 are consistent with treatment effects increasing linearly in $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$, which suggests the following two parsimonious regression specifications:

$$\Delta\widetilde{w_{i,j,j'}} = \gamma_1 \, T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}) + \gamma_2 \, (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}) + e_{i,j,j'} \tag{2}$$

$$\Delta\pi_{i,j,j'} = \theta_1 \, T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}) + \theta_2 \, (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}) + u_{i,j,j'} \tag{3}$$

The coefficients on the interaction terms, γ_1 and θ_1, capture the (linear) effects of the information treatment and have a simple interpretation: if the information treatment reveals to a job seeker that they underestimated a particular prior cohort wage gap by $X\%$, equations (2) and (3) imply that the job seeker's perceived gap in potential wages increases by $\gamma_1 X\%$ and that their relative likelihood of applying changes by $\theta_1 X\%$. Since we control for the underestimation of the prior cohort wage gap, $(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$, identification of γ_1 and θ_1 comes from treatment-control comparisons *within* different groups of individuals and job

pairs, each with a given level of $(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$ (as in Figure 4). Random assignment therefore ensures that γ_1 and θ_1 have a causal interpretation.

To treat all jobs symmetrically, we estimate (2) and (3) on the full sample of all possible pairwise job comparisons.[17] For the same reason, equations (2) and (3) do not include constant terms (γ_0, θ_0) or an uninteracted treatment dummy (T_i) because the left- and right-hand-side variables are mean zero by construction in this sample, both for the treatment and control group.[18] We continue to use standard errors clustered on individual. Our inference thus accounts fully for the fact that a given job (and job pair) enters the sample several times for the same individual.

Column (1) of Table 5 presents OLS estimates of (2) (Panel A) and (3) (Panel B). The estimate for γ_1 is 0.3: If informed that they underestimated the wage gap by $X\%$ for prior cohorts, job seekers adjust their perceived gap in potential wages by $0.3\,X\%$. This is in line with typical estimates from the literature on belief updating for information treatments (Haaland et al., 2023). In particular, our estimated learning rate of 30% is similar to the 12–34% learning rates for beliefs about re-employment wages in Denmark found by Altmann et al. (2025). The estimate for θ_1 is 1.2: if informed that they underestimated the wage gap between job j and j' by $X\%$ for prior cohorts, job seekers respond by increasing their relative likelihood of applying for j by $1.2\,X\%$. Columns (2)-(4) demonstrate the robustness of the estimates to controlling for pre-treatment covariates interacted with the underestimation of the prior cohort wage gap, and/or restricting only to the sample of currently active job seekers.

[17] Each job seeker thus contributes six job pairs to this sample since a given unique pair of jobs A and B enters twice, both for $(j, j') = (A, B)$ and for $(j, j') = (B, A)$. Avoiding this duplicity would require imposing some (arbitrary) rule for determining whether to include (A, B) or (B, A) in the sample. Deterministic rules (e.g. requiring $\left(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right) > 0$) would imply that individual jobs are not treated symmetrically in estimation. Picking the ordering at random simply introduces additional noise in estimates. As a robustness check, Appendix Table A5 shows results using various different rules that restrict to unique job pairs. Results are similar throughout.

[18] As is easily verified, introducing a constant term and/or an uninteracted treatment dummy therefore generates exact zero estimates for the constant term and the coefficient on the treatment dummy, while leaving other coefficients unchanged.

4.3 Applications' responsiveness to wage beliefs: 2SLS estimate

The results above confirm that our information treatment shifted relative wage beliefs and job application behavior as expected. To collapse these results into an estimate of how wage beliefs affect job applications, we turn to a parametric 2SLS framework. Formally, this corresponds to treating (2) and (3) as the first stage and reduced form in a 2SLS framework. The excluded instrument is the interacted treatment dummy, $T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$, and the second stage equation of interest is

$$\Delta\pi_{i,j,j'} = \beta_1 \Delta\widetilde{w_{i,j,j'}} + \beta_2 (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}) + \varepsilon_{i,j,j'} \tag{4}$$

The coefficient β_1 corresponds to our key causal elasticity of interest: if the perceived wage gap between job j and j' increases by 1 percent, the relative likelihood of applying to j increases by β_1 percent. Consistent 2SLS estimation of β_1 requires the usual instrument exogeneity assumption that the instrument, $T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$, is uncorrelated with the error term, $\varepsilon_{i,j,j'}$. As above, random assignment implies that the treatment and control groups are similar at baseline, meaning that instrument exogeneity will not be violated due to differences in baseline unobservables. This estimate can be interpreted as capturing the pure effect of wage beliefs only under the exclusion restriction that the instrument *only* affects application behavior through its effect on perceived wages. The information treatment therefore should not affect any other decision-relevant beliefs. Our survey design allows us to directly test this assumption and we present the results in Section 4.4.

Panel C of Table 5 shows 2SLS estimates of (4). The estimate for β_1 in Column (1) is 4.2. Increasing the perceived wage gap between job j and j' by 1 percent increases the relative likelihood of applying for j by 4.2%. This is a substantial effect and implies that wage misperceptions can generate important distortions in job search (as we flesh out in Section 5). As above, Columns (2)-(4) of Table 5 show robustness to the inclusion of pre-treatment covariates and/or restricting to active job seekers. Moreover, Appendix Table A10 shows that we obtain similar results when estimating a 2SLS specification with T_i as the excluded instrument in the reordered sample approach of Section 4.1—which identifies a LATE without the linearity assumption by scaling the average effect on applications by

the average effect on perceived own wage. Appendix Table A4 show that results are also robust to considering only job seekers with a good understanding of the three job types and reweighing to correct for survey non-participation. Finally, Column (1) of Appendix Table A11 shows that our estimates are robust to letting the prior-cohort wage gap misperception, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$, enter the specification flexibly rather than linearly.

4.4 Belief spillovers and interpretation of the 2SLS estimate

Our 2SLS estimates quantify how relative application behavior responds to perceived relative wages, when changes in perceived wages are driven by information about prior cohort wages—an object that is informative in its own right about how early career job seekers respond to wage information. Going further, under the assumption that the information treatment affects applications only through perceived wages, the 2SLS estimate helps recover structural preference parameters (see Section 5). A potential concern with this stronger interpretation is belief spillovers—which are typically thought to attenuate estimated wage effects: learning that a job offers a higher wage could also lead workers to perceive the job as harder to get or as offering worse non-wage amenities.[19] We test for such spillovers using several post-treatment questions designed to detect them, and find no economically meaningful spillovers: the information treatment moves wage beliefs but leaves beliefs about hiring probabilities and non-wage amenities unchanged.

Test of belief spillovers We test for spillovers along two dimensions. First, consider beliefs about how hard each job is to get. Recall that, alongside applications, we asked respondents how likely they were to accept an offer from each job (Section 2.5). If learning that a job pays more led workers to perceive it as harder to get, acceptances would respond by *less* than applications. Panel A of Table 6 estimates the reduced form (3) with the gap in log likelihood of accepting an offer as outcome. The coefficients (1.02–1.19) are very close to the corresponding application estimates in Table 5 (1.04–1.28), indicating little

[19] In a Bayesian learner framework, it is straightforward to write down information structures in which wage signals lead to spillovers on beliefs about other characteristics, and structures in which they do not, regardless of whether job seekers perceive that actual wages and other job characteristics are correlated. What matters for spillovers is only the correlation between *misperceptions* about wages and *misperceptions* about other job characteristics.

role for offer-probability spillovers. If anything, they are slightly *smaller*—the reverse of the classic concern: workers respond marginally more to wages when applying than when accepting offers, suggesting that higher-wage jobs are, if anything, perceived as *easier* to get. Panel B confirms this directly, using respondents' self-reported gap in the log probability of receiving an offer if applying: the estimates are small (0.36–0.44), positive, and statistically insignificant. The picture across both panels is that any offer-probability spillover is small and indistinguishable from zero.

Second, we test whether the treatment shifted perceived non-wage amenities, using the "equal wage" follow-up to our offer-acceptance question. As described in Section 2.5, the original question asked respondents how likely they were to accept an offer in each of the three jobs; the follow-up posed an identical question in the counterfactual scenario in which the three starting wages were equal.[20] With wages held equal, any remaining response to the treatment must operate through perceived non-wage amenities. Panel C of Table 6 reports this result. The coefficients are statistically insignificant and much smaller than the corresponding actual-wage acceptance coefficients in Panel A—the two are directly comparable, since Panel C is the equal-wage version of the very same question. Because Panel A reflects both wage beliefs and any non-wage spillover while Panel C isolates the non-wage component, the gap between them indicates that the acceptance response operates overwhelmingly through wage beliefs rather than spillovers.

In addition, we directly asked respondents about four specific job characteristics: the wage they would expect to earn in five years if starting today, expected weekly hours, expected performance, and expected fit with colleagues (6-item Likert scales). In Table 7, we use cross-job gaps in these dimensions as outcomes. We see a significant effect only for the expected wage in five years (Panel A), corroborating that the treatment affects wage perceptions but does not spill over onto non-wage amenities.

Interpretation Taken together, these results support the stronger interpretation of our 2SLS coefficient as capturing the pure effect of own wage beliefs on applications. Appendix

[20] Specifically, each job's wage was set to the middle of the three offered wages the job seeker expected to face.

Table A11 further shows that our 2SLS estimates are robust to controlling for gaps in other perceived job characteristics, including the "equal wage" acceptance likelihood. We adopt this structural interpretation throughout the rest of the paper.

The fact that we isolate the wage-belief channel may also explain why our estimate is large relative to prior work. A natural benchmark for our 2SLS estimate is the own wage elasticity of job applications (Holzer et al., 1991; Dal Bó et al., 2013; Banfi and Villena-Roldán, 2019; Marinescu and Wolthoff, 2020; Dube et al., 2020; Belot et al., 2022; Azar et al., 2022)—the percentage increase in likelihood of applying to a given job when that job increases its offered wage by 1%. Our estimate captures a slightly different parameter,[21] but can be converted into an own wage elasticity using the link between our framework and a standard discrete choice application model (see Appendix Section B.4). Applying this conversion at the bottom of Table 5, we obtain an own wage elasticity of about 3—substantially higher than prior estimates, which typically find elasticities below 1. Our spillover analysis suggests an explanation. Most prior estimates exploit variation in posted wages on online job-ad platforms, where wages may be perceived as signals about competition for the jobs and non-wage amenities; the estimated elasticity therefore combines the pure wage response with offsetting responses through other channels. Our information treatment, by contrast, conveys information about realized past wages to a small number of job seekers, so it is unsurprising that it is not perceived as a signal for firm-level changes in other amenities or shifts in application competition. Our setting allows us to isolate the wage-belief channel that prior estimates may capture only as part of a bundled response.

4.5 Treatment effects on post-survey wages

The previous sections established that our information treatment changes reported beliefs and planned applications in the survey. We now leverage our individual linkage to administrative data to verify that these effects also translate into actual labor market behavior. This analysis is crucial for establishing that our estimated effects are not confined to survey

[21] To see the difference, consider a case where job j increases its offered wage while wages are unchanged in job j' and all other jobs. The likelihood of applying to job j increases, and the own wage elasticity measures the extent of this increase. If focusing instead on the *relative* likelihood of applying to j vs. j', there is an additional effect because the likelihood of applying to job j' also decreases.

responses, a common concern with survey experiments that often goes unaddressed.

Two features of this analysis warrant emphasis. First, unlike the analyses above, the unit of observation is now the individual rather than the individual-by-job-type pair: we analyze wages in each job seeker's first post-survey job, restricting attention to those who found a new job within our administrative data window (1,170 observations).[22] Second, our survey and information treatment were designed to identify how wage perceptions affect job search, rather than to evaluate any policy intervention. The purpose of this analysis is therefore to test directional predictions about post-survey behavior, not to estimate precise effect sizes. For the majority of job seekers, who tend to underestimate true wage gaps, misperceptions make lower-paying jobs appear relatively more attractive; the information treatment should therefore shift these job seekers toward higher-paying jobs. The reverse should hold for those who overestimate true wage gaps. We further expect both predictions to be sharper among job seekers actively searching at the time of the survey, who can adjust their behavior without incurring switching or search costs.

Table 8 estimates treatment effects on starting wages in job seekers' first post-survey job. Column (1) ignores heterogeneity by initial misperceptions and simply regresses log monthly wages on a treatment dummy. Consistent with most job seekers being underestimators, the average treatment effect is positive but not statistically significant. Column (3) instead estimates heterogeneous effects by initial misperceptions: we classify each job seeker as either an *underestimator* or *overestimator* based on whether they underestimate or overestimate the log gap between the highest-paying job and the average of the two other jobs (as in Section 3), and interact this classification with the treatment dummy. As predicted, the information treatment significantly increases post-survey wages for underestimators (3.1 percent on average) and the interaction term is negative and statistically significant at the 10% level. The implied effect for overestimators, reported at the bottom of the table, is −1.1 percent, though not statistically significant. Columns (5) and (7) re-

[22]Our available administrative data runs until December 2024. Job seekers' new jobs are measured as the first post-survey job spell at a firm that the job seeker did not work at prior to the survey. To be counted, we require job spells to last more than one month and to be the worker's highest-paid job. To avoid measurement errors from jobs that de facto started later in a month, we exclude the first month of employment but average over all remaining months of the employment when computing monthly wages in the new jobs. As we show in Appendix Table A14, we see no effect of the treatment on the likelihood of finding a new job within our sample period.

strict attention to respondents actively searching at the time of the survey. As anticipated, all effects increase in magnitude in this subsample, and the interaction term is significant at the 5% level throughout.

As robustness checks, Table 8 also reports estimates with pre-treatment controls included: as expected given random assignment, estimates change little, although the interaction term loses significance once controls are added. Refining the classification, we split respondents into three categories—relatively accurate, large overestimators, and large underestimators—and find that treatment effects are largest in magnitude among those with large misperceptions and negligible among those with accurate perceptions (Table A13). Finally, because Table 8 conditions on having found a new job, a natural concern is that the estimated wage effects could reflect composition changes. Appendix Table A14 shows that the information treatment does not affect the likelihood of finding a new job, consistent with the treatment changing beliefs about relative wages across jobs—which matter for *which* jobs to target but not for job-finding rates.[23]

Finally, we examine whether the wage effects in administrative data are consistent with the applications respondents reported in the survey. We first construct, for each respondent, an *applied-for wage*: the average of prior cohorts' wages across the three job types, weighting each by the stated likelihood of applying times the perceived likelihood of receiving an offer. This is the wage a job seeker could expect if they acted on their stated applications and received the same wages as prior cohorts. Realized wages need not match it, since they also depend on which offers employers extend and on how far an individual's own wage departs from prior cohorts'; absent these wedges, however, the two should track each other closely. Indeed, Panel A of Table A15 recovers the same pattern for the applied-for wage as for realized wages: a positive effect for underestimators, a negative interaction for overestimators, and sharper effects among active searchers. Panel B takes the difference between the log realized wage and the log applied-for wage as outcome, isolating the part of the realized-wage effect not already accounted for by stated applications; the estimated

[23] In standard models of job search, job-finding rates respond primarily to beliefs about the overall level of wages, since these affect the relative attractiveness of unemployment vs. employment (see e.g. Altmann et al. (2025)). Given that beliefs about wage levels are much closer to correct on average (cf. Figure 1), our information treatment should have a much more limited systematic impact on these beliefs.

effects are small and statistically insignificant throughout. Realized wages thus move with the treatment by about as much as stated applications imply, indicating that respondents act on the choices they report—and that the sorting mechanism we study materializes in actual labor market outcomes.

5 The costs of relative wage misperceptions

We close our analysis by gauging the costs that misperceptions impose on workers, incorporating our estimates into a simple discrete-choice model of job search. This step takes us beyond the reduced-form analysis in three ways. First, the reduced-form results show that wage information shifts workers' relative willingness to apply across pairs of jobs; the model lets us aggregate these pairwise shifts into the total volume of applications *reallocated*. Second, the reduced-form analysis captures effects on wages but misses effects along non-wage dimensions which are difficult to measure; the model structure lets us incorporate these non-wage effects into a full welfare assessment. Third, our information treatment cleanly identifies the determinants of job applications by providing exogenous shifts in perceptions, but it is small in magnitude and so cannot speak to the full scope of misperception; the model lets us extrapolate from these local estimates to the costs of misperception at full scale. In the rest of this section, we present our discrete-choice model, how we use it in our counterfactual analysis, and the estimates we obtain for the full costs of misperceptions in terms of application reallocation and welfare.

5.1 Model outline

Our survey and reduced form analysis were designed to identify a simple partial equilibrium discrete choice model of job search. Here, we outline the main features of our model which builds on Harmon et al. (2025); Appendix Section B gives full details and derivations.

Matching the setup in our survey, we consider a job seeker i who at a point in time has the option of applying for one of three different jobs indexed by $j = 1, 2, 3$. Alternatively, she can decide to apply for none of them and simply continue searching. We assume that the continuation value of being hired into job j is $Y_{i,j}$ and that the continuation value of

continuing to search is U_i. Accordingly, the surplus value of job j over continued search can be defined as $S_{i,j} = Y_{i,j} - U_i$. We will assume that this surplus value depends on the wage the worker will earn, $W_{i,j}$, a set of non-wage amenities summarized by the scalar $Z_{i,j}$ and an idiosyncratic taste shock, $\xi_{i,j}$, capturing other utility-relevant characteristics: [24]

$$S_{i,j} = \Psi_i W_{i,j}^{\delta} Z_{i,j} \xi_{i,j} \tag{5}$$

Here Ψ_i is an individual-specific parameter allowing for heterogeneity in the general returns to employment, while δ is a parameter governing the curvature of surplus with respect to wages.

Optimal application behavior can be characterized by considering the surplus value of an application relative to the value of continued search. We let $A_{i,j}$ denote the surplus value of an application to job j. This will simply equal the likelihood of getting hired times the surplus value from being hired. Letting $P_{i,j}$ denote the probability of being hired at job j, we have:

$$A_{i,j} = P_{i,j} S_{i,j} \tag{6}$$

Importantly, as (5) and (6) make clear, the attractiveness of applying to job j depends not only on the wage it offers but also on its non-wage amenities and the likelihood of being hired. In what follows we let $H_{i,j}$ denote the total *non-pecuniary value* of applying to job j, which we define as

$$H_{i,j} = P_{i,j} Z_{i,j} \xi_{i,j}$$

The surplus value of an application to job j can then be written as $A_{i,j} = \Psi_i W_{i,j}^{\delta} H_{i,j}$.

To allow for misperceptions, we assume that job seekers do not necessarily base their decisions on the true surplus value, $A_{i,j}$, but on their *perceived* surplus value, $\widetilde{A_{i,j}}$. This depends on their perceptions about the potential wage they would earn, $\widetilde{W_{i,j}}$, their prob-

[24] Treating $Z_{i,j}$ as scalar is not restrictive. For example, if we were to instead assume that jobs are characterized by a K-dimensional vector $\left(Z_{i,j}^1, Z_{i,j}^2, ..., Z_{i,j}^K\right)$ and that surplus is given by $\Psi_i W_{i,j}^{\delta} \Pi_k \left(Z_{i,j}^k\right)^{\delta_k}$, we would still arrive at equation (5) by simply defining $Z_{i,j} = \Pi_k \left(Z_{i,j}^k\right)^{\delta_k}$.

ability of being hired $\widetilde{P_{i,j}}$ and the non-wage amenities $\widetilde{Z_{i,j}}$:

$$\widetilde{A_{i,j}} = \widetilde{P_{i,j}}\widetilde{S_{i,j}} \tag{7}$$

$$\widetilde{S_{i,j}} = \Psi_i \widetilde{W_{i,j}}^{\delta} \widetilde{Z_{i,j}} \xi_{i,j} \tag{8}$$

Finally, we assume that sending an application incurs a cost of c_i.

Letting $\widetilde{A_{i,0}}$ denote the surplus value of not applying anywhere, job seeker i's application choice then solves:

$$j_i^* = \text{argmax}_{j \in \{0,1,2,3\}} \widetilde{A_{i,j}} - \mathbf{1}_{\{j \neq 0\}} c_i$$

where $\mathbf{1}_{\{j \neq 0\}}$ is an indicator for applying to any job (and thus incurring the application cost)

After imposing a standard extreme value distribution on the taste shocks, optimal behavior for job seekers implies that the likelihood of applying to job j obeys the following equation:[25]

$$P(j_i^* = j) = \frac{\exp((\alpha_i + \widetilde{p_{i,j}} + \delta \widetilde{w_{i,j}} + \widetilde{z_{i,j}})/\sigma)}{1 + \sum_{j'=1,2,3} \exp((\alpha_i + \widetilde{p_{i,j'}} + \delta \widetilde{w_{i,j'}} + \widetilde{z_{i,j'}})/\sigma)} \tag{9}$$

Here, σ is a scale parameter for the taste shocks and α_i is a constant capturing the overall perceived attractiveness of the three jobs. As in previous sections, lowercase letters refer to logged values.

Equation (9) motivates our survey design and reduced form regression analyses. Prior to actually making an application decision, the left hand side of Equation (9) shows the expected likelihood that job seeker i will choose to apply to job j. This is exactly what job seekers report in our main survey question on application probabilities. The right hand side of (9) relates this likelihood to perceived characteristics of the different jobs, including particularly the potential wage they expect to face in the job, $\widetilde{w_{i,j}}$, which job seekers also report in our survey. This equation also implies a direct link between the

[25] Specifically, we assume that the taste shocks, $\xi_{i,j}$, are i.i.d. Frechet-distributed with shape parameter σ^{-1}.

structural parameters of the model and our 2SLS estimate of the effect of wage beliefs on applications from equation (4)—in the absence of spillover effects of our information treatment on other beliefs: $\beta_1 = \frac{1}{\sigma}\delta$. Below, we use this to compute a model counterfactual based on the 2SLS estimates.

5.2 Counterfactual analysis

Our counterfactual analysis aims to quantify the cost of misperception for application allocation and welfare. In practice, we compare observed applications—formed under current misperceptions—to counterfactual applications—formed under a hypothetical scenario without misperception. We compute these counterfactual applications for every individual in our control group using the formula derived from our discrete-choice model, combining survey measures with results from our empirical analysis. Specifically, we draw on the survey-reported propensities to apply, the measure of misperceptions (Section 3.2), and the 2SLS estimate of the responsiveness of job applications to wage information (Section 4.3). We focus on the control group so that the survey-reported propensities to apply are not contaminated by our information treatment; thanks to randomization, the results remain representative of the full set of respondents. We additionally make a few assumptions that depend on the specific cost considered.

Applications misallocation Our survey provides direct measures of misperception about average prior-cohort wages. But search behavior depends on the potential wages workers expect to face themselves, not on the average wage of prior cohorts. Computing the counterfactual therefore requires us to take a stand on workers' relative misperceptions about their own potential wages. We adopt the simple assumption that the measured relative misperceptions about wage gaps for prior cohorts coincide with the relative misperceptions about own potential wage gaps across jobs. Since our information treatment moved beliefs about respondents' own wages by about one third (an elasticity of 0.3 in Panel A of Table 5), this assumption implies that fully removing misperception would be about three times as large as our information treatment. This single assumption suffices to characterize counterfactual application behavior and to measure how many applications

would be reallocated across jobs in the absence of relative wage misperceptions.

Welfare cost We also use the model to assess welfare. Specifically, for each job seeker i, we benchmark the welfare costs of relative wage misperceptions against an across-the-board wage change. This amounts to asking how much lower all wages would have to be in order for i to remain at their current utility level in the absence of relative wage misperceptions. This welfare counterfactual requires three additional assumptions: First, we need to take a stand on misperceptions about the non-pecuniary value of applying to each job, i.e. hiring likelihoods and non-wage amenities. Since the focus of our article is on wage misperceptions, we perform our counterfactual under the simple benchmark that job seekers have correct beliefs about these non-pecuniary values. Second, we need to take a stand on what happens to job seekers for whom the first application decision does not result in employment. Mirroring standard job search models, we assume that these workers continue searching with some individual-specific flow utility and time discount rate, while sending additional applications at some exogenously fixed Poisson rate. Finally, we need to impose a parameter restriction on the parameter governing the curvature of utility wrt. wages, δ. This reflects the usual challenge that choice behavior does not separately identify this parameter from the scale parameter of the taste shocks.[26]. For our main results, we impose $\delta = 1$ corresponding to risk-neutrality. In Appendix Tables A17 and A18 we show results for the risk-averse case, $\delta < 1$. This leads to slightly higher welfare costs of misperceptions.

5.3 Misperception costs for applications allocation and welfare

The results of our counterfactual analysis, which assesses the consequences of relative wage misperceptions for application allocation and welfare, are presented in Table 9. The first row of Panel A shows results for all job seekers; the next two rows split them into underestimators and overestimators, depending on whether they under- or overestimate the gap between the job offering them the highest wage and the average of the two others.

[26]Since $\beta_1 = \frac{1}{\sigma}\delta$, scaling up σ and δ by the same amount leaves application behavior unchanged.

Main results Relative wage misperceptions misallocate a substantial share of applications: for the average job seeker, 9.5% would go to a different job under correct beliefs (Column (1)), and the misallocation is large for underestimators and overestimators alike. This generalizes our reduced-form analysis (Panel B of Table 5) under minimal additional assumptions, showing concretely that the misperceptions we highlight can substantially misallocate jobs among labor market entrants.

We then turn to welfare costs. The average cost of this misallocation is equivalent to a 0.8% drop in all wages (Column (2)). Columns (3) and (4) show how this comes about: misperceptions depress the average applied-for wage by 1.2% (Column (3)), but partially offset this by distorting applications towards jobs with better non-wage amenities and/or higher hiring probabilities, raising the average non-pecuniary value of an application by 0.14% (Column (4)). This average masks opposing patterns across groups, however, and the two types bear costs through different channels. For underestimators, misperceptions distort applications away from high-paying jobs, so the cost is essentially a wage cost. For overestimators, the wage channel matters less; their cost comes primarily from misperceptions distorting applications *toward* jobs with worse non-wage amenities and/or lower hiring probabilities.

Heterogeneity Panels B and C focus on job seekers in the top and bottom decile of overall misperception about relative wages, measured as the absolute value of their misperceptions about log wage gaps across all job pairs. The costs are highly heterogeneous. Among those with the largest misperceptions, almost one-in-five applications are misallocated and welfare costs are equivalent to a decrease in all wages of more than 2.8%; among those with the smallest, fewer than one-in-fifty applications are misallocated and the welfare cost is negligible.

Comparison of counterfactual and reduced-form results The goal of the counterfactual is not to pin down precise magnitudes but to translate our reduced-form estimates into concrete, interpretable quantities, so we begin by checking that it is consistent with our model-free evidence. Qualitatively, Column (3)—the effect of misperceptions on applied-

for wages—is consistent with the reduced-form effects of the information treatment on realized wages (Table 8) and on applied-for wages (Table A15), and it is the counterfactual quantity closest to what the reduced-form analysis can deliver. The magnitudes cannot be compared directly, for two reasons. First, by assumption, removing misperception amounts to a change in beliefs about three times as large as that induced by our information treatment, so a simple rule of thumb suggests the counterfactual effect should be roughly three times larger. Second, we implement the counterfactual analysis in the control group—which random assignment makes representative of the full sample—whereas the reduced-form effects on realized and applied-for wages are estimated on the subsample of respondents who started a new job before the end of 2024. The cleanest comparison is therefore with the reduced-form effect of the information treatment on applied-for wages for the full sample (Table A16). Multiplying the estimates from Columns (2) and (4) by three implies that fully removing misperception would shift applied-for wages by about 1.8% overall, 3% for underestimators, and -0.6% for overestimators. These magnitudes—though slightly larger—line up well with the applied-wage effects in our counterfactual analysis—suggestive evidence that the assumptions underlying our counterfactual are sound.

This consistency on wages is reassuring precisely because the counterfactual delivers more than wages alone. The other columns show that wages do not capture the full impact of misperceptions: The welfare cost is lower (resp. higher) than wage losses alone would imply for underestimators (resp. overestimators), because the non-wage channel offsets the wage cost for the former and compounds it for the latter. Going beyond the reduced-form analysis is therefore necessary to grasp the full welfare consequences of misperceptions. Taken together, these results tell a richer story than wages alone. Wage information moves not just the wages people apply for but every dimension of their choices—amenities and hiring odds included—and by accounting for these margins explicitly, we recover a fuller, more nuanced picture of what misperceptions cost than wages alone could give.

6 Conclusion

Using linked administrative and survey data in Denmark, this paper shows that job seekers have large misperceptions about wage differences across jobs. While there is substantial heterogeneity, most job seekers underestimate the true wage gap between jobs such that lower-paying jobs look relatively more attractive. Providing information that corrects these misperceptions has large effects on job search decisions and hiring outcomes, on average shifting workers towards higher-paying jobs. Crucially, these effects are not confined to stated intentions: linking respondents to administrative records, we find that correcting wage beliefs raises the wages workers actually earn in their next job. Overall, our findings show that relative wage misperceptions impose substantial costs by distorting the allocation of workers to jobs. Early career job seekers appear both responsive to such information and readily reachable at the point of labor market entry, making labor market entry a particularly promising moment for policy to intervene.

Bibliography

Altmann, Steffen, Anita Glenny, Robert Mahlstedt, and Alexander Sebald, "The Direct and Indirect Effects of Online Job Search Advice," IZA Discussion Papers 15830, Institute of Labor Economics (IZA) 2022.

_ , **Armin Falk, Simon Jäger, and Florian Zimmermann**, "Learning about Job Search: A Field Experiment with Job Seekers in Germany," *Journal of Public Economics*, August 2018, *164*, 33–49.

_ , **Robert Mahlstedt, Malte Rattenborg, Alexander Sebald, Sonja Settele, and Johannes Wohlfart**, "Wage Expectations and Job Search," 2025. .

Ansolabehere, Stephen, Marc Meredith, and Erik Snowberg, "Asking about Numbers: Why and How," *Political Analysis*, 2013, *21* (1), 48–69.

Arellano-Bover, Jaime, "Career Consequences of Firm Heterogeneity for Young Workers: First Job and Firm Size," *Journal of Labor Economics*, April 2024, *42* (2), 549–589. Publisher: The University of Chicago Press.

Azar, José A., Steven T. Berry, and Ioana Marinescu, "Estimating Labor Market Power," August 2022. .

Baker, Rachel, Eric Bettinger, Brian Jacob, and Ioana Marinescu, "The Effect of Labor Market Information on Community College Students' Major Choice," *Economics of Education Review*, 2018, *65*, 18–30.

Banfi, Stefano and Benjamín Villena-Roldán, "Do High-Wage Jobs Attract More Applicants? Directed Search Evidence from the Online Labor Market," *Journal of Labor Economics*, 2019.

Beffy, Magali, Denis Fougère, and Arnaud Maurel, "Choosing the Field of Study in Postsecondary Education: Do Expected Earnings Matter?," *The Review of Economics and Statistics*, 2012, *94* (1), 334–347.

Behaghel, Luc, Sofia Dromundo, Marc Gurgand, Yagan Hazard, and Thomas Zuber, "The Potential of Recommender Systems for Directing Job Search: A Large-Scale Experiment," IZA Discussion Papers 16781, Institute of Labor Economics (IZA) February 2024.

Belot, Michèle, Philipp Kircher, and Paul Muller, "Providing Advice to Jobseekers at Low Cost: An Experimental Study on Online Advice," *The Review of Economic Studies*, July 2019, *86* (4), 1411–1447.

_ , _ , and _ , "How Wage Announcements Affect Job Search—A Field Experiment," *American Economic Journal: Macroeconomics*, October 2022, *14* (4), 1–67.

_ , _ , and _ , "Do the Long-Term Unemployed Benefit from Automated Occupational Advice during Online Job Search?," *The Economic Journal*, May 2025, *136* (673), 184–206.

Bó, Ernesto Dal, Frederico Finan, and Martín A. Rossi, "Strengthening State Capabilities: The Role of Financial Incentives in the Call to Public Service*," *The Quarterly Journal of Economics*, August 2013, *128* (3), 1169–1218.

Caldwell, Sydnee, Ingrid Haegele, and Jörg Heining, "Firm Pay and Worker Search," February 2025.

Cockx, Bart and Corinna Ghirelli, "Scars of Recessions in a Rigid Labor Market," *Labour Economics*, August 2016, *41*, 162–176.

Dahl, Christian, Daniel le Maire, and Jakob Munch, "Wage Dispersion and Decentralization of Wage Bargaining," *Journal of Labor Economics*, 2013, *31* (3), 501 – 533. Publisher: University of Chicago Press.

Dhia, Aïcha Ben, Bruno Crépon, Esther Mbih, Louise Paul-Delvaux, Bertille Picard, and Vincent Pons, "Can a Website Bring Unemployment Down? Experimental Evidence from France," Working Paper 29914, National Bureau of Economic Research April 2022.

Dube, Arindrajit, Jeff Jacobs, Suresh Naidu, and Siddharth Suri, "Monopsony in Online Labor Markets," *American Economic Review: Insights*, March 2020, *2* (1), 33–46.

Haaland, Ingar, Christopher Roth, and Johannes Wohlfart, "Designing Information Provision Experiments," *Journal of Economic Literature*, March 2023, *61* (1), 3–40.

Harmon, Nikolaj A, Marie Pietraszek, and Jonas Maibom, "Why Do Coworker Networks Affect Job Search Outcomes?," Study Paper No. 262, The ROCKWOOL Foundation 2025.

_ **, Robert Mahlstedt, and Mette Rasmussen**, "Job Search, Overoptimism and Statistical Profiling: Can Information Provision Improve Job Search Outcomes?," 2024.

Holzer, H. J., L. F. Katz, and A. B. Krueger, "Job Queues and Wages," *The Quarterly Journal of Economics*, August 1991, *106* (3), 739–768.

Jäger, Simon, Christopher Roth, Nina Roussille, and Benjamin Schoefer, "Worker Beliefs About Outside Options," *The Quarterly Journal of Economics*, August 2024, *193* (3), 1505–1556.

Kahn, Lisa B., "The Long-Term Labor Market Consequences of Graduating from College in a Bad Economy," *Labour Economics*, April 2010, *17* (2), 303–316.

Kerr, Sari Pekkala, Tuomas Pekkarinen, Matti Sarvimäki, and Roope Uusitalo, "Post-secondary Education and Information on Labor Market Prospects: A Randomized Field Experiment," *Labour Economics*, 2020, *66*, 101888.

Krueger, Alan B. and Andreas I. Mueller, "A Contribution to the Empirics of Reservation Wages," *American Economic Journal: Economic Policy*, February 2016, *8* (1), 142–179.

Marinescu, Ioana and Ronald Wolthoff, "Opening the Black Box of the Matching Function: The Power of Words," *Journal of Labor Economics*, April 2020, *38* (2), 535–568.

Mueller, Andreas I., Johannes Spinnewijn, and Giorgio Topa, "Job Seekers' Perceptions and Employment Prospects: Heterogeneity, Duration Dependence, and Bias," *American Economic Review*, January 2021, *111* (1), 324–363.

Spinnewijn, Johannes, "Unemployed but Optimistic: Optimal Insurance Design with Biased Beliefs," *Journal of the European Economic Association*, February 2015, *13* (1), 130–167.

von Wachter, Till, "The Persistent Effects of Initial Labor Market Conditions for Young Adults and Their Sources," *Journal of Economic Perspectives*, November 2020, *34* (4), 168–194.

von Wachter, Till and Stefan Bender, "In the Right Place at the Wrong Time: The Role of Firms and Luck in Young Workers' Careers," *American Economic Review*, December 2006, *96* (5), 1679–1705.

Wiswall, M. and B. Zafar, "Determinants of College Major Choice: Identification Using an Information Experiment," *The Review of Economic Studies*, April 2015, *82* (2), 791–824.

Wiswall, Matthew and Basit Zafar, "Preference for the Workplace, Investment in Human Capital, and Gender*," *The Quarterly Journal of Economics*, February 2018, *133* (1), 457–507.

Økonomi og Indenrigsministeriet, "Dagpenge til Nyuddannede," Økonomisk Analyse Nr. 33, Økonomi og Indenrigsministeriet 2018.

7 Tables and Figures

Table 1: Covariate balance and descriptive statistics, study sample individuals

	(1) All	(2) Control group	(3) Treatment group	(4) p-value (2) = (3)
Age	28.07	28.15	27.98	0.326
Female	0.62	0.61	0.63	0.420
Months between survey and expected/actual graduation	16.61	17.58	15.60	0.195
Employed long-term	0.11	0.12	0.11	0.270
Employed short-term, without job offer	0.05	0.06	0.04	0.117
Employed short-term, with job offer	0.03	0.04	0.02	0.122
Unemployed, with job offer	0.25	0.25	0.25	0.972
Unemployed, without job offer	0.56	0.53	0.58	0.060*
Currently studying	0.07	0.07	0.07	0.957
Actively searching for a job	0.65	0.63	0.67	0.130
Currently searching or about to start new job	0.87	0.86	0.88	0.112
Currently searching, about to start new job or planning to search in the next 3 months	0.89	0.88	0.89	0.270
Good understanding of:				
Most common job	0.82	0.81	0.82	0.564
Second most common job	0.73	0.73	0.73	0.940
Least common job	0.72	0.73	0.71	0.388
Close to someone:				
In most common job	0.59	0.60	0.58	0.394
In second most common job	0.49	0.48	0.49	0.666
In least common job	0.51	0.53	0.49	0.092*
Perceived avg. wage of prior cohorts (1000s):				
In most common job	32.27	32.13	32.42	0.259
In second most common job	31.21	31.16	31.26	0.717
In least common job	32.55	32.37	32.73	0.180
Perceived avg. wage of prior cohorts after 5 years (1000s):				
After starting at most common job	39.51	39.51	39.52	0.968
At second most common job	38.15	38.10	38.21	0.784
At least common job	39.89	39.91	39.88	0.943
Perceived avg. probability to receive offer, if applying:				
In most common job	0.17	0.17	0.16	0.263
In second most common job	0.15	0.15	0.14	0.481
In least common job	0.14	0.15	0.14	0.639
Part of STAR sample	0.91	0.91	0.91	0.971
Individuals	1,902	976	926	

Notes: The first three columns of the table show sample means for all pre-treatment survey variables, respectively for the full study sample (Column (1)), only for the control group (Columns (2)) and only for the treatment group (Column (3)). Column (4) shows p-values from a test of equal means between the treatment and control groups. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table 2: Descriptive statistics, survey job types

	(1) Share of first jobs, prior cohorts	(2) Stated likelihood of applying	(3) Avg. wage, prior cohorts	(4) Perceived avg. wage, prior cohorts	(5) Perceived own potential wage
Data source:	Admin	Survey	Admin	Survey	Survey
Panel A: *Grouped by share of first jobs in prior cohorts*					
Highest	25.64%	37.80%	30,883	32,269	33,312
Middle	14.81%	24.11%	30,735	31,208	32,385
Lowest	09.62%	28.75%	31,674	32,548	33,503
Panel B: *Grouped by stated likelihood of applying*					
Highest	18.62%	61.94%	31,310	33,080	34,322
Middle	16.01%	21.73%	31,164	32,021	33,061
Lowest	15.44%	07.00%	30,817	30,923	31,816
Panel C: *Grouped by avg. wage of prior cohorts*					
Highest	14.75%	31.46%	33,148	32,763	34,091
Middle	17.93%	31.04%	31,176	31,901	33,075
Lowest	17.39%	28.16%	28,968	31,360	32,033

Notes: The table shows study sample means for various characteristics of the three job types asked about in the survey. Columns correspond to different variables, either from administrative data or the survey. Panels correspond to different ways of grouping the three jobs into highest, middle and lowest. In Panel A jobs are grouped in terms of how large a share of entry level jobs they cover in past cohorts. In Panel B, jobs are grouped in terms of the respondent's stated likelihood of applying to them. In Panel C, jobs are grouped by their average wage for prior cohorts.

Table 3: Relative wage misperceptions, summary

	Wage gap underestimation rate between job pairs $\frac{\Delta V_{i,j,j'} - \Delta\widetilde{V_{i,j,j'}}}{\Delta V_{i,j,j'}}$				Underestimation of log wage gap between job pairs $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$		Individuals who are underestimators	Sample size:	
	$> 100\%$	$> 50\%$	$> 0\%$	$< -50\%$	Mean abs. value	Std. Dev.	Share	Individuals	Job pairs
Full sample	36.1%	55.2%	66.7%	25.1%	0.129	0.167	66.6%	1,902	5,706
Actual log wage gap > 0.05	30.0%	51.1%	68.6%	20.2%	0.142	0.164	68.8%	1,583	2,456
Actual log wage gap > 0.1	20.9%	45.7%	69.2%	17.3%	0.146	0.166	70.3%	1,046	1,471
Two most frequent job types	36.2%	55.4%	66.5%	25.3%	0.123	0.162	66.6%	1,902	3,804
Likelihood of applying $> 15\%$ for both jobs, control group	31.3%	54.9%	66.3%	24.8%	0.114	0.151	65.7%	623	1,392
Full sample, correcting reporting errors	36.2%	55.1%	66.8%	24.9%	0.126	0.162	66.5%	1,901	5,703
Currently active job seekers	35.9%	55.3%	65.9%	26.5%	0.129	0.166	66.1%	1,237	3,711
Reweighted for non-participation	35.9%	55.3%	67.0%	24.4%	0.130	0.170	65.6%	1,826	5,478
Good understanding of all job types	34.5%	54.2%	66.7%	24.5%	0.125	0.163	66.3%	1,019	3,057
Correct answer to job type validation question	35.0%	52.4%	63.8%	27.6%	0.123	0.159	65.8%	1,135	3,405

Notes: The table summarizes the distribution of relative wage misperceptions. Rows correspond to different subsamples. The first four columns shows the share of individual-by-job pairs (i, j, j') for which the wage gap underestimation rate exceeds 100%, 50%, 0% or falls below -50% respectively. The next two columns report the mean absolute value and standard deviation of the underestimation of the log wage gap. The third last column instead uses individual-level data (i) and shows what fraction of individuals are underestimators in the sense that they underestimate the typical wage in the highest-paying job relative to the average of the two other jobs. The last two columns show the number of individuals and unique individual-by-job pairs in each subsample. The shown subsamples are as follows: *Actual log wage gap* restricts to job pairs where the actual log gap in prior cohort wages is above some cutoff. *Two most frequent job types* restricts to the two jobs for each major that cover the largest share of starting jobs for past cohorts. *Likelihood of applying>15% for both jobs* only considers the control group and restricts the sample based on the respondent's reported likelihood of applying for the job types. *Correcting reporting errors* uses a simple procedure to correct likely reporting errors (see Appendix D.2). *Currently active job seekers* are respondents who report being actively searching at the time of the survey. *Reweighted for non-participation* reweights the study sample to match the invited population using propensity score weighting. *Good understanding of all job types* restricts to respondents who reported having a "Good" or "Very good" understanding of the survey job types. *Correct answer to job type validation question* restricts to respondents who answered correctly job type validation question (see Appendix C.1.2).

Table 4: Overall treatment effect of the information experiment

	(1)	(2)	(3)	(4)
A): Perceived gap in log own potential wages $\Delta\widetilde{w_{i,j,j'}}$				
Treatment, T_i	0.042***	0.042***	0.043***	0.042***
	(0.005)	(0.005)	(0.006)	(0.006)
B): Gap in log likelihood of applying $\Delta\pi_{i,j,j'}$				
Treatment, T_i	0.195***	0.196***	0.199**	0.192**
	(0.073)	(0.071)	(0.089)	(0.088)
Pre-treatment controls		✓		✓
Currently active job seekers			✓	✓
Individuals	1,902	1,902	1,237	1,237
N	5,706	5,706	3,711	3,711

Notes: The table shows OLS estimates over individual-by-job pairs (i, j, j') in the study sample, ordered so that the wage gap for prior cohorts is underestimated in all cases, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}} > 0$. Columns (3) and (4) restrict the sample to individuals reporting that they are actively searching at the time of the survey. Columns (2) and (4) include as controls all pre-treatment survey variables from Table 1. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table 5: Effect of relative misperceptions on search, 2SLS

	(1)	(2)	(3)	(4)
A) First stage: Perceived gap in log own potential wages $\Delta\widetilde{w_{i,j,j'}}$				
$T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$	0.294***	0.298***	0.302***	0.297***
	(0.032)	(0.031)	(0.040)	(0.039)
B) Reduced form: Gap in log likelihood of applying $\Delta\pi_{i,j,j'}$				
$T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$	1.235***	1.275***	1.151*	1.039*
	(0.468)	(0.450)	(0.596)	(0.577)
C) Second stage: Gap in log likelihood of applying $\Delta\pi_{i,j,j'}$				
$\Delta\widetilde{w_{i,j,j'}}$	4.197***	4.276***	3.813**	3.500*
	(1.539)	(1.452)	(1.909)	(1.862)
Controls:				
$\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$	✓	✓	✓	✓
Pre-treatment controls		✓		✓
Currently active job seekers			✓	✓
First stage F-stat	85.48	90.92	56.72	57.44
Own wage elasticity	3.04	3.09	2.76	2.53
Individuals	1,902	1,902	1,237	1,237
N	11,412	11,412	7,422	7,422

Notes: The table shows estimates over individual-by-job pairs (i, j, j') in the study sample. Panel A and B show OLS estimates, while Panel C shows 2SLS estimates using $T_i \times \left(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right)$ as the excluded instrument. All specifications include the underestimation of the log wage gap for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$, as a control. Columns (3) and (4) restrict the sample to individuals reporting that they are actively searching at the time of the survey. Columns (2) and (4) include as additional controls all pre-treatment survey variables from Table 1 interacted with the underestimation of the log wage gap for prior cohorts. The implied own wage elasticity of applications to a job is computed at the bottom as explained in Appendix Section B.4. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table 6: Belief spillovers and the exclusion restriction

	(1)	(2)	(3)	(4)
A): Gap in log likelihood of accepting job offer				
$T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$	1.022**	1.111**	1.188**	1.165**
	(0.475)	(0.457)	(0.599)	(0.576)
Pre-treatment controls		✓		✓
Currently active job seekers			✓	✓
Individuals	1,902	1,902	1,237	1,237
N	11,412	11,412	7,422	7,422
B): Perceived gap in log probability of offer, if applying				
$T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$	0.361	0.371	0.389	0.440
	(0.256)	(0.254)	(0.331)	(0.325)
Individuals	1,859	1,859	1,204	1,204
N	10,762	10,762	6,936	6,936
C): Gap in log likelihood of accepting job offer, if wages were equal				
$T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$	0.355	0.457	0.758	0.825
	(0.461)	(0.456)	(0.588)	(0.579)
Individuals	1,902	1,902	1,237	1,237
N	11,412	11,412	7,422	7,422
Controls: $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$	✓	✓	✓	✓
Pre-treatment controls		✓		✓
Currently active job seekers			✓	✓

Notes: The table shows OLS estimates over individual-by-job pairs (i, j, j') in the study sample. The outcome is the gap across the two jobs in: the log reported likelihood of accepting an offer (Panel A); the perceived log probability of receiving an offer conditional on applying (Panel B); and the log reported likelihood of accepting an offer when both jobs offer the same starting wage (Panel C). Changes in observation totals across panels stem from missing reports and zeros in the raw outcome variables. All specifications control for the underestimation of the log wage gap for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$. Columns (3) and (4) restrict the sample to individuals reporting that they are actively searching at the time of the survey. Columns (2) and (4) include as additional controls all pre-treatment survey variables from Table 1 interacted with the underestimation of the log wage gap for prior cohorts. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table 7: Belief spillovers on additional job characteristics

	(1)	(2)	(3)	(4)
A): Perceived gap in log wage after 5 years				
$T_i \times (\Delta v_{i,j,j'} - \Delta \widetilde{v_{i,j,j'}})$	0.234***	0.239***	0.270***	0.265***
	(0.038)	(0.038)	(0.047)	(0.046)
Individuals	1,887	1,887	1,228	1,228
N	11,314	11,314	7,364	7,364
B): Perceived gap in log weekly hours worked				
$T_i \times (\Delta v_{i,j,j'} - \Delta \widetilde{v_{i,j,j'}})$	0.017	0.025	0.029	0.027
	(0.026)	(0.026)	(0.032)	(0.031)
Individuals	1,855	1,855	1,208	1,208
N	10,998	10,998	7,160	7,160
C): Gap in how well they think they would get on with their colleagues				
$T_i \times (\Delta v_{i,j,j'} - \Delta \widetilde{v_{i,j,j'}})$	0.041	-0.016	0.374	0.284
	(0.237)	(0.238)	(0.300)	(0.295)
Individuals	1,902	1,902	1,237	1,237
N	11,412	11,412	7,422	7,422
D): Gap in how well they think they would perform in the job type				
$T_i \times (\Delta v_{i,j,j'} - \Delta \widetilde{v_{i,j,j'}})$	0.139	0.128	0.318	0.291
	(0.329)	(0.322)	(0.416)	(0.403)
Individuals	1,902	1,902	1,237	1,237
N	11,412	11,412	7,422	7,422
Controls:				
$\Delta v_{i,j,j'} - \Delta \widetilde{v_{i,j,j'}}$	✓	✓	✓	✓
Pre-treatment controls		✓		✓
Currently active job seekers			✓	✓

Notes: The table shows OLS estimates over individual-by-job pairs (i, j, j') in the study sample. In Panels A-B, outcome variables are gaps in logged values of perceived job characteristics. In Panel C-D, outcome variables are the gap in how well the job seeker think they would get on with colleagues in the jobs and in how well the job seeker think they would perform in the jobs (6-item Likert scales). Changes in observation totals across panels stem from missing reports and zeros in the raw outcome variables. Besides the reported regressor, all specifications include the underestimation of the log wage gap for prior cohorts, $\Delta v_{i,j,j'} - \Delta \widetilde{v_{i,j,j'}}$ as a control. Columns (3) and (4) restrict the sample to individuals reporting that they are actively searching at the time of the survey. Columns (2) and (4) include as additional controls all pre-treatment survey variables from Table 1 interacted with the underestimation of the log wage gap for prior cohorts. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table 8: Effect of information treatment on post-survey wages

	Full sample				Currently active job seekers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment, T_i	0.017	0.019	0.031*	0.029*	0.026	0.022	0.051**	0.050**
	(0.015)	(0.014)	(0.018)	(0.016)	(0.019)	(0.018)	(0.023)	(0.021)
$T_i \times$ Overestimator			-0.043	-0.031			-0.071*	-0.080**
			(0.032)	(0.029)			(0.040)	(0.039)
Overestimator			0.035	0.030			0.040	0.046*
			(0.022)	(0.020)			(0.028)	(0.026)
Treatment Effect for Overestimators			-0.011	-0.002			-0.020	-0.030
			(0.027)	(0.025)			(0.033)	(0.033)
Pre-treatment controls		✓		✓		✓		✓
N	1,171	1,171	1,171	1,171	802	802	802	802

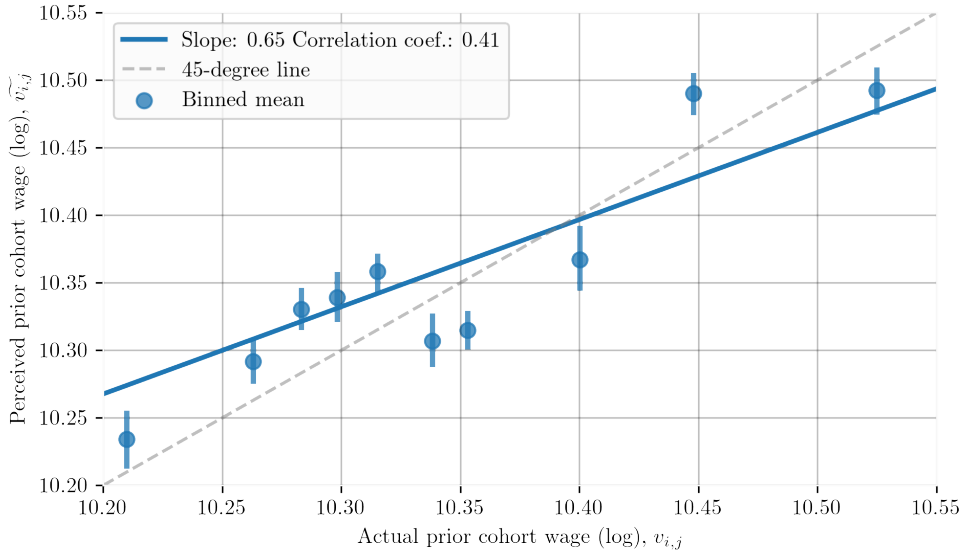
Notes: The table shows OLS estimates over individuals (i) in the study sample who are observed finding a new job post-survey. The outcome variable is the log monthly wage in the new job, measured in administrative data. Columns (1) and (5) contains the most parsimonious regression specifications, including only a constant term and treatment dummy. Other columns additionally add a dummy for the job seeker being an overestimator along with the corresponding interaction with the treatment dummy, and/or add all pre-treatment survey variables from Table 1 as controls. Columns (5)-(8) restrict the sample to individuals reporting that they are actively searching at the time of the survey. In Columns (3), (4), (7) and (8), the implied treatment effect for overestimators is reported in the row below the coefficients along with the standard error. A person is an overestimator if they overestimate the log gap in average wages between the the highest-paying job and the average of the two other jobs. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table 9: Model Counterfactual: The costs of relative wage misperceptions

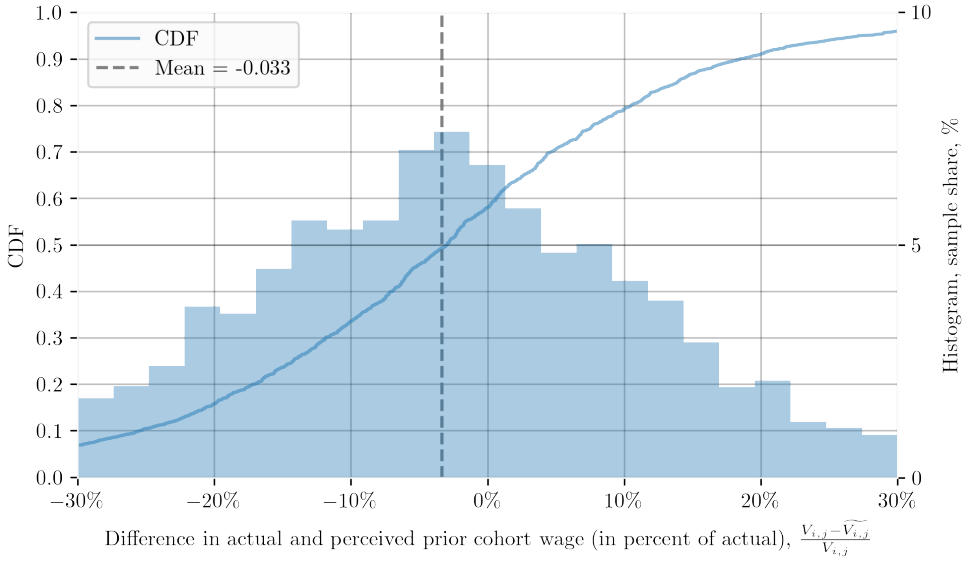
	(1) Share of applications misallocated	(2) Welfare cost (wage growth equivalent)	(3) Effect on avg. applied-for wage	(4) Effect on avg. non-pecuniary application value
Panel A: No restrictions				
Everyone	9.47%	-0.83%	-1.20%	0.14%
	(3.49)	(0.40)	(0.42)	(0.08)
Underestimators	10.4%	-0.95%	-1.76%	0.46%
	(3.80)	(0.46)	(0.62)	(0.22)
Overestimators	6.74%	-0.49%	0.38%	-0.75%
	(2.66)	(0.25)	(0.19)	(0.39)
Panel B: Relative misperceptions in top decile				
Everyone	19.2%	-2.88%	-3.74%	0.14%
	(7.33)	(1.42)	(1.48)	(0.44)
Underestimators	19.1%	-2.84%	-4.80%	0.96%
	(7.25)	(1.41)	(1.82)	(0.53)
Overestimators	19.6%	-3.22%	3.78%	-5.65%
	(9.06)	(1.80)	(2.45)	(3.19)
Panel C: Relative misperceptions in bottom decile				
Everyone	1.92%	-0.03%	-0.08%	0.03%
	(0.76)	(0.01)	(0.05)	(0.03)
Underestimators	1.92%	-0.03%	-0.13%	0.07%
	(0.79)	(0.02)	(0.06)	(0.04)
Overestimators	1.93%	-0.03%	-0.02%	-0.01%
	(0.77)	(0.02)	(0.06)	(0.05)

Notes: The table shows the average effects of current relative wage misperceptions computed from a discrete-choice model of job search. The base data are individuals (i) in the control group of the study sample. For each individual, model outcomes are computed both under current beliefs, $\widetilde{\boldsymbol{W}}_i^0$, and under a counterfactual set of beliefs, $\widetilde{\boldsymbol{W}}_i^1$, that removes all relative wage misperceptions. Details of the model and counterfactual calculations are given in Section 5 and Appendix B. Column (1) reports the share of an individual's applications that would go to a different job in the absence of relative wage misperceptions. Column (2) reports the welfare cost of relative wage misperceptions, expressed as the equivalent across-the-board proportional change in all wages: the amount by which all wages would have to change to leave the job seeker at their current utility level under the counterfactual. Column (3) reports the effect of misperceptions on the average applied-for wage, and Column (4) reports the effect on the average non-pecuniary application value (the value of hiring likelihoods and non-wage amenities, holding wages fixed). Both are expressed as the deviation from the counterfactual in percent. Panels restrict the sample by the size of relative misperceptions, defined as the mean of the absolute value of misperceptions about log wage gaps across all job pairs. Panel A imposes no restriction. Panel B restricts to job seekers in the top decile, and Panel C to job seekers in the bottom decile. Within each panel, the row *Everyone* averages over all individuals in the panel's sample, while *Underestimators* and *Overestimators* are individuals who, respectively, underestimate or overestimate the log gap in average wages between the highest-paying job and the average of the two other jobs. Results in this table normalize the wage-utility curvature parameter to $\delta = 1$, corresponding to risk neutrality. Standard errors in parentheses are obtained by a bootstrap procedure that resamples individuals and recomputes both the 2SLS estimate and the model counterfactuals within each bootstrap sample.

Figure 1: Misperceptions about wages in individual jobs



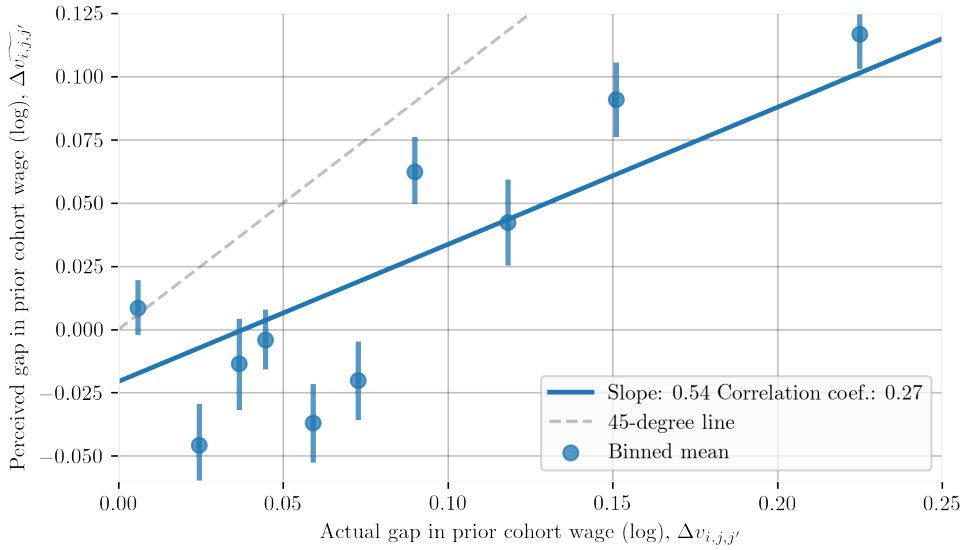
(a) Binscatter, perceived vs. actual prior-cohort wages



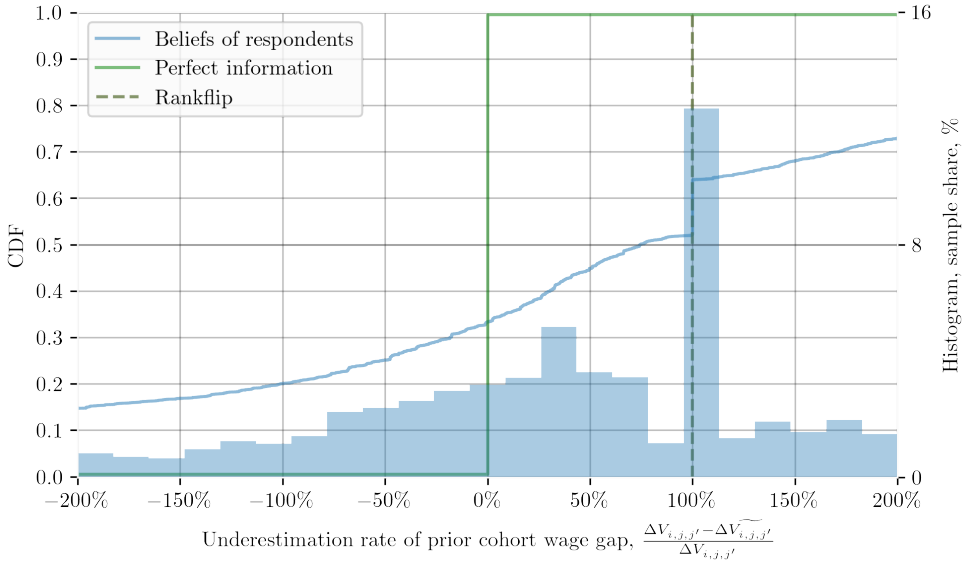
(b) Distribution of misperceptions about individual jobs

Note: The figures are based on all individual-by-jobs (i, j) in the study sample. In Panel (a), the bin scatter splits the data in ten equal-sized bins according to the actual prior cohort wage and then plots within-bin means against each other. Error bars show 95% confidence intervals for the within-bin mean of the perceived (log) prior cohort wage, clustering on individuals. The line shows the fit of an OLS regression. The legend reports the regression line slope as well as the pairwise correlation coefficient. Panel (b) shows a histogram and empirical CDF for the difference in the actual vs. perceived prior cohort wage (in percent of the actual prior cohort wage).

Figure 2: Relative wage misperceptions across job pairs



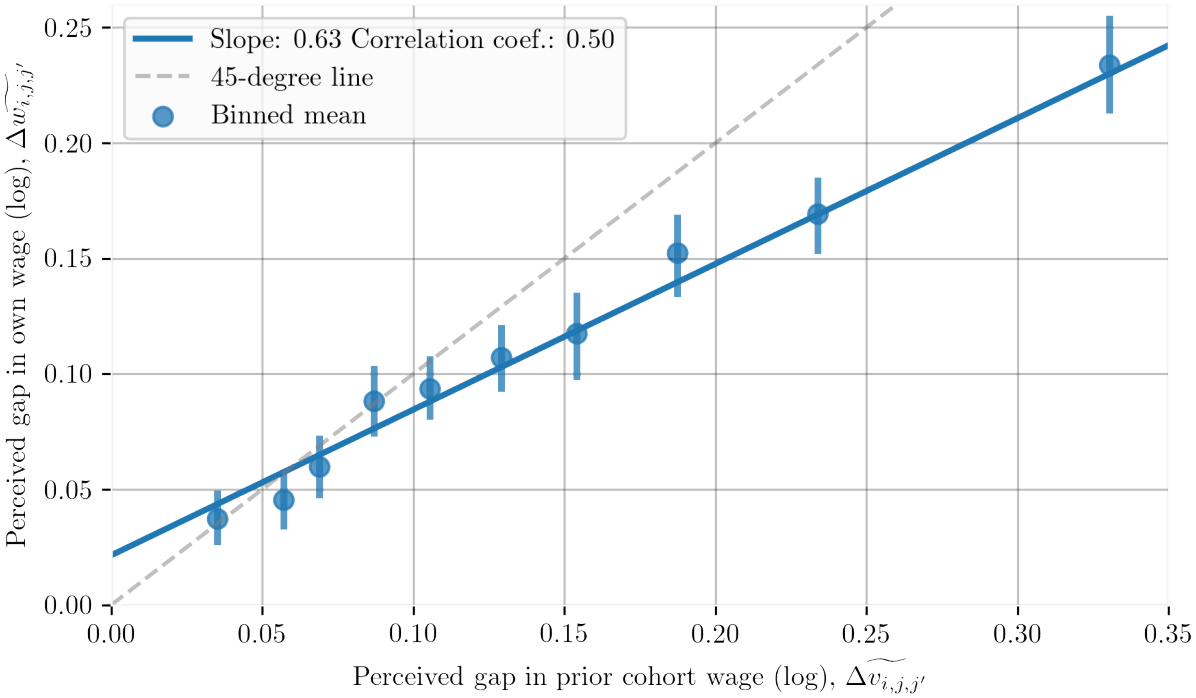
(a) Bin scatter perceived vs actual prior-cohort wage gaps



(b) Distribution of misperceptions about wage gaps

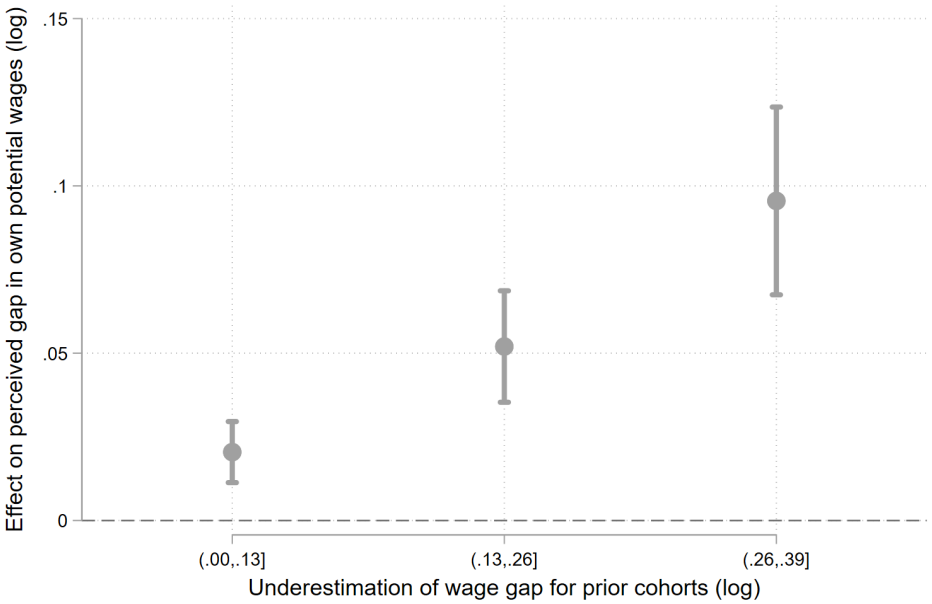
Note: The figure is based on individual-by-job pairs (i, j, j') in the study sample, ordered so that the wage gap for prior cohorts is positive in all cases, $\Delta v_{i,j,j'} > 0$. In Panel (a), the bin scatter splits the data in ten equal-sized bins according to the actual prior cohort wage gap and then plots within-bin means against each other. Error bars show 95% confidence intervals for the within-bin mean of the perceived (log) gap in prior cohort wages, clustering on individuals. The line shows the fit of an OLS regression. The legend reports the regression line slope as well as the pairwise correlation coefficient. Panel (b) shows a histogram and empirical CDF for the underestimation rate of the prior cohort wage gap. The observed mass point at 1 reflects that job seekers report two jobs paying the exact same wage in 11.7% of cases.

Figure 3: Perceived own potential wages vs. perceived prior cohort wages

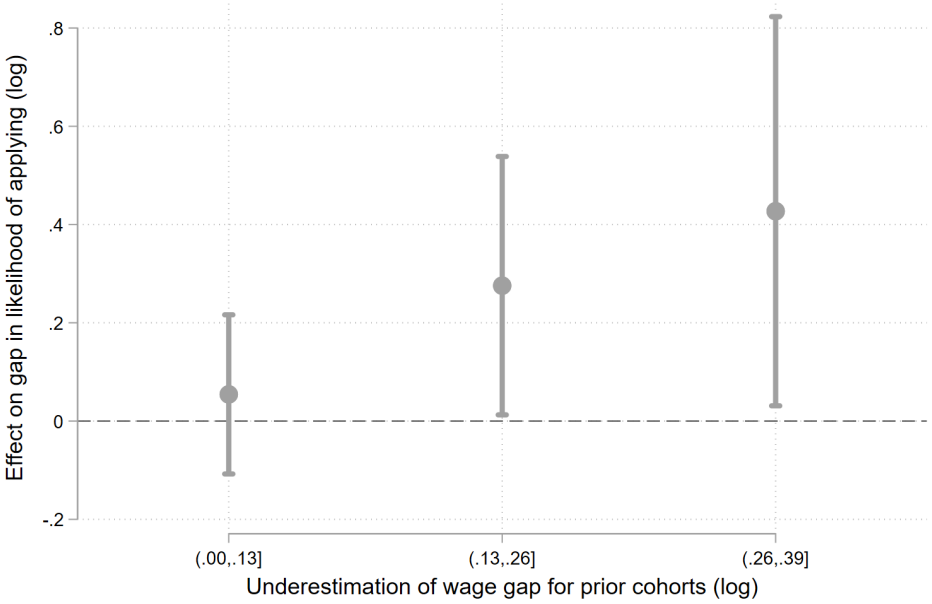


Note: The figure is based on individual-by-job pairs (i, j, j') in the study sample, ordered so that the wage gap for prior cohorts is positive in all cases, $\Delta v_{i,j,j'} > 0$. The bin scatter splits the data in ten equal-sized bins according to the perceived prior cohort wage gap and then plots within-bin means against each other. Error bars show 95% confidence intervals for the within-bin mean of the perceived (log) gap in own potential wages, clustering on individuals. The line shows the fit of an OLS regression. The legend reports the regression line slope as well as the pairwise correlation coefficient.

Figure 4: Heterogeneous effects by size of initial misperceptions



(a) Treatment effect on $\Delta\widetilde{w_{i,j,j'}}$



(b) Treatment effect on $\Delta\pi_{i,j,j'}$

Note: The figure is based on all individual-by-job pairs (i, j, j') in the study sample, ordered such that $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}} > 0$. The data is split into three equal-width bins according to the underestimation of the log wage gap for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$. The figures plot the effect of the information treatment within each bin, estimated via a linear regression that includes a constant and the treatment dummy. In Panel A, the outcome variable is the perceived gap in (log) own potential wages between the jobs, $\Delta\widetilde{w_{i,j,j'}}$. In Panel B, the outcome variable is the (log) relative likelihood of applying to the two jobs, $\Delta\pi_{i,j,j'}$. Error bars show 95% confidence intervals using clustering on individuals.

Appendices

A Additional tables and figures

Table A1: Descriptive statistics, graduates from higher education 2010-2018

	All (%)	Selected educations (%)
Education level		
Less than a bachelor's degree	12.2	12.2
Bachelor's degrees	49.3	47.0
Master's degree	35.1	36.5
Ph.d. and research programmes	3.4	4.3
Education field		
Teaching and learning	6.9	8.2
Human sciences	6.9	4.3
Arts	3.0	2.7
Social sciences	9.7	10.6
Business economics, administration and law	22.2	23.7
Science	4.4	2.9
Information and communication technology (ICT)	4.4	5.6
Engineering, technology and industrial production	7.3	7.7
Building and civil engineering	3.8	4.2
Social and health	25.1	24.6
Services	1.6	1.7
Education region		
Copenhagen	43.4	41.8
Zealand	8.6	10.2
Southern Denmark	14.2	13.4
Central Jutland	23.6	24.3
North Jutland	10.2	10.4
Educations	287	88
Individuals	469,961	359,413

Notes: The table shows the composition of all higher education graduates in Denmark 2010-2018, as well as the composition only for graduates from the selected educations included in the study. The first panel shows the shares with different education levels (based on Danish ISCED groupings), the second panel shows the shares in different education fields (based on Danish ISCED groupings), the third panel shows the shares in each geographical region of Denmark.

Table A2: Descriptive statistics, invited sample, study sample and all young job seekers

	Sample (1) completed survey	Sample (2) invited to survey	Population (3) of young job seekers
Demographic characteristics			
Age	27.97	28.15	28.44
Woman	0.62	0.62	0.59
Danish citizen	0.92	0.91	0.88
Education level			
Less than a Bachelor's degree	0.07	0.11	0.14
Bachelor's degree	0.30	0.42	0.49
Master's degree	0.63	0.46	0.37
Education field			
Teaching and learning	0.09	0.11	0.07
Social sciences	0.18	0.13	0.12
Business economics, administration and law	0.14	0.20	0.25
Science	0.07	0.04	0.03
Information and communication technology (ICT)	0.09	0.07	0.07
Engineering, technology and industrial production	0.08	0.06	0.07
Building and civil engineering	0.03	0.04	0.04
Social and health	0.15	0.22	0.20
Status in 2023			
Graduated less than 1 year ago	0.66	0.44	0.27
Graduated 1-2 years ago	0.06	0.08	0.07
Graduated 2-5 years ago	0.11	0.14	0.14
Graduated less than 1 year after survey	0.04	0.22	0.36
Work experience below 1 year	0.70	0.67	0.62
Work experience of 1-2 years	0.10	0.12	0.13
Work experience of 2-5 years	0.14	0.15	0.17
Work experience above 5 years	0.05	0.06	0.09
Received UI in 2023	0.90	0.89	0.47
Cumulated weeks of UI in 2023	13.39	13.33	7.04
Monthly wage in last job (1,000 DKK)	34.35	34.46	34.47
Lives in Copenhagen	0.27	0.28	0.28
Individuals	1,902	22,461	89,978

Notes: Using a range of variables from administrative data, the table compares the study sample to both the invited sample and the corresponding population of all young job seekers. Column (1) shows variable means for the study sample. Column (2) shows variable means for all individuals invited to the survey. Column (3) shows variable means for all young job seekers from the selected educations, defined as individuals less than 40 years old who either graduated from their degree or were unemployed in 2023.

Table A3: Covariate balance and descriptive statistics, currently active job seekers

	(1) All	(2) Control group	(3) Treatment group	(4) p-value (2) = (3)
Age	28.41	28.49	28.32	0.471
Female	0.62	0.60	0.64	0.185
Months between survey and expected/actual graduation	19.63	20.61	18.65	0.334
Employed long-term	0.04	0.04	0.04	0.889
Employed short-term, without job offer	0.05	0.06	0.05	0.376
Employed short-term, with job offer	0.01	0.01	0.01	0.405
Unemployed, with job offer	0.09	0.09	0.09	0.772
Unemployed, without job offer	0.81	0.80	0.82	0.320
Currently studying	0.05	0.05	0.05	0.792
Good understanding of:				
Most common job	0.80	0.80	0.79	0.766
Second most common job	0.73	0.73	0.74	0.713
Least common job	0.72	0.72	0.72	0.914
Close to someone:				
In most common job	0.57	0.59	0.55	0.115
In second most common job	0.48	0.47	0.49	0.514
In least common job	0.50	0.52	0.48	0.132
Perceived avg. wage of prior cohorts (1000s):				
In most common job	32.01	31.89	32.12	0.469
In second most common job	31.08	30.98	31.18	0.539
In least common job	32.25	32.02	32.48	0.170
Perceived avg. wage of prior cohorts after 5 years (1000s):				
After starting at most common job	39.14	39.33	38.94	0.372
At second most common job	37.81	37.75	37.87	0.807
At least common job	39.44	39.48	39.39	0.853
Perceived avg. probability to receive offer, if applying:				
In most common job	0.15	0.15	0.15	0.648
In second most common job	0.13	0.13	0.13	0.975
In least common job	0.13	0.13	0.13	0.976
Part of STAR sample	0.95	0.95	0.94	0.520
Individuals	1,237	619	618	

Notes: The first three columns of the table show sample means for all pre-treatment survey variables, respectively for the currently active job seekers in the study sample (Column (1)), only for the control group (Columns (2)) and only for the treatment group (Column (3)). Column (4) shows p-values from a test of equal means between the treatment and control groups among the currently active job seekers. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A4: Effect of relative misperceptions on search, additional robustness

	Reweighted for non-participation		Received and passed validation test		Good understanding of all job types	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\widetilde{w_{i,j,j'}}$	3.762*	3.675**	5.446**	5.384**	2.838	3.636**
	(2.025)	(1.856)	(2.368)	(2.188)	(1.972)	(1.818)
Controls:						
$\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$	✓	✓	✓	✓	✓	✓
Pre-treatment controls		✓		✓		✓
First stage F-stat	48.91	70.42	36.31	38.57	52.17	58.10
Own wage elasticity	2.72	2.66	3.93	3.88	2.05	2.62
Individuals	1,826	1,826	1,019	1,019	1,135	1,135
N	10,956	10,956	6,114	6,114	6,810	6,810

Notes: The table shows additional 2SLS estimates, over individual-by-job pairs (i, j, j') in the study sample. The outcome variable is the (log) relative likelihood of applying to the two jobs, $\Delta\pi_{i,j,j'}$. The excluded instrument is $T_i \times \left(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right)$. Besides the reported regressors, all specifications include the underestimation of the log wage gap for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$ as a control. Columns (1) and (2) reweight the study sample to match the invited population using propensity score reweighting. Columns (3) and (4) restrict attention to individuals who were shown the validation test regarding understanding of the survey job types and answered it correctly. Columns (5) and (6) restrict attention to individuals who reported having a good understanding of all the survey job types. Columns (2), (4) and (6) include as additional controls all pre-treatment survey variables from Table 1 interacted with the underestimation of the log wage gap for prior cohorts. The implied own wage elasticity of applications to a job is computed at the bottom via a standard discrete choice framework (see Appendix Section B.4 for details). Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A5: Effect of relative misperceptions on search, results when restricting to unique job pairs

	$\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}} > 0$		$\Delta v_{i,j,j'} > 0$		Randomly selected ordering	
	(1)	(2)	(3)	(4)	(5)	(6)
A) First stage: Perceived gap in log own potential wages $\Delta\widetilde{w_{i,j,j'}}$						
Constant		0.000		0.061***		0.004
		(0.004)		(0.003)		(0.003)
Treatment, T_i		0.003		0.006		-0.004
		(0.006)		(0.004)		(0.004)
$T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$	0.294***	0.281***	0.294***	0.273***	0.294***	0.294***
	(0.032)	(0.045)	(0.032)	(0.032)	(0.032)	(0.032)
B) Reduced form: Gap in log likelihood of applying $\Delta\pi_{i,j,j'}$						
Constant		0.030		0.313***		0.029
		(0.071)		(0.052)		(0.043)
Treatment, T_i		0.006		0.093		-0.081
		(0.102)		(0.075)		(0.061)
$T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$	1.235***	1.198*	1.235***	0.992**	1.235***	1.229***
	(0.468)	(0.680)	(0.468)	(0.491)	(0.468)	(0.468)
C) Second stage: Gap in log likelihood of applying $\Delta\pi_{i,j,j'}$						
Constant		0.029		0.089		0.013
		(0.069)		(0.108)		(0.041)
Treatment, T_i		-0.005		0.072		-0.063
		(0.102)		(0.076)		(0.059)
$\Delta\widetilde{w_{i,j,j'}}$	4.197***	4.260*	4.197***	3.639**	4.197***	4.181***
	(1.539)	(2.351)	(1.539)	(1.747)	(1.539)	(1.540)
Control for $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$	✓	✓	✓	✓	✓	✓
First stage F-stat	85.48	39.58	85.48	70.90	85.48	85.43
Own wage elasticity	3.04	3.08	3.04	2.63	3.04	3.03
Individuals	1,902	1,902	1,902	1,902	1,902	1,902
N	5,706	5,706	5,706	5,706	5,706	5,706

Notes: The table shows estimates over individual-by-job pairs (i, j, j') in the study sample. Each pair of columns uses a different restriction to impose that a unique job pair enters the sample only once per individual: Columns (1) and (2) impose that the underestimation of the log wage gap for prior cohorts is positive. Columns (3) and (4) impose that the true wage gap is positive. Finally, Columns (5) and (6) randomly pick an ordering (for a given person i and unique pair of jobs A, B, it is chosen at random whether to include i, A, B or i, B, A in the data). Panel A and B show OLS estimates, while Panel C shows 2SLS estimates using $T_i \times \left(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right)$ as the excluded instrument. In addition to the reported regressors, all specifications include the underestimation of the log wage gap for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$, as a control. Columns (1), (3) and (5) use the specifications from the main text and omit constant terms and uninteracted treatment dummies. This always leads to numerically identical results. Columns (2), (4) and (6) show results including a constant term and uninteracted treatment dummy. The implied own wage elasticity of applications to a job is computed at the bottom via a standard discrete choice framework (see Appendix Section B.4 for details). Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A6: Relationship between jobs' perceived starting wages and other perceived characteristics

	(1) $\widetilde{w_{i,j,t+5}}$	(2) Hours$\leq$ 35	(3) Hours $\in (35, 40)$	(4) Hours$\geq$ 40	(5) Log hours	(6) Log probability of offer, if applying	(7) Colleagues	(8) Performance
A): Full sample								
Log perceived wage, $\widetilde{w_{i,j}}$	0.837***	-0.363***	-0.099**	0.461***	0.148***	0.385**	0.311***	0.448***
	(0.020)	(0.034)	(0.039)	(0.039)	(0.012)	(0.157)	(0.115)	(0.099)
Constant	✓	✓	✓	✓	✓	✓	✓	✓
Log perceived wage, $\widetilde{w_{i,j}}$	0.818***	-0.333***	-0.557***	0.890***	0.227***	0.101	0.437***	0.737***
	(0.031)	(0.042)	(0.068)	(0.066)	(0.017)	(0.143)	(0.143)	(0.174)
Individual FE	✓	✓	✓	✓	✓	✓	✓	✓
Individuals	1,887	1,902	1,902	1,902	1,855	1,859	1,902	1,902
N	5,659	5,706	5,706	5,706	5,532	5,479	5,706	5,706
B): Control sample								
Log perceived wage, $\widetilde{w_{i,j}}$	0.865***	-0.378***	-0.108**	0.485***	0.153***	0.145	0.379**	0.363***
	(0.024)	(0.048)	(0.053)	(0.052)	(0.016)	(0.215)	(0.160)	(0.137)
Constant	✓	✓	✓	✓	✓	✓	✓	✓
Log perceived wage, $\widetilde{w_{i,j}}$	0.837***	-0.388***	-0.605***	0.993***	0.250***	0.168	0.558***	0.761***
	(0.044)	(0.058)	(0.090)	(0.089)	(0.023)	(0.197)	(0.183)	(0.233)
Individual FE	✓	✓	✓	✓	✓	✓	✓	✓
Individuals	966	976	976	976	946	949	976	976
N	2,897	2,928	2,928	2,928	2,823	2,797	2,928	2,928

Notes: The table shows OLS estimates from regressing various job characteristics on the log perceived starting wage. It is based on individual-by-jobs (i, j) in the study sample. Panel A shows results for this full sample, while Panel B restricts attention to the control group. Within each of these panels, the first subpanel shows a specification with just the regressor and a constant term, while the second subpanel includes individual fixed effects. In Column (1) the outcome variable is the perceived log monthly wage 5 years after starting in the job. In Columns (2), (3) and (4) the outcome variables are dummies for perceived weekly hours being below 35, being 35-40 or being above 40 respectively. In Column (5) the outcome is perceived log weekly hours. In Column (6) the outcome is the perceived log probability of receiving an offer, if applying. In Column (7) the outcome is how well the job seeker expects to get along with their colleagues in the job (6-item Likert scale). In Column (8) the outcome is how well the job seeker expects to perform in the job (6-item Likert scale). Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A7: Relationship between likelihood of applying and perceived characteristics of jobs

	Log likelihood of applying, $\pi_{i,j}$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A): Full sample							
Log perceived wage, $\widetilde{w_{i,j}}$	5.132***						1.868***
	(0.282)						(0.355)
Log perceived wage after 5 years, $\widetilde{w_{i,j,t+5}}$		4.744***					2.802***
		(0.215)					(0.291)
Hours$\in (35, 40)$			1.198***				0.538***
			(0.125)				(0.106)
Hours≥ 40			2.039***				0.904***
			(0.127)				(0.113)
Log probability of offer, if applying				0.534***			0.371***
				(0.033)			(0.029)
Colleagues					0.542***		0.265***
					(0.032)		(0.030)
Performance						0.611***	0.362***
						(0.022)	(0.025)
Individual FE	✓	✓	✓	✓	✓	✓	✓
Individuals	1,902	1,887	1,902	1,859	1,902	1,902	1,850
N	5,706	5,659	5,706	5,479	5,706	5,706	5,453
B): Control sample							
Log perceived wage, $\widetilde{w_{i,j}}$	5.166***						1.994***
	(0.357)						(0.489)
Log perceived wage after 5 years, $\widetilde{w_{i,j,t+5}}$		4.552***					2.471***
		(0.287)					(0.399)
Hours$\in (35, 40)$			1.472***				0.619***
			(0.165)				(0.148)
Hours≥ 40			2.328***				0.930***
			(0.163)				(0.155)
Log probability of offer, if applying				0.533***			0.361***
				(0.044)			(0.038)
Colleagues					0.577***		0.273***
					(0.042)		(0.041)
Performance						0.616***	0.344***
						(0.029)	(0.032)
Individual FE	✓	✓	✓	✓	✓	✓	✓
Individuals	976	966	976	949	976	976	943
N	2,928	2,897	2,928	2,797	2,928	2,928	2,781

Notes: The table shows OLS estimates from regressing various job characteristics on the log likelihood of applying to the job with individual fixed effects. It is based on individual-by-jobs (i, j) in the study sample. Panel A shows results for this full sample, while Panel B restricts attention to the control group. Row 1 is the perceived log monthly starting wage in the job. Row 2 is the perceived log monthly wage 5 years after starting in the job. Row 3 and 4 are dummies for perceived weekly hours being between 35-40 and being above 40, respectively. Row 5 is the perceived log probability of receiving an offer, if applying. Row 6 is how well the job seeker expects to get along with their colleagues in the job (6-item Likert scale). Row 7 is how well the job seeker expects to perform in the job (6-item Likert scale). Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A8: Likelihood of transitioning into a job type post-survey vs. planned likelihood of applying for it in the survey, regressions

	First new job is in the jobtype			
	(1)	(2)	(3)	(4)
Reported likelihood of applying	0.281***	0.254***	0.784***	0.669***
	(0.018)	(0.022)	(0.039)	(0.050)
Constant	0.030***		0.097***	
	(0.005)		(0.012)	
Education-by-job type FE		✓		✓
First job is in one of the job types			✓	✓
Individuals	1,902	1,899	646	635
N	5,706	5,697	1,938	1,905

Notes: The table shows OLS estimates over individual-by-jobs (i, j) in the study sample. The outcome variable is a dummy for whether the individual found a new job of the corresponding type post-survey, measured in administrative data. The regressor of interest is the reported likelihood of applying to the corresponding job type in the survey. Columns (2) and (4) add fixed effects defined at the level of the respondents' educational background-by-job type (j). Columns (3) and (4) drop individuals who did not find a new job in any of the three survey job types. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A9: Overall treatment effect on reported job acceptance

	Gap in log likelihood of accepting job offer			
	(1)	(2)	(3)	(4)
Treatment, T_i	0.222***	0.230***	0.255***	0.255***
	(0.075)	(0.074)	(0.093)	(0.092)
Pre-treatment controls		✓		✓
Currently active job seekers			✓	✓
Individuals	1,902	1,902	1,237	1,237
N	5,706	5,706	3,711	3,711

Notes: The table shows OLS estimates from a regression including a constant term, the treatment dummy and possibly controls. The base data are individual-by-job pairs (i, j, j') in the study sample, ordered so that the wage gap for prior cohorts is underestimated in all cases, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}} > 0$. The outcome variable is the gap in the reported log likelihood of accepting an offer from the jobs. Columns (3) and (4) restrict attention only to individuals reporting that they are actively searching at the time of the survey. Columns (2) and (4) include as controls all pre-treatment survey variables from Table 1. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A10: Overall local average treatment effect

	Gap in log likelihood of applying $\Delta\pi_{i,j,j'}$			
	(1)	(2)	(3)	(4)
$\Delta\widetilde{w_{i,j,j'}}$	4.634***	4.647***	4.635**	4.596**
	(1.633)	(1.598)	(1.968)	(2.005)
Pre-treatment controls		✓		✓
Currently active job seekers			✓	✓
First stage F-stat	82.23	86.06	56.71	56.69
Individuals	1,902	1,902	1,237	1,237
N	5,706	5,706	3,711	3,711

Notes: The table shows 2SLS estimates using T_i as the excluded instrument and perceived gap in log own potential wages, $\Delta\widetilde{w_{i,j,j'}}$, as the endogenous variable. Besides the reported regressors all specifications include a constant. The base data are individual-by-job pairs (i, j, j') in the study sample, ordered so that the wage gap for prior cohorts is underestimated in all cases, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}} > 0$. Columns (3) and (4) restrict attention only to individuals reporting that they are actively searching at the time of the survey. Columns (2) and (4) include as controls all pre-treatment survey variables from Table 1. The first stage and reduced form are reported in Table 4. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A11: Effect of relative wage perceptions on search, non-linear controls and controls for differences in other perceived job characteristics

	Gap in log likelihood of applying $\Delta\pi_{i,j,j'}$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\Delta\widetilde{w_{i,j,j'}}$	4.285***	4.531***	3.365***	4.052**	4.130***	3.927***	3.445***
	(1.542)	(1.584)	(1.147)	(1.597)	(1.505)	(1.500)	(1.206)
$\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$	-2.705***	-2.229***	-0.800*	-1.804***	-2.047***	-2.161***	-0.710*
	(0.651)	(0.606)	(0.418)	(0.550)	(0.588)	(0.577)	(0.401)
$(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})^2$	-0.000**						
	(0.000)						
$(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})^3$	4.783						
	(4.110)						
Δ Log probability of offer, if applying		0.602***					0.243***
		(0.034)					(0.026)
Δ Non-pecuniary attractiveness of job			0.690***				0.591***
			(0.016)				(0.017)
Δ Log hours				2.492***			1.233***
				(0.614)			(0.423)
Δ Colleagues					0.475***		0.080***
					(0.033)		(0.024)
Δ Performance						0.570***	0.110***
						(0.024)	(0.019)
First stage F-stat	85.15	75.13	84.80	76.72	85.32	84.91	71.30
Own wage elasticity	3.10	3.27	2.44	2.93	2.99	2.84	2.48
Individuals	1,902	1,859	1,902	1,855	1,902	1,902	1,824
N	11,412	10,762	11,412	10,998	11,412	11,412	10,500

Notes: The table shows 2SLS estimates with additional controls. It is based on individual-by-job pairs (i, j, j') in the study sample. The outcome variable is the (log) relative likelihood of applying to the two jobs, $\Delta\pi_{i,j,j'}$. The excluded instrument is $T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$. Column (1) controls non-linearly for the underestimation of the log wage gap for prior cohorts. Column (2) controls for the perceived gap in the (log) probability of receiving a job offer if applying to the jobs. Column (3) controls for the perceived gap in non-pecuniary attractiveness, defined as the gap in the log reported likelihood of accepting an offer from the jobs in the situation where the jobs offer the exact same starting wage. Column (4) controls for the perceived gap in log weekly hours of the jobs. Columns (5) and (6) control for the gap in how well the respondent thinks they would get along with colleagues or perform in the job (6-item Likert scales). Column (7) combines all job characteristic controls. The implied own wage elasticity of applications to a job is computed at the bottom via a standard discrete choice framework (see Appendix Section B.4 for details). Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A12: Effect of relative misperceptions on job acceptance likelihood

	Gap in log likelihood of accepting job offer			
	(1)	(2)	(3)	(4)
$T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$	1.022**	1.111**	1.188**	1.165**
	(0.475)	(0.457)	(0.599)	(0.576)
Pre-treatment controls		✓		✓
Currently active job seekers			✓	✓
Individuals	1,902	1,902	1,237	1,237
N	11,412	11,412	7,422	7,422

Notes: This table shows OLS estimates over individual-by-job pairs (i, j, j') in the study sample. The outcome variables is the gap in the log likelihood of accepting a job offer from job j given that the respondent is offered a job in all 3 job types. Besides the reported regressor, all specifications include the underestimation of the log wage gap for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$ as a control. Columns (3) and (4) restrict attention only to individuals reporting that they are actively searching at the time of the survey. Columns (2) and (4) include as additional controls all pre-treatment survey variables from Table 1 interacted with the underestimation of the log wage gap for prior cohorts. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A13: Effect of information treatment on post-survey wages, alternative grouping

	Full sample		Currently active job seekers	
	(1)	(2)	(3)	(4)
Treatment, T_i	0.032	0.030	0.056**	0.047*
	(0.020)	(0.018)	(0.026)	(0.025)
$T_i \times$ Close estimator	-0.038	-0.020	-0.048	-0.024
	(0.036)	(0.032)	(0.045)	(0.042)
$T_i \times$ Large overestimator	-0.023	-0.027	-0.074	-0.080*
	(0.038)	(0.036)	(0.047)	(0.046)
Close estimator	0.054**	0.018	0.055*	0.018
	(0.026)	(0.023)	(0.033)	(0.030)
Large overestimator	0.042	0.042	0.073**	0.066**
	(0.026)	(0.025)	(0.033)	(0.033)
Treatment Effect for Close estimators	-0.006	0.010	0.009	0.023
	(0.029)	(0.026)	(0.037)	(0.034)
Treatment Effect for Large overestimators	0.009	0.003	-0.018	-0.033
	(0.032)	(0.030)	(0.039)	(0.039)
Pre-treatment controls		✓		✓
N	1,171	1,171	802	802

Notes: The table shows OLS estimates over individuals (i) in the study sample who are observed finding a new job post-survey. The outcome variable is the log monthly wage in the new job, measured in administrative data. All specifications include a constant term, a treatment dummy, and dummies for being a close estimator and a large overestimator, both also interacted with the treatment dummy. The dummies are based on the difference between the perceived and the true gap in logs between the highest-paying job and the average of the two other jobs for the averages wages of prior cohorts. If the absolute value of this measure is less than 0.05, the respondent is a close estimator, and if the measure is larger than 0.05 the respondent is a large overestimator. Columns (3) and (4) restrict attention only to individuals reporting that they are actively searching at the time of the survey. Columns (2) and (4) include as controls all pre-treatment survey variables from Table 1. The implied treatment effects for close estimators and large overestimators is reported below the coefficient estimates along with the standard errors. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A14: Effect of information treatment on the likelihood of finding a new job

	Full sample				Currently active job seekers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment, T_i	0.027	0.014	0.026	0.013	0.030	0.028	0.031	0.025
	(0.022)	(0.021)	(0.027)	(0.027)	(0.027)	(0.027)	(0.034)	(0.033)
$T_i \times$ Overestimator			0.004	0.004			-0.002	0.008
			(0.047)	(0.045)			(0.057)	(0.056)
Overestimator			0.025	0.015			0.045	0.046
			(0.033)	(0.032)			(0.041)	(0.040)
Treatment Effect for Overestimators			0.029	0.017			0.028	0.033
			(0.038)	(0.036)			(0.046)	(0.045)
Pre-treatment controls		✓		✓		✓		✓
N	1,902	1,902	1,902	1,902	1,237	1,237	1,237	1,237

Notes: The table shows OLS estimates over individuals (i) in the study sample. The outcome variable is a dummy for whether the individual is observed finding a new job post-survey, measured in administrative data. Columns (1) and (5) contains the most parsimonious regression specifications, including only a constant term and treatment dummy. Other columns additionally add a dummy for the job seeker being an overestimator along with the corresponding interaction with the treatment dummy, and/or add all pre-treatment survey variables from Table 1 as controls. Columns (5)-(8) restrict attention only to individuals reporting that they are actively searching at the time of the survey. In Columns (3), (4), (7) and (8), the implied treatment effect for overestimators is reported in the row below the coefficients along with the standard error. A person is an overestimator if they overestimate the log gap in average wages between the the highest-paying job and the average of the two other jobs. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A15: Effect of information treatment on applied-for wage for persons who start a new job

	Full sample				Currently active job seekers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A): Log stated applied for wage, adjusted for perceived avg. likelihood of job offer								
Treatment, T_i	0.011*	0.009*	0.019**	0.015**	0.011	0.008	0.019**	0.018**
	(0.006)	(0.005)	(0.008)	(0.007)	(0.007)	(0.006)	(0.009)	(0.008)
$T_i \times$ Overestimator			-0.023*	-0.017			-0.022	-0.028**
			(0.013)	(0.010)			(0.015)	(0.013)
Overestimator			0.020**	0.015**			0.024**	0.027***
			(0.008)	(0.007)			(0.010)	(0.009)
Treatment Effect for Overestimators			-0.004	-0.002			-0.003	-0.011
			(0.010)	(0.008)			(0.012)	(0.010)
B): Log post-survey wage minus log stated applied for wage								
Treatment, T_i	0.006	0.010	0.013	0.014	0.015	0.014	0.032	0.032
	(0.014)	(0.013)	(0.016)	(0.015)	(0.018)	(0.018)	(0.021)	(0.020)
$T_i \times$ Overestimator			-0.020	-0.014			-0.049	-0.051
			(0.030)	(0.029)			(0.038)	(0.039)
Overestimator			0.015	0.014			0.016	0.019
			(0.019)	(0.019)			(0.025)	(0.025)
Treatment Effect for Overestimators			-0.007	0.000			-0.017	-0.019
			(0.025)	(0.025)			(0.032)	(0.034)
Pre-treatment controls		✓		✓		✓		✓
N	1,171	1,171	1,171	1,171	802	802	802	802

Notes: The table shows OLS estimates of the information treatment effect, restricted to respondents in our study sample who start a new job after the survey. In Panel A, the outcome is the log *applied-for wage*: the wage a respondent can expect to obtain given her stated application plans. For each respondent, we construct it as a weighted average of prior cohorts' wages across the three job types, weighting each job's wage by the respondent's stated likelihood of applying to it times her perceived likelihood of being hired if she applies. In Panel B, the outcome is the log realized wage in the administrative data minus the log applied-for wage. Columns (1) and (5) contains the most parsimonious regression specifications, including only a constant term and treatment dummy. Other columns additionally add a dummy for the job seeker being an overestimator along with the corresponding interaction with the treatment dummy, and/or add all pre-treatment survey variables from Table 1 as controls. Columns (5)-(8) restrict the sample to individuals reporting that they are actively searching at the time of the survey. In Columns (3), (4), (7) and (8), the implied treatment effect for overestimators is reported in the row below the coefficients along with the standard error. A person is an overestimator if they overestimate the log gap in average wages between the the highest-paying job and the average of the two other jobs. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A16: Effect of information treatment on applied-for wage

	Full sample				Currently active job seekers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment, T_i	0.006	0.006	0.012**	0.010**	0.007	0.007	0.014*	0.014**
	(0.005)	(0.004)	(0.006)	(0.005)	(0.006)	(0.005)	(0.008)	(0.006)
$T_i \times$ Overestimator			-0.018*	-0.012			-0.021*	-0.023**
			(0.010)	(0.008)			(0.012)	(0.010)
Overestimator			0.017**	0.013**			0.022***	0.025***
			(0.007)	(0.006)			(0.008)	(0.007)
Treatment Effect for Overestimators			-0.006	-0.002			-0.007	-0.008
			(0.008)	(0.007)			(0.010)	(0.008)
Pre-treatment controls		✓		✓		✓		✓
N	1,901	1,901	1,901	1,901	1,236	1,236	1,236	1,236

Notes: The table shows OLS estimates of the information treatment effect for respondents in our study sample. The outcome is the log *applied-for wage*: the wage a respondent can expect to obtain given her stated application plans. For each respondent, we construct it as a weighted average of prior cohorts' wages across the three job types, weighting each job's wage by the respondent's stated likelihood of applying to it times her perceived likelihood of being hired if she applies. Columns (1) and (5) contains the most parsimonious regression specifications, including only a constant term and treatment dummy. Other columns additionally add a dummy for the job seeker being an overestimator along with the corresponding interaction with the treatment dummy, and/or add all pre-treatment survey variables from Table 1 as controls. Columns (5)-(8) restrict the sample to individuals reporting that they are actively searching at the time of the survey. In Columns (3), (4), (7) and (8), the implied treatment effect for overestimators is reported in the row below the coefficients along with the standard error. A person is an overestimator if they overestimate the log gap in average wages between the the highest-paying job and the average of the two other jobs. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$.

Table A17: The costs of relative wage misperceptions, $\delta = 0.5$

	(1) Share of applications misallocated	(2) Welfare cost (wage growth equivalent)	(3) Effect on avg. applied-for wage	(4) Effect on avg. non-pecuniary application value
Panel A: No restrictions				
Everyone	9.47%	-0.95%	-1.20%	0.08%
	(3.49)	(0.40)	(0.42)	(0.04)
Underestimators	10.4%	-1.10%	-1.76%	0.26%
	(3.80)	(0.46)	(0.62)	(0.10)
Overestimators	6.74%	-0.55%	0.38%	-0.44%
	(2.66)	(0.25)	(0.19)	(0.20)
Panel B: Relative misperceptions in top decile				
Everyone	19.2%	-3.30%	-3.74%	0.07%
	(7.33)	(1.42)	(1.48)	(0.26)
Underestimators	19.1%	-3.25%	-4.80%	0.55%
	(7.25)	(1.40)	(1.82)	(0.27)
Overestimators	19.6%	-3.65%	3.78%	-3.34%
	(9.06)	(1.86)	(2.45)	(1.81)
Panel C: Relative misperceptions in bottom decile				
Everyone	1.92%	-0.03%	-0.08%	0.02%
	(0.76)	(0.02)	(0.05)	(0.02)
Underestimators	1.92%	-0.03%	-0.13%	0.04%
	(0.79)	(0.02)	(0.06)	(0.02)
Overestimators	1.93%	-0.03%	-0.02%	-0.01%
	(0.77)	(0.02)	(0.06)	(0.03)

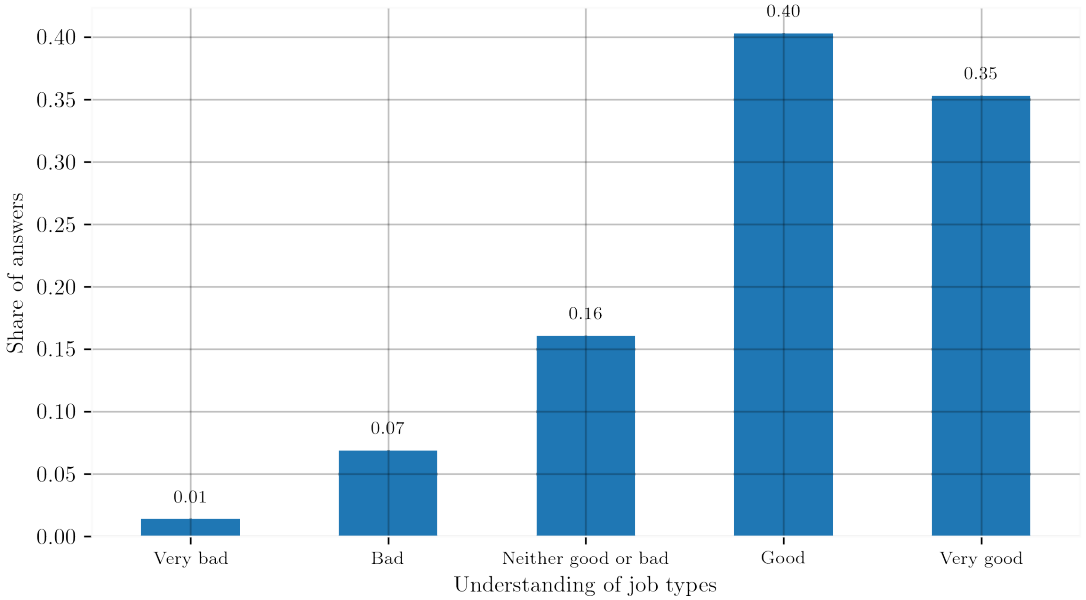
Notes: The table shows the average effects of current relative wage misperceptions computed from a discrete-choice model of job search. The base data are individuals (i) in the control group of the study sample. For each individual, model outcomes are computed both under current beliefs, $\widetilde{\boldsymbol{W}}_i^0$, and under a counterfactual set of beliefs, $\widetilde{\boldsymbol{W}}_i^1$, that removes all relative wage misperceptions. Details of the model and counterfactual calculations are given in Section 5 and Appendix B. Column (1) reports the share of an individual's applications that would go to a different job in the absence of relative wage misperceptions. Column (2) reports the welfare cost of relative wage misperceptions, expressed as the equivalent across-the-board proportional change in all wages: the amount by which all wages would have to change to leave the job seeker at their current utility level under the counterfactual. Column (3) reports the effect of misperceptions on the average applied-for wage, and Column (4) reports the effect on the average non-pecuniary application value (the value of hiring likelihoods and non-wage amenities, holding wages fixed). Both are expressed as the deviation from the counterfactual in percent. Panels restrict the sample by the size of relative misperceptions, defined as the mean of the absolute value of misperceptions about log wage gaps across all job pairs. Panel A imposes no restriction. Panel B restricts to job seekers in the top decile, and Panel C to job seekers in the bottom decile. Within each panel, the row *Everyone* averages over all individuals in the panel's sample, while *Underestimators* and *Overestimators* are individuals who, respectively, underestimate or overestimate the log gap in average wages between the highest-paying job and the average of the two other jobs. Results in this table normalize the wage-utility curvature parameter to $\delta = 0.5$. Standard errors in parentheses are obtained by a bootstrap procedure that resamples individuals and recomputes both the 2SLS estimate and the model counterfactuals within each bootstrap sample.

Table A18: The costs of relative wage misperceptions, $\delta = 0.25$

	(1) Share of applications misallocated	(2) Welfare cost (wage growth equivalent)	(3) Effect on avg. applied-for wage	(4) Effect on avg. non-pecuniary application value
Panel A: No restrictions				
Everyone	9.47%	-1.02%	-1.20%	0.04%
	(3.49)	(0.40)	(0.42)	(0.02)
Underestimators	10.4%	-1.17%	-1.76%	0.14%
	(3.80)	(0.46)	(0.62)	(0.05)
Overestimators	6.74%	-0.59%	0.38%	-0.23%
	(2.66)	(0.25)	(0.19)	(0.10)
Panel B: Relative misperceptions in top decile				
Everyone	19.2%	-3.50%	-3.74%	0.04%
	(7.33)	(1.42)	(1.48)	(0.14)
Underestimators	19.1%	-3.45%	-4.80%	0.30%
	(7.25)	(1.40)	(1.82)	(0.14)
Overestimators	19.6%	-3.86%	3.78%	-1.81%
	(9.06)	(1.89)	(2.45)	(0.97)
Panel C: Relative misperceptions in bottom decile				
Everyone	1.92%	-0.03%	-0.08%	0.01%
	(0.76)	(0.02)	(0.05)	(0.01)
Underestimators	1.92%	-0.03%	-0.13%	0.02%
	(0.79)	(0.02)	(0.06)	(0.01)
Overestimators	1.93%	-0.04%	-0.02%	0.00%
	(0.77)	(0.02)	(0.06)	(0.02)

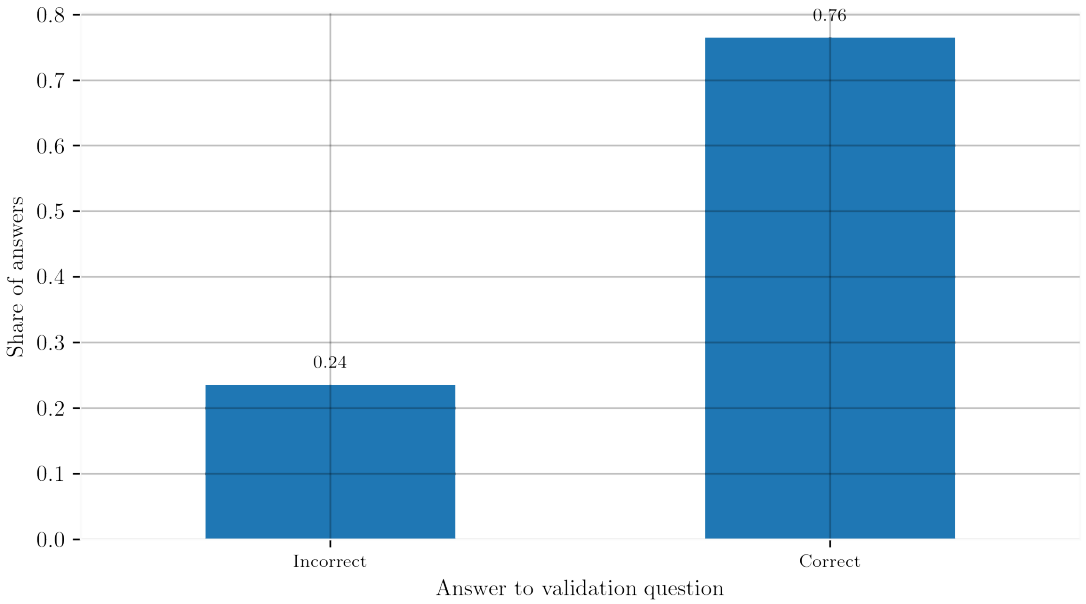
Notes: The table shows the average effects of current relative wage misperceptions computed from a discrete-choice model of job search. The base data are individuals (i) in the control group of the study sample. For each individual, model outcomes are computed both under current beliefs, $\widetilde{\boldsymbol{W}}_i^0$, and under a counterfactual set of beliefs, $\widetilde{\boldsymbol{W}}_i^1$, that removes all relative wage misperceptions. Details of the model and counterfactual calculations are given in Section 5 and Appendix B. Column (1) reports the share of an individual's applications that would go to a different job in the absence of relative wage misperceptions. Column (2) reports the welfare cost of relative wage misperceptions, expressed as the equivalent across-the-board proportional change in all wages: the amount by which all wages would have to change to leave the job seeker at their current utility level under the counterfactual. Column (3) reports the effect of misperceptions on the average applied-for wage, and Column (4) reports the effect on the average non-pecuniary application value (the value of hiring likelihoods and non-wage amenities, holding wages fixed). Both are expressed as the deviation from the counterfactual in percent. Panels restrict the sample by the size of relative misperceptions, defined as the mean of the absolute value of misperceptions about log wage gaps across all job pairs. Panel A imposes no restriction. Panel B restricts to job seekers in the top decile, and Panel C to job seekers in the bottom decile. Within each panel, the row *Everyone* averages over all individuals in the panel's sample, while *Underestimators* and *Overestimators* are individuals who, respectively, underestimate or overestimate the log gap in average wages between the highest-paying job and the average of the two other jobs. Results in this table normalize the wage-utility curvature parameter to $\delta = 0.25$. Standard errors in parentheses are obtained by a bootstrap procedure that resamples individuals and recomputes both the 2SLS estimate and the model counterfactuals within each bootstrap sample.

Figure A1: Answers to validation question; 'How good is your understanding of what each of these job types means?'



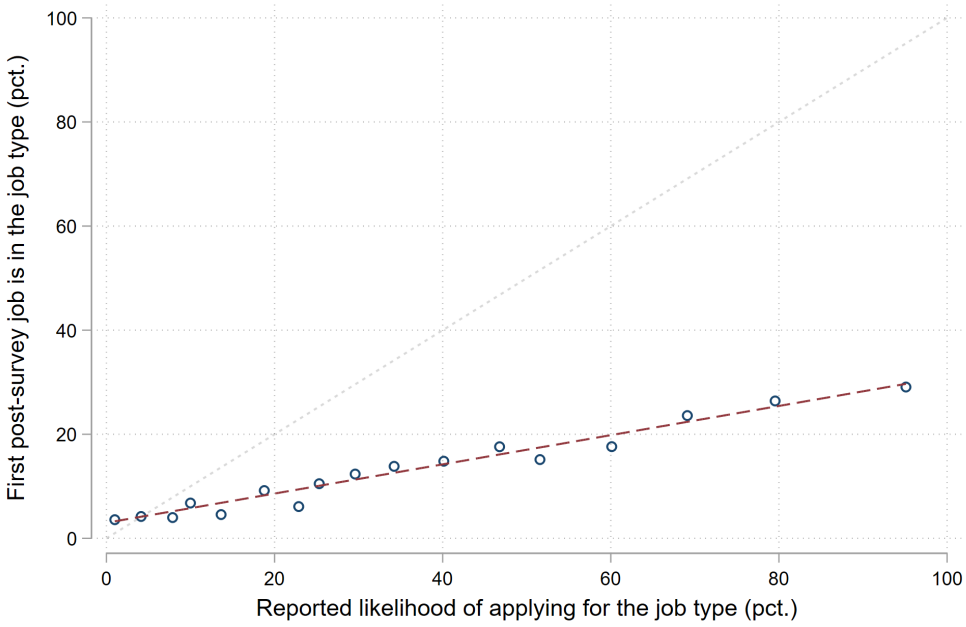
Note: This figure shows a histogram of the answers to the question 'How good is your understanding of what each of these job types means?'.

Figure A2: Answers to validation question; placing an example job in the right job type

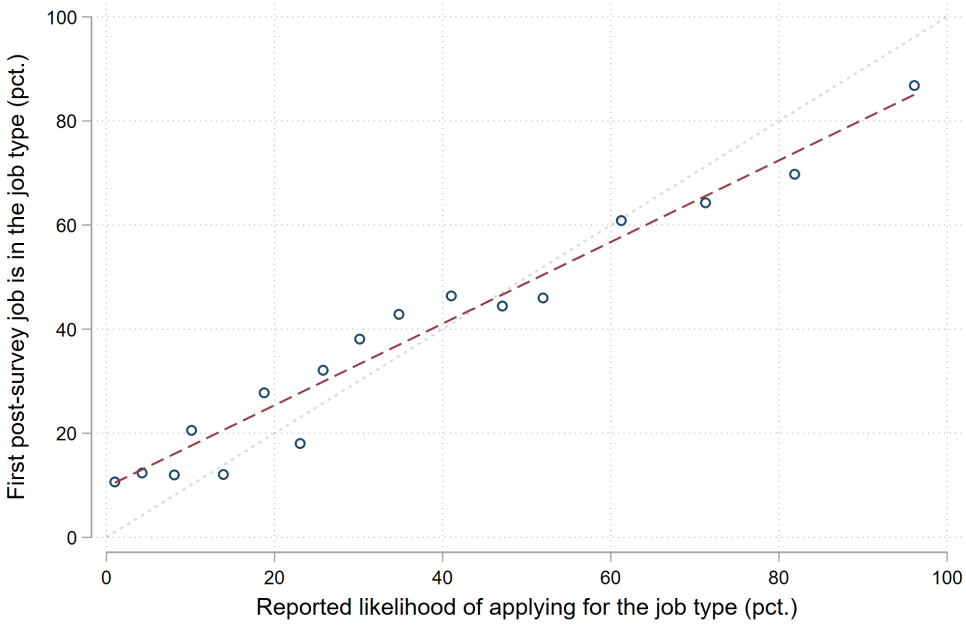


Note: A subset of respondents were asked to place a more detailed example job into the correct survey job type (see Section C.1.2). The histogram shows the shares of respondent getting this correct vs. incorrect.

Figure A3: Likelihood of transitioning into a job type post-survey vs. planned likelihood of applying for it in the survey, binscatter



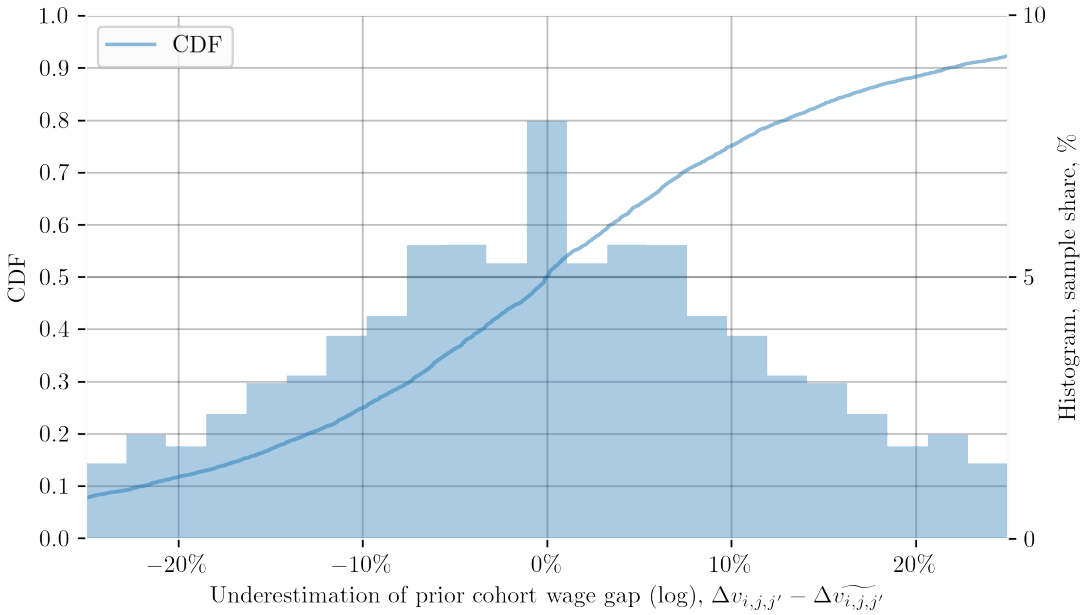
(a) All job seekers



(b) Conditional on finding a job within one of the three job types post-survey

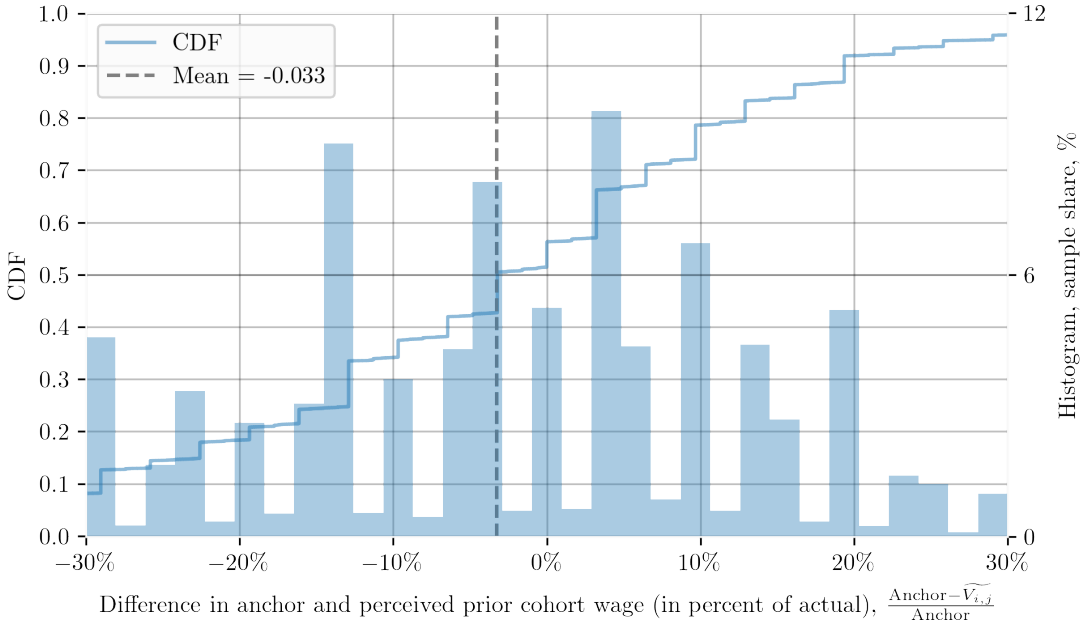
Note: The figures plots the likelihood of finding a new job post-survey in a given job type against the reported likelihood of applying for this job type in the survey. It is based on all individual-by-job observations (i, j) in the study sample. Panel A examines this full sample, while Panel B restricts attention to individuals who are in fact observed finding a new job within one of the three survey job types. The circle plot splits the data into bins according to the reported likelihood of applying for the job type in question and then plots the within-bin share actually finding a new job within this type against the mean reported likelihood of applying for the job type. The line plot shows the fit of a corresponding OLS regression (see Table A8 for corresponding OLS estimates).

Figure A4: Distribution of misperceptions about wage gaps; log wage gaps



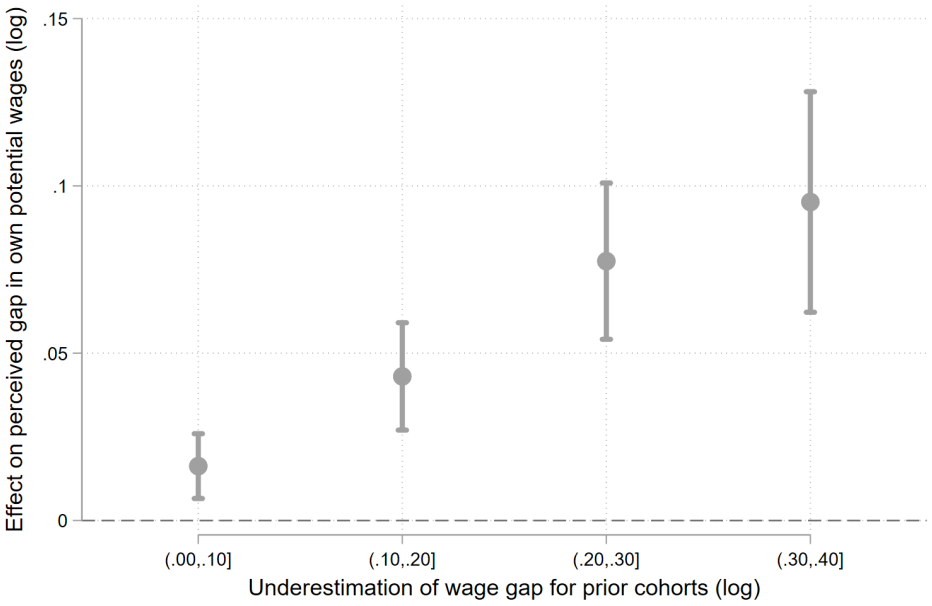
Note: The figure is based on all individual-by-job pairs (i, j, j') in the study sample. The figure shows a histogram and empirical CDF for the underestimation of the prior cohort wage gap in logs, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$.

Figure A5: Distribution of perceived wages about individual jobs relative to anchor

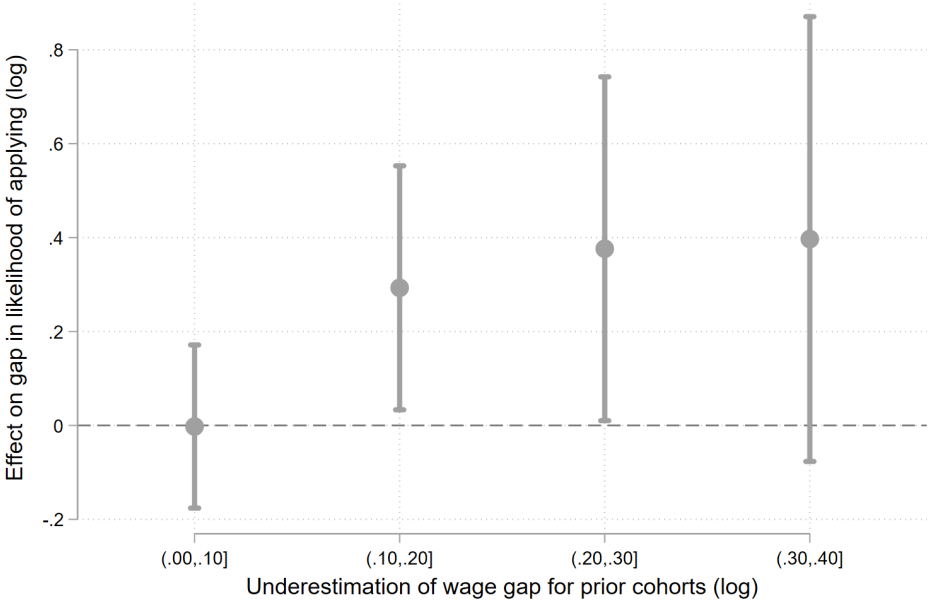


Note: The figure is based on all individual-by-jobs (i, j) in the study sample. The figure shows the histogram and empirical CDF for the difference in the anchor vs. perceived prior cohort wage (in percent of the anchor). The same anchor of DKK31,000 was shown to all respondents.

Figure A6: Heterogeneous effects by size of initial misperceptions, alternative bins 1



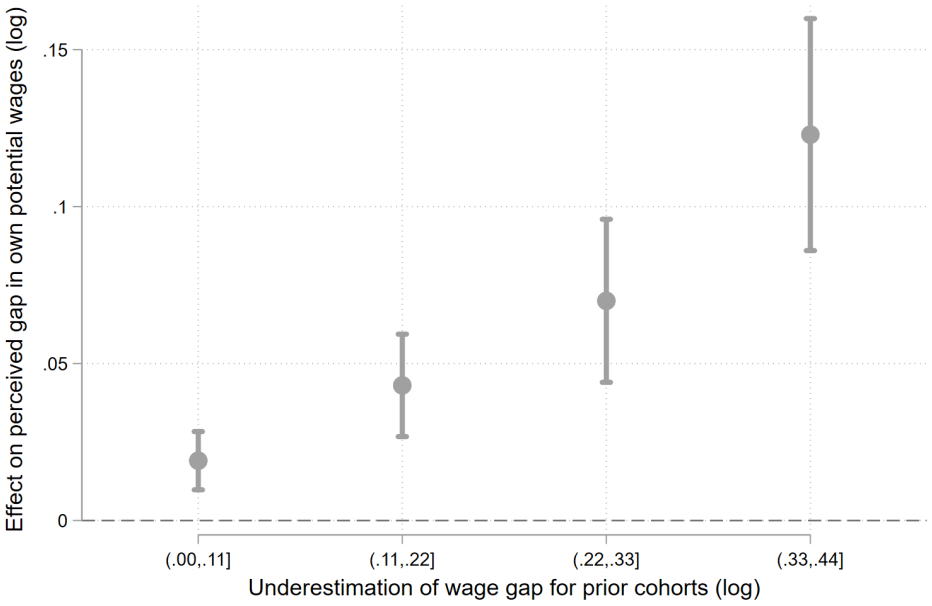
(a) Treatment effect on $\Delta\widetilde{w_{i,j,j'}}$



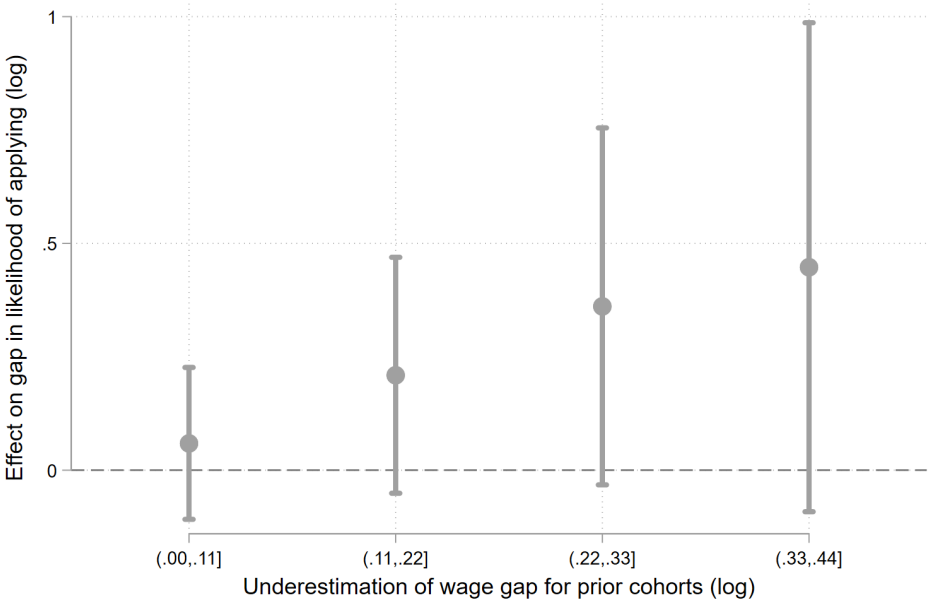
(b) Treatment effect on $\Delta\pi_{i,j,j'}$

Note: These figures examine the sensitivity of Figure 4 to alternative choices of bins. The figure is based on all individual-by-job pairs (i, j, j') in the study sample. The data is split into four equal-width bins according to the underestimation of the log wage gap for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$, with the job pairs ordered such that $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}} > 0$. The figures plot the effect of the information treatment within each bin, estimated via a linear regression that includes a constant and the treatment dummy. In Panel A, the outcome variable is the perceived gap in (log) own potential wages between the jobs, $\Delta\widetilde{w_{i,j,j'}}$. In Panel B, the outcome variable is the (log) relative likelihood of applying to the two jobs, $\Delta\pi_{i,j,j'}$. Error bars show 95% confidence intervals using clustering on individuals.

Figure A7: Heterogeneous effects by size of initial misperceptions, alternative bins 2



(a) Treatment effect on $\Delta\widetilde{w_{i,j,j'}}$



(b) Treatment effect on $\Delta\pi_{i,j,j'}$

Note: These figures examine the sensitivity of Figure 4 to alternative choices of bins. The figure is based on all individual-by-job pairs (i, j, j') in the study sample. The data is split into four equal-width bins according to the underestimation of the log wage gap for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$, with the job pairs ordered such that $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}} > 0$. The figures plot the effect of the information treatment within each bin, estimated via a linear regression that includes a constant and the treatment dummy. In Panel A, the outcome variable is the perceived gap in (log) own potential wages between the jobs, $\Delta\widetilde{w_{i,j,j'}}$. In Panel B, the outcome variable is the (log) relative likelihood of applying to the two jobs, $\Delta\pi_{i,j,j'}$. Error bars show 95% confidence intervals using clustering on individuals.

B Discrete choice job search framework

This section contains additional details and derivations regarding the discrete choice job search framework used in the main text.

B.1 Model setup, recap

The full model setup and notation is discussed in Section 5 of the main text. We briefly recap the key elements here for completeness. At a point in time, job seeker i has the opportunity to apply for one of three different jobs indexed by $j = 1, 2, 3$. Alternatively, she can apply to none of them and continue searching which we will treat as option $j = 0$. Let $Y_{i,j}$ denote the continuation value of being hired into job j and let U_i denote the continuation value of continued search. The surplus value of being hired into job j is assumed to be:

$$S_{i,j} = Y_{i,j} - U_i = \Psi_i W_{i,j}^{\delta} Z_{i,j} \xi_{i,j}. \tag{B.1}$$

Here $W_{i,j}$ is the wage worker i would earn in job j, $Z_{i,j}$ summarizes non-wage amenities, and $\xi_{i,j}$ is an idiosyncratic taste shock capturing other utility-relevant characteristics of the job. The parameter Ψ_i allows for individual heterogeneity in the general surplus from employment, while δ governs the curvature of surplus with respect to wages.

The surplus value of an application to job j relative to continued search, $A_{i,j}$, will equal the probability of being hired times the surplus from employment. Letting $P_{i,j}$ denote the probability of being hired at job j, we have

$$A_{i,j} = P_{i,j} S_{i,j}. \tag{B.2}$$

In what follows it is useful to collect the non-wage components of the surplus value of applying in the term

$$H_{i,j} = P_{i,j} Z_{i,j} \xi_{i,j}, \tag{B.3}$$

which we refer to as the non-pecuniary value of applying to job j. With this notation, $A_{i,j} = \Psi_i W_{i,j}^{\delta} H_{i,j}$.

When making application decisions, job seekers need not base choices on the actual application surplus value $A_{i,j}$. Instead, they act on the perceived surplus value $\widetilde{A_{i,j}}$, determined by perceived wages $\widetilde{W_{i,j}}$, perceived hiring probabilities $\widetilde{P_{i,j}}$ and perceived non-wage amenities $\widetilde{Z_{i,j}}$:

$$\widetilde{S_{i,j}} = \Psi_i \widetilde{W_{i,j}}^{\delta} \widetilde{Z_{i,j}} \xi_{i,j}, \tag{B.4}$$

$$\widetilde{A_{i,j}} = \widetilde{P_{i,j}} \widetilde{S_{i,j}}. \tag{B.5}$$

Additionally, we assume that applying for some job imposes an application cost of c_i.

Finally, we adopt the convenient notation that $A_{i,0}$ and $\widetilde{A_{i,0}}$ denote the actual and perceived surplus value of choosing not to apply anywhere. Since not applying anywhere guarantees that the worker continues searching, the surpluses $A_{i,0}$ and $\widetilde{A_{i,0}}$ are both mechanically 0.

B.2 Choice probabilities and the value of an application

Let j_i^* denote the actual application choice that the worker makes, with $j_i^* = 0$ corresponding to the choice to not apply anywhere. Optimizing behavior implies that the worker chooses the option that offers the highest perceived surplus:

$$j_i^* = \text{argmax}_{j \in \{0,1,2,3\}} \widetilde{A_{i,j}} - \mathbf{1}_{\{j \neq 0\}} c_i \tag{B.6}$$

Since the optimal choice is unaffected by monotone transformations of the objective, we can transform the objective in (B.6) by adding the application cost and taking logs: Defining

$$Q_{i,j} = \log\Big(\Big(\widetilde{A_{i,j}} - \mathbf{1}_{\{j \neq 0\}} c_i\Big) + c_i\Big)$$

the problem (B.6) becomes equivalent to

$$j_i^* = \text{argmax}_{j \in \{0,1,2,3\}} \quad Q_{i,j} \tag{B.7}$$

To see why this is useful, note that for $j = 1, 2, 3$ we have (using lowercase to denote logged values as in the main text)

$$Q_{i,j} = \psi_i + \widetilde{p_{i,j}} + \delta \widetilde{w_{i,j}} + \widetilde{z_{i,j}} + \log \xi_{i,j} \tag{B.8}$$

while for $j = 0$

$$Q_{i,0} = \log c_i \tag{B.9}$$

Under the assumed Frechet distribution, $\log \xi_{i,j}$ follows a Type I extreme value distribution with scale parameter σ. Equations (B.7), (B.8) and (B.9) therefore describe a standard discrete choice logit framework, albeit with a deterministic outside option (the option of not applying anywhere). Standard derivations for discrete choice logit models therefore yield familiar expressions for the choice probabilities, which we denote by $\Pi_{i,j}^* = \Pr(j_i^* = j)$. For $j = 1, 2, 3$ we can write this as in equation (9) from the main text:

$$\Pi_{i,j}^* = \frac{\exp((\alpha_i + \widetilde{p_{i,j}} + \delta \widetilde{w_{i,j}} + \widetilde{z_{i,j}})/\sigma)}{1 + \sum_{j'=1,2,3} \exp((\alpha_i + \widetilde{p_{i,j'}} + \delta \widetilde{w_{i,j'}} + \widetilde{z_{i,j'}})/\sigma)} \tag{B.10}$$

Here α_i is an alternative-invariant constant that indexes the overall attractiveness of applying to one of the three jobs relative to the deterministic outside option of not applying. It is defined as:

$$\alpha_i = \psi_i - \log c_i + \sigma \log \left(\frac{\exp\left[\sum_{j'=1}^3 \exp((\psi_i - \log c_i + \widetilde{p_{i,j'}} + \delta \widetilde{w_{i,j'}} + \widetilde{z_{i,j'}})/\sigma)\right] - 1}{\sum_{j'=1}^3 \exp((\psi_i - \log c_i + \widetilde{p_{i,j'}} + \delta \widetilde{w_{i,j'}} + \widetilde{z_{i,j'}})/\sigma)} \right).$$

For $j = 0$ we have

$$\Pi^*_{i,0} = \frac{1}{1+\sum_{j'=1,2,3} \exp((\alpha_i+\widetilde{p_{i,j'}}+\delta\widetilde{w_{i,j'}}+\widetilde{z_{i,j'}})/\sigma)} \tag{B.11}$$

For much of our analysis, we will be focusing on the likelihood that a given application goes to a particular job, i.e., the likelihood of applying to a given job j conditional on applying *somewhere*. We denote this by $\overline{\Pi}^*_{i,j}$:

$$\overline{\Pi}^*_{i,j} = \frac{\Pi^*_{i,j}}{1-\Pi^*_{i,0}} = \frac{\exp((\widetilde{p_{i,j}}+\delta\widetilde{w_{i,j}}+\widetilde{z_{i,j}})/\sigma)}{\sum_{j'=1,2,3} \exp((\widetilde{p_{i,j'}}+\delta\widetilde{w_{i,j'}}+\widetilde{z_{i,j'}})/\sigma)} \tag{B.12}$$

Finally, for the purpose of the welfare counterfactual later, it will be convenient to let $A^E_{i,j}$ denote the expected value of an application to job j, conditional on choosing to apply to job j:

$$\begin{aligned} A^E_{i,j} =& E[A_{i,j} \mid j^*_i = j] \\ =& \Psi_i P_{i,j} W^{\delta}_{i,j} Z_{i,j} E[\xi_{i,j} \mid j^*_i = j]. \end{aligned}$$

Letting Ω_i be the expected value of making an application, we then have:

$$\Omega_i = \sum_{j=1,2,3} \overline{\Pi}^*_{i,j} \times \left(A^E_{i,j} - c_i + U_i \right) \tag{B.13}$$

B.3 Link to data, 2SLS analysis and exclusion restriction

The model framework above links naturally to the survey data we use in our reduced form analysis. Elicited beliefs about the potential wage each job in the survey would offer the respondent is a direct measure of $\widetilde{W_{i,j}}$. Similarly, the reported probabilities of applying to each job in the survey, corresponds to the choice probabilities in equation (B.10) and (B.11). As convenient notation, we let $\Pi_{i,j}$ (without an asterisk) denote the reported probabilities in the data.[27]

[27] We do this to distinguish the data from the endogenous objects in the model. The distinction becomes relevant when we turn to consider counterfactuals further below.

Taking logs and differencing (B.10) across two jobs j and j' now yields the equation (with Δ denoting differences across pairs of jobs, as in the main text):

$$\Delta\pi_{i,j,j'} = \beta_1 \Delta\widetilde{w_{i,j,j'}} + \Delta u_{i,j,j'} \tag{B.14}$$

Here we have defined the random variable (error term) $\Delta u_{i,j,j'}$ as

$$\Delta u_{i,j,j'} = \frac{1}{\sigma}\left(\Delta\widetilde{p_{i,j,j'}} + \Delta\widetilde{z_{i,j,j'}}\right)$$

Additionally, we have defined the key reduced form parameter β_1 as

$$\beta_1 = \frac{\delta}{\sigma} \tag{B.15}$$

As emphasized in the main text, (B.14) implies that β_1 has the reduced form interpretation of being the causal effect of the wage gap between two jobs on the relative likelihood of applying to them.[28] Specifically, β_1 is the elasticity of the relative application likelihood between two jobs ($\Pi_{i,j}/\Pi_{i,j'}$) with respect to the perceived relative wage offers ($\widetilde{W}_{i,j}/\widetilde{W}_{i,j'}$). As shown in (B.15), however, β_1, also has a direct, formal mapping to key primitive parameters in the model framework. We use this further below to compute model counterfactuals from the reduced form estimates.

Equation (B.14) and the model framework can also be used to formally discuss the role played by the random information treatment and instrumental variable assumptions in our 2SLS results. The error term in (B.14), $\Delta u_{i,j,j'}$, reflects beliefs about all non-pecuniary differences that affect the attractiveness of applying to job j versus j', specifically the perceived difference in application success probability, $\Delta\widetilde{p_{i,j,j'}}$ and in the non-wage amenities, $\Delta\widetilde{z_{i,j,j'}}$. In general, we expect job seekers with particular beliefs about wage differences to likely also have particular beliefs about differences in these other dimensions. This leads to a classic identification problem because the regressor in (B.14) is correlated with the error term.

[28] To be precise here, under the model, β_1 is a causal effect in the sense that it measures the effect of changes in perceived wages, keeping all other non-random variables constant and the distribution of the taste shocks unchanged.

We address this identification problem by leveraging our randomized treatment assignment, T_i, in a 2SLS framework. As covered in Section 4 of the main text, for a given level of the wage gap underestimation for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$, our information treatment generates variation in perceived own potential wages, $\Delta\widetilde{w_{i,j,j'}}$. This means that it can be used to construct a relevant instrument. Moreover, due to random assignment, treated and untreated job seekers cannot have systematically different beliefs prior to treatment. Under the additional exclusion restriction that the information treatment also does not change beliefs about non-wage amenities or the application success probability, treatment will be independent of non-pecuniary beliefs, both overall and for a given level of the wage gap underestimation for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$:

$$T_i \perp \left(\Delta\widetilde{p_{i,j,j'}}, \Delta\widetilde{z_{i,j,j'}}\right) \Big| \left(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right) \tag{B.16}$$

As covered in Section 4.4, we use a range of additional questions in our survey to test this exclusion restriction directly and find it to be well-supported by the data. If we finally impose a standard 'linearity-of-controls' functional form assumption, we arrive at the second stage equation and instrument exogeneity condition used in our 2SLS analysis:[29]

$$\Delta\pi_{i,j,j'} = \beta_1 \Delta\widetilde{w_{i,j,j'}} + \beta_2 \left(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right) + \varepsilon_{i,j,j'} \tag{B.17}$$

$$E[\varepsilon_{i,j,j'} | T_i, \Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}] = 0 \tag{B.18}$$

[29] The usual functional form assumption imposes that in the absence of treatment, the conditional expectation of the error term is linear in the conditioning variable:

$$E\left[\Delta u_{i,j,j'} | T_i = 0, \left(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right)\right] = \beta_2\left(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right)$$

With this, the second stage equation then arises simply by appropriately defining its error term:

$$\varepsilon_{i,j,j'} = \Delta u_{i,j,j'} - E\left[\Delta u_{i,j,j'} | \left(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right), T_i = 0\right]$$

The instrument exogeneity condition follows from (B.16) because $\Delta u_{i,j,j'} = \frac{1}{\sigma}\left(\Delta\widetilde{p_{i,j,j'}} + \Delta\widetilde{z_{i,j,j'}}\right)$:

$$\begin{aligned}
&E\left[\varepsilon_{i,j,j'} | T_i, \Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right] = \\
&E\left[\Delta u_{i,j,j'} | T_i, \Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right] - E\left[\Delta u_{i,j,j'} | T_i = 0, \Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right] = \\
&E\left[\Delta u_{i,j,j'} | \Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right] - E\left[\Delta u_{i,j,j'} | \Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}\right] = 0
\end{aligned}$$

B.4 Own Wage Elasticity of Applications

As discussed above, the parameter of interest in our 2SLS analysis, β_1, measures the effect of the perceived *wage gap* between two jobs on the *relative likelihood* of applying to them. Using the discrete choice framework above, however, we can convert this into an own wage elasticity of applications to a given job, i.e., a measure of how much more likely job seekers are to apply for a given job when this job unilaterally increases its wage. This is a key parameter in the literature on job search and employer market power.

Differentiating the logged choice probability in (B.10) (including the definition of α_i) shows that for a given person i and job j the own wage elasticity is:[30]

$$\mathcal{E}_{i,j} = \frac{\partial \ln \Pi^*_{i,j}}{\partial \widetilde{w_{i,j}}} = \beta_1 \left[1 - \Pi^*_{i,j} \left(\frac{1 - \Pi^*_{i,0}\left(1 - \log \Pi^*_{i,0}\right)}{\left(1 - \Pi^*_{i,0}\right)^2} \right) \right] \tag{B.19}$$

To arrive at an aggregate elasticity over the sample, we plug in our estimate for β_1 as well as the observed choice probabilities $\Pi_{i,j}$, and then average over the N job seekers in our data and the three jobs they each face:

$$\widehat{\mathcal{E}}^a = \frac{1}{N} \sum_i \left(\frac{1}{3} \sum_{j=1,2,3} \hat{\beta}_1 \left[1 - \Pi_{i,j} \left(\frac{1 - \Pi_{i,0}(1 - \log \Pi_{i,0})}{(1 - \Pi_{i,0})^2} \right) \right] \right) \tag{B.20}$$

B.5 Counterfactuals without relative wage misperceptions

In Section 5 of the main text, we combine our reduced form estimates and the discrete choice framework to assess the overall costs of relative wage misperceptions. Specifically, for our control group of untreated individuals ($T_i = 0$), we compare actual current outcomes to a counterfactual in which all relative wage misperceptions are removed. The subsections below go through the key derivations and additional assumptions underlying this counterfactual.

For the purpose of describing the counterfactual, we first modify and introduce ad-

[30] Note that the expression for the elasticity here differs slightly from the standard logit elasticity formula $\mathcal{E}_{i,j} = \beta_1(1 - \Pi^*_{i,j})$. The difference reflects the deterministic outside option since the value of not applying anywhere is not subject to a random taste shock. The difference is quantitatively unimportant however because the likelihood of not applying anywhere is small in our data and $\frac{1 - \Pi^*_{i,0}\left(1 - \log \Pi^*_{i,0}\right)}{\left(1 - \Pi^*_{i,0}\right)^2} \to 1$ as $\Pi^*_{i,0} \to 0$.

ditional notation relative to the preceding sections. Let $\boldsymbol{X} = (X_1, X_2, X_3)$ and $\widetilde{\boldsymbol{X}} = (\widetilde{X_1}, \widetilde{X_2}, \widetilde{X_3})$ be arbitrary vectors of wages for the three jobs a given worker faces. Chiefly, we are interested in examining how application choices would be different under different beliefs. We thus let $j_i^*(\widetilde{\boldsymbol{X}})$ be the workers optimal application choice if the job seeker perceives the wages to be $\widetilde{\boldsymbol{X}}$ and let $\Pi_{i,j}^*\left(\widetilde{\boldsymbol{X}}\right)$ and $\overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{X}}\right)$ denote the corresponding application choice probabilities. Additionally, we consider counterfactuals for the utility value of making an application. We let $\Omega_i(\boldsymbol{X}, \widetilde{\boldsymbol{X}})$ denote the utility value of making an application if the wages are actually $\boldsymbol{X}$ but the worker perceives the wages to be $\widetilde{\boldsymbol{X}}$. In what follows, we additionally use $U_i\left(\boldsymbol{X}, \widetilde{\boldsymbol{X}}\right)$ and $A_{i,j}^E\left(\boldsymbol{X}, \widetilde{\boldsymbol{X}}\right)$ for the corresponding continuation value of unemployment and conditional surplus value of applying to a given job.

In terms of the specific counterfactual comparison we are interested in, we let $\boldsymbol{W_i} = (W_{i,1}, W_{i,2}, W_{i,3})$ be the vector of actual wages in the three jobs for person i and $\widetilde{\boldsymbol{W}_i^0} = (\widetilde{W_{i,1}^0}, \widetilde{W_{i,2}^0}, \widetilde{W_{i,3}^0})$ be their vector of currently perceived wages. We are then interested in computing outcomes under a counterfactual set of wage beliefs $\widetilde{\boldsymbol{W}_i^1} = (\widetilde{W_{i,1}^1}, \widetilde{W_{i,2}^1}, \widetilde{W_{i,3}^1})$ in which there is no relative wage misperception. This corresponds to $\widetilde{\boldsymbol{W}_i^1}$ satisfying:

$$\widetilde{\boldsymbol{W}_i^1} = \iota_i \boldsymbol{W}_i \tag{B.21}$$

for some $\iota_i > 0$. Note that the constant ι_i here can be interpreted as the extent of overestimation/underestimation of the general level of wages. As will become clear below, our counterfactual for application behavior calculation is invariant to ι_i, reflecting that these application counterfactuals only capture the effects of removing *relative* wage misperceptions, not misperceptions about the overall level of wages.

To make sure that our welfare counterfactual also isolates the costs of relative wage misperceptions distorting applications across jobs, our welfare counterfactual will impose that the likelihood of applying to some job (i.e. the rate of making applications) is unchanged. This corresponds to choosing ι_i such that

$$\Pi_{i,0}^*\left(\widetilde{\boldsymbol{W}_i^1}\right) = \Pi_{i,0}^*\left(\widetilde{\boldsymbol{W}_i^0}\right)$$

This restriction ensures that our welfare counterfactual is not affected by misperceptions about the overall wage level that lead workers to apply to more or fewer jobs overall.

B.5.1 Counterfactual application behavior

The first counterfactual of interest regards how application behavior would change if job seeker i did not have any misperceptions about relative wages. Specifically, we want to compare the likelihood that an application currently goes to a particular job, $\overline{\Pi}^*_{i,j}\left(\widetilde{\boldsymbol{W}_i^0}\right)$ to the counterfactual likelihood without relative wage misperceptions $\overline{\Pi}^*_{i,j}\left(\widetilde{\boldsymbol{W}_i^1}\right)$. Given that we are focusing on individuals in the control group, $\overline{\Pi}^*_{i,j}\left(\widetilde{\boldsymbol{W}_i^0}\right)$ can be directly computed from the survey responses about likelihood of applying to the different jobs:

$$\overline{\Pi}^*_{i,j}\left(\widetilde{\boldsymbol{W}_i^0}\right) = \frac{\Pi_{i,j}}{1-\Pi_{i,0}}$$

Further note that *if* we knew the current perceived and actual wages, $\widetilde{\boldsymbol{W}_i^0}$ and $\boldsymbol{W_i}$, the only thing necessary to compute $\overline{\Pi}^*_{i,j}\left(\widetilde{\boldsymbol{W}_i^1}\right)$ is the reduced form parameter β_1 since (B.12) and (B.21) imply:

$$\overline{\Pi}^*_{i,j}\left(\widetilde{\boldsymbol{W}_i^1}\right) = \frac{\left(\frac{W_{i,j}}{\widetilde{W^0_{i,j}}}\right)^{\beta_1 \frac{\Pi_{i,j}}{1-\Pi_{i,0}}}}{\sum_{j'=1,2,3}\left(\frac{W_{i,j'}}{\widetilde{W^0_{i,j'}}}\right)^{\beta_1 \frac{\Pi_{i,j'}}{1-\Pi_{i,0}}}}$$

For the control group we in fact observe $\widetilde{W^0_{i,j}}$ directly from the survey questions:

$$\widetilde{W^0_{i,j}} = \widetilde{W_{i,j}}$$

To compute the counterfactual however, we still need to take a stand on how actual potential wage differences differ from perceptions to get a handle on $\boldsymbol{W_i}$. Given the content of our survey, our benchmark will be to simply equate misperceptions about own potential (log) wage gaps to the measured misperceptions about wage gaps for prior cohorts from the survey, i.e. we assume:

$$\Delta v_{i,j,j'} - \widetilde{\Delta v_{i,j,j'}} = \Delta w_{i,j,j'} - \widetilde{\Delta w^0_{i,j,j'}} \tag{B.22}$$

Since this assumption additionally allows us to infer $\boldsymbol{W}_i$, imposing (B.22) is enough for us to compute the counterfactual application likelihood $\overline{\Pi}^*_{i,j}\left(\widetilde{\boldsymbol{W}_i^1}\right)$ using our estimate for β_1 and the equation further above.

Having computed the counterfactual application probabilities, our first measure of distortions is the share of i's applications that would go to different jobs in the absence of relative wage misperceptions. We summarize this by computing the predicted total change in the shares of applications going to the different jobs if relative misperceptions were removed:[31]

$$\frac{1}{2}\sum_{j=1,2,3}\left|\overline{\Pi}^*_{i,j}\left(\widetilde{\boldsymbol{W}_i^1}\right)-\overline{\Pi}^*_{i,j}\left(\widetilde{\boldsymbol{W}_i^0}\right)\right|$$

B.5.2 Welfare costs

Finally, leaning further into the model structure, we can also use this framework to assess the welfare costs of relative wage misperceptions.

Specifically, we consider what the total utility value of sending an application is with and without relative wage misperceptions, i.e. how much does $\Omega_i(\boldsymbol{W}_i, \widetilde{\boldsymbol{W}_i^0})$ differ from $\Omega_i(\boldsymbol{W}_i, \widetilde{\boldsymbol{W}_i^1})$. To make this comparison on a meaningful scale, we ask how much all wages would have to decrease in order to deliver the same welfare loss as the introduction of relative wage misperceptions. In the absence of relative wage misperceptions, if all wages for i are scaled down by k_i, the value of making application will be $\Omega_i(k_i\boldsymbol{W}_i, \widetilde{\boldsymbol{W}_i^1})$. Accordingly, we compute the value k_i that satisfies:

$$\Omega_i(k_i\boldsymbol{W}_i, \widetilde{\boldsymbol{W}_i^1}) = \Omega_i(\boldsymbol{W}_i, \widetilde{\boldsymbol{W}_i^0}) \tag{B.23}$$

To do this requires us to impose three additional assumptions. First, we need to take a stand on the continuation value if workers do not get a job but continue searching, U_i. Mirroring standard job search models, we simply assume that such job seekers earn some exogenous flow utility b_i and get the opportunity to make an application again at

[31] Invoking a law of large numbers, this measure formally measures the share of applications that will have gone to a different job, after i have made a large number of applications. A different summary measure would be to focus on the probability that a given single application goes to a different job, e.g. $Pr(j_i^*(\widetilde{W_{i,j}^0}) \neq j_i^*(\widetilde{W_{i,j}^1}))$. Experimenting with this type of measure gives virtually identical results.

some fixed, individual-specific Poisson arrival rate, λ_i. Assuming also that the application decision problem is stationary and a time discounting rate of ρ_i, this implies the following continuous time Bellman equation:

$$\rho_i U_i(\boldsymbol{X}, \widetilde{\boldsymbol{X}}) = b_i + \lambda_i \Big(1 - \Pi^*_{i,0}(\widetilde{\boldsymbol{X}})\Big) \Big(\Omega_i(\boldsymbol{X}, \widetilde{\boldsymbol{X}}) - U_i(\boldsymbol{X}, \widetilde{\boldsymbol{X}})\Big) \tag{B.24}$$

Second, to assess actual utilities, we also need to take a stand on the relationship between perceived and actual non-pecuniary values, i.e. hiring likelihoods and non-wage amenities. Here we simply adopt the benchmark assumption that workers have correct beliefs about non-pecuniary values:

$$P_{i,j} = \widetilde{P_{i,j}} \tag{B.25}$$

$$Z_{i,j} = \widetilde{Z_{i,j}} \tag{B.26}$$

Finally, we need to impose a parameter restriction on the surplus utility from jobs. This reflects the usual challenge that choice behavior does not separately identify the scale of the taste shocks.[32] For our main results we impose that surplus is linear in the wage, corresponding to risk neutrality with respect to wages

$$\delta = 1 \tag{B.27}$$

As we show in Tables A17 and A18, setting $\delta < 1$ to introduce risk aversion leads to slightly larger estimated welfare costs of misperceptions.

With the additional assumptions (B.24), (B.25), and (B.26) and a normalization like (B.27), we are able to compute the utility cost of relative wage misperceptions, k_i for each individual in our control group. Subsection B.5.4 gives the details of this derivation. The key insight however is that for given σ, the conditional choice probabilities without relative

[32] From our reduced form equations, the identified parameter governing application behavior is $\beta_1 = \frac{1}{\sigma}\delta$. Scaling up σ and δ by the same amount thus leaves application behavior unchanged.

misperceptions identify the deterministic application values up to scale since

$$\eta_i \left(\overline{\Pi}^*_{i,j} \left(\widetilde{\boldsymbol{W}_i^1} \right) \right)^{\sigma} = \Psi_i P_{i,j} W_{i,j}^{\delta} Z_{i,j} \tag{B.28}$$

with η_i an alternative-invariant constant.

Finally, to expand on how the welfare costs arise, it is instructive to examine how distortions lead the job seekers to target jobs with systematically different wage-levels or non-pecuniary values. Under some wage beliefs $\widetilde{\boldsymbol{X}}$, we can consider the average applied-for wage, $E\left[W_{i,j_i^*(\widetilde{\boldsymbol{X}})} \mid j_i^*(\widetilde{\boldsymbol{X}}) \neq 0\right]$, as well as the average non-pecuniary value of a sent application, $E\left[H_{i,j_i^*(\widetilde{\boldsymbol{X}})} \mid j_i^*(\widetilde{\boldsymbol{X}}) \neq 0\right]$. As we detail in Subsection B.5.4, the assumptions above allow us to compute the effect of wage misperceptions on both these quantities:

$$\frac{E\left[W_{i,j_i^*(\widetilde{\boldsymbol{W}_i^0})} \mid j_i^*(\widetilde{\boldsymbol{W}_i^0}) \neq 0\right] - E\left[W_{i,j_i^*(\widetilde{\boldsymbol{W}_i^1})} \mid j_i^*(\widetilde{\boldsymbol{W}_i^1}) \neq 0\right]}{E\left[W_{i,j_i^*(\widetilde{\boldsymbol{W}_i^1})} \mid j_i^*(\widetilde{\boldsymbol{W}_i^1}) \neq 0\right]} \tag{B.29}$$

and

$$\frac{E\left[H_{i,j_i^*(\widetilde{\boldsymbol{W}_i^0})} \mid j_i^*(\widetilde{\boldsymbol{W}_i^0}) \neq 0\right] - E\left[H_{i,j_i^*(\widetilde{\boldsymbol{W}_i^1})} \mid j_i^*(\widetilde{\boldsymbol{W}_i^1}) \neq 0\right]}{E\left[H_{i,j_i^*(\widetilde{\boldsymbol{W}_i^1})} \mid j_i^*(\widetilde{\boldsymbol{W}_i^1}) \neq 0\right]} \tag{B.30}$$

B.5.3 Implementation and inference

To generate the counterfactual results presented in the main text, we compute the different quantities of interest described above for each individual in the control group, while replacing β_1 with its estimated value $\hat{\beta}_1$ from the baseline 2SLS specification (Table 5, Panel C, Column 1). We then compute averages or percentiles over the sample as relevant. To quantify the sampling uncertainty around these estimated counterfactuals, we produce standard errors via a bootstrap procedure that resamples individuals and then recomputes both the 2SLS estimate and the counterfactual quantities of interest within each bootstrap sample. For the welfare counterfactual, we use the normalization in (B.27).

B.5.4 Detailed derivation of welfare counterfactuals

To compute our measure of the welfare cost k_i, we first reorganize the assumed Bellman equation for the value of continued search, (B.24), to arrive at:

$$U_i(\boldsymbol{X}, \widetilde{\boldsymbol{X}}) = \frac{b_i + \lambda_i \Big(1 - \Pi^*_{i,0}(\widetilde{\boldsymbol{X}})\Big) \Omega_i(\boldsymbol{X}, \widetilde{\boldsymbol{X}})}{\rho_i + \lambda_i \Big(1 - \Pi^*_{i,0}(\widetilde{\boldsymbol{X}})\Big)}.$$

Plugging this into the equation for the value of an application, (B.13), and rearranging yields:

$$\Omega_i(\boldsymbol{X}, \widetilde{\boldsymbol{X}}) = \frac{b_i}{\rho_i} + \frac{\rho_i + \lambda_i \Big(1 - \Pi^*_{i,0}(\widetilde{\boldsymbol{X}})\Big)}{\rho_i} \left(\sum_{j=1,2,3} \overline{\Pi}^*_{i,j}(\widetilde{\boldsymbol{X}}) A^E_{i,j}(\boldsymbol{X}, \widetilde{\boldsymbol{X}}) - c_i \right).$$

The normalization imposed above fixes $\Pi^*_{i,0}\Big(\widetilde{\boldsymbol{W}^1_i}\Big) = \Pi^*_{i,0}\Big(\widetilde{\boldsymbol{W}^0_i}\Big)$. This means that the terms involving b_i, c_i, λ_i and ρ_i all cancel when comparing the two sides of (B.23). Therefore, the equation defining k_i is equivalent to:

$$\sum_{j=1,2,3} \overline{\Pi}^*_{i,j}(\widetilde{\boldsymbol{W}^1_i}) A^E_{i,j}(k_i \boldsymbol{W}_i, \widetilde{\boldsymbol{W}^1_i}) = \sum_{j=1,2,3} \overline{\Pi}^*_{i,j}(\widetilde{\boldsymbol{W}^0_i}) A^E_{i,j}(\boldsymbol{W}_i, \widetilde{\boldsymbol{W}^0_i}).$$

Since scaling actual wages by k_i scales the surplus value of every application by k_i^δ, this gives:

$$k_i = \left(\frac{\sum_{j=1,2,3} \overline{\Pi}^*_{i,j}(\widetilde{\boldsymbol{W}^0_i}) A^E_{i,j}(\boldsymbol{W}_i, \widetilde{\boldsymbol{W}^0_i})}{\sum_{j=1,2,3} \overline{\Pi}^*_{i,j}(\widetilde{\boldsymbol{W}^1_i}) A^E_{i,j}(\boldsymbol{W}_i, \widetilde{\boldsymbol{W}^1_i})} \right)^{1/\delta} \tag{B.31}$$

Next, using (B.28), we have for any wage beliefs $\widetilde{\boldsymbol{X}}$:

$$\begin{aligned} A^E_{i,j}(\boldsymbol{W}_i, \widetilde{\boldsymbol{X}}) &= \Psi_i W^\delta_{i,j} P_{i,j} Z_{i,j} E\Big[\xi_{i,j} \mid j^*_i(\widetilde{\boldsymbol{X}}) = j\Big] \\ &= \eta_i \Big(\overline{\Pi}^*_{i,j}\Big(\widetilde{\boldsymbol{W}^1_i}\Big)\Big)^\sigma E\Big[\xi_{i,j} \mid j^*_i(\widetilde{\boldsymbol{X}}) = j\Big]. \end{aligned}$$

Moreover, from the assumed Frechet distribution (assuming $\sigma < 1$) we get:

$$E\left[\xi_{i,j} \mid j_i^*(\widetilde{\boldsymbol{X}}) = j\right] = \left(\overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{X}}\right)\right)^{-\sigma}\left[\frac{\gamma\left(1-\sigma, -\log \Pi_{i,0}^*\left(\widetilde{\boldsymbol{X}}\right)\right)}{1-\Pi_{i,0}^*\left(\widetilde{\boldsymbol{X}}\right)}\right],$$

where $\gamma(\cdot,\cdot)$ is the lower incomplete gamma function. Plugging this into (B.31), note that since $\Pi_{i,0}^*$ is unchanged in our counterfactual, both the fraction in square brackets and the constant η_i cancel, implying:

$$k_i = \left(\frac{\sum_{j=1,2,3} \overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^0}\right)\left(\overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^1}\right)\right)^{\sigma}\left(\overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^0}\right)\right)^{-\sigma}}{\sum_{j=1,2,3} \overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^1}\right)\left(\overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^1}\right)\right)^{\sigma}\left(\overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^1}\right)\right)^{-\sigma}}\right)^{1/\delta}.$$

With our assumptions above, this formula can be implemented given our data and reduced form results: Given an estimate of β_1, the actual and counterfactual application probabilities, $\overline{\Pi}_{i,j}^*(\widetilde{\boldsymbol{W}_i^0})$ and $\overline{\Pi}_{i,j}^*(\widetilde{\boldsymbol{W}_i^1})$, can be computed under assumption (B.22). Additionally, under the normalization in (B.27), the values for δ and σ can also be inferred from the reduced form estimates because $\beta_1 = \frac{\delta}{\sigma}$.

Finally, we consider computation of the effect of wage misperceptions on average applied-for wages and non-pecuniary values. The quantity in (B.29) can be computed directly via:

$$\frac{\sum_{j=1,2,3} \overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^0}\right) W_{i,j} - \sum_{j=1,2,3} \overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^1}\right) W_{i,j}}{\sum_{j=1,2,3} \overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^1}\right) W_{i,j}}.$$

For the quantity in (B.30), we have the analogous expression:

$$\frac{\sum_{j=1,2,3} \overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^0}\right) E\left[H_{i,j} \mid j_i^*(\widetilde{\boldsymbol{W}_i^0}) = j\right] - \sum_{j=1,2,3} \overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^1}\right) E\left[H_{i,j} \mid j_i^*(\widetilde{\boldsymbol{W}_i^1}) = j\right]}{\sum_{j=1,2,3} \overline{\Pi}_{i,j}^*\left(\widetilde{\boldsymbol{W}_i^1}\right) E\left[H_{i,j} \mid j_i^*(\widetilde{\boldsymbol{W}_i^1}) = j\right]}.$$

To compute this, we combine (B.28) with the definition $H_{i,j} = P_{i,j} Z_{i,j} \xi_{i,j}$ and the condi-

tional expectation of $\xi_{i,j}$ above. These imply:

$$E\Big[H_{i,j} \mid j_i^*(\widetilde{\boldsymbol{X}}) = j\Big] = \frac{\eta_i}{\Psi_i} W_{i,j}^{-\delta} \Big(\overline{\Pi}_{i,j}^*\Big(\widetilde{\boldsymbol{W}_i^1}\Big)\Big)^{\sigma} \Big(\overline{\Pi}_{i,j}^*\Big(\widetilde{\boldsymbol{X}}\Big)\Big)^{-\sigma} \times \left[\frac{\gamma\Big(1-\sigma, -\log \Pi_{i,0}^*\Big(\widetilde{\boldsymbol{X}}\Big)\Big)}{1-\Pi_{i,0}^*\Big(\widetilde{\boldsymbol{X}}\Big)}\right].$$

Plugging this in, the constant η_i/Ψ_i and the outside-option term in square brackets cancel in (B.30), and we arrive at:

$$\frac{\sum\limits_{j=1,2,3} W_{i,j}^{-\delta} \Big(\overline{\Pi}_{i,j}^*\Big(\widetilde{\boldsymbol{W}_i^1}\Big)\Big)^{\sigma} \Big(\overline{\Pi}_{i,j}^*\Big(\widetilde{\boldsymbol{W}_i^0}\Big)\Big)^{1-\sigma} - \sum\limits_{j=1,2,3} W_{i,j}^{-\delta} \overline{\Pi}_{i,j}^*\Big(\widetilde{\boldsymbol{W}_i^1}\Big)}{\sum\limits_{j=1,2,3} W_{i,j}^{-\delta} \overline{\Pi}_{i,j}^*\Big(\widetilde{\boldsymbol{W}_i^1}\Big)}.$$

C Additional details of the survey

This section i) outlines the data-driven procedure used to create the major-specific job types in the survey and also includes a full list of job types translated to English (Section C.1), ii) presents the full survey instructions translated into English (Section C.2), and iii) shows screenshots of the interactive figures shown in the survey (Section C.3). The original Danish survey instructions can be found at https://tinyurl.com/zjw6dc83.

C.1 Details on creation of job types

Creation of job types This section of the appendix provides additional details on the data-driven procedure used to construct job types described in Section 2.3. For each major (measured using 6-digit Danish ISCED codes), we construct these job types following three objectives: First, they must be relevant and capture a substantial share of respondents' realistic employment opportunities. Second, they should differ in wages to ensure meaningful trade-offs. Third, they must correspond directly to categories in administrative records, enabling us to build objective benchmarks for typical wages using administrative data.

In practice, for each major, we define the three job types by grouping jobs based on some combination of occupation, industry, firm sector, and firm size. We implement a data-driven procedure to select the appropriate job groupings using administrative data on the first jobs of graduates from the same major between 2010-2018, starting in each graduate's month of graduation and only including graduates who find a job within 12 months. In order to avoid issues with multiple jobs in the same month, we prioritize jobs that have at least 3 months of positive earnings in the job spell and the highest mean earnings of the job spell. Job spells are defined as person-firm connections with gaps of at most 3 months between consecutive months of employment.

For each major, we consider all possible ways of grouping jobs by firm sector and size (public/private, above or below 50 employees), occupation codes (using either 3, 4, or 6 digit Danish ISCO codes), and/or industry codes (using either 1- or 2-digit Danish NACE codes). Since the concept of firm size is not well-defined for public sector jobs, we only allow groupings on firm size when also grouping into private sector jobs. Among all such possible

groupings, we then select those that satisfy two conditions: i) the 3 most common job types must cover more than 40% of the transitions into first jobs for prior graduates and ii) the 3rd most common job type must cover at least 5% of transitions. These conditions ensure that the three most common job types altogether cover a substantial share of relevant jobs and that none of them are too narrow.

If multiple groupings satisfy the two conditions, we choose the grouping that best predicts entry-level wages. For each grouping, we calculate the out-of-sample mean squared error of the prediction from a regression of log wages on dummies for the job types using 5-fold cross-validation. The grouping that has the lowest average mean squared error is then chosen. Wages are here based on the average monthly earnings for the first 12 months after the hiring month for any employment. The hiring month is dropped due to instability, e.g. if a person is hired in the middle of the month.

After hand-checking the resulting job types for each major and making marginal adjustments, this procedure gives us three job types based on basic categories that we can intuitively describe to each survey respondent. The final paragraph in this section describes the few manual changes we made to labels of occupations and industries to ensure clarity. Additionally, for two majors, Philosophy PhD (DISCED: 802520) and Information science bachelor (DISCED: 602535) graduates, we found an error in a particular 2010 occupation code that was part of one of the three most common job types in our selected grouping. For Information science bachelor we addressed this by choosing a different grouping satisfying condition i) and ii) since more were available (specifically the grouping with the second best out-of-sample mean squared error for predicting wages). For Philosophy PhD no other groupings satisfied the conditions however. Accordingly, we reran our procedure using data on only the 2011-2018 cohorts to produce the grouping for this major.

Selection of major programs into the experiment For very specialized major programs, where all graduates take similar jobs, the procedure above does not deliver any job types because no grouping satisfies conditions i) and ii). This reflects the fact that graduates in these programs de facto make their career-defining decisions before entering the labor market and thus are not affected by the potential wage misperceptions we study

in this paper. Accordingly, we exclude these majors from our analysis. To avoid asking only about broad sectors, we also exclude majors where only coarse grouping into public vs. small/large private sector jobs satisfies conditions i) and ii). Finally, we exclude major programs where we observe very few graduates in past administrative data, since we cannot calculate precise measures of typical wages for those. We exclude majors where fewer than 50 people transitioned to a job type in 2010-2018. These restrictions lead us to focus on 88 higher-education programs, covering 76% of higher-education graduates in 2010-2018 (see Appendix Table A1). We refer to these as the "selected" majors. Individuals from non-selected majors could participate in the survey but were ineligible for the randomized information treatment and are excluded from our analysis.

Table C1: Examples of job types asked about in the survey for different majors

Degree	**Job types**	**Categories**
Ms. Economics DISCED: 703950	- Job in the private sector, in the industry: banking, financial, or insurance activities - Job in the public sector, in the industry: public administration, defense and social security - Job in the private sector, in the industry: specialized services, incl. business administration	- Public/private - Industry, 1-digit
Ms. Physics DISCED: 703530	- Job in the public sector, with the occupation: teaching and research at higher education - Job in the public sector, with the occupation: teaching at upper secondary level - Job in the private sector (>50 employees), with the occupation: software and application development	- Public/private/size - Occupation, 3-digit

Notes: Examples of the job types, that were showed to the respondents, for two different majors. They are translated from Danish.

Manual changes made to job types After creating the job types using the approach explained in section 2.3, all job types were examined manually to ensure consistency and interpretability. Relative to the default labels for industry and occupation, given by Statistics Denmark, we changed 18 occupation labels and 19 industry labels, most of which were smaller word changes for clarity or other typesetting changes. The remaining changes consisted of correcting instances where the default label described an industry or occupation as the residual of the other categories. For example ISCO/DISCO code 2149 is named "Other engineering work (except Electrical engineering)", referring to the other categories under the 21 code. But as our respondents would not be able to see all other categories, we created a new label, spelling out the content of this residual.

C.1.1 Full list of job types for major that were eligible for treatment

Table C2: Educations eligible for treatment and associated job types

Education level	Education narrow field	Jobtype 1	Jobtype 2	Jobtype 3
Short-cycle higher education	Multimedia design	Job in the industry: wholesale and retail trade	Job in the industry: information and communication	Job in the industry: administrative services
Short-cycle higher education	Financial advisor etc.	Job in the private sector (>50 employees), in the industry: financial services excluding insurance	Job in the private sector ($\leq$ 50 employees), in the industry: real estate	Job in the private sector (>50 employees), in the industry: retail trade excluding motor vehicles
Short-cycle higher education	Trade and marketing economist, etc.	Job in the private sector (>50 employees), in the industry: wholesale and retail trade	Job in the private sector ($\leq$ 50 employees), in the industry: wholesale and retail trade	Job in the private sector ($\leq$ 50 employees), in the industry: accommodation and food service
Short-cycle higher education	Service economist etc.	Job in the private sector, in the industry: food service activities	Job in the private sector, in the industry: accommodation	Job in the private sector, in the industry: retail trade excluding motor vehicles
Short-cycle higher education	Construction and construction engineering	Job in the industry: architecture and engineering, technical testing and analysis	Job in the industry: specialized construction	Job in the industry: residential construction and building projects
Short-cycle higher education	Electronics and IT	Job in the private sector ($\leq$ 50 employees), in the industry: information and communication	Job in the private sector (>50 employees), in the industry: information and communication	Job in the private sector (>50 employees), in the industry: wholesale and retail trade
Short-cycle higher education	Installer of high voltage and plumbing technology	Job in the private sector, in the industry: construction	Job in the private sector, in the industry: specialized services, incl. business administration	Job in the private sector, in the industry: manufacturing
Short-cycle higher education	Technical, production and product development	Job in the private sector, in the industry: retail trade excluding motor vehicles	Job in the private sector, in the industry: wholesale trade excluding motor vehicles	Job in the private sector, in the industry: manufacture of machinery and equipment
Short-cycle higher education	Laboratory, food and process technology	Job in the industry: manufacturing	Job in the industry: wholesale and retail trade	Job in the industry: administrative services
Short-cycle higher education	Laboratory Technician	Job in the private sector, in the industry: manufacturing, with the occupation: laboratory technician work	Job in the public sector, in the industry: education, with the occupation: laboratory technician work	Job in the private sector, in the industry: specialized services, with the occupation: laboratory technician work
Short-cycle higher education	Agricultural technology	Job in the private sector, in the industry: agriculture, hunting, forestry and fishing	Job in the private sector, in the industry: administrative services	Job in the public sector, in the industry: education
Medium-cycle higher education	Pedagogue	Job in the public sector, with the occupation: childcare/child educator	Job in the public sector, with the occupation: childcare/child educator, assistant	Job in the public sector, with the occupation: special needs education
Medium-cycle higher education	Elementary school teacher	Job in the public sector, in the industry: education, with the occupation: teaching at elementary school level	Job in the public sector, in the industry: residential care, with the occupation: teaching at elementary school level	Job in the private sector, in the industry: education, with the occupation: teaching at elementary school level

Continued on next page

Table C2: Educations eligible for treatment and associated job types

Education level	Education narrow field	Jobtype 1	Jobtype 2	Jobtype 3
Medium-cycle higher education	Graphical design and communication	Job in the industry: advertising and market research	Job in the industry: services in design, photography, translation, agricultural consultancy etc.	Job in the industry: wholesale trade excluding motor vehicles
Medium-cycle higher education	Journalist and journalistic work	Job in the private sector (>50 employees), in the industry: information and communication, with the occupation: writing, journalism and linguistics	Job in the public sector, in the industry: information and communication, with the occupation: writing, journalism and linguistics	Job in the private sector ($\leq$ 50 employees), in the industry: information and communication, with the occupation: writing, journalism and linguistics
Medium-cycle higher education	Media and communication, other	Job in the industry: information and communication	Job in the industry: specialized services, incl. business administration	Job in the industry: wholesale and retail trade
Medium-cycle higher education	Actor	Job in the private sector (>50 employees), in the industry: creative activities, arts and entertainment, with the occupation: acting	Job in the private sector ($\leq$ 50 employees), in the industry: creative activities, arts and entertainment, with the occupation: acting	Job in the private sector ($\leq$ 50 employees), in the industry: creative activities, arts and entertainment, with the occupation: film, theater direction and production
Medium-cycle higher education	Design	Job in the private sector (>50 employees), in the industry: wholesale and retail trade	Job in the private sector ($\leq$ 50 employees), in the industry: wholesale and retail trade	Job in the private sector ($\leq$ 50 employees), in the industry: specialized services, incl. business administration
Medium-cycle higher education	Administration etc.	Job with the occupation: general office work not requiring advanced degree	Job with the occupation: retail sales work	Job with the occupation: customer information
Medium-cycle higher education	Finance etc.	Job in the private sector (>50 employees), in the industry: financial services excluding insurance	Job in the private sector ($\leq$ 50 employees), in the industry: real estate	Job in the private sector (>50 employees), in the industry: insurance, reinsurance and pension funding
Medium-cycle higher education	Trade and marketing, etc.	Job in the industry: wholesale and retail trade	Job in the industry: administrative services	Job in the industry: accommodation and food service
Medium-cycle higher education	Social counseling and mediation, etc.	Job in the public sector, in the industry: education, with the occupation: social counseling	Job in the public sector, in the industry: health and social services, with the occupation: social counseling	Job in the public sector, in the industry: education, with the occupation: general office work not requiring advanced degree
Medium-cycle higher education	Construction and construction engineering	Job in the industry: architecture and engineering, technical testing and analysis	Job in the industry: residential construction and building projects	Job in the industry: specialized construction
Medium-cycle higher education	Electronics and IT, technical	Job with the occupation: software and application development	Job with the occupation: engineering work (except electrical engineering)	Job with the occupation: electrical engineering work
Medium-cycle higher education	Mechanical engineering	Job in the private sector (>50 employees), in the industry: manufacturing	Job in the private sector (>50 employees), in the industry: specialized services, incl. business administration	Job in the private sector (>50 employees), in the industry: transport and freight handling
Medium-cycle higher education	Technical, production and product development	Job in the industry: manufacturing	Job in the industry: wholesale and retail trade	Job in the industry: specialized services, incl. business administration

Continued on next page

Table C2: Educations eligible for treatment and associated job types

Education level	Education narrow field	Jobtype 1	Jobtype 2	Jobtype 3
Medium-cycle higher education	Other technical educations	Job in the industry: manufacturing	Job in the industry: wholesale and retail trade	Job in the industry: specialized services, incl. business administration
Medium-cycle higher education	Nutrition and Health	Job in the industry: health and social services	Job in the industry: education	Job in the industry: wholesale and retail trade
Medium-cycle higher education	Occupational and physiotherapist	Job in the public sector, with the occupation: occupational therapy	Job in the public sector, with the occupation: physiotherapy	Job in the private sector, with the occupation: physiotherapy
Medium-cycle higher education	Nursing and health care	Job in the public sector, in the industry: health care, with the occupation: nursing, basic functions	Job in the public sector, in the industry: health care, with the occupation: social and health work at institutions and hospitals, helpers	Job in the public sector, in the industry: health care, with the occupation: social and health work at institutions and hospitals, assistants
Medium-cycle higher education	Officer in the defense	Job in the industry: defense, with the occupation: military work at officer level	Job in the industry: defense, with the occupation: military work, neither officer nor non-commissioned officer level	Job in the industry: defense, with the occupation: military work at non-commissioned officer level
Bachelor degree	Humanities without further specification	Job in the industry: health and social services	Job in the industry: wholesale and retail trade	Job in the industry: education
Bachelor degree	History	Job in the industry: education	Job in the industry: culture, entertainment and sports	Job in the industry: health and social services
Bachelor degree	Film, theater and musicology, etc.	Job in the industry: information and communication	Job in the industry: education	Job in the industry: wholesale and retail trade
Bachelor degree	Communication and dissemination	Job in the private sector, in the industry: wholesale and retail trade	Job in the public sector, in the industry: education	Job in the private sector, in the industry: information and communication
Bachelor degree	Business language bachelors	Job in the industry: wholesale and retail trade	Job in the industry: education	Job in the industry: information and communication
Bachelor degree	Business language, combined	Job in the industry: wholesale and retail trade	Job in the industry: specialized services, incl. business administration	Job in the industry: information and communication
Bachelor degree	Mathematics	Job in the industry: education	Job in the industry: banking, financial, or insurance activities	Job in the industry: information and communication
Bachelor degree	Natural science IT educations	Job in the private sector, with the occupation: software and application development	Job in the public sector, with the occupation: teaching and research at higher education	Job in the private sector, with the occupation: general office work not requiring advanced degree
Bachelor degree	Social sciences without further specification	Job in the industry: health and social services	Job in the industry: education	Job in the industry: wholesale and retail trade
Bachelor degree	Law	Job in the industry: specialized services, incl. business administration, with the occupation: legal work	Job in the industry: public administration, defense and social security, with the occupation: general office work not requiring advanced degree	Job in the industry: public administration, defense and social security, with the occupation: legal work

Continued on next page

Table C2: Educations eligible for treatment and associated job types

Education level	Education narrow field	Jobtype 1	Jobtype 2	Jobtype 3
Bachelor degree	Political science	Job in the public sector, in the industry: education	Job in the public sector, in the industry: public administration, defense and social security	Job in the private sector, in the industry: organizations and associations
Bachelor degree	Economics	Job with the occupation: general office work not requiring advanced degree	Job with the occupation: teaching and research at long-cycle higher education	Job with the occupation: work in economics
Bachelor degree	Business Economics, HA	Job in the industry: specialized services, incl. business administration	Job in the industry: wholesale and retail trade	Job in the industry: banking, financial, or insurance activities
Bachelor degree	Technical science without further specification	Job in the industry: education	Job in the industry: specialized services, incl. business administration	Job in the industry: wholesale and retail trade
Bachelor degree	Electronics and IT	Job in the private sector, in the industry: information and communication	Job in the public sector, in the industry: education	Job in the private sector, in the industry: wholesale and retail trade
Bachelor degree	Mechanical engineering	Job in the public sector, in the industry: education	Job in the private sector (>50 employees), in the industry: manufacturing	Job in the private sector (>50 employees), in the industry: specialized services, incl. business administration
Bachelor degree	Pharmacy and Pharmaceutical Sciences	Job in the private sector ($\leq$ 50 employees), in the industry: wholesale and retail trade	Job in the private sector (>50 employees), in the industry: manufacturing	Job in the private sector (>50 employees), in the industry: wholesale and retail trade
Long-cycle higher education	Pedagogy	Job in the public sector, with the occupation: teaching at elementary school level or pedagogical work	Job in the public sector, with the occupation: special needs education and research	Job in the public sector, with the occupation: teaching and research at higher education
Long-cycle higher education	Film, theater and musicology, etc.	Job in the public sector, in the industry: education	Job in the private sector, in the industry: information and communication	Job in the public sector, in the industry: health and social services
Long-cycle higher education	Communication and dissemination	Job in the industry: education	Job in the industry: health and social services	Job in the industry: information and communication
Long-cycle higher education	Journalism and rhetoric	Job in the private sector (>50 employees), with the occupation: writing, journalism and linguistics	Job in the public sector, with the occupation: writing, journalism and linguistics	Job in the private sector ($\leq$ 50 employees), with the occupation: writing, journalism and linguistics
Long-cycle higher education	Digital design and interaction design	Job in the industry: information and communication	Job in the industry: education	Job in the industry: specialized services, incl. business administration
Long-cycle higher education	Experience Design	Job in the industry: health and social services	Job in the industry: education	Job in the industry: specialized services, incl. business administration
Long-cycle higher education	Danish-Nordic language and literature	Job with the occupation: teaching at upper secondary level	Job with the occupation: general office work not requiring advanced degree	Job with the occupation: teaching at elementary school level
Long-cycle higher education	West Germanic languages (English, German or Dutch)	Job with the occupation: teaching at upper secondary level	Job with the occupation: general office work not requiring advanced degree	Job with the occupation: teaching at elementary school level

Continued on next page

Table C2: Educations eligible for treatment and associated job types

Education level	Education narrow field	Jobtype 1	Jobtype 2	Jobtype 3
Long-cycle higher education	Business Language, MA in International Business Communication	Job in the private sector, in the industry: wholesale and retail trade	Job in the private sector, in the industry: specialized services, incl. business administration	Job in the private sector, in the industry: information and communication
Long-cycle higher education	Humanities, other	Job in the public sector, in the industry: education	Job in the public sector, in the industry: health and social services	Job in the private sector, in the industry: education
Long-cycle higher education	Architecture	Job in the private sector ($\leq$ 50 employees), in the industry: specialized services, with the occupation: architecture, infrastructure and design work	Job in the private sector (>50 employees), in the industry: specialized services, with the occupation: architecture, infrastructure and design work	Job in the public sector, in the industry: health and social services, with the occupation: architecture, infrastructure and design work requiring advanced degree
Long-cycle higher education	Designer	Job in the industry: education	Job in the industry: wholesale trade excluding motor vehicles	Job in the industry: services in design, photography, translation, agricultural consultancy etc.
Long-cycle higher education	Biology	Job in the public sector, in the industry: education	Job in the public sector, in the industry: health and social services	Job in the private sector, in the industry: specialized services, incl. business administration
Long-cycle higher education	Physics and physical subjects	Job in the public sector, with the occupation: teaching and research at higher education	Job in the public sector, with the occupation: teaching at upper secondary level	Job in the private sector (>50 employees), with the occupation: software and application development
Long-cycle higher education	Sports	Job with the occupation: teaching at upper secondary level	Job with the occupation: teaching and research at higher education	Job with the occupation: teaching at elementary school level or pedagogical work for children
Long-cycle higher education	Mathematics	Job in the industry: education	Job in the industry: banking, financial, or insurance activities	Job in the industry: information and communication
Long-cycle higher education	Natural science IT educations	Job in the private sector, with the occupation: software and application development	Job in the public sector, with the occupation: teaching and research at higher education	Job in the private sector, with the occupation: general office work not requiring advanced degree
Long-cycle higher education	Administration, governance and management	Job in the private sector, with the occupation: finance, accounting and math work not requiring advanced degree	Job in the private sector, with the occupation: work in finance and economics	Job in the private sector, with the occupation: general office work not requiring advanced degree
Long-cycle higher education	Anthropology	Job in the industry: health and social services	Job in the industry: education	Job in the industry: specialized services, incl. business administration
Long-cycle higher education	Globalization and international social studies	Job in the industry: social work activities without accommodation	Job in the industry: education	Job in the industry: public administration, defense and social security
Long-cycle higher education	Law	Job in the public sector, with the occupation: other legal work besides judge and lawyer work	Job in the private sector (>50 employees), with the occupation: associate lawyer work	Job in the private sector (>50 employees), with the occupation: other legal work besides judge and lawyer work

Continued on next page

Table C2: Educations eligible for treatment and associated job types

Education level	Education narrow field	Jobtype 1	Jobtype 2	Jobtype 3
Long-cycle higher education	Political science	Job with the occupation: work in social sciences and religion	Job with the occupation: general office work not requiring advanced degree	Job with the occupation: legal work
Long-cycle higher education	Psychology	Job with the occupation: work in social sciences and religion	Job with the occupation: general office work not requiring advanced degree	Job with the occupation: care work in health industry
Long-cycle higher education	Sociology	Job in the public sector, in the industry: health and social services	Job in the public sector, in the industry: education	Job in the public sector, in the industry: public administration, defense and social security
Long-cycle higher education	Economics	Job in the private sector, in the industry: banking, financial, or insurance activities	Job in the public sector, in the industry: public administration, defense and social security	Job in the private sector, in the industry: specialized services, incl. business administration
Long-cycle higher education	Business Economics, (Cand.merc.)	Job in the industry: specialized services, incl. business administration	Job in the industry: wholesale and retail trade	Job in the industry: banking, financial, or insurance activities
Long-cycle higher education	Technical science without further specification	Job with the occupation: engineering work (except electrical engineering)	Job with the occupation: software and application development	Job with the occupation: teaching and research at higher education
Long-cycle higher education	Biotechnology and chemical technology	Job in the public sector, with the occupation: teaching and research at higher education	Job in the private sector, with the occupation: engineering work (except electrical)	Job in the public sector, with the occupation: engineering work (except electrical)
Long-cycle higher education	Building and construction engineering	Job in the private sector, with the occupation: engineering work (except electrical)	Job in the public sector, with the occupation: engineering work (except electrical)	Job in the public sector, with the occupation: teaching and research at higher education
Long-cycle higher education	Electronics and IT	Job in the private sector, in the industry: information and communication	Job in the public sector, in the industry: education	Job in the private sector, in the industry: manufacturing
Long-cycle higher education	Health and welfare technology, etc.	Job in the public sector, in the industry: education	Job in the public sector, in the industry: health care	Job in the private sector (>50 employees), in the industry: manufacture of pharmaceuticals
Long-cycle higher education	Technology-natural sciences, combined	Job in the public sector, with the occupation: teaching and research at long-cycle higher education	Job in the private sector, with the occupation: engineering work (except electrical)	Job in the private sector, with the occupation: software and application development
Long-cycle higher education	Technology, production and product development	Job in the private sector (>50 employees), in the industry: manufacturing	Job in the private sector ($\leq$ 50 employees), in the industry: specialized services, incl. business administration	Job in the private sector (>50 employees), in the industry: specialized services, incl. business administration
Long-cycle higher education	Technical science, other	Job in the industry: manufacturing	Job in the industry: specialized services, incl. business administration	Job in the industry: information and communication
Long-cycle higher education	Veterinary medicine	Job with the occupation: veterinary work	Job with the occupation: teaching and research at long-cycle higher education	Job with the occupation: general office work not requiring advanced degree
Long-cycle higher education	Natural resources and environment	Job in the industry: education	Job in the industry: health and social services	Job in the industry: specialized services, incl. business administration

Continued on next page

Table C2: Educations eligible for treatment and associated job types

Education level	Education narrow field	Jobtype 1	Jobtype 2	Jobtype 3
Long-cycle higher education	Pharmacy and Pharmaceutical Sciences	Job in the private sector ($\leq$ 50 employees), in the industry: wholesale and retail trade	Job in the private sector (>50 employees), in the industry: manufacturing	Job in the public sector, in the industry: education
Long-cycle higher education	Public health science	Job in the public sector, in the industry: education	Job in the public sector, in the industry: health care	Job in the private sector, in the industry: social work without accommodation
Long-cycle higher education	Odontology	Job in the private sector ($\leq$ 50 employees), in the industry: health care, with the occupation: dental work	Job in the public sector, in the industry: education, with the occupation: dental work	Job in the public sector, in the industry: residential care, with the occupation: dental work
Long-cycle higher education	Healthcare sciences, other	Job in the public sector, with the occupation: nursing and midwifery	Job in the public sector, with the occupation: dentist, pharmacist, physiotherapy and health work in occupational health, nutrition, audiology, optics and chiropractic	Job in the public sector, with the occupation: teaching and research at higher education
PhD and research education	Philosophy	Job in the public sector, with the occupation: teaching and research at higher education	Job in the private sector ($\leq$ 50 employees), with the occupation: teaching and research at higher education	Job in the public sector, with the occupation: work in social sciences and religion
PhD and research education	Natural sciences without further specification	Job in the industry: education	Job in the industry: specialized services, incl. business administration	Job in the industry: manufacturing
PhD and research education	Political science	Job in the public sector, in the industry: education	Job in the private sector, in the industry: specialized services, incl. business administration	Job in the public sector, in the industry: public administration, defense and social security
PhD and research education	Technical science without further specification	Job in the industry: education	Job in the industry: specialized services, incl. business administration	Job in the industry: manufacturing
PhD and research education	Agriculture, nature and environment without further specification	Job in the public sector, in the industry: education	Job in the private sector, in the industry: manufacturing	Job in the private sector, in the industry: specialized services, incl. business administration
PhD and research education	Medicine	Job in the industry: health and social services, with the occupation: general medical practice	Job in the industry: education, with the occupation: teaching and research at higher education	Job in the industry: health and social services, with the occupation: work in biochemistry, biology, botany, zoology and related fields

C.1.2 Creation of job type examples for explanation and validation question

To create more concrete examples used for explaining and validating understanding of the three job types in the survey, we used more granular industry and occupation codes when available. For job types shown in the survey which were defined in terms of industry, we used 4-digit NACE industries as examples of industries that fall within the given survey job type. Similarly, for job types defined in terms of occupation at the DISCO 3 or 4-digit level, we used 6-digit DISCO occupations as examples. These examples were checked manually to make sure they were interpretable and were not described using the exact same text labels as the survey job type.[33] When available, the two most common example occupations and/or industries for a given major were shown in question D1 of the survey (with 'common' again being defined in terms of the hiring outcomes of past cohorts from the same major upon entering the labor market).

We also used an analogous approach to construct a validation question at the end of the survey. For job types defined in terms of industry at the 1 or 2-digit level, or occupation at the DISCO 3 or 4-digit level, we constructed example jobs as combinations of 4-digit NACE industries and 6-digit DISCO occupation labels, using the same corrections as above. All respondents from the corresponding major were then shown one of the two most common combinations at random as an example job, and were asked to place this example within the three job types shown in the survey.

[33] This involved changing 41 labels to make them interpretable and distinct from the upper category, removing 10 labels that were not distinguishable from the upper category, and adding 11 examples where none was usable in the existing labels, but could easily be constructed.

C.2 Full survey questionnaire (in English)

A Thank you very much for participating in this survey about job search and career choices - we greatly appreciate it!

As a token of our gratitude for your time, we are giving away a total of **20 gift cards to GoGift** among those who complete the questionnaire. These gift cards can be used for many things and are very easy to use in both stores and online. Each gift card is worth **1,000 DKK**.

Half of the gift cards are distributed at random to respondents who complete the entire questionnaire (it is expected that this will take approximately **15 minutes**).

The rest of the gift cards are given to respondents who complete the questionnaire and do especially well in the exercises that make up part of the survey (e.g., by guessing the answer to some factual questions correctly). Keep an eye out for questions marked with an emoji. In these questions, you can win additional gift cards!

[*Emoji*]

If you win one (or more) gift cards, you will receive a direct notification in your e-Boks [e-Boks is a Danish state-official email] no later than September 31st, 2023. During the survey, you are able to disconnect and return to complete the survey later if you need to. Your answers are saved and when you click the link again, you will automatically return to where you left off.

B1 We are now going to ask you some questions about your education and current employment circumstances.

B2 Are you currently undertaking education (e.g., at a university or elsewhere)?

- Yes
- No

B3 [**if B2==Yes**] Which of the following educations are you currently undertaking?
[**if B2==No**] Which of the following educations correspond to the highest level of education that you have completed?

[**3-layer dropdown with first layer corresponding to the following**]

- Primary or lower secondary education
- Upper secondary education (High school)
- Vocational basic course
- Vocational education
- Labor market education
- Short-cycle higher education
- Medium-cycle higher education
- Bachelor's degree
- Long-cycle higher education (e.g., Master's degree)
- PhD programme and research education

[**The education categories are based on DISCED, the Danish implementation of ISCED. The second layer corresponds to the broad field, and the third layer corresponds to the narrow field. See** https://www.dst.dk/da/Statistik/dokumentation/nomenklaturer/disced15-audd]

B4 [**if B2==No**] When did you finish studying this education?

- MM/YYYY

[**If B2==Yes**] When do you expect to finish studying this education?

- MM/YYYY

[Create custom variables that take on values depending on responses to question B3: *custom_educ*: **Detailed education (all three layers)** *jobtype*1, *jobtype*2, *jobtype*3: **three specific types of jobs that are relevant to the respondent's education. The three job types are specific to the respondent's education and based on combinations of sector, industry, occupation and firm size. See Appendix Section C.1 for details on how the job types are generated.]**

C1 Which of the following statements best describes your current labor market situation position?

- **I have a job** and my contract is **not expiring in the immediate future.**
- **I have a job**, but my contract **expires soon** and I have **not** accepted another job offer.
- **I have a job**. My contract **expires soon**, but I have **accepted** another job offer.
- **I do not have a job**, but I have **accepted** a job offer which is starting soon.
- **I do not have a job** and I have not accepted a job offer.

C2 [if C1==5] Are you currently searching for a job?

[if C1==1 or 2 or 3 or 4] Are you currently searching for another job?

Note: Searching for a job means e.g., asking employers about potential job opportunities, gathering information about vacant positions, or sending out job applications.

- Yes
- No

C3 [if C2==Yes & C1==4 or 5] When did you start searching for a job?

[If C2==Yes and C1==1 or 2 or 3] When did you start searching for another job?

[If C2==No and C1==1] When did you start searching for your current job?

[If C2==No and C1==2] When do you plan on starting to search for another job?

[If C2==No and C1==3 or 4] When did you start searching for the job you have accepted?

[If C2==No and C1==5] When do you plan on starting to search for a job?

Note: Searching for a job means e.g., asking employers about potential job opportunities, gathering information about vacant positions, or sending out job applications.

- MM/YYYY

D1 [The Industri_example* and Funktion_example* variables are job type specific variables that give examples of the industry and occupations mentioned in the job types. These variables are empty if the job types does not include industry or occupation, or examples where deemed unnecessary.]

We would now like for you to answer some questions about three categories of jobs people with your educational background ("$custom_educ$") could have:

- **$jobtype_1$**

  [If Industri_example1a !="" and Industri_example1b ==”” show:]

  An example of this industry could be:

  - *Industri_example1a*

  [Else if Industri_example1a != “” and Industri_example1b !="" show:]

  Examples of this industry could be:

  - *Industri_example1a* - *Industri_example1b*

  [Do the same for Funktion instead of Industri:]

  [If Funktion_example1a !="" and Funktion_example1b ==”” show:]

  An example of this type of occupation could be:

  - *Funktion_example1a*

  [Else if Funktion_example1a != “” and Funktion_example1b !="" show:]

  Examples of this type of occupation could be:

  - *Funktion_example1a* - *Funktion_example1b*

  [If none of the conditions are met, don't show anything.]

- **$jobtype_2$**

[Same as for jobtype_1 just using Industri_example2a−b and Funktion_example2a−b instead of 1]

- **$jobtype_3$**

  [Same as for jobtype_1 just using Industri_example3a−b and Funktion_example3a−b instead of 1]

D2 Before you continue, use a moment to read and think about the three categories of jobs listed above.

How well do you understand what we mean by a **$Jobtype_1$**?

• Very poorly • Poorly • Neither well nor poorly • Well • Very well

How well do you understand what we mean by a **$Jobtype_2$**?

• Very poorly • Poorly • Neither well nor poorly • Well • Very well

How well do you understand what we mean by a **$Jobtype_3$**?

• Very poorly • Poorly • Neither well nor poorly • Well • Very well

D4 Do you know someone close to you who has had a job in this job category, such as a close friend or family member? If you are close to multiple people, think of the person who is closest to you.

$Jobtype_1$:

• Very close • Close • Not so close • I am not close to anybody who has had a job in this field.

$Jobtype_2$:

• Very close • Close • Not so close • I am not close to anybody who has had a job in this field.

$Jobtype_3$:

• Very close • Close • Not so close • I am not close to anybody who has had a job in this field.

E1 In the following, you are asked to guess the real answers to some **factual** questions about the labor market in recent years. You will be asked a few questions about things that persons with **the same educational background as you ("$custom_educ$")** have experienced in the three job categories above.

Take your time and try to guess the correct answer!

[*Emoji*]

In the questions marked with this emoji, the 10 participants with the most correct answers will receive **a gift card of 1,000 DKK** in their e-Boks.

E2 [**Salary expectations**]

Some jobs pay better than others. This question concerns monthly **gross salary** (i.e. **before tax** and **including** contribution to pension savings) of people in full-time employment. Full-time employment refers to contracts of employment of at least 37 hours of work a week.

Consider persons with the **same** educational background as you ([**"$custom_educ$"**]), who are newly graduated and completed their education in the 2010s whereafter they began working full-time.

What do you think were the **average monthly gross earnings during the first year of work for persons with jobs** in the following job types in the 2010s?

[*Emoji*]

As a reference, the average was approximately 31,000 DKK if **ALL** newly graduated workers of the period are considered (all positions and all educational backgrounds).

- Full-time salary of **$Jobtype_1$**: _____ DKK
- Full-time salary of **$Jobtype_2$**: _____ DKK
- Full-time salary of **$Jobtype_3$**: _____ DKK

If you are one of 3 respondents with answers closest to the actual numbers calculated from data from the Danish labor market of the years 2010 to 2018, you will receive a gift card of 1,000 DKK directly to your e-Boks. Remember that your answer should be in Danish "Kroner" (DKK) per month (before tax and including pension savings).

E3 [Expectations of job application success]

Some jobs are harder to get than others. This question concerns how often the application of unemployed people leads to an actual hiring in the firm.

Consider a person with the **same** educational background as you (**["$custom_educ$"]**), who is newly graduated and completed their education in the 2010s whereafter they began receiving unemployment benefits and did not get a job for at least 6 weeks.

When the average person who is newly graduated sent out a job application for a position in the following job types, **what was the applicant's chance of employment at that firm**?

[*Emoji*]

As a reference, the average success rate was approximately 27 out of 1000 if considering **ALL** newly graduated (all positions and all educational backgrounds).
- Rate of success when applying for **$Jobtype_1$**: ____ out of 1000
- Rate of success when applying for **$Jobtype_2$**: ____ out of 1000
- Rate of success when applying for**$Jobtype_3$**: ____ out of 1000

If you are one of 3 respondents with answers closest to the actual success rate calculated from data on application gathered from applications logged in joblog.dk and administrative data on hirings in the years 2016 to 2018, you will receive a gift card of 1,000 DKK directly to your e-Boks. Remember that your answer should be given as a share of 1000. If you answer 10, this means that you think 10 out of 1000 applications lead to employment for the applicant. If you answer 45, this means that you think 45 out of 1000 applications lead to employment instead.

E4 [Salary after 5 years]

Some jobs offer better opportunities when it comes to wage growth than others. This question concerns the monthly **gross salary** (i.e. **before tax** and **including** pension savings) **5 years after having started a job.**

Consider again a person with the **same** educational background as you ([**"$custom_educ$"**]), who is newly graduated and completed their education in the 2010s.

[*Emoji*]

What do you think was the monthly full-time gross salary after 5 years, for persons who started working in the following job types?

As a reference, you are informed that the average total salary was approximately 41,000

DKK 5 years after employment if **ALL** newly graduated workers of the period are considered (all positions and all educational backgrounds).

- Full-time salary 5 years after first employment in **$Jobtype_1$**: _____DKK
- Full-time salary 5 years after first employment in **$Jobtype_2$**: _____DKK
- Full-time salary 5 years after first employment in **$Jobtype_3$**: _____DKK

If you are one of 4 respondents with answers closest to the actual numbers calculated from data for the Danish labor market of the years 2010 to 2014, you will receive a gift card of 1,000 DKK directly to your e-Boks. Remember that your answer should be in Danish "Kroner" (DKK) per month (before tax and including pension savings).

F1.A [Treatment status T1, is determined in the following way.
It is 0 if not eligible, or eligible and randomly chosen for control.
It is 1 if eligible and chosen for treatment.
Eligibility is determined on an education basis:
Eligible educations have both a data driven specific job type classification, and the number of observed education-to-job transitions with salary information in the least common job type was at least 30.]

[wage treatment] [show if T1==1]

Earlier, we asked you about the average monthly gross salary of full-time employees in different types of jobs.

We have researched the **actual** monthly gross salary of people with the same education as you ("**$custom_educ$**") who completed their education and worked full-time (at least 37 hours) in different job categories in the 2010s. The following graph depicts the **actual average monthly full-time gross salary in the first year of employment after graduating (the three bars) compared to your estimates (the three dots)**.

Figure: Actual data concerning salaries vs. your estimates (see Figure C1)

Note: When proceeding you are not able to return to this graph. You can continue from this page by clicking "next" after 15 seconds.

F1.B [Reminder wages] [show if T1==0]

Previously, we asked you about the **full-time salaries of newly graduated** in different types of jobs.

One of the questions was about the monthly gross salary of people with the same education as you ([**"$custom_educ$"**]) who completed their education and worked full-time (at least 37 hours) in different fields in the 2010's. The following graph depicts **your estimates** regarding the **average monthly full-time gross salary in the first year of employment after graduation.**

Figure: Your estimates regarding salaries (see Figure C2)

Note: You can continue from this page by clicking "next" after 15 seconds.

G1 We now ask you to **consider three distinct jobs** - one from each category listed below (the same that you were asked about earlier). The jobs you consider should be relevant to a person with the same educational background as you. The positions should also be **representative** of what you think the category of jobs in question would offer **you personally** in **every aspect** (pay, probability of receiving an offer if you apply, working conditions, etc.).

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

When you have thought about each of the three jobs and what they offer, we would like

for you to answer some questions about the three positions. Some of the questions can be hard to answer, but we are interested in your best guess.

Click "next" to start answering the questions.

G2 Imagine that you are looking for a job and find postings corresponding to **the three positions we asked you to think about before** (i.e. three job postings in total). You only have the time to apply to exactly one of the positions. Alternatively, you do not apply to any of the three jobs.

It is hard to know exactly how you would act in this situation, but what do you think is the probability of each of the possibilities listed below?

Please enter your answers as percentages. An answer of 10 percent means that you would apply for the position in 10 out of 100 instances while 50 percent means that you would apply for the job about half of the time. Please make sure that your answers add up to 100 percent.

[Here the respondents answer by moving indicators on three sliders from 1 to 97%, the default value is 25%. The sliders are set up, such that they always sum to 100. In practice this is done as follows: When respondents move one of the sliders, the other sliders adjust to make them sum to a 100. The adjustment is done with priority, so, if possible, only the 4th slider adjusts, then the 3rd, then 2nd then 1st.]

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

- Probability that I will apply for **Job A**: [**slider**]
- Probability that I will apply for **Job B**: [**slider**]
- Probability that I will apply for **Job C**: [**slider**]
- Probability that I will not apply to any of the jobs: [**slider**]

[If respondents do not move the indicator, they cannot go to the next screen; error message "You have not moved any of the sliders. You need to move at least one, to show that you have answered the question."]

G3 Imagine that you are offered **the three positions we asked you to think about before** (i.e. three job offers in total). You can accept exactly one offer. Alternatively, you do not accept any of the three job offers.

It is hard to know exactly how you would act in this situation, but what do you think is the probability of each of the possibilities listed below?

Please enter your answers as percentages. An answer of 10 percent means that you would accept a job offer in 10 out of 100 instances while 50 percent means that you would accept a job offer about half of the time. Please make sure that your answers add up to 100 percent.

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

[Here the respondents answer by moving indicators on three sliders from 1 to 97%, the default value is 25%. The sliders are set up, such that they always sum to 100. In practice this is done as follows: When respondents move one of the sliders, the other sliders adjust to make them sum to a 100. The adjustment is done with priority, so, if possible, only the 4th slider adjusts, then the 3rd, then 2nd then 1st.]

- Probability that I would accept **Job A**: [**slider**]
- Probability that I would accept **Job B**: [**slider**]
- Probability that I would accept **Job C**: [**slider**]
- Probability that I would not accept any of the three jobs offered: [**slider**]

[If respondents do not move the indicator, they cannot go to the next screen;

error message "You have not moved any of the sliders. You need to move at least one, to show that you have answered the question."]

G4 Consider again the three jobs from earlier. How many hours a week would you end up working in each of these positions?

*Note: We are interested in **actual working hours**. A job that is stated to be 37 hours could require more or less than 37 hours a week in practice. The position does not need to be stated as full time.*

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

- Weekly hours in **Job A**: _____ hours
- Weekly hours in **Job B**: _____ hours
- Weekly hours in **Job C**: _____ hours

G5 Consider once again the three jobs from earlier. What would be your starting salary in each of these positions?

*Please enter the monthly gross salary **before tax** and **including pension savings**.*

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

- Starting salary in **Job A**: _____ DKK
- Starting salary in **Job B**: _____ DKK
- Starting salary in **Job C**: _____ DKK

G6 Consider again the three jobs from earlier. If you applied for these, how likely is it that you would receive a job offer?

Please enter your answers as a share of 1000. If you answer 10, this means that you think you would receive an offer 10 out of 1000 times. If you answer 45, this means that you think you would receive an offer 45 out of 1000 times.

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

- The chance that I would receive an offer if I applied for **Job A**: _____ out of 1000
- The chance that I would receive an offer if I applied for **Job B**: _____ out of 1000
- The chance that I would receive an offer if I applied for **Job C**: _____ out of 1000

G7 Consider again the three positions we told you to imagine earlier. **In 5 years**, how many hours a week would you spend working if you started working in each of these positions tomorrow?

*Note: We are interested in **actual working hours**. A job that is stated to be 37 hours could require more or less than 37 hours a week in practice. The position does not need to be stated as full time.*

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

- Weekly hours of work in 5 years if I started working in **Job A** tomorrow: _____ hours
- Weekly hours of work in 5 years if I started working in **Job B** tomorrow: _____ hours
- Weekly hours of work in 5 years if I started working in **Job C** tomorrow: _____ hours

G8 Consider again the three positions from earlier. What would your monthly gross salary be **in 5 years** if you started in each of these positions tomorrow?

*Please enter the monthly gross salary **before tax** and **including pension savings** after 5 years.*

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

- Monthly gross salary in 5 years if I started working in **Job A** tomorrow: _____ DKK

- Monthly gross salary in 5 years if I started working in **Job B** tomorrow: _____ DKK

- Monthly gross salary in 5 years if I started working in **Job C** tomorrow: _____ DKK

G9 Consider again the three positions from earlier. Imagine the experiences in the workplace that these three jobs would entail. How likely is it that you would get along really well with your coworkers in these positions?

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

- The probability that I would get along really well with my colleagues at **Job A**:

- **Very unlikely** • **Unlikely** • **A little unlikely** • **A little likely**
- **Likely** • **Very likely**

- The probability that I would get along really well with my colleagues at **Job B**:

[**Same dropdown**]

- The probability that I would get along really well with my colleagues at **Job C**:

[**Same dropdown**]

G10 Consider again the three jobs from earlier. Think about how challenging you would find the work in these positions. How likely is it that you would perform excellently in each of the three positions?

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

- Probability that I would perform really well at **Job A**:

- **Very unlikely** • **Unlikely** • **A little unlikely** • **A little likely**

- **Likely** • **Very likely**

- Probability that I would perform really well at **Job B**:

[**Same dropdown**]

- Probability that I would perform really well at **Job C**:

[**Same dropdown**]

G11 Consider again the hypothetical situation where you receive **job offers regarding the three positions** from earlier questions. Earlier, you gave the following probabilities when answering how likely you were to accept exactly one of the offers.

- Probability that I would accept **Job A**:

[**Previous answer from G3 is inserted here**]

- Probability that I would accept **Job B**:

[**Previous answer from G3 is inserted here**]

- Probability that I would accept **Job C**:

[**Previous answer from G3 is inserted here**]

- Probability that I would not accept any of the three jobs offered:

[**Previous answer from G3 is inserted here**]

Job A:	**Job B:**	**Job C:**
$Jobtype_1$	$Jobtype_2$	$Jobtype_3$

Imagine that you again receive the same job offers, but that **the offers are atypical**, such that the starting salary is [**median answer from G5, rounded to 1000s, is inserted**] in all three of the jobs. Besides the starting salary, the jobs are exactly as before (including the weekly hours, relative wage growth and so on). How would this change your previous probabilities stated above?

Please enter your answers as percentages. An answer of 10 percent means that you would accept a job offer in 10 out of 100 instances while 50 percent means that you would accept a job offer about half of the time. Please make sure that your answers add up to 100 percent.

[**Here the respondents answer by moving indicators on three sliders from 1 to 97%, the default value is 25%. The sliders are set up, such that they always sum to 100. In practice this is done as follows: When respondents move one of the sliders, the other sliders adjust to make the sum correspond to 100. The adjustment is done with priority, so, if possible, only the 4th slider adjusts, then the 3rd, then 2nd then 1st.**]

- Probability that I would accept **Job A** when the starting salary of all three of the positions is [**median answer from G5, rounded to 1000s, is inserted**]:
[**slider**]

- Probability that I would accept **Job B** when the starting salary of all three of the positions is [**median answer from G5, rounded to 1000s, is inserted**]:
[**slider**]

- Probability that I would accept **Job C** when the starting salary of all three of the positions is [**median answer from G5, rounded to 1000s, is inserted**]:
[**slider**]

- The probability that I would turn down all three job offers is:
[**slider**]

[**If respondents do not move the indicator, they cannot go to the next screen; error message "You have not moved any of the sliders. You need to move at least one, to show that you have answered the question."**]

G12 [$Jobtype_example_chosen$ is an example of one of the three education-specific job types. Only show this question when the job types examples are available.]

Throughout this survey we have asked you about 3 different types of jobs. To better understand how you understand these, we pose the following question:

In which job category would you place the more concrete example:
$Jobtype_example_chosen$

- It is an example of **$Jobtype_1$**.
- It is an example of **$Jobtype_2$**.
- It is an example of **$jobtype_3$**.
- It does not fit into any of the three types of jobs.

H1 Finally, we would like for you to answer some questions about yourself.

How willing are you, in general to take a risk?

Please answer on a scale from 0 to 10[34] where 0 means that you are "completely unwilling to run any risks" and 10 means that you are "very willing to run a risk". You can also use any integer between 0 and 10 to indicate where you are situated on the scale.

[The respondents can choose between the integers 1-10]

Think of all the job offers you have received in the last 2 years.

How many job offers have you received in total?

[34]Note: In the survey there was a mistake, and the text stated 0 to 10, but is was only possible to choose between 1 to 10

___

How many of these did you reject?

___

How many of these did you reject because you received an offer for another job at the same time that you liked better?

___

How many job offers did you reject although you had no other jobs available?

___

How many children would you like to have throughout your entire life? (your best estimate is fine)
_____ children

What do you think is the probability that you will have a child within 4 years?
_____ percent (%)

I1 [wage treatment] [T1==1]

On a previous page of this survey, we showed you some information about the **actual** monthly gross salary of people with your educational background ([**"$custom_educ$"**]) who graduated and worked full-time (at least 37 hours) in different job categories between 2010 and 2018.

The same graph that you saw earlier is depicted below. It shows the **actual** **average monthly full-time gross salary in the first year of employment after graduating (the three columns) compared to your estimates (the three dots)**.

Figure: Actual data concerning salaries vs. your estimates (see Figure C1)

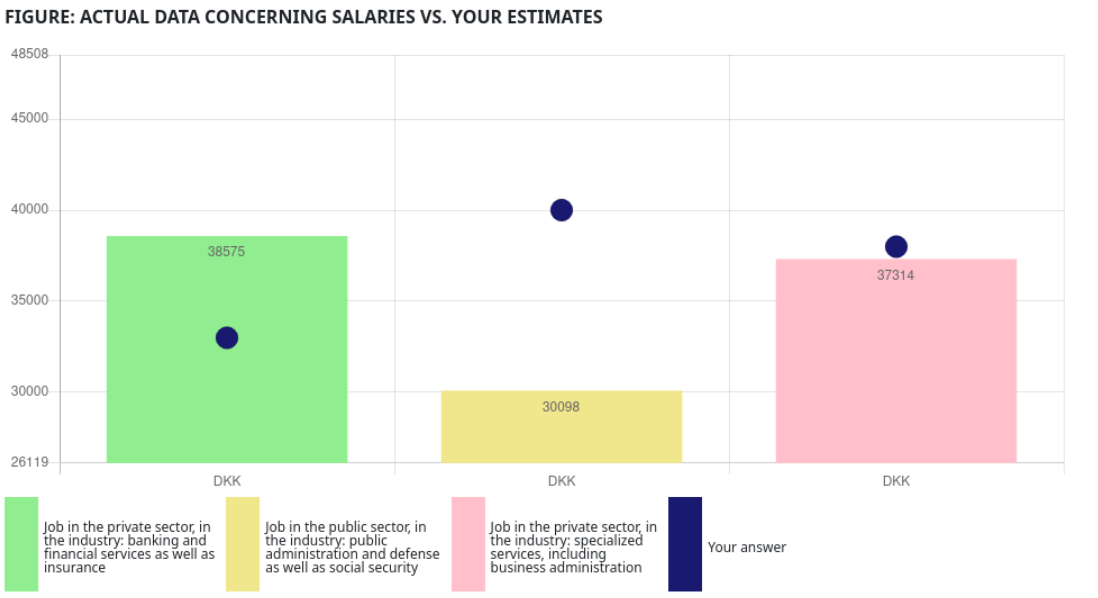
[**if T1==0 just go to next question**]

I2 If you have any thoughts or input, you would like to share with us about job search or the survey you have just completed, please write them down below:
[**Text entry box**]

Thank you for taking part in this survey!

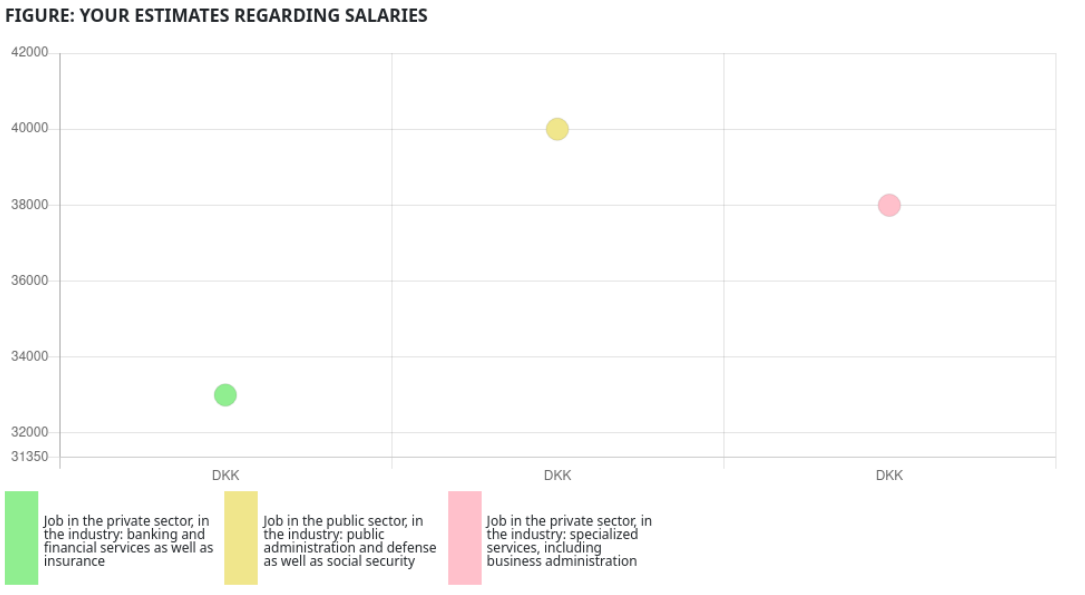
C.3 Survey screenshots and examples

Figure C1: Example of treatment slide shown to the treatment group in the survey



Note: This figure shows an example of the interactive graph (translated into English) shown to the treatment group (translated into English), reminding them of their stated answers about the average wage in each job, and informing them of the actual averages in the register data.

Figure C2: Example of treatment slide shown to the control group in the survey



Note: This figure shows an example of the interactive graph (translated into English) shown to the control group, reminding them of their stated answers about the average wage in each job.

D Post survey data handling

D.1 Lost observations due to send-out error in the second wave

Due to an error introduced by our "e-Boks" e-mail provider in the second wave, the same link(s) were initially sent to multiple people, making their answers invalid and making it impossible for us to link them to register data. The mistake was found and the correct links were sent out within 4 hours. Since we can not be certain that links sent and filled out in the first 4 hours of the second wave were correct and unique, however, we have been forced to remove all answers registered in this 4 hour period (corresponding to 151 distinct online survey links, answered by multiple persons).

D.2 Procedure to correct reporting mistakes

To assess how simple reporting errors may be contributing to our measured misperceptions, Table 3 shows how measured misperceptions change if we apply a simple procedure to correct obvious reporting errors in the raw survey data.

The specific procedure we consider works as follows. All reported monthly wages outside 7,000–120,000 DKK were flagged as being implausible. For these reports, we examined the number of digits in the raw responses. If all flagged values for the same respondent had the same number of digits, we interpret it as a systematic reporting issue. In that case, the value was rescaled according to the most plausible interpretation:

- six-digit values were treated as annual earnings and divided by 12
- two-digit values were treated as monthly earnings reported in thousands and multiplied by 1,000
- three-digit values were interpreted as hourly wages and multiplied by 160.33 (approximate full-time hours per month)

If instead the number of digits varied across flagged responses, we interpreted this as an idiosyncratic reporting error and corrected each report individually as follows:

- six-digit values were assumed to contain an extra digit and divided by 10

- four-digit values were assumed to be missing a final digit and multiplied by 10
- two-digit values were again treated as thousands and multiplied by 1,000.

Finally after applying the corrections above, values within the acceptable range of 7,000–120,000 DKK were retained, while all others were coded as missing.

Chapter 3

Parental Transmission of Jobs and Non-Wage Amenities

Chapter 3

Parental Transmission of Jobs and Non-Wage Amenities

Asker Nygaard David Nikolaj Arpe Harmon
Sonja Settele Daphné Skandalis

Abstract

Workers tend to follow their parents into similar kinds of jobs. Whether this is a problem depends on why: if children inherit privileged access to their parents' line of work or the pay that comes with it, transmission may reflect a misallocation of talent; if instead they inherit a taste for the work itself—its non-wage amenities—the same sorting may carry welfare gains. The amenity channel is the hardest to observe and the least studied. To fill this gap, we analyse a survey of early career job seekers in Denmark, linked to administrative data. For each respondent we define three job types relevant to them and elicit, for each, the likelihood of (i) applying; (ii) accepting an offer; and (iii) accepting an offer when all jobs pay the same wage. These questions map to a discrete-choice model of job search that lets us distinguish the role of perceived access, pay, and non-wage amenities in shaping applications. Respondents are 29% more likely to apply to a job type where a parent has worked, and 25% more likely to accept an offer from it at equal wages. Perceived access and pay leave little trace on intended parental transmission; meaning preferences over non-wage amenities are an important factor.

JEL-Codes: J31, J62, J64, D83

Contact: Asker Nygaard David, University of Copenhagen, anc@econ.ku.dk. Nikolaj Harmon, University of Copenhagen, nikolaj.harmon@econ.ku.dk. Sonja Settele, University of Cologne, ECONtribute and MPI for Research on Collective Goods Bonn settele@wiso.uni-koeln.de. Daphné Skandalis, University of Copenhagen, daphne.skandalis@econ.ku.dk. **Acknowledgements**: Annasofie Marckstrøm Olesen, Sophie Laura Bune, Dánjal Hátún Augustinussen, Carl Benedikt Huels, and Emma Thevissen provided excellent research assistance. **Funding**: Financial support through the Rockwool Foundation is gratefully acknowledged. Sonja Settele additionally acknowledges funding from the Danish National Research Foundation (CEBI, Grant DNRF134) and the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy – EXC 2126/1-390838866. **Ethics approval:** Ethics approval was received from the IRB of the University of Copenhagen. **Pre-registration:** The data collection was pre-registered at the AEA RCT Registry (https://www.socialscienceregistry.org/trials/11592).

1 Introduction

Children learn from what you do, not what you say. This advice is often given to parents of young children, but it appears to stay relevant as the children grow to be adults and enter the labour market: workers are more likely to work in the same occupations, industries, and even firms as their parents.[1] This phenomenon has spurred a large literature documenting it and discussing the mechanisms behind it. Parental transmission of employment outcomes can reflect a market failure if workers are not allocated to the jobs which best suit their innate abilities, e.g. due to preferential access, but it can also reflect optimal allocation of labour based on inherited abilities and preferences over amenities. This amenity channel is the hardest to observe and the least studied, and this paper provides new evidence for its importance.

This paper asks why workers follow their parents into the same *types of jobs*. We conduct a survey of 1,902 Danish early career job seekers designed to map to a discrete-choice model, which we use to decompose the intended parental transmission of job types at the beginning of job search into different channels and quantify their magnitudes. We can link the survey to both the respondents' future employment outcomes and their parents' employment history, allowing us to document both realised and intended parental transmission of job types.

The job types are bundles of occupation, industry, and sector, specifically constructed for the major of the job seeker based on the transitions from graduation to the job market for prior graduates. This ensures that our job types are relevant to early career job seekers. It also confines our study to the role of parental influence *conditional* on major.

We focus on the job seeker side of the intended parental transmission and separate it into three channels: *access*, the job is easier to get; *pay*, it pays more; and *preferences*, it is more attractive for any other reason, such as preferences over amenities, familiarity, or a sense of fit. The first two are the channels that would make the transmission a concern on efficiency grounds, unless they are driven by ability associated with parental background; the third is the amenity channel, which need not signal any misallocation.

[1] Among others, discussed later in this section, see Lentz and Laband (1989); Nandi (2016); Corak and Piraino (2011); Kramarz and Skans (2014).

The survey allows us to separate these channels by eliciting the likelihood of applying to different jobs, accepting an existing job offer, and accepting an offer when all jobs pay the same wage. The survey is designed to elicit the job seeker's valuation of the different job types, and to map these valuations to a discrete-choice model of job search, so that we can shut down the channels one at a time by design, and see how parental transmission differs across these channels.

We document in administrative data that parental transmission of our job types is present in our setting and more broadly in the Danish labour market. We then use the survey to ask which channel it runs through. The survey speaks to the job seeker side: intended application behaviour, based on the values job seekers perceive at the start of their search. If anything, parental transmission seems stronger here than when looking at realised outcomes. Our decomposition then shows that it is not primarily driven by access or pay but by preferences: When offered all three job types at the same wage, respondents remain far more likely to choose the one a parent has worked in. In the decomposition, the preference channel accounts for an estimated 84% of the transmission in applications, and the 95% confidence interval rejects that it accounts for less than 46%.

We support this finding with directly elicited perceptions of job type characteristics. Respondents with parental experience report no higher expected wages, at entry or five years from starting the job; and we find, at most, weak evidence of higher perceived hiring likelihood (statistically significant only at the 10% level and fragile across robustness checks). Adding the full set of elicited perceptions as controls does not absorb the parental transmission effect on applications.

Our survey data speaks to what job seekers perceive and act on; not to the employer side of hiring, and not to the effect of a specific parental firm connection. In our data, realised parental transmission in the administrative data is smaller than the intended parental transmission in the survey data, confirming that the job seeker side is important. However, our results are still consistent with an important firm side, and we highlight a specific angle our survey fails to capture: in realised outcomes, a parent *currently* working in a job type predicts entry far more strongly than a parent with only *prior* experience, more than doubling the realised effect. This result is not replicated in the survey data, where

the parental transmission of intended applications and acceptances is similar regardless of whether the parent is currently working in the job type. And respondents also do not *perceive* a higher likelihood of their application being successful if their parent is currently working in the job type.

Related literature and contributions Children have a tendency to follow their parents' employment history. The pattern holds in many facets of the labour market: their field of study (Altmejd, 2024; Brunello et al., 2025; Birkelund, 2024); their occupation (Lentz and Laband, 1989; Lo Bello and Morchio, 2022); their industry (Nandi, 2016); the specific employer a parent works for (Corak and Piraino, 2011; Bingley et al., 2012; Kramarz and Skans, 2014; Staiger, 2025; Engzell and Wilmers, 2025); and even the choice to be self-employed (Lindquist et al., 2015). Along with documenting the phenomenon, the literature has also tried to uncover the underlying causes with a number of likely suspects.

The first is access—a parent might make *getting* the job more likely. The intergenerational employer literature traces much of the transmission to the doors a parent opens: Kramarz and Skans (2014) show that in Sweden graduates entering the labour market are much more likely to work at their parents' employers than their comparable peers, and that this effect is larger for graduates with less education, poor grades, and those graduating in years of high unemployment. They also show that this practice seems optimal for the firm, as it appears able to pay both the graduate and the parent lower wages when hiring connected graduates. Staiger (2025) and Engzell and Wilmers (2025) both document parental transmission of firms, and further document that parental connection appears to give access to higher-paying firms that would otherwise be unavailable. The seminal paper of Lentz and Laband (1989) documents that children of doctors are about 14 percentage points more likely to be admitted to medical school than comparable applicants, and argues the underlying cause of this is nepotism.

Another explanation is that families transmit preferences and familiarity, which causes realised labour market outcomes to be correlated across generations. Greenberg et al. (2024) shows that parental transmission is strong in army recruitment, where recruitment happens via standardised testing, which limits the scope for an access channel through

nepotism and points to transmitted preferences. In related work, Dohmen et al. (2012) show that parents pass on risk and trust attitudes, and Schüle (2026) show that career preferences differ by socio-economic background in adolescence and can account for 7–28% of intergenerational income persistence. Müller (2024) show in an experiment that preferences can also be imposed on children: German high school students were twice as likely to report an aspiration to go to college if they knew their parents could view their answers, and their reported preferred field also became more likely to be the same as the one their parents preferred.

Quantifying these channels is difficult. Lo Bello and Morchio (2022) fits a structural model of occupational choice to father–son transitions and decomposes the persistence into parental networks, comparative advantages, and preferences. Their findings suggest parental networks are the main driver, with preferences a distant second, and in their model this leads to an inefficient allocation of labour. We develop a novel approach to decompose the channels that is model consistent but estimated by reduced form: We elicit choices in scenarios that shut down the channels by design, allowing us to decompose them directly. We limit the scope to one side of the market—the job seekers' choices and their perceptions—abstracting from the employer side of hiring. Our approach builds on a tradition of using stated choice probabilities to recover preferences (Manski, 1999; Blass et al., 2010), applied to the value of job attributes by Wiswall and Zafar (2018), Mas and Pallais (2017), and Maestas et al. (2023), and to occupational choice by Arcidiacono et al. (2020).

Roadmap Section 2 describes the job types, the administrative data, the measure of parental experience, the survey, and how the questions can be mapped to a discrete-choice model that decomposes the stated choice transmission into an access, pay, and non-pecuniary value channel. Section 3 documents realised parental transmission in the administrative data. Section 4 shows our main decomposition results and corroborates the results with the elicited perceptions of job characteristics. Section 5 concludes.

2 Data and methodology

Our analysis combines a survey of early career job seekers in Denmark fielded in the summer of 2023 with linked Danish administrative data. This is the same survey described in Chapter 2, so information relevant to both papers is repeated here. The survey was designed to target early career job seekers and elicit their perceptions and valuation of job types relevant to their major. As disclosed in the pre-analysis plan, the survey data is then linked to administrative data to identify the employment history of parents prior to the survey and the employment of respondents after the survey.

In this section we first define the job types, the unit at which we measure parental transmission. We then describe the administrative data and the measure of parental experience, before turning to the survey and the sample and, finally, to the elicited choices, which we map to a discrete-choice model of job search.

2.1 Job types

A key concept in our analysis is the *job type*: a major specific bundle of occupation, industry, and firm sector. The construction has three objectives: the job types should capture a substantial share of graduates' employment opportunities; be meaningful and relevant for the graduates; and be possible to map in the administrative data so that we can link them to realised employment.

For each major we construct three job types based on the most common transitions from graduation to employment in the administrative data in 2010–2018. We consider all ways of grouping jobs by firm sector and size (public/private, above or below 50 employees), occupation (3-, 4-, or 6-digit Danish ISCO codes), and industry (1- or 2-digit NACE codes).[2] For each major we keep the groupings in which the three most common job types cover more than 40% of prior graduates' first job transitions and the third most common type alone covers at least 5%. Among qualifying groupings we select the one that best predicts entry wages. Majors for which no grouping qualifies are excluded from the study; the procedure yields three job types for each of 88 majors, covering 76% of higher education

[2]Including grouping only on one or two of the categories, except for only grouping on firm sector.

graduates in 2010–2018.

The procedure ensures that each respondent is asked about job types that are relevant for their specific major. For example, for economists, jobs are grouped by sector and 1-digit industry, giving: private sector jobs in finance and insurance, public sector jobs in public administration, and private sector jobs in business services; for physicists, they are instead grouped by sector, firm size, and 3-digit occupation, giving: research and teaching in higher education, upper-secondary teaching, and software development in large private firms. The full list of majors and job types is provided in the appendix of Chapter 2 along with more details on their construction. In Chapter 2 we also show that survey respondents understood the job types well: 75% report a good or very good understanding of their job types, and 76% correctly classify a concrete example job into the right type.

Table 2 shows that the job types are often split on industry, occupation, and sector, but also that most are not split on all of categories, nor all on the same, which emphasises the need to make the categorisation major specific.

Two features of this construction are important for interpreting our results. First, the coverage condition focuses the analysis on the three most common job types for each major. These are by construction the most relevant jobs for the population, at the cost that we ignore effects from having parents in less common or less obvious job types that these persons might be encouraged to transition to. Within the job types we focus on, this restriction does not appear to mute the pattern: restricting the administrative samples to persons whose first job is in one of the three types strengthens the estimated transmission (see Section 3, Table 3). Second, the job types are constructed to maximise wage differences across types when the other conditions are met. If the parental transmission of job types runs through pay, a partition designed around pay differences should be well suited to detect it; this design strengthens our finding of its absence.

2.2 Administrative data and parental experience

We supplement the survey data with Danish administrative registers to link parents and children and measure actual employment experience in the job types for both of them.

We also show the representativeness of our sample by comparing the parental transmission for survey participants with parental transmission in realised job outcomes for the survey sample and all graduates and unemployment benefit recipients in 2023.

We use Danish administrative registers from 2008 to 2024. Employment histories come from the monthly employer-reported employment records (BFL), which record each worker's employer, sector, and occupation; we add industry from the firm statistics register (FIRM), allowing us to classify each employment spell into the job type cells defined above. We further use the education register (UDDA) for completed majors and graduation dates, the DREAM register for government transfers including unemployment benefits, and the population register (BEF) to create parental links.

Our key regressor measures parental experience in a job type. We define $\text{ParExp}_{i,j}$ as a dummy equal to one if either parent of individual i has worked in a job falling within the combination of sector, occupation, and industry codes that defines job type j for i's major, between 2008 and the time of the survey. When we apply the same definition in the administrative samples of Section 3, the window instead ends at the event defining each sample: graduation, the first month of unemployment benefit receipt, or the survey invitation. This definition ignores the major of the parents, which is consistent with the earlier literature, none of which interacts *having a parent with the same education* with *having a parent with the same work experience*. Further, zooming in on persons whose parents have the same education or even field severely limits the sample, as shown in Table 1. Table 2 shows that the job types where we identify parental experience are more likely to have been split on industry than occupation,[3] meaning the effects we identify can be viewed primarily through the lens of parental transmission of industry.

Table 1 shows how common parental experience is among the survey respondents. At least one parent can be linked to the administrative data for 92% of respondents, and to the employment records for 90%.[4] For 38% of respondents, either parent has worked in at least one of their three relevant job types since 2008. Much of this experience is recent: for 29% of respondents the experience falls within the five years before the survey, and for

[3]This pattern is likely a feature of the industry codes being coarser than the occupation codes.

[4]Respondents whose parents cannot be linked to the employment records are coded as having no parental job type experience.

20% a parent works in one of the job types at the time of the survey. Our baseline measure pools current and prior experience; Section 4.4 separates the two.

2.3 Survey and sample

The survey was conducted in Denmark in the summer of 2023. To reach early career job seekers around the time they start searching, we used two sampling sources: all individuals below age 40 who initiated a claim for unemployment benefits during the summer of 2023 spread over three three-week waves, most of whom recently graduated[5]; and students at the University of Copenhagen close to finishing their Master's degree. Sampled individuals were invited in three waves through "e-Boks", the official channel through which government agencies in Denmark communicate with citizens. In total, 22,461 individuals from the 88 selected higher education majors covered by our design (Section 2.1) were invited, and 1,902 completed the survey. We refer to these respondents as the *survey sample*. The sampling worked as intended: 65% of respondents report actively searching for a job at the time of the survey, and a further 24% were about to start a new job or job search.

For each respondent, the survey asks about the three types of jobs most relevant for graduates with their educational background, defined in Section 2.1. It first elicits beliefs about the average wages earned by prior cohorts in each job type, then presents a randomised information treatment, and finally elicits planned search behaviour—the likelihood of applying to each job type, of accepting an offer from it, and of accepting an offer when all three job types pay the same wage—along with perceived characteristics for the respondent if they were to work in each job type: own potential starting wages, own wages five years after starting, the probability of receiving an offer if applying, work hours, and how well the respondent expects to perform and to get along with colleagues. Typed numerical survey responses can generate outliers, and we therefore winsorise all such variables at the 2.5th and 97.5th percentiles. This includes reported wages, hours, and application success probabilities.

Section 2.5 shows how the three search behaviour questions map to a discrete-choice

[5]Graduates in Denmark can claim unemployment benefits immediately upon graduation and most tend to graduate in the summer.

model of job search, and Chapter 2 validates these stated measures against administrative outcomes: a respondent's reported likelihood of applying to a job type strongly predicts their realised probability of being hired into it after the survey.

One scope restriction should be emphasised: we study graduates *after* they have completed their major. Parental background also shapes the choice of field of study itself (Altmejd, 2024; Brunello et al., 2025). Our analysis therefore captures only the part of parental influence on careers that operates after the choice of major, understating the total parental influence on workers' careers.

Midway through the survey, respondents were randomly assigned an information treatment, which is integral to Chapter 2 but not a focus of this chapter. The treatment group was shown the true average wages earned by prior cohorts in each of their three job types. To keep our variables uncontaminated by the experiment, we restrict all analyses using questions elicited after the treatment to the control group. The treatment itself shows no relation to parental background: its effects on perceived own wages and on planned applications are similar for respondents with and without parental job type experience (Appendix Table A14). Appendix Section A.4 also shows that our main results are similar when we include the treatment group.

2.4 Survey questions

A parental connection to a job type could increase the likelihood that a worker wants to apply for a job for many reasons. The likelihood of a successful application could be higher, an *access* channel; it could pay the worker more, a *pay* channel; or it could simply be more attractive for reasons unrelated to pay, such as preferences for amenities, familiarity with the work, or a sense of fit, a *preference* channel. This section dissects the three search behaviour questions in our survey that let us separate these channels. The full survey is provided in the Appendix of Chapter 2.

The relevant block for the decomposition comes after the respondents have been introduced to the job types and have been asked questions about the average for prior cohorts who have worked in them. The block then starts out by reminding them of the job types

and asking them to consider a position that is *representative* of what they would experience in each job type. The exact formulation, translated from Danish, was:

Instructions (G1)

We now ask you to **consider three distinct jobs** - one from each category listed below (the same that you were asked about earlier). The jobs you consider should be relevant to a person with the same educational background as you. The positions should also be **representative** of what you think the category of jobs in question would offer **you personally** in **every aspect** (pay, probability of receiving an offer if you apply, working conditions, etc.).

[The names of the three job types are displayed here in boxes.]

This focuses the respondent on a specific job representative of each job type. The wording prevents the respondent from thinking about the specific firm that their parent works or has worked for, focusing instead on the job types in general.

On the next page respondents are then asked how likely they would be to apply for each of the three jobs:

Application question (G2) eliciting $\Pi_{i,j}$

Imagine that you are looking for a job and find postings corresponding to **the three positions we asked you to think about before** (i.e. three job postings in total). You only have the time to apply to exactly one of the positions. Alternatively, you do not apply to any of the three jobs.
It is hard to know exactly how you would act in this situation, but what do you think is the probability of each of the possibilities listed below?

[Answers are recorded using four sliders that range from 0.01-0.97 and automatically adjust to make the probabilities sum to 1.]

This question incorporates all three channels of the employee side of parental transmission: whether a respondent applies to a job type depends on how likely the application is to succeed, as it adjusts the expected payoff of sending an application, the *access* channel;

on how much the job pays, the *pay* channel; and on how attractive the job is for any other reason, the *preference* channel.

Because applying is worthwhile *only* if it results in an offer, a job type the respondent expects to be hard to get into draws fewer applications conditional on its attractiveness, since the respondent can apply to only one job. The next question eliminates the access channel by allowing the respondent to choose between offered positions:

Acceptance question (G3) eliciting $\Phi_{i,j}$

Imagine that you are offered all **the positions we asked you to think about before** (i.e. three job offers in total). You can accept exactly one offer. Alternatively, you do not accept any of the three job offers.

It is hard to know exactly how you would act in this situation, but what do you think is the probability of each of the possibilities listed below?

[Answers are recorded using four sliders that range from 0.01-0.97 and automatically adjust to make the probabilities sum to 1.]

This question examines the attractiveness of the jobs irrespective of the difficulties of getting a job offer, meaning the factors that influence their answers should be the utility they think they would get from working in the job.

The next question shuts down the pay channel by offering all three jobs at the same wage, namely the median of the wages the respondent reported they thought they would earn in each job in a previous question. The question has the following wording:

Equal-wage acceptance question (G11) eliciting $\Phi_{i,j}^{\text{Eq}}$

Consider again the hypothetical situation where you receive **job offers regarding the three positions** from earlier questions. Earlier, you gave the following probabilities when answering how likely you were to accept exactly one of the offers.

[The respondent's four probabilities from the acceptance question (G3) are displayed here.]

Imagine that you again receive the same job offers, but that **the offers are atypical**, such that the starting salary is *[Median, rounded to 1000s, of what they stated would be their starting salary in each of the 3 job types (G5).]* in all three of the jobs. Besides the starting salary, the jobs are exactly as before (including the weekly hours, relative wage growth and so on). How would this change your previous probabilities stated above?

[Answers are recorded using four sliders that range from 0.01-0.97 and automatically adjust to make the probabilities sum to 1.]

This question isolates the non-pecuniary value of the jobs. All jobs are fully available and pay the same, so the only remaining reason for preferring one job over another is preferences over amenities, familiarity, or a sense of fit. We allow for differential wage growth to keep the question tractable, but show in Table 6 that expected future wages are not influenced by parental experience.

2.5 Decomposition of access, pay, and preference value

This section shows how the questions can be mapped to a discrete-choice model of job search similar to the one in Chapter 2, building on Harmon et al. (2025). We state a compact version here and relegate the full model to Appendix B.

A job seeker i chooses among the three jobs relevant for their major, denoted by j. The surplus from being hired into job j, relative to continued search, is

$$S_{i,j} = \Psi_i \, \Psi_j \, W_{i,j}^{\delta} \, Z_{i,j} \tag{1}$$

where $W_{i,j}$ is the wage they would make in the job, $Z_{i,j}$ is a scalar summarising their non-

pecuniary value, and δ is the wage elasticity of the surplus. The terms Ψ_i and Ψ_j capture heterogeneity in the value of employment across job seekers and across jobs.

The three survey questions then differ only in which components are active in the value of choosing job j. When deciding where to *apply* in G2, getting the job is uncertain, so the value of applying for job j, relative to not applying at all, is the surplus weighted by the probability of receiving an offer, $P_{i,j}\, S_{i,j}$. When deciding which offer to *accept* in G3 the offers are in hand, so the relative value is the surplus itself, $S_{i,j}$. In the *equal-wage* acceptance question, G11, all three jobs offer the same wage $\overline{W_i}$, so the surplus value varies across jobs only through the common job value Ψ_j and the non-pecuniary term $Z_{i,j}$.

When answering the survey, respondents act on *perceived* values, which we denote with tildes: $\widetilde{W_{i,j}}$, $\widetilde{P_{i,j}}$, and $\widetilde{Z_{i,j}}$. Further, we allow for uncertainty in the choice process through taste shocks scaled by σ, as respondents choose the likelihood that they would apply to or accept a job, or reject all of them.

The likelihood of choosing each job in questions G2, G3 and G11 can then be derived as logit choice probabilities and their logs are linear in the perceived (log) components in the following form:[6]

$$\pi_{i,j} = c_i^{\pi} + \sigma^{-1}\psi_j + \sigma^{-1}(\widetilde{p_{i,j}} + \delta\,\widetilde{w_{i,j}} + \widetilde{z_{i,j}}) \tag{2}$$

$$\phi_{i,j} = c_i^{\phi} + \sigma^{-1}\psi_j + \sigma^{-1}(\delta\,\widetilde{w_{i,j}} + \widetilde{z_{i,j}}) \tag{3}$$

$$\phi_{i,j}^{\text{Eq}} = c_i^{\phi^{\text{Eq}}} + \sigma^{-1}\delta\overline{w_i} + \sigma^{-1}\psi_j + \sigma^{-1}(\widetilde{z_{i,j}}) \tag{4}$$

where $\pi_{i,j}$, $\phi_{i,j}$ and $\phi_{i,j}^{\text{Eq}}$ are the logs of the reported likelihoods of applying, accepting, and accepting at equal wages. The structure of the three questions shows up directly in the equations: the application equation (2) loads on all three perceived components; the acceptance equation (3) drops the hiring term $\widetilde{p_{i,j}}$; and the equal-wage equation (4) additionally makes the wage term constant across jobs, leaving only the job value ψ_j and non-pecuniary value, $\sigma^{-1}(\widetilde{z_{i,j}})$, to vary within persons. The person specific constants absorb both the individual surplus value of working and the likelihood of not choosing any of the

[6] Throughout, lowercase letters denote logs (e.g., $w_{i,j} = \log W_{i,j}$)

three jobs.[7] Differencing the equations shows how the channels can be isolated when person and job fixed effects are controlled for: (2)–(3) isolates the access channel, $\sigma^{-1}\widetilde{p_{i,j}}$; (3)–(4) isolates the pay channel, $\sigma^{-1}\delta\,\widetilde{w_{i,j}}$; and (4) isolates the preference channel, $\sigma^{-1}\widetilde{z_{i,j}}$.

Our main empirical specifications regress each log-likelihood on a dummy for parental experience in the job type, $\text{ParExp}_{i,j}$, with person and job type fixed effects:

$$y_{i,j} = \mu_i^y + \mu_j^y + \beta^y\,\text{ParExp}_{i,j} + u_{i,j}^y, \qquad y \in \{\pi,\, \phi,\, \phi^{\text{Eq}}\}. \tag{5}$$

The person and job type fixed effects μ_i^y and μ_j^y are motivated by the model, and β^y is thus the percentage increase in the likelihood that the respondent chooses the job type if their parent has worked in it, net of how attractive each job type is to respondents on average, and net of how likely they are to choose any job. Note also that respondents whose parents worked in none of their three job types contribute no within-person variation in $\text{ParExp}_{i,j}$. Our main estimates therefore restrict the sample to the 38% of respondents with parental experience in at least one job type. We return to this restriction and the resulting sample sizes in Section 4.1.

To interpret the coefficients β^y as measuring the different channels through which parental experience can affect choices, we deliberately do not control for $\widetilde{p_{i,j}}$, $\widetilde{w_{i,j}}$, nor variables that could be included in $\widetilde{z_{i,j}}$, allowing them to enter the error terms. For the different likelihood questions, $y \in \{\pi,\, \phi,\, \phi^{\text{Eq}}\}$, β^y thus captures the effect of parental experience that operates through these variables.

These coefficients are differences in log-likelihoods that hold the person fixed effect constant, so they do not directly give the percentage change in the choice probability associated with parental experience: because a respondent's three choice probabilities and the outside option must sum to one, raising one likelihood mechanically lowers the others, meaning β^y does not estimate the unconditional percentage change in the choice probability. To recover this, we convert each coefficient into a model-consistent semi-elasticity by scaling it by 1 minus the reported probability and averaging over respondents and job types (derived in Appendix B.1).

[7] Formally $c_i^y = \sigma^{-1}\psi_i + y_{i,0}\ \forall y \in \{\pi, \phi, \phi^{\text{Eq}}\}$

Two remarks on interpretation are in order. First, the decomposition does not uncover why persons with parental experience would have different perceptions about their employment, nor whether the perceptions are correct; rather, it identifies which perceptions matter for the parental transmission of job types. Second, the non-pecuniary term $\widetilde{Z_{i,j}}$ is a broad category: it collects everything that makes a job attractive other than its starting wage and the likelihood of getting an offer. This includes preferences and the non-pecuniary value they gain from working in the job. We return to the content of $\widetilde{Z_{i,j}}$ when discussing the results in Section 4.2.

3 Parental transmission in administrative data

Before using the survey to decompose the channels of parental transmission of job types, we first show that the phenomenon exists in realised labour market outcomes for our sample and in the broader Danish labour market. We estimate the transmission in a linear probability model with job type fixed effects across four samples. Table 3 shows that early career job seekers are 3 to 4 percentage points more likely to enter a job type as their first job if their parents have experience in it, relative to other persons with the same major.

The unit of observation is job type by person, as each person has three job types constructed for their major (Section 2.1). The outcome is a dummy for the person's first job after the sample event being in the job type.[8] We regress this dummy on the regressor, $\text{ParExp}_{i,j}$, the indicator for either parent having worked in job type j between 2008 and the event, in a linear probability model with job type fixed effects, α_j. That is, we estimate separately for each sample,

$$\text{FirstJob}_{i,j} = \alpha_j + \gamma\, \text{ParExp}_{i,j} + \varepsilon_{i,j} \tag{6}$$

where $\varepsilon_{i,j}$ is the error term. Because the job types are major specific, the fixed effects absorb the average entry rate into each job type for persons with the same major, so the coefficient, γ, compares the likelihood of working in that job type for persons with and

[8]The sample event is the relevant event for each sample denoted in the columns: graduation, the first month of unemployment insurance (UI) receipt, the survey invitation date, and the survey response date.

without parental experience in that job type.

Table 3 reports estimates for four samples, allowing us to document parental transmission in general and show that the population pattern survives in the selected sample that carries the mechanism analysis in Section 4. All columns restrict to individuals younger than 40 with one of the selected majors included in the study. The first two samples are based on the administrative data: all persons graduating in 2023 and all persons who receive UI benefits in 2023, mirroring the sample conditions for the survey. The remaining two directly reflect the later analysis data: all persons invited to the survey and the survey respondents themselves.

Parental experience is a strong predictor of entry, as shown in Panel A. In the administrative samples, having a parent who has worked in a job type is associated with a 3.4 to 3.9 percentage point higher probability of entering that job type (Columns (1) and (3)), against mean entry rates of 15–17%. The effect is of similar magnitude for the persons invited to the survey (Column (5)) and for the survey sample itself (Column (7)), so the population pattern carries over to the sample we use for the mechanism analysis. The estimate for the survey sample is somewhat smaller and statistically significant only at the 10% level; however, the job type fixed effects are demanding in a sample this size, so here we lean on its consistency with the three larger samples.

In the survey, respondents are asked only about the three job types constructed for their major. Columns (2), (4), (6) and (8) therefore restrict each sample to persons whose first job is in one of the three job types, so that the mean entry rate is 33.3% by construction; these columns are closer to the choice setting in the survey. The association is, if anything, stronger in the restricted samples, at 4.4 to 4.9 percentage points. The focus on the most common job types of each major does not appear to mute the transmission among the workers we study. A further difference between this setting and the survey setting is that the administrative data measures the final market outcome, while the section on mechanisms in the survey isolates the job seeker's beliefs and planned behaviour, meaning the survey abstracts from whether firm labour demand is shaped by parental experience.

The magnitudes mirror the intergenerational transmission of employers documented in the literature. The approach is the regression equivalent of the share-based approach

in Kramarz and Skans (2014) but for job types instead of firms. Appendix Table A1 shows similar estimates are found using the share-based approach. For university students, Kramarz and Skans (2014) find an effect of 1.8% in the transmission of parental employers, which is somewhat smaller than our estimates for the much more coarsely defined job types. In Appendix Table A2 we recreate the share-based approach and find even larger estimates at the firm level in our data, 6.2% for graduates.

To alleviate the worry that the parental transmission in job types is driven by transmission of firms, Panel B in Table 3 re-estimates the specification but excludes persons whose first job is at a firm where either parent is working or has worked. This is a blunt approach that conditions on the outcome variable, but it shows that the documented patterns cannot be entirely explained by the transmission of firms: The estimated coefficients of parental transmission of job types are reduced by 35–55% but remain large and statistically significant. As noted in Section 2.4 the respondents are asked to consider specific jobs representative of the job type, which limits the role of parents' firm-specific experience.

4 The mechanisms of parental transmission of job types

4.1 Main results: the parental transmission in elicited choices

Having established a strong parental transmission of job types in realised outcomes, we now turn to the survey to decompose the mechanisms through which parental experience shapes job seekers' choices. Table 4 reports our estimates of equation (5) for the different likelihood questions. Parental experience is associated with a higher likelihood of applying to and accepting offers from job types. Respondents are 42% more likely to apply to job types where a parent has worked (Column (1)) and 34% more likely to accept an offer from them (Column (2)). The equal-wage acceptance effect in Column (3) is 35%, meaning that even when the wage is held constant across job types, respondents are still much more likely to accept an offer from a job type where a parent has worked. The model-consistent semi-elasticities, which account for the likelihoods having to sum to 100%, are about 29% for applications, 23% for acceptance, and 25% for equal-wage acceptance.

The evidence for a strong access channel is weak. The gap between applying and accepting (Column (4)) is 0.081 log points, about a fifth of the total application effect, but not statistically significant, and the estimate is not larger in alternative specifications: it falls to 0.030 without person fixed effects (Appendix Table A5) and to 0.061 when the treated group is included (Appendix Table A8).

The pay channel leaves no discernible trace. Column (3) repeats the acceptance regression for the scenario in which all three job types offer the same wage; the estimate is slightly smaller but essentially unchanged from the acceptance question in Column (2). Column (5) estimates the pay component directly, using the difference between the two acceptance outcomes: the estimate is slightly negative and not statistically significant.

What remains is the preference channel. The equal-wage acceptance effect in Column (3) isolates the non-pecuniary value $\sigma^{-1}\widetilde{z_{i,j}}$ of Section 2.5 and accounts for an estimated 84% of the total application effect.

Table 5 bounds the channels by estimating the coefficients in Columns (1)–(3) of Table 4 jointly in a stacked dataset, allowing us to calculate each channel's share of the total application effect, along with 95% confidence intervals that account for the correlation across the three estimates using the delta method. Due to our modest sample size, the bounds are quite wide, but still informative: We can reject that the access channel accounts for more than 52% of the application effect, and that the non-pecuniary preference channel accounts for less than 46%. Our limited power means we cannot rule out a sizeable access channel, but we can establish that the preference channel is important for job seekers when choosing among job types.

A natural concern is that anchoring between the different likelihood questions could make the stability of the coefficients across columns mechanical rather than informative. We do not believe this to be a cause for concern for two reasons. First, answers to the questions do differ: for example, Chapter 2 shows that the randomised wage information treatment affects application and acceptance likelihoods without affecting equal-wage acceptance. Second, the perceived characteristics, elicited separately, display the same pattern as the choice questions: no difference in perceived wages, but differences on non-pecuniary dimensions, as we discuss in Sections 4.2 and 4.3.

The sample is restricted in two ways. First, because only respondents with parental experience in at least one job type contribute identifying variation, we restrict to this group. Second, because the choice questions are elicited after the information treatment, we use only the control group. Table A8 shows the results are similar when the treated group is included, and Appendix Table A5 shows the estimates are similar without person fixed effects.

These stated choice semi-elasticities are larger than the realised transmission of Section 3. The closest administrative analogue in realised outcomes is the choice among the three job types in Columns (2), (4) and (6) of Table 3, which restrict the first job to one of the three types; dividing those coefficients by their 33% mean entry rate gives semi-elasticities of about 13–15%, roughly half the survey semi-elasticity of about 29% for applications.[9] The two are not measuring the same object: the survey captures perceived characteristics and planned behaviour at the start of search, while the administrative data captures realised entry at any point after the event. The most apparent difference is that realised outcomes interact with employer behaviour which is not captured in the survey, but the timing could also matter, as changes during the unemployment spell could attenuate the realised transmission. Because the survey measures are imprecisely estimated, we do not put too much emphasis on the size of the difference, but note that the employee side of the transmission is important and that, if anything, our estimates would suggest that the employer side mitigates the transmission.

The next two sections corroborate the decomposition of parental transmission in the survey with the directly elicited perceptions of characteristics of the job types.

4.2 The association between parental experience and perceived job characteristics

The decomposition in Section 4.1 points to non-pecuniary value as the primary driver of transmission. Alongside the choice questions, respondents reported what they expected each job type's characteristics to be for them personally, which lets us corroborate this

[9] Appendix Table A3 shows that similar estimates are found with a logit model in the administrative data.

reading directly. Table 6 regresses each perceived outcome on $\text{ParExp}_{i,j}$ using the same within-person design as our main results.

The perceptions reproduce the choice decomposition. The perceived probability of receiving an offer (Column (1)), the most direct measure of access, is 13% higher where a parent has experience, but statistically significant only at the 10% level and fragile: it shrinks and is not statistically significant without person fixed effects or when the treated group is included (Appendix Tables A6 and A9). Pecuniary rewards (Columns (2) and (3)) are small and not statistically significant, reinforcing the equal-wage result that transmission does not operate through expected pay. This includes perceived future wages, which are not equalised in the equal-wage acceptance question if respondents perceive they will have higher wage growth in the job type where they have parental experience. This further emphasises that the equal-wage question captures non-pecuniary aspects of the job. Among the non-pecuniary outcomes, getting on with colleagues (Column (4)) shows no link, but expected own performance (Column (5)) is positive and statistically significant at the 10% level despite no additional pay being expected, and it survives both excluding person fixed effects and including the treatment group (Appendix Tables A6 and A9). This variable however captures only part of the non-pecuniary value: Section 4.3 shows parental transmission is not primarily driven by perceived performance.

Because these regressions use within-person variation among respondents with parental experience in some job type, the associations are not a general family background effect on general perceptions of job characteristics. Appendix Table A10 includes persons without parental experience, drops the individual fixed effect and adds an individual level indicator for having a parent with experience in any of the three job types, and shows that having any parental experience is positively correlated with wage expectations but not other perceived job characteristics. Inspired by Chapter 2, we also investigate whether parental experience decreases misperceptions about job type characteristics, which could be a channel for the transmission. We find no evidence for this, as shown in Appendix Section C. Persons who have parents with experience in one of the job types seem to be generally better informed about all three job types, but not relatively better informed about the specific job type their parent has experience in. Further, Appendix Table A14 shows that the

wage information treatment does not have a differential effect by parental background. A related consideration is whether parental experience is merely a proxy for knowing *someone* who works in the job type, or whether there is something distinct about the parental connection. The parental transmission persists when controlling for whether and how closely respondents know someone in the job type, even though parental experience is strongly correlated with both (Appendix Table A4).

Taken together, the elicited perceptions reproduce the decomposition: no difference in perceived pay, at most weak evidence on perceived access, and a residual difference on the non-pecuniary dimension of expected fit. The next section asks whether the perceptions we measure can jointly account for the transmission in applications.

4.3 Perceived characteristics do not explain the parental transmission

If the perceptions we measure accounted for the transmission, controlling for them should absorb the parental experience effect on applications. Table 7 adds the perceived characteristics of each job type to the application regression of Section 4.1 sequentially, in the spirit of a mediation analysis. The parental transmission of job types through application behaviour survives both the sequential controls and their joint inclusion. Appendix Table A7 shows this also holds without individual fixed effects. We view this as supportive evidence for the conclusion of Section 4.1.

4.4 Comparing past and current parental experience

So far we have treated parental experience without regard to contemporaneity, pooling parents who worked in a job type in the past with those still employed in it at the time of the survey. The timing turns out to matter for the transmission, however. For realised outcomes, contemporaneous parental experience is a stronger predictor of entry into a job type than prior experience; however, for our main results in the survey data there is no differential effect. This pattern could be interpreted as supportive of an access channel that operates not through the job seeker side that our survey measures, but through the

employer side on factors that are not necessarily perceived by the job seeker, or through firm-specific factors that are correlated with job type but not considered in the survey, as the respondents were asked to consider generally representative jobs.

The administrative results are shown in Appendix Table A11, which follows the approach of Table 3 but adds a dummy for the parents still working in the job type at the time of the sample event (graduation, UI receipt, survey invitation). They show that while having a parent who previously worked in the job type is associated with a 1.5 to 2.4 percentage point higher probability of entry, having a parent currently working in the job type increases the effect by 2.6 to 4.5 percentage points, more than doubling the transmission. The same split in the survey sample is too demanding for its size: The corresponding coefficients go in the same direction but are not statistically significant.

The same difficulty arises when we zoom in on the survey variables. Tables A12 and A13 also add a dummy for parental experience in the job type at the time of responding to the survey, and estimate noisy zeros for the added effect on all outcomes, including application, acceptance, and perceived likelihood of receiving an offer.

While these estimates are noisy, they are consistent with the idea that the parental transmission effects uncovered in our survey operate mainly through non-pecuniary preferences, which should be independent of the current working status of the parent. The results in the administrative data, however, suggest that the job seeker side factors that our survey is designed to capture do not tell the full story of the realised transmission, and that there could be an employer side access channel that operates at the firm-specific level or is not perceived by the job seeker a priori, which still shapes realised entry into job types, consistent with the broader literature.

5 Conclusion

Using a survey of early career job seekers in Denmark linked to administrative data, this paper asks why children tend to mirror their parents' labour market choices. The welfare implications of the transmission depend on why it happens: inherited access or pay could point to a misallocation of talent, while an inherited taste for the work itself need not.

We use a survey built around a discrete-choice model of job search to separate the job seeker side into these channels—access, pay, and non-pecuniary preferences. In our setting, preferences are the primary channel for the transmission, suggesting that they have a non-negligible importance for the intergenerational transmission of employment outcomes.

Respondents are 29% more likely to apply to a job type where a parent has worked, and 25% more likely to accept an offer from it even when all three job types pay the same wage. The access and pay channels leave little trace on the job seeker side: respondents expect no higher pay where a parent has worked, and their perceived advantage in the odds of being hired is relatively small and fragile to robustness checks. The directly elicited perceptions tell the same story, and adding them as controls does not absorb the effect of parental experience on the likelihood of applying for a job.

Our survey results focus on the job seeker side, their perceptions and planned behaviour, abstracting from the employer side of the parental transmission. Using administrative data, we show that parental experience is a strong predictor of realised entry into a job type, with a similar magnitude to the survey results for a broader population of graduates and young job seekers. However, we also show that there are facets of the parental transmission in realised outcomes that are not captured by the survey. If a parent is currently working in the job type, the parental transmission on realised likelihood of entry is more than doubled relative to having parents who used to work in the job type. This pattern is absent from the survey, where the timing of parental experience does not matter for the likelihood of applying to or accepting an offer from a job type. This reconciles our results with the broader literature which has emphasised the role of access and networks, suggesting that access primarily operates through the employer side, and is not initially perceived by the job seeker or present in their planned behaviour, but still shapes realised entry into job types.

These findings suggest a more prominent role for preferences and amenities in the intergenerational transmission of employment outcomes than the literature currently emphasises. They also informally speak to the efficiency of parental transmission when considering the allocation of labour. If transmission is primarily driven by access, theory suggests it represents market failures that lead to misallocation of talent. On the other hand, if

it is driven by preferences and a sense of fit in the job, the implication for efficiency is ambiguous and depends on the source of differential expectations. Differential preferences and comparative advantage could point towards an efficient allocation, although the malleability of preferences in this setting and the possibility of differential information regarding amenities leave the efficiency question open for future research.

6 Tables and Figures

Table 1: Parental experience shares

	Share
Both parents in data	90.43%
Either parent in data	92.06%
Either parent in education data	88.75%
Either parent in work data	90.12%
Either parent has experience in a job type	38.22%
- Within 5 years before survey	29.13%
- At time of survey	20.45%
Either parent in same education	3.31%
Either parent has experience in a job type and same disced	2.26%
Either parent in same field	19.35%
Either parent has experience in a job type and same field	7.94%
Persons	1902

Notes: The table shows the shares of respondents in the survey sample with different types of parental background, measured in administrative data. The first four rows show the shares for whom both or either parent can be linked to the administrative data at all, to the education registers and to the employment registers. *Either parent has experience in a job type* is the share for whom either parent has worked in at least one of the respondent's three job types between 2008 and the time of the survey; the two indented rows restrict this experience to the five years before the survey and to the time of the survey, respectively. *Either parent in same education* is the share for whom either parent has completed the same education (DISCED code) as the respondent, while *Either parent in same field* uses the broader ISCED field of education. The rows combining job type experience with the same education or field require both conditions to hold.

Table 2: Survey respondents by parental background group, and the dimensions job types split on

	Job type is split on			
	Industry	Occupation	Sector	N
Full survey population	66.04%	47.37%	64.04%	1902
Has parents with same field	61.14%	52.99%	69.57%	368
Has parents with job type experience	79.78%	22.83%	52.54%	727
Same field and job type experience	74.83%	35.10%	54.97%	151

Notes: The table describes how the three job types are constructed for different groups of respondents. The base data are the individuals in the survey sample, with rows corresponding to (partly overlapping) groups defined by parental background and N reporting the number of respondents in each group. For each group, the first three columns report the share of the respondents' job types that are distinguished from the other job types of the same education along industry, occupation and sector, respectively. *Has parents with same field* are respondents where either parent has completed an education in the same ISCED field as the respondent. *Has parents with job type experience* are respondents where either parent has worked in at least one of the three job types between 2008 and the time of the survey. *Same field and job type experience* requires both conditions to hold.

Table 3: Parental transmission of job types

	Graduated in 2023		Received UI in 2023		Invited to survey		Survey sample	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A): Full sample								
$\text{ParExp}_{i,j}$	0.039***	0.049***	0.034***	0.044***	0.033***	0.048***	0.023*	0.044*
	(0.003)	(0.005)	(0.003)	(0.005)	(0.005)	(0.008)	(0.014)	(0.024)
Mean likelihood	16.53%	33.33%	14.90%	33.33%	15.84%	33.33%	16.41%	33.33%
Unique job types	240	240	264	264	264	261	240	192
Persons	46,643	23,130	55,160	24,650	19,940	9,476	1,751	852
N	139,929	69,390	165,480	73,950	59,820	28,428	5,253	2,556
B): Excluding persons where first job is at a parental firm								
$\text{ParExp}_{i,j}$	0.020***	0.022***	0.022***	0.026***	0.019***	0.025***	0.014	0.031
	(0.003)	(0.005)	(0.003)	(0.005)	(0.005)	(0.008)	(0.014)	(0.026)
Mean likelihood	16.49%	33.33%	14.76%	33.33%	15.75%	33.33%	16.33%	33.33%
Unique job types	240	240	264	264	264	261	237	192
Persons	43,407	21,471	51,404	22,756	18,518	8,749	1,653	800
N	130,221	64,413	154,212	68,268	55,554	26,247	4,959	2,400
Job type FE	✓	✓	✓	✓	✓	✓	✓	✓
First job is in one of the 3 types		✓		✓		✓		✓

Notes: The table shows estimates from a linear probability model in administrative data. The base data are person-by-job types (i, j), where each person has three job types, constructed for their major, so that each person appears three times. The outcome variable is a dummy for the person's first job after the event defining the sample being in job type j. The regressor, $\text{ParExp}_{i,j}$, is a dummy for either parent having worked in job type j between 2008 and the event. Each pair of columns corresponds to a different sample, all restricted to individuals younger than 40 with one of the selected majors included in the study: persons graduating in 2023 (the event being graduation), persons receiving UI benefits in 2023 (the event being the first month of UI receipt), all persons invited to the survey (the event being the survey invitation) and the survey sample (the event being the survey response). Columns (2), (4), (6) and (8) further restrict each sample to persons whose first job is within one of the three job types, so that the mean of the outcome variable is 33.3% by construction. Panel A shows estimates for the full sample, while Panel B repeats the same specifications but excludes persons whose first job after the event is in a firm where either parent has worked between 2008 and the event. All specifications include job type fixed effects. Differences between the number of persons shown and the full sample sizes stem from singleton observations dropped in estimation. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Job choice: applications and acceptance by parental experience

	(1) Log likelihood of applying, $\pi_{i,j}$	(2) Log likelihood of accepting, $\phi_{i,j}$	(3) Log likelihood of accepting with equal wages, $\phi_{i,j}^{\text{Eq}}$	(4) $\pi_{i,j} - \phi_{i,j}$	(5) $\phi_{i,j}$ - $\phi_{i,j}^{\text{Eq}}$
$\text{ParExp}_{i,j}$	0.417***	0.336***	0.350***	0.081	-0.014
	(0.093)	(0.102)	(0.107)	(0.071)	(0.070)
Job type FE	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓
Semi-elasticity	0.290	0.232	0.245		
Individuals	389	389	389	389	389
Job types	168	168	168	168	168
N	1,167	1,167	1,167	1,167	1,167

Notes: The table shows OLS estimates from the survey. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents in the control group whose parents have work experience in at least one of the three job types. The regressor, $\text{ParExp}_{i,j}$, is a dummy for either parent having worked in job type j between 2008 and the time of the survey. The outcome variable in Column (1) is the (log) reported likelihood of applying to the job type, $\pi_{i,j}$. In Column (2) it is the (log) reported likelihood of accepting an offer from the job type, $\phi_{i,j}$, and in Column (3) the (log) reported likelihood of accepting an offer in the situation where all three job types offer the exact same starting wage. Columns (4) and (5) use the differences between these outcomes: respectively the access gap, $\pi_{i,j} - \phi_{i,j}$, and the pay gap, $\phi_{i,j}$ minus its equal-wage counterpart. All specifications include individual and job type fixed effects. The *Semi-elasticity* row converts the coefficient in Columns (1)–(3) into the implied effect on the level of the choice probability, correcting for the fact that the person fixed effects hold the outside option likelihood fixed while the likelihoods must sum to one; it is computed as $\frac{1}{N}\sum_i \frac{1}{3}\sum_j \hat{\beta}^y(1-Y_{i,j})$, where $Y_{i,j}$ is the respondent's elicited likelihood, as derived in Appendix B.1. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Job choice: decomposing the effect of parental experience on stated applications

	(1) Level effect	(2) Share of effect on application
Access channel	0.081	0.195
	[-0.058,0.221]	[-0.130,0.520]
Pay channel	-0.014	-0.033
	[-0.150,0.123]	[-0.361,0.295]
Preference channel	0.350	0.838
	[0.140,0.559]	[0.464,1.212]
Job type FE	✓	✓
Individual FE	✓	✓
Individuals	389	389
Job types	168	168
N	1,167	1,167

Notes: The table decomposes the effect of parental experience on the stated likelihood of applying into the access, pay, and preference channels of Section 2.5. The base data are the individual-by-job (i, j) observations of Table 4, restricted to respondents in the control group whose parents have work experience in at least one of the three job types. To account for correlation between the estimates, the regression data are stacked so that each observation appears three times, once for each of the three choice outcomes. A single OLS regression is estimated on the stacked data with outcome-specific slopes on $\text{ParExp}_{i,j}$ and outcome-specific individual and job type fixed effects; this recovers the application, acceptance, and equal-wage acceptance coefficients of Columns (1)–(3) of Table 4. The access channel is the difference between the application and acceptance coefficients; the pay channel is the difference between the acceptance and equal-wage acceptance coefficients; and the preference channel is the equal-wage acceptance coefficient. The three channels sum to the total effect of parental experience on applications. Column (1) reports the level effect of each channel, mirroring the last three columns of Table 4 (reordered). Column (2) reports its share of the total application effect, i.e., the level effect divided by the application coefficient. Brackets report 95% confidence intervals obtained by the delta method from the joint variance–covariance matrix of the coefficients, based on standard errors clustered at the level of the individual. All specifications include individual and job type fixed effects.

Table 6: Perceived own outcomes by parental experience

	(1) Log perceived probability of offer, if applying	(2) Log perceived wage	(3) Log perceived wage after 5 years	(4) Getting on with colleagues	(5) Own performance
$\text{ParExp}_{i,j}$	0.136*	-0.002	0.004	-0.011	0.147*
	(0.081)	(0.006)	(0.008)	(0.055)	(0.075)
Job type FE	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓
Individuals	359	389	385	389	389
Job types	165	168	165	168	168
N	1,077	1,167	1,155	1,167	1,167

Notes: The table shows OLS estimates from the survey. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents in the control group whose parents have work experience in at least one of the three job types. The regressor, $\text{ParExp}_{i,j}$, is a dummy for either parent having worked in job type j between 2008 and the time of the survey. The outcome variables are the perceived (log) probability of receiving an offer if applying (Column (1)), the perceived (log) own starting wage in the job type (Column (2)), the perceived (log) own wage five years after starting (Column (3)), and how well the respondent expects to get on with colleagues and to perform in the job type (6-item Likert scales, Columns (4) and (5)). All specifications include individual and job type fixed effects. Changes in observation totals across columns stem from missing reports and zeros in the raw outcome variables. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Application likelihood with perceived characteristics, job type FE

	Log likelihood of applying, $\pi_{i,j}$					
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{ParExp}_{i,j}$	0.417***	0.409***	0.365***	0.366***	0.368***	0.303***
	(0.093)	(0.089)	(0.089)	(0.092)	(0.089)	(0.085)
Log perceived wage, $\widetilde{w_{i,j}}$		3.096***				1.913**
		(0.721)				(0.774)
Log perceived wage after 5 years, $\widetilde{w_{i,j,t+5}}$		0.314				0.635
		(0.527)				(0.519)
Log probability of offer, if applying			0.475***			0.357***
			(0.055)			(0.057)
Hours$\in (35, 40)$				0.659***		0.356*
				(0.219)		(0.202)
Hours≥ 40				1.185***		0.651***
				(0.232)		(0.214)
Colleagues					0.280***	0.225***
					(0.062)	(0.061)
Performance					0.359***	0.228***
					(0.053)	(0.056)
Job type FE	✓	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓	✓
Individuals	389	385	359	389	389	356
Job types	168	165	165	168	168	162
N	1,167	1,155	1,077	1,167	1,167	1,068

Notes: The table shows OLS estimates from the survey. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents in the control group whose parents have work experience in at least one of the three job types. The outcome variable in all columns is the (log) reported likelihood of applying to the job type, $\pi_{i,j}$. The regressor, $\text{ParExp}_{i,j}$, is a dummy for either parent having worked in job type j between 2008 and the time of the survey. Column (1) includes no further controls. The remaining columns sequentially add the respondent's perceived characteristics of the job types as controls: the perceived (log) own starting wage and own wage five years after starting (Column (2)), the perceived (log) probability of receiving an offer if applying (Column (3)), dummies for perceived weekly hours being 35–40 or above 40 (Column (4)), and how well the respondent expects to get on with colleagues and to perform in the job type (6-item Likert scales, Column (5)). Column (6) includes all of these controls jointly. All specifications include individual and job type fixed effects. Changes in observation totals across columns stem from missing reports and zeros in the raw control variables. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Bibliography

Altmejd, Adam, "Inheritance of Fields of Study," SSRN Scholarly Paper 4827105, Social Science Research Network, Rochester, NY 2024.

Arcidiacono, Peter, V. Joseph Hotz, Arnaud Maurel, and Teresa Romano, "Ex Ante Returns and Occupational Choice," *Journal of Political Economy*, December 2020, *128* (12), 4475–4522.

Bello, Salvatore Lo and Iacopo Morchio, "Like Father, like Son: Occupational Choice, Intergenerational Persistence and Misallocation," *Quantitative Economics*, 2022, *13* (2), 629–679.

Bingley, Paul, Miles Corak, and Niels Westergård-Nielsen, "Equality of Opportunity and Intergenerational Transmission of Employers," in John Ermisch, Markus Jäntti, and Timothy Smeeding, eds., *From Parents to Children*, The Intergenerational Transmission of Advantage, Russell Sage Foundation, 2012, pp. 441–460.

Birkelund, Jesper Fels, "Economic Returns to Reproducing Parents' Field of Study," *The British Journal of Sociology*, 2024, *75* (3), 303–321.

Blass, Asher A., Saul Lach, and Charles F. Manski, "Using Elicited Choice Probabilities to Estimate Random Utility Models: Preferences for Electricity Reliability*," *International Economic Review*, 2010, *51* (2), 421–440.

Brunello, Giorgio, Francesco Campo, Elisabetta Lodigiani, Martina Miotto, and Lorenzo Rocco, "Intended College Major Choice and the Inheritance of Majors," 2025.

Corak, Miles and Patrizio Piraino, "The Intergenerational Transmission of Employers," *Journal of Labor Economics*, January 2011, *29* (1), 37–68.

Dohmen, Thomas, Armin Falk, David Huffman, and Uwe Sunde, "The Intergenerational Transmission of Risk and Trust Attitudes," *The Review of Economic Studies*, April 2012, *79* (2), 645–677.

Engzell, Per and Nathan Wilmers, "Firms and the Intergenerational Transmission of Labor Market Advantage," *American Journal of Sociology*, September 2025, *131* (2), 322–370.

Greenberg, Kyle, Matthew Gudgeon, Adam Isen, Corbin L. Miller, and Richard W. Patterson, "Intergenerational Transmission of Occupation: Lessons from the United States Army," Working Paper 33009, National Bureau of Economic Research September 2024.

Harmon, Nikolaj A, Marie Pietraszek, and Jonas Maibom, "Why Do Coworker Networks Affect Job Search Outcomes?," Study Paper No. 262, The ROCKWOOL Foundation 2025.

Kramarz, Francis and Oskar Nordström Skans, "When Strong Ties Are Strong: Networks and Youth Labour Market Entry," *The Review of Economic Studies*, July 2014, *81* (3), 1164–1200.

Lentz, Bernard F. and David N. Laband, "Why So Many Children of Doctors Become Doctors: Nepotism vs. Human Capital Transfers," *The Journal of Human Resources*, 1989, *24* (3), 396.

Lindquist, Matthew J., Joeri Sol, and Mirjam van Praag, "Why Do Entrepreneurial Parents Have Entrepreneurial Children," *Journal of Labor Economics*, June 2015, *33* (2), 269–296.

Maestas, Nicole, Kathleen J. Mullen, David Powell, Till von Wachter, and Jeffrey B. Wenger, "The Value of Working Conditions in the United States and the Implications for the Structure of Wages," *American Economic Review*, July 2023, *113* (7), 2007–2047.

Manski, Charles, "Analysis of Choice Expectations in Incomplete Scenarios," *Journal of Risk and Uncertainty*, 1999, *19* (1-3), 49–66.

Mas, Alexandre and Amanda Pallais, "Valuing Alternative Work Arrangements," *American Economic Review*, December 2017, *107* (12), 3722–3759.

Müller, Maximilian W, "Parental Pressure and Educational Choices," *Working paper*, 2024.

Nandi, Tushar K, "Intergenerational Persistence of Industry in India," *The European Journal of Development Research*, July 2016, *28* (3), 495–511.

Schüle, Paul, "Career Preferences and Socio-Economic Background," *Labour Economics*, February 2026, *98*, 102814.

Staiger, Matthew, "The Intergenerational Transmission of Employers and the Earnings of Young Workers," *Working paper*, 2025.

Train, Kenneth, *Discrete Choice Methods with Simulation*, 2nd ed ed., Cambridge ; New York: Cambridge University Press, 2009.

Wiswall, Matthew and Basit Zafar, "Preference for the Workplace, Investment in Human Capital, and Gender*," *The Quarterly Journal of Economics*, February 2018, *133* (1), 457–507.

Chapter 3

Appendices

A Additional tables and figures

A.1 Parental transmission shares following Kramarz and Skans (2014)

Table A1: Parental transmission shares by job type

	Graduated in 2023	Received UI in 2023	Invited to survey	Survey sample
Association with parent's job type	19.72%	18.07%	18.75%	18.42%
Counterfactual for job type	15.92%	14.67%	15.52%	15.91%
Difference	3.79%	3.40%	3.24%	2.51%
Major-job type combinations	236	259	251	169
Persons	46,643	55,160	19,902	1,677

Notes: The table shows the association between parental job types and first jobs in linked administrative data, together with the counterfactual association expected absent parental experience, following the approach in Kramarz and Skans (2014), who use classmates to measure the counterfactual probability that a graduate would have started working in the parent's plant even if the parent had not worked there. The base data are individual-by-job type observations (i, j), with three observations per person covering the three job types associated with the person's major, restricted to individuals younger than 40 with one of the selected majors included in the study. The samples in the columns are: persons graduating in 2023 (the event being graduation), persons receiving UI benefits in 2023 (the event being the first month of UI receipt), all persons invited to the survey (the event being the survey invitation) and the survey sample (the event being the survey response). *Association with parent's job type* is, among observations where either parent worked in the job type between 2008 and the event, the share where the person's first job after the event is in that job type. *Counterfactual for job type* is the corresponding share among observations from persons with the same major whose parents did not work in the job type, so that persons with the same major take the role of classmates in measuring the counterfactual. Both shares are computed within major-by-job type cells and averaged across cells, weighting each cell by the number of persons with parental experience in the job type. *Difference* is the association minus the counterfactual; it nets out major specific propensities to enter particular job types (the class-by-plant match effects in their equation (1)) and corresponds to their within-class estimate of the strong-tie effect (their equation (4)). *Major-job type combinations* reports the number of unique major-by-job type cells used in the computation. *Persons* reports the number of individuals underlying the shares; it varies across samples with the availability of an observed first job that can be classified into a job type.

Table A2: Parental transmission shares by firm

	Graduated in 2023	Received UI in 2023	Invited to survey	Survey sample
Association with parent's firm	7.01%	6.03%	8.47%	17.35%
Counterfactual for firm	0.82%	0.73%	1.05%	3.31%
Difference	6.19%	5.30%	7.42%	14.04%
Major-firm combinations	7,305	8,902	3,289	270
Persons	46,643	55,151	19,887	1,513

Notes: The table repeats the computation in Table A1 at the level of the firm, paralleling the plant level analysis in Kramarz and Skans (2014). The samples in the columns are defined as in Table A1. *Association with parent's firm* is, among persons where either parent worked at the firm between 2008 and the event, the share whose first job after the event is at that firm. *Counterfactual for firm* is the corresponding share among persons with the same major whose parents did not work at the firm. Both shares are computed within major-by-firm cells and averaged across cells, weighting each cell by the number of persons with a parent at the firm. *Difference* is the association minus the counterfactual; it nets out major specific propensities to enter particular firms and corresponds to the within-class estimate of the strong-tie effect in Kramarz and Skans (2014), with the caveat that firms are coarser than plants, that our majors are not institution specific so are not as similar to classes, and that we include prior parental experience. *Major-firm combinations* reports the number of unique major-by-firm cells used in the computation. *Persons* reports the number of individuals underlying the shares; it varies across samples as each major-firm cell needs to include at least one person with parental experience at the firm and at least one person without such experience to be included in the computation.

Table A3: Parental transmission of job types estimated with logit

	Graduated in 2023		Received UI in 2023		Invited to survey		Survey sample	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\text{ParExp}_{i,j}$	0.290***	0.244***	0.270***	0.222***	0.253***	0.249***	0.190*	0.253*
	(0.021)	(0.023)	(0.021)	(0.024)	(0.034)	(0.039)	(0.110)	(0.132)
Job type dummies	✓	✓	✓	✓	✓	✓	✓	✓
First job is in one of the 3 types		✓		✓		✓		✓
Mean likelihood	16.53%	33.34%	14.90%	33.34%	15.88%	33.37%	18.36%	35.28%
Unique job types	239	239	262	262	256	254	166	148
Persons	46,643	23,130	55,160	24,650	19,940	9,471	1,735	832
N	139,900	69,386	165,439	73,942	59,693	28,386	4,683	2,358

Notes: The table repeats the analysis of Table 3, estimating an unconditional logit model with job type dummies instead of a linear probability model; the reported figures are the logit coefficients. The base data are person-by-job types (i, j), where each person has three job types, constructed for their major, so that each person appears three times. The outcome variable is a dummy for the person's first job after the event defining the sample being in job type j. The regressor, $\text{ParExp}_{i,j}$, is a dummy for either parent having worked in job type j between 2008 and the event. Each pair of columns corresponds to a different sample, all restricted to individuals younger than 40 with one of the selected majors included in the study: persons graduating in 2023 (the event being graduation), persons receiving UI benefits in 2023 (the event being the first month of UI receipt), all persons invited to the survey (the event being the survey invitation) and the survey sample (the event being the survey response). Columns (2), (4), (6) and (8) further restrict each sample to persons whose first job is within one of the three job types. The procedure drops observations from job types in which the outcome does not vary, which lowers the job type and observation counts relative to Table 3. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.2 Additional survey regressions

Table A4: Parental effect not driven by network

	Log likelihood of applying, $\pi_{i,j}$			Network	Very close
	(1)	(2)	(3)	(4)	(5)
$\text{ParExp}_{i,j}$	0.417***	0.324***	0.339***	0.391***	0.167***
	(0.093)	(0.094)	(0.093)	(0.052)	(0.023)
Network in the job type		0.296***			
		(0.053)			
Very close with someone			0.553***		
			(0.119)		
Job type FE	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓
Control sample	✓	✓	✓		
Individuals	389	389	389	718	718
Job types	168	168	168	183	183
N	1,167	1,167	1,167	2,154	2,154

Notes: The table shows OLS estimates from the survey. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents whose parents have work experience in at least one of the three job types. In Columns (1)–(3) the outcome variable is the (log) reported likelihood of accepting an offer from the job type, $\phi_{i,j}$, and the sample is restricted to the control group; Column (2) adds the network dummy and Column (3) the very close dummy as a control. In Columns (4) and (5) the outcome variables are dummies for the respondent reporting to know someone working in the job type and for reporting to be very close with someone working in the job type, respectively, and both the treatment and control groups are included. The regressor of interest, $\text{ParExp}_{i,j}$, is a dummy for either parent having worked in job type j between 2008 and the time of the survey. All specifications include individual and job type fixed effects. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.3 Primary survey regressions repeated without person fixed effects

Table A5: Job choice: applications and acceptance by parental experience

	(1) Log likelihood of applying, $\pi_{i,j}$	(2) Log likelihood of accepting, $\phi_{i,j}$	(3) Log likelihood of accepting with equal wages, $\phi_{i,j}^{\text{Eq}}$	(4) $\pi_{i,j} - \phi_{i,j}$	(5) $\phi_{i,j}$ - $\phi_{i,j}^{\text{Eq}}$
$\text{ParExp}_{i,j}$	0.371***	0.341***	0.338***	0.030	0.003
	(0.081)	(0.091)	(0.094)	(0.065)	(0.060)
Job type FE	✓	✓	✓	✓	✓
Semi-elasticity	0.258	0.235	0.237		
Individuals	389	389	389	389	389
Job types	168	168	168	168	168
N	1,167	1,167	1,167	1,167	1,167

Notes: As in Table 4, but including only job type fixed effects and no individual fixed effects. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents in the control group whose parents have work experience in at least one of the three job types. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Perceived own outcomes by parental experience

	(1) Log perceived probability of offer, if applying	(2) Log perceived wage	(3) Log perceived wage after 5 years	(4) Getting on with colleagues	(5) Own performance
$\text{ParExp}_{i,j}$	0.036	-0.001	0.002	0.040	0.160**
	(0.090)	(0.008)	(0.010)	(0.066)	(0.073)
Job type FE	✓	✓	✓	✓	✓
Individuals	359	389	385	389	389
Job types	165	168	165	168	168
N	1,077	1,167	1,155	1,167	1,167

Notes: As in Table 6, but including only job type fixed effects and no individual fixed effects. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents in the control group whose parents have work experience in at least one of the three job types. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Application likelihood with perceived characteristics, job type FE

	Log likelihood of applying, $\pi_{i,j}$					
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{ParExp}_{i,j}$	0.371***	0.355***	0.365***	0.340***	0.336***	0.299***
	(0.081)	(0.081)	(0.082)	(0.081)	(0.079)	(0.081)
Log perceived wage, $\widetilde{w_{i,j}}$		0.287				-0.064
		(0.327)				(0.325)
Log perceived wage after 5 years, $\widetilde{w_{i,j,t+5}}$		0.118				0.160
		(0.265)				(0.260)
Log probability of offer, if applying			0.152***			0.125***
			(0.028)			(0.027)
Hours$\in (35, 40)$				0.200		0.108
				(0.132)		(0.142)
Hours≥ 40				0.599***		0.420***
				(0.133)		(0.140)
Colleagues					0.045	0.052
					(0.034)	(0.034)
Performance					0.205***	0.136***
					(0.039)	(0.041)
Job type FE	✓	✓	✓	✓	✓	✓
Individuals	389	385	359	389	389	356
Job types	168	165	165	168	168	162
N	1,167	1,155	1,077	1,167	1,167	1,068

Notes: As in Table 7, but including only job type fixed effects and no individual fixed effects. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents in the control group whose parents have work experience in at least one of the three job types. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.4 Primary survey regressions repeated with control and treatment sample

Table A8: Job choice: applications and acceptance by parental experience

	(1) Log likelihood of applying, $\pi_{i,j}$	(2) Log likelihood of accepting, $\phi_{i,j}$	(3) Log likelihood of accepting with equal wages, $\phi_{i,j}^{\text{Eq}}$	(4) $\pi_{i,j} - \phi_{i,j}$	(5) $\phi_{i,j}$ - $\phi_{i,j}^{\text{Eq}}$
$\text{ParExp}_{i,j}$	0.322***	0.261***	0.270***	0.061	-0.009
	(0.070)	(0.074)	(0.075)	(0.052)	(0.053)
Job type FE	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓
Semi-elasticity	0.225	0.181	0.189		
Individuals	718	718	718	718	718
Job types	183	183	183	183	183
N	2,154	2,154	2,154	2,154	2,154

Notes: As in Table 4, but including the treated sample as well. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents whose parents have work experience in at least one of the three job types. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Perceived own outcomes by parental experience

	(1) Log perceived probability of offer, if applying	(2) Log perceived wage	(3) Log perceived wage after 5 years	(4) Getting on with colleagues	(5) Own performance
$\text{ParExp}_{i,j}$	0.058	0.004	0.009	0.076*	0.133**
	(0.051)	(0.005)	(0.006)	(0.042)	(0.054)
Job type FE	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓
Individuals	675	718	711	718	718
Job types	183	183	180	183	183
N	2,025	2,154	2,133	2,154	2,154

Notes: As in Table 6, but including the treated sample as well. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents whose parents have work experience in at least one of the three job types. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.5 Association between beliefs and general and job specific parental experience

Table A10: Perceived job type averages by general and job specific parental experience

	(1) Log perceived probability of offer, if applying	(2) Log perceived wage	(3) Log perceived wage after 5 years	(4) Getting on with colleagues	(5) Own performance
ParExp_i	0.145	0.028**	0.012	-0.019	0.007
	(0.109)	(0.012)	(0.013)	(0.092)	(0.080)
$\text{ParExp}_{i,j}$	0.019	-0.000	0.008	0.006	0.109
	(0.079)	(0.007)	(0.009)	(0.062)	(0.067)
Job type FE	✓	✓	✓	✓	✓
Individuals	893	970	958	970	970
Job types	213	213	210	213	213
N	2,679	2,910	2,874	2,910	2,910

Notes: The table shows OLS estimates from the survey. The base data are all individual-by-jobs (i, j) for respondents in the control group of the survey sample, including respondents whose parents have no work experience in any of the three job types. The regressors are ParExp_i, a dummy for either parent having work experience in at least one of the respondent's three job types between 2008 and the time of the survey, and $\text{ParExp}_{i,j}$, a dummy for either parent having such experience in job type j specifically. The outcome variables are the perceived (log) probability of receiving an offer if applying (Column (1)), the perceived (log) own starting wage in the job type (Column (2)), the perceived (log) own wage five years after starting (Column (3)), and how well the respondent expects to get on with colleagues and to perform in the job type (6-item Likert scales, Columns (4) and (5)). All specifications include job type fixed effects; individual fixed effects are excluded since ParExp_i only varies across individuals. Changes in observation totals across columns stem from missing reports and zeros in the raw outcome variables. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.6 Accounting for current relative to prior parental experience

Table A11: Parental transmission of job types, current versus prior parental experience

	Graduated in 2023	Received UI in 2023	Invited to survey	Survey sample
	(1)	(2)	(3)	(4)
$\text{ParExp}_{i,j}$	0.022***	0.024***	0.015***	0.018
	(0.004)	(0.003)	(0.005)	(0.018)
$\text{ParExp}_{i,j}$ Presently	0.040***	0.026***	0.045***	0.012
	(0.006)	(0.005)	(0.009)	(0.026)
Job type FE	✓	✓	✓	✓
Mean likelihood	16.53%	14.90%	15.84%	16.41%
Unique job types	240	264	264	240
Persons	46,643	55,160	19,940	1,751
N	139,929	165,480	59,820	5,253

Notes: The table shows estimates from linear probability models in administrative data, splitting parental experience by timing. The base data are person-by-job types (i, j), where each person has three job types, constructed for their major. The outcome variable is a dummy for the person's first job after the event defining the sample being in job type j. $\text{ParExp}_{i,j}$ is a dummy for either parent having worked in job type j between 2008 and the event, while $\text{ParExp}_{i,j}$ *Presently* indicates that a parent works in the job type at the time of the event, so that its coefficient measures the additional transmission from current parental employment. The samples in the columns are all restricted to individuals younger than 40 with one of the selected majors included in the study: persons graduating in 2023 (the event being graduation), persons receiving UI benefits in 2023 (the event being the first month of UI receipt), all persons invited to the survey (the event being the survey invitation) and the survey sample (the event being the survey response). All specifications include job type fixed effects. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Job choice outcomes on current parental experience, any job type sample

	(1) Log likelihood of applying, $\pi_{i,j}$	(2) Log likelihood of accepting, $\phi_{i,j}$	(3) Log likelihood of accepting with equal wages, $\phi_{i,j}^{\text{Eq}}$	(4) $\pi_{i,j} - \phi_{i,j}$	(5) $\phi_{i,j}$ - $\phi_{i,j}^{\text{Eq}}$
$\text{ParExp}_{i,j}$	0.420***	0.407***	0.453***	0.013	-0.046
	(0.124)	(0.134)	(0.131)	(0.092)	(0.091)
$\text{ParExp}_{i,j}$ Presently	-0.006	-0.156	-0.227	0.150	0.071
	(0.169)	(0.176)	(0.180)	(0.124)	(0.119)
Job type FE	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓
Individuals	389	389	389	389	389
Job types	168	168	168	168	168
N	1,167	1,167	1,167	1,167	1,167

Notes: The table shows OLS estimates from the survey, splitting parental experience by timing. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents in the control group whose parents have work experience in at least one of the three job types. $\text{ParExp}_{i,j}$ is a dummy for either parent having worked in job type j between 2008 and the time of the survey, while $\text{ParExp}_{i,j}$ *Presently* indicates that a parent works in the job type at the time of the survey, so that its coefficient measures the additional effect of current parental employment. The outcome variable in Column (1) is the (log) reported likelihood of applying to the job type, $\pi_{i,j}$. In Column (2) it is the (log) reported likelihood of accepting an offer from the job type, $\phi_{i,j}$, and in Column (3) the (log) reported likelihood of accepting an offer in the situation where all three job types offer the exact same starting wage. Columns (4) and (5) use the differences between these outcomes: respectively the access gap, $\pi_{i,j} - \phi_{i,j}$, and the pay gap, $\phi_{i,j}$ minus its equal-wage counterpart. All specifications include job type fixed effects. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Post-match subjective outcomes on current parental experience, any job type sample

	(1) Log perceived probability of offer, if applying	(2) Log perceived wage	(3) Log perceived wage after 5 years	(4) Getting on with colleagues	(5) Own performance
$\text{ParExp}_{i,j}$	0.164	0.002	0.006	-0.023	0.124
	(0.101)	(0.007)	(0.010)	(0.074)	(0.090)
$\text{ParExp}_{i,j}$ Presently	-0.062	-0.010	-0.006	0.026	0.050
	(0.134)	(0.010)	(0.012)	(0.094)	(0.124)
Job type FE	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓
Individuals	359	389	385	389	389
Job types	165	168	165	168	168
N	1,077	1,167	1,155	1,167	1,167

Notes: The table shows OLS estimates from the survey, splitting parental experience by timing. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents in the control group whose parents have work experience in at least one of the three job types. $\text{ParExp}_{i,j}$ is a dummy for either parent having worked in job type j between 2008 and the time of the survey, while $\text{ParExp}_{i,j}$ *Presently* indicates that a parent works in the job type at the time of the survey, so that its coefficient measures the additional effect of current parental employment. The outcome variables are the perceived (log) probability of receiving an offer if applying (Column (1)), the perceived (log) own starting wage in the job type (Column (2)), the perceived (log) own wage five years after starting (Column (3)), and how well the respondent expects to get on with colleagues and to perform in the job type (6-item Likert scales, Columns (4) and (5)). All specifications include job type fixed effects. Changes in observation totals across columns stem from missing reports and zeros in the raw outcome variables. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.7 Wage information treatment effect

Table A14: Wage information treatment effect on perceived wages and likelihood of applying split by parental experience

	$\Delta\widetilde{w_{i,j,j'}}$		$\Delta\pi_{i,j,j'}$	
	(1)	(2)	(3)	(4)
$T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}})$	0.289***	0.292***	1.186**	1.069**
	(0.039)	(0.039)	(0.574)	(0.541)
$T_i \times (\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}) \times \text{ParExp}_i$	0.008	0.013	0.065	0.503
	(0.067)	(0.067)	(0.992)	(0.965)
$\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$	-0.466***	-0.389**	-4.156***	-7.441***
	(0.026)	(0.194)	(0.379)	(2.867)
$(\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}) \times \text{ParExp}_i$	-0.050	-0.043	-0.592	-1.061*
	(0.042)	(0.041)	(0.642)	(0.608)
Pre-treatment controls		✓		✓
Individuals	1,902	1,902	1,902	1,902
N	11,412	11,412	11,412	11,412

Notes: The table shows OLS estimates of the effect of the wage information treatment, allowing the effect to vary with parental background. The base data are all individual-by-job pairs (i, j, j') in the survey sample. The outcome variable in Columns (1) and (2) is the perceived gap in (log) own potential wages between the jobs, $\Delta\widetilde{w_{i,j,j'}}$ (the first stage in Chapter 2's 2SLS framework), and in Columns (3) and (4) the (log) relative likelihood of applying to the two jobs, $\Delta\pi_{i,j,j'}$ (the reduced form). The reported regressors are the treatment dummy T_i interacted with the underestimation of the log wage gap for prior cohorts, $\Delta v_{i,j,j'} - \Delta\widetilde{v_{i,j,j'}}$, this interaction further interacted with ParExp_i (a dummy for either parent having work experience in at least one of the respondent's three job types between 2008 and the time of the survey), the underestimation term itself and its interaction with ParExp_i. Columns (2) and (4) additionally include as controls the pre-treatment survey variables (age, gender, months to expected/actual graduation, labour market status, job search status, reported understanding of and closeness to each job type, perceived prior cohort wages at entry and after five years, perceived offer probabilities, and an indicator for being part of the STAR sample) interacted with the underestimation of the log wage gap. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

B A model of application and acceptance choices

This appendix derives the estimating equations used in Section 2.5. The model is similar to the discrete-choice model of job search from Chapter 2, which builds on Harmon et al. (2025), but here we outline how the application and acceptance questions map to the model together.

Values and the outside option A job seeker i chooses among three jobs denoted by j. We denote the set of three jobs specific to each job seeker's major as $\mathcal{J}_i$. Write $Y_{i,j}$ for the value of being hired into job j and U_i for the value of continued search. The surplus of job j over continued search is

$$S_{i,j} \equiv Y_{i,j} - U_i = \Psi_i \, \Psi_j \, W_{i,j}^{\delta} \, Z_{i,j} > 0 \tag{B.1}$$

where Ψ_i and Ψ_j capture heterogeneity in the returns to employment across job seekers and across jobs, δ governs the role of the wage $W_{i,j}$, and $Z_{i,j}$ collects non-pecuniary value. With $P_{i,j}$ denoting the probability that an application to job j results in an offer, the values of the alternatives in the two choice scenarios posed by the survey are:

$$\text{apply to one job type, or none:} \quad A_{i,j}^{\pi} = \begin{cases} U_i + P_{i,j} \, S_{i,j}, & j \in \mathcal{J}_i \\ U_i, & j = 0 \quad \text{(apply nowhere)} \end{cases} \tag{B.2}$$

$$\text{accept one of three offers, or none:} \quad A_{i,j}^{\phi} = \begin{cases} U_i + S_{i,j}, & j \in \mathcal{J}_i \\ U_i, & j = 0 \quad \text{(reject all offers)} \end{cases} \tag{B.3}$$

The residual option is worth the same in both scenarios: the worker keeps searching and obtains U_i. The only difference between applying and accepting is that the surplus in the application scenario is weighted by the hiring probability $P_{i,j}$; an offer in hand is certain, so $P_{i,j} = 1$ for the acceptance values. The equal-wage acceptance scenario replaces the job specific wage by a common offered wage $\overline{W_i}$, so the relevant surplus becomes

$S_{i,j}^{\text{Eq}} = \Psi_i \, \Psi_j \, \overline{W_i}^{\delta} \, Z_{i,j}$.[10]

Perceived values and taste shocks Job seekers act on perceived magnitudes, which we denote with tildes. The perceived hiring probability is $\widetilde{P_{i,j}}$. The perceived surplus replaces the wage and the non-pecuniary value in (B.1) by their perceived counterparts and includes a multiplicative taste shock $\xi_{i,j}$ that allows for uncertainty about the true values:

$$\widetilde{S_{i,j}} = \Psi_i \, \Psi_j \, \widetilde{W_{i,j}}^{\delta} \, \widetilde{Z_{i,j}} \, \xi_{i,j} \tag{B.4}$$

The residual option carries its own shock, $\xi_{i,0}$. Net of the common continuation value U_i, the perceived values of choosing one of the three jobs in the three scenarios are:

$$\text{apply to } j: \qquad \widetilde{A_{i,j}^{\pi}} - U_i = \widetilde{P_{i,j}} \, \widetilde{S_{i,j}} \tag{B.5}$$

$$\text{accept } j: \qquad \widetilde{A_{i,j}^{\phi}} - U_i = \widetilde{S_{i,j}} \tag{B.6}$$

$$\text{accept } j \text{ at equal wages}: \qquad \widetilde{A_{i,j}^{\phi^{\text{Eq}}}} - U_i = \widetilde{S_{i,j}^{\text{Eq}}} \equiv \Psi_i \, \Psi_j \, \overline{W_i}^{\delta} \, \widetilde{Z_{i,j}} \, \xi_{i,j} \tag{B.7}$$

while the perceived residual option net of U_i is the taste shock:

$$\widetilde{A_{i,0}^{y}} - U_i = \xi_{i,0} \qquad \forall y \in \left\{\pi, \phi, \phi^{\text{Eq}}\right\} \tag{B.8}$$

We impose the same shock distribution in all three scenarios; this is what makes the coefficients comparable across scenarios in (B.18) below.[11]

Two features of (B.5)–(B.7) warrant emphasis. First, the perceived hiring probability $\widetilde{P_{i,j}}$ multiplies the surplus *only* in the application value: once an offer is in hand, getting it is certain, so it disappears from the two acceptance values. Second, the equal-wage scenario replaces the job specific perceived wage $\widetilde{W_{i,j}}$ by $\overline{W_i}$ which is constant within persons, so the deterministic part of $\widetilde{S_{i,j}^{\text{Eq}}}$ varies across jobs only through the job specific return Ψ_j and the non-pecuniary term $\widetilde{Z_{i,j}}$.

[10] In the survey, $\overline{W_i}$ is set to the median of the respondent's stated beliefs about what they would earn in the three jobs.

[11] With scenario specific shock scales, each parameter in the estimation equations would be divided by a different scale parameter, and differences between coefficients would mix channels and scales.

Choice probabilities The shocks $\xi_{i,j}$ and $\xi_{i,0}$ are i.i.d. across alternatives and respondents, drawn from a Fréchet distribution with shape parameter σ^{-1}, so that their logs $\log \xi_{i,j}$ are type-I extreme value with scale σ. In each scenario, the respondent reports the probability that job j yields the highest perceived value:

$$Y_{i,j} = \Pr\left(j = \underset{j' \in \mathcal{J}_i \cup \{0\}}{\operatorname{argmax}} \widetilde{A^{y}_{i,j'}} \right) \quad \forall\, Y \in \{\Pi, \Phi, \Phi^{\mathrm{Eq}}\} \tag{B.9}$$

With a monotone log transformation this yields the logit choice probabilities. Writing lowercase letters for logs ($\psi_i \equiv \log \Psi_i$, $\psi_j \equiv \log \Psi_j$, $\widetilde{p_{i,j}} \equiv \log \widetilde{P_{i,j}}$, and so on), the model equivalents of the three elicited job specific likelihoods are

$$\Pi_{i,j} = \frac{\exp\!\left((\psi_i + \psi_j + \widetilde{p_{i,j}} + \delta\, \widetilde{w_{i,j}} + \widetilde{z_{i,j}})/\sigma\right)}{1 + \sum_{j' \in \mathcal{J}_i} \exp\!\left((\psi_i + \psi_{j'} + \widetilde{p_{i,j'}} + \delta\, \widetilde{w_{i,j'}} + \widetilde{z_{i,j'}})/\sigma\right)} \tag{B.10}$$

$$\Phi_{i,j} = \frac{\exp\!\left((\psi_i + \psi_j + \delta\, \widetilde{w_{i,j}} + \widetilde{z_{i,j}})/\sigma\right)}{1 + \sum_{j' \in \mathcal{J}_i} \exp\!\left((\psi_i + \psi_{j'} + \delta\, \widetilde{w_{i,j'}} + \widetilde{z_{i,j'}})/\sigma\right)} \tag{B.11}$$

$$\Phi^{\mathrm{Eq}}_{i,j} = \frac{\exp\!\left((\psi_i + \psi_j + \delta\, \overline{w_i} + \widetilde{z_{i,j}})/\sigma\right)}{1 + \sum_{j' \in \mathcal{J}_i} \exp\!\left((\psi_i + \psi_{j'} + \delta\, \overline{w_i} + \widetilde{z_{i,j'}})/\sigma\right)}. \tag{B.12}$$

The application likelihood (B.10) loads on all three perceived components; the acceptance likelihood (B.11) drops the hiring term $\widetilde{p_{i,j}}$ because the offer is certain; and the equal-wage likelihood (B.12) additionally collapses the wage term to the common $\overline{w_i}$, leaving only the common job value ψ_j and the non-pecuniary term $\widetilde{z_{i,j}}$ to vary across jobs. Equation (B.8) implies that the likelihood of the outside options will always be 1 divided by the denominators in (B.10)–(B.12).

Log-linearity and the fixed effects Taking logs in (B.10)–(B.12)[12], subtracting the log of the likelihood of the outside option and rearranging, yields a linear expression for

[12] With the notation $\pi_{i,j} \equiv \log \Pi_{i,j}$, and likewise for $\phi_{i,j}$ and $\phi^{\mathrm{Eq}}_{i,j}$.

each scenario:

$$\pi_{i,j} = c_i^{\pi} + \sigma^{-1}\psi_j \qquad + \sigma^{-1}(\widetilde{p_{i,j}} + \delta\,\widetilde{w_{i,j}} + \widetilde{z_{i,j}}) \tag{B.13}$$

$$\phi_{i,j} = c_i^{\phi} + \sigma^{-1}\psi_j \qquad + \sigma^{-1}(\delta\,\widetilde{w_{i,j}} + \widetilde{z_{i,j}}) \tag{B.14}$$

$$\phi_{i,j}^{\text{Eq}} = c_i^{\phi^{\text{Eq}}} + \sigma^{-1}\delta\overline{w_i} + \sigma^{-1}\psi_j + \sigma^{-1}(\widetilde{z_{i,j}}) \tag{B.15}$$

The individual return ψ_i, the common wage term $\delta\,\overline{w_i}$ in the equal-wage scenario, and the log of the logit denominator are constant within individuals, so only the job value ψ_j and the job specific perceived components vary within individuals. The constants are defined as:

$$c_i^y = \sigma^{-1}\psi_i + y_{i,0}\ \forall y \in \{\pi, \phi, \phi^{\text{Eq}}\} \tag{B.16}$$

Chapter 2 eliminates these constants by differencing the application equation across job pairs, so that outcomes are relative likelihoods. Here we instead absorb the constants with person fixed effects. This is what allows the elicited likelihoods themselves to serve as outcome variables: conditional on a person fixed effect, equations (B.13)–(B.15) are linear in the job value ψ_j and the perceived (log) components, so no transformation of the outcomes beyond taking logs is required.

The estimating equations Including a dummy for parental experience in equations (B.13)–(B.15) yields the equation from Section 2.5:

$$y_{i,j} = \mu_i^y + \mu_j^y + \beta^y\,\text{ParExp}_{i,j} + u_{i,j}^y, \qquad y \in \{\pi,\ \phi,\ \phi^{\text{Eq}}\}. \tag{B.17}$$

Section 2.5 interprets each β^y as the parental experience effect operating through the perceived components that are left in the error term. To make this mapping explicit, let parental experience shift each perceived component, $\widetilde{p_{i,j}} = \rho\,\text{ParExp}_{i,j} + \cdots$, $\widetilde{w_{i,j}} = \omega\,\text{ParExp}_{i,j} + \cdots$ and $\widetilde{z_{i,j}} = \zeta\,\text{ParExp}_{i,j} + \cdots$, with the remaining variation in each component that is not correlated with $\text{ParExp}_{i,j}$ collected in the error terms. Substituting into (B.13)–(B.15) and regressing each log-likelihood on $\text{ParExp}_{i,j}$, with person fixed effects absorbing the constants c_i and job type fixed effects absorbing the job type values ψ_j/σ

together with the average perceived components of each job type, recovers

$$\beta^{\pi} = \sigma^{-1}(\rho+\delta\omega+\zeta), \qquad \beta^{\phi} = \sigma^{-1}(\delta\omega+\zeta), \qquad \beta^{\phi^{\text{Eq}}} = \sigma^{-1}\,\zeta, \tag{B.18}$$

matching the channel mapping stated in Section 2.5. The differences between the coefficients isolate the individual channels: $\beta^{\pi}-\beta^{\phi} = \sigma^{-1}\rho$ (access), $\beta^{\phi}-\beta^{\phi^{\text{Eq}}} = \dfrac{\delta}{\sigma}\omega$ (pay), and $\beta^{\phi^{\text{Eq}}} = \sigma^{-1}\zeta$ (preference). Table 4 reports the corresponding estimates, with the differences $\pi_{i,j}-\phi_{i,j}$ and $\phi_{i,j}-\phi^{\text{Eq}}_{i,j}$ used directly as outcomes in Columns (4) and (5).

B.1 From coefficients to semi-elasticities

The coefficients β^{y} in (B.18) are the effects of parental experience on the log-likelihoods keeping the person fixed effect constant. However, because the likelihoods must sum to one, this constraint inflates the interpretation of β^{y} as a semi-elasticity. Similarly to how it was done in Chapter 2, we recover the semi-elasticity of the choice probability by restoring the response of the logit denominator to changes in the likelihoods. The resulting formula is a simple scaling of β^{y} by $1-Y_{i,j}$, which we average over the sample to obtain an overall semi-elasticity.

The semi-elasticity is defined as:

$$\mathcal{E}^{y}_{i,j} \equiv \frac{\partial y_{i,j}}{\partial\,\text{ParExp}_{i,j}} \forall y \in \{\pi,\, \phi,\, \phi^{\text{Eq}}\}. \tag{B.19}$$

Using $\sigma^{-1}(\widetilde{p_{i,j}}+\delta\,\widetilde{w_{i,j}}+\widetilde{z_{i,j}}) = \beta^{\pi}\,\text{ParExp}_{i,j}+u^{\pi}_{i,j}$, $\sigma^{-1}(\delta\,\widetilde{w_{i,j}}+\widetilde{z_{i,j}}) = \beta^{\phi}\,\text{ParExp}_{i,j}+u^{\phi}_{i,j}$ and $\sigma^{-1}\,\widetilde{z_{i,j}} = \beta^{\phi^{\text{Eq}}}\,\text{ParExp}_{i,j}+u^{\phi^{\text{Eq}}}_{i,j}$ implied from equations (B.13)–(B.15) and (B.17), and inserting them into the choice probability formulas (B.10)–(B.12), standard derivations yield the following formula for the semi-elasticity:

$$\mathcal{E}^{y}_{i,j} \equiv \frac{d\log Y_{i,j}}{d\,\text{ParExp}_{i,j}} = \beta^{y}-Y_{i,j}\,\beta^{y} = (1-Y_{i,j})\,\beta^{y} \qquad \forall Y \in \{\Pi, \Phi, \Phi^{\text{Eq}}\}. \tag{B.20}$$

The factor $1-Y_{i,j}$ deflates the fixed-effects coefficient toward the probability response of the model. We summarise this by sample enumeration (Train, 2009), plugging the elicited

likelihood $Y_{i,j}$ into (B.20) and averaging over the N respondents and their three job types:

$$\widehat{\mathcal{E}}^{y} = \frac{1}{N}\sum_{i}\frac{1}{3}\sum_{j\in\mathcal{J}_i}\hat{\beta}^{y}(1-Y_{i,j}) \qquad \forall Y \in \{\Pi, \Phi, \Phi^{\text{Eq}}\}. \tag{B.21}$$

C Parental experience and belief accuracy

Chapter 2 shows that wage misperceptions are large and that correcting them shifts where job seekers apply. This raises a concern for our decomposition: if a parent's experience makes beliefs about their job type more accurate—or merely more confident—the added certainty could make that job type more attractive, which our mapping would record as non-pecuniary preference value. We find little support for this. Respondents with parental experience are, if anything, better informed about *all three* of their job types, but no better informed about the one their parent worked in. We measure (in)accuracy as the absolute log gap between belief and actual prior cohort value, separating a general margin, ParExp_i (either parent worked in any of the three job types), from the job type specific margin, $\text{ParExp}_{i,j}$ (or $\text{ParExp}_{i,j,j'}$ at the job pair level[13]).

Distribution of wage gap misperceptions Table C1 compares the distribution of relative wage misperceptions across respondents with and without parental experience in any job type. The two groups look alike: two-thirds underestimate in both (66.6%), and the share underestimating the log wage gap by more than 50% differs by about four percentage points (56.9% versus 52.4%). Reweighting the parental experience group to the comparison group's distribution of majors renders these tail share differences not statistically significant (Rows (5) and (7)), though a small mean advantage survives (0.122 versus 0.134 log points)—a general accuracy advantage between people, not between a respondent's own job types.

Accuracy by general versus job type specific experience Table C2 separates the two margins. Regressing the absolute prior cohort wage misperception on both dummies with job type fixed effects, the general margin is negative and statistically significant, while the job type specific margin is small and not statistically significant. The five-year wage and offer probability[14] perceived levels and misperceptions show no differential accuracy by parental experience on either margin. Table C3 shows the job type specific margin, by

[13] $\text{ParExp}_{i,j,j'}$ equals one if either $\text{ParExp}_{i,j}$ or $\text{ParExp}_{i,j'}$ is one.

[14] The actual prior cohort offer probability $\bar{p}_{i,j}$ is constructed from administrative job log (vacancy) data, following Harmon et al. (2025).

including only persons with any parental experience and including individual fixed effects.

Relative misperceptions at the job pair level Because applications respond to perceived wage *gaps*, Tables C4 and C5 repeat the analysis at the job pair level (i, j, j'). The general margin seems most relevant for entry wage perceptions: ParExp_i lowers the absolute underestimation of the log wage gap. The pair specific margin is weaker and mixed—it leaves average underestimation unchanged but raises the probability of being within 50% of the true gap by about four percentage points. For five-year wage beliefs the pair specific margin seems to improve accuracy somewhat, while the offer probability gap shows little improvement.

Table C1: Underestimation gaps: parental job type experience vs. none

	Wage gap underestimation rate between job pairs $\frac{\Delta V_{i,j,j'} - \widehat{\Delta V_{i,j,j'}}}{\Delta V_{i,j,j'}}$				Underestimation of log wage gap between job pairs $\Delta v_{i,j,j'} - \widehat{\Delta v_{i,j,j'}}$		Wage underestimation $\frac{V_{i,j_i} - \widehat{V_{i,j}}}{V_{i,j}}$	Individuals who are underestimators	Sample size:	
	$> 100\%$	$> 50\%$	$> 0\%$	$< -50\%$	Mean abs. value	Std. Dev.	Mean	Share	Individuals	Job pairs
(1) Neither parent has experience in a job type	37.1%	56.9%	68.1%	23.7%	0.134	0.173	−2.82%	66.6%	1,175	3,525
	(1.0)	(1.1)	(0.9)	(0.8)	(0.003)	(0.003)	(0.43)	(1.4)		
(2) - (1) weightable	37.0%	56.1%	67.4%	24.3%	0.132	0.171	−2.84%	65.0%	1,098	3,294
	(1.6)	(1.5)	(1.2)	(0.9)	(0.003)	(0.004)	(0.57)	(1.6)		
(3) Either parent has experience in a job type	34.5%	52.4%	64.5%	27.4%	0.122	0.157	−4.17%	66.6%	727	2,181
	(1.2)	(1.3)	(1.3)	(1.1)	(0.003)	(0.003)	(0.49)	(1.7)		
(4) - (3) weightable	34.8%	52.4%	64.5%	27.3%	0.122	0.157	−4.11%	66.6%	716	2,148
	(1.2)	(1.3)	(1.3)	(1.2)	(0.003)	(0.003)	(0.50)	(1.8)		
(5) - (3) reweighted to (2)	35.5%	52.5%	66.0%	24.3%	0.120	0.160	−2.02%	62.0%	716	2,148
	(2.3)	(2.7)	(2.2)	(1.5)	(0.004)	(0.005)	(0.81)	(3.4)		
(6) p-val for difference (1)-(3)	0.116	0.008	0.028	0.012	0.002	0.000	0.034	1.000		
(7) p-val for reweighted difference (2)-(5)	0.482	0.190	0.622	0.934	0.000	0.010	0.630	0.568		

Notes: The table summarises the distribution of wage misperceptions by parental background. The base data are individual-by-job pairs (i, j, j') in the survey sample. The first four columns show the share of pairs for which the wage gap underestimation rate exceeds 100%, 50%, 0% or falls below -50% respectively. The next two columns report the mean absolute value and standard deviation of the underestimation of the log wage gap for prior cohorts. The seventh column instead uses individual-by-jobs data (i, j) and shows the mean underestimation of the average prior cohort wage in individual job types (in percent of the actual wage). The eighth column uses individual-level data (i) and shows what fraction of individuals are underestimators in the sense that they underestimate the typical wage in the highest paying job relative to the average of the two other jobs. The last two columns show the number of individuals and unique individual-by-job pairs in each subsample. Row (1) restricts to respondents where neither parent has work experience in any of the three job types (between 2008 and the time of the survey), and Row (3) to respondents where either parent has such experience. Rows (2) and (4) further restrict these groups to the reweightable respondents, i.e. those with a propensity score between 0.01 and 0.99 from a propensity score model based on majors. Row (5) reweights the respondents from Row (4) to match the major composition of the group in Row (2) using these propensity scores. Rows (6) and (7) report p-values for tests of equality between the groups in Rows (1) and (3) and between the groups in Rows (2) and (5), respectively. Standard errors in parentheses are clustered at the level of the individual.

Table C2: Perceived job type averages by general and job specific parental experience

	(1) $\widetilde{v_{i,j}}$	(2) $\widetilde{v_{i,j}^{t+5}}$	(3) $\widetilde{\bar{p}_{i,j}}$	(4) $\lvert v_{i,j} - \widetilde{v_{i,j}} \rvert$	(5) $\lvert v_{i,j}^{t+5} - \widetilde{v_{i,j}^{t+5}} \rvert$	(6) $\lvert \bar{p}_{i,j} - \widetilde{\bar{p}_{i,j}} \rvert$
ParExp_i	0.008	-0.000	0.001	-0.009**	-0.005	-0.029
	(0.007)	(0.008)	(0.064)	(0.004)	(0.005)	(0.051)
$\text{ParExp}_{i,j}$	0.003	0.008	0.037	-0.001	-0.004	0.055
	(0.005)	(0.006)	(0.042)	(0.004)	(0.004)	(0.036)
Job type FE	✓	✓	✓	✓	✓	✓
Individuals	1,899	1,898	1,894	1,899	1,891	1,494
Job types	240	240	240	240	237	204
N	5,697	5,694	5,682	5,697	5,673	4,482

Notes: The table shows OLS estimates from the survey. The base data are all individual-by-jobs (i, j) in the survey sample, including respondents whose parents have no work experience in any of the three job types; since the outcomes are elicited before the information treatment, both the treatment and control groups are included. The regressors are ParExp_i, a dummy for either parent having work experience in at least one of the respondent's three job types between 2008 and the time of the survey, and $\text{ParExp}_{i,j}$, a dummy for either parent having such experience in job type j specifically. In Columns (1)–(3) the outcomes are the perceived levels: the perceived (log) average wage of prior cohorts in the job type, $\widetilde{v_{i,j}}$, and the corresponding perceived values for the (log) average wage of prior cohorts five years after starting and for the (log) average probability of receiving an offer when applying. In Columns (4)–(6) the outcomes are the corresponding absolute misperceptions about these job type averages: the absolute difference between the perceived and the actual value, $|v_{i,j} - \widetilde{v_{i,j}}|$ for the prior cohort wage and analogously for the five-year wage and the offer probability. All specifications include job type fixed effects; individual fixed effects are excluded since ParExp_i only varies across individuals. Changes in observation totals across columns stem from missing reports and zeros in the raw outcome variables. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C3: Perceived own outcomes by parental experience

	(1) $\widetilde{v_{i,j}}$	(2) $\widetilde{v_{i,j}^{t+5}}$	(3) $\widetilde{\bar{p}_{i,j}}$	(4) $\lvert v_{i,j}-\widetilde{v_{i,j}}\rvert$	(5) $\lvert v_{i,j}^{t+5}-\widetilde{v_{i,j}^{t+5}}\rvert$	(6) $\lvert \bar{p}_{i,j}-\widetilde{\bar{p}_{i,j}}\rvert$
$\text{ParExp}_{i,j}$	0.004	0.005	-0.001	-0.001	-0.002	0.036
	(0.005)	(0.006)	(0.031)	(0.004)	(0.004)	(0.029)
Job type FE	✓	✓	✓	✓	✓	✓
Individual FE	✓	✓	✓	✓	✓	✓
Individuals	718	718	715	718	718	628
Job types	183	183	183	183	183	159
N	2,154	2,154	2,145	2,154	2,154	1,884

Notes: The table shows OLS estimates from the survey. The base data are all individual-by-jobs (i, j) in the survey sample, restricted to respondents whose parents have work experience in at least one of the three job types; since the outcomes are elicited before the information treatment, both the treatment and control groups are included. In Columns (1)–(3) the outcomes are the perceived levels: the perceived (log) average wage of prior cohorts in the job type, $\widetilde{v_{i,j}}$, and the corresponding perceived values for the (log) average wage of prior cohorts five years after starting and for the (log) average probability of receiving an offer when applying. In Columns (4)–(6) the outcomes are the corresponding absolute misperceptions about these job type averages: the absolute difference between the perceived and the actual value, $|v_{i,j}-\widetilde{v_{i,j}}|$ for the prior cohort wage and analogously for the five-year wage and the offer probability. The regressor of interest, $\text{ParExp}_{i,j}$, is a dummy for either parent having worked in job type j between 2008 and the time of the survey. All specifications include individual and job type fixed effects. Changes in observation totals across columns stem from missing reports and zeros in the raw outcome variables. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C4: Job value misperception on general and job pair specific parental experience

	$\lvert\Delta v_{i,j,j'}-\Delta\widetilde{v_{i,j,j'}}\rvert$		$\left\lvert\frac{\Delta V_{i,j,j'}-\Delta\widetilde{V_{i,j,j'}}}{\Delta V_{i,j,j'}}\right\rvert < 0.5$		$\frac{\Delta V_{i,j,j'}-\Delta\widetilde{V_{i,j,j'}}}{\Delta V_{i,j,j'}} < -0.5$		$\frac{\Delta V_{i,j,j'}-\Delta\widetilde{V_{i,j,j'}}}{\Delta V_{i,j,j'}} > 0.5$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ParExp_i	-0.010**	-0.012**	-0.026	-0.011	0.067***	0.030	-0.040*	-0.019
	(0.005)	(0.005)	(0.018)	(0.018)	(0.022)	(0.021)	(0.024)	(0.022)
$\text{ParExp}_{i,j,j'}$	-0.002	0.002	0.046**	0.043**	-0.039*	-0.045**	-0.006	0.002
	(0.005)	(0.004)	(0.018)	(0.018)	(0.021)	(0.020)	(0.023)	(0.022)
Constant	✓		✓		✓		✓	
Major FE		✓		✓		✓		✓
Individuals	1,902	1,902	1,902	1,902	1,902	1,902	1,902	1,902
Job pairs	249	249	249	249	249	249	249	249
N	11,412	11,412	11,412	11,412	11,412	11,412	11,412	11,412

Notes: The table shows OLS estimates from the survey. The base data are all individual-by-job pairs (i, j, j') in the survey sample; since the perceptions are elicited before the information treatment, both the treatment and control groups are included. The outcome variables measure misperceptions of the gap in average prior cohort wages between the two job types. In Columns (1) and (2) the outcome is the absolute underestimation of the log wage gap, $|\Delta v_{i,j,j'}-\Delta\widetilde{v_{i,j,j'}}|$. In Columns (3) and (4) it is a dummy for the wage gap underestimation rate being below 50% in absolute value, while in Columns (5) and (6) and Columns (7) and (8) it is a dummy for the underestimation rate falling below -50% (overestimation) or exceeding 50% (underestimation), respectively. The regressors are ParExp_i, a dummy for either parent having work experience in at least one of the respondent's three job types between 2008 and the time of the survey, and $\text{ParExp}_{i,j,j'}$, a dummy for either parent having such experience in one of the two job types in the pair. Odd-numbered columns include a constant term, while even-numbered columns instead include fixed effects for the respondent's major. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C5: Post-5y wage and offer probability misperception on general and job pair specific parental experience

	$\left\lvert \Delta v^{t+5}_{i,j,j'} - \widetilde{\Delta v^{t+5}_{i,j,j'}} \right\rvert$		$\left\lvert \frac{\Delta v^{t+5}_{i,j,j'} - \widetilde{\Delta v^{t+5}_{i,j,j'}}}{\Delta v^{t+5}_{i,j,j'}} \right\rvert < 0.5$		$\left\lvert \Delta \bar{p}_{i,j,j'} - \widetilde{\Delta \bar{p}_{i,j,j'}} \right\rvert$		$\left\lvert \frac{\Delta \bar{p}_{i,j,j'} - \widetilde{\Delta \bar{p}_{i,j,j'}}}{\Delta \bar{p}_{i,j,j'}} \right\rvert < 0.5$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ParExp_i	0.006	-0.000	-0.028	-0.030	0.011	0.028	0.002	-0.003
	(0.006)	(0.006)	(0.018)	(0.018)	(0.042)	(0.040)	(0.015)	(0.016)
$\text{ParExp}_{i,j,j'}$	-0.013***	-0.010**	0.056***	0.052***	-0.071*	-0.054	0.011	0.005
	(0.005)	(0.005)	(0.019)	(0.019)	(0.037)	(0.035)	(0.016)	(0.016)
Constant	✓		✓		✓		✓	
Major FE		✓		✓		✓		✓
Individuals	1,894	1,894	1,895	1,895	1,496	1,496	1,499	1,499
Job pairs	246	246	246	246	210	210	210	210
N	11,364	11,364	11,370	11,370	8,976	8,976	8,994	8,994

Notes: The table shows OLS estimates from the survey. The base data are all individual-by-job pairs (i, j, j') in the survey sample; since the perceptions are elicited before the information treatment, both the treatment and control groups are included. In Columns (1)–(4) the outcomes measure misperceptions of the gap in average prior cohort wages five years after starting in the job types: the absolute underestimation of the log gap (Columns (1) and (2)) and a dummy for the underestimation rate being below 50% in absolute value (Columns (3) and (4)). In Columns (5)–(8) the outcomes are defined analogously for the gap in the (log) average probability of receiving an offer when applying to the job types. The regressors are ParExp_i, a dummy for either parent having work experience in at least one of the respondent's three job types between 2008 and the time of the survey, and $\text{ParExp}_{i,j,j'}$, a dummy for either parent having such experience in one of the two job types in the pair. Odd-numbered columns include a constant term, while even-numbered columns instead include fixed effects for the respondent's major. Changes in observation totals across columns stem from missing reports and zeros in the raw outcome variables. Standard errors in parentheses are clustered at the level of the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.